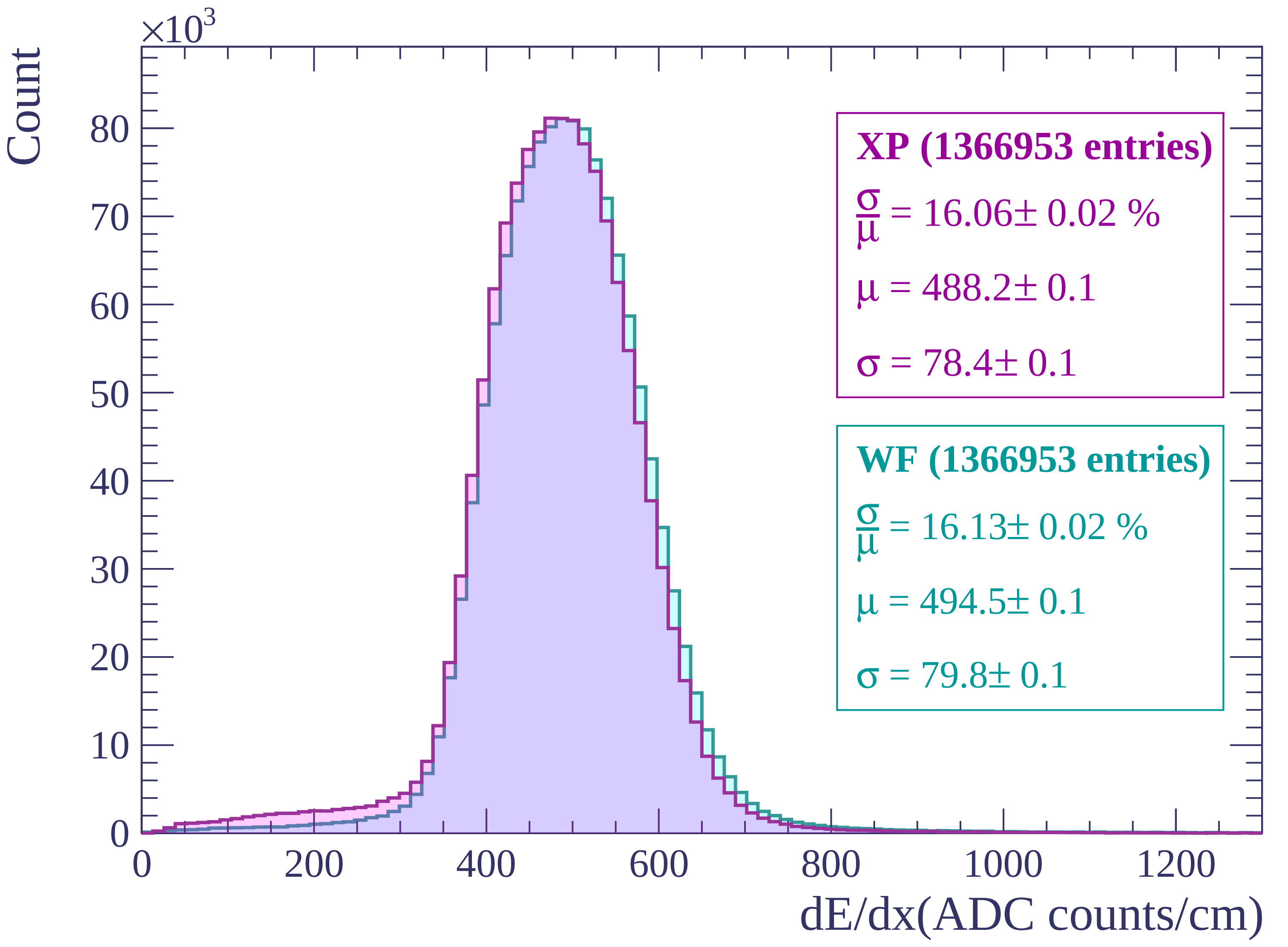
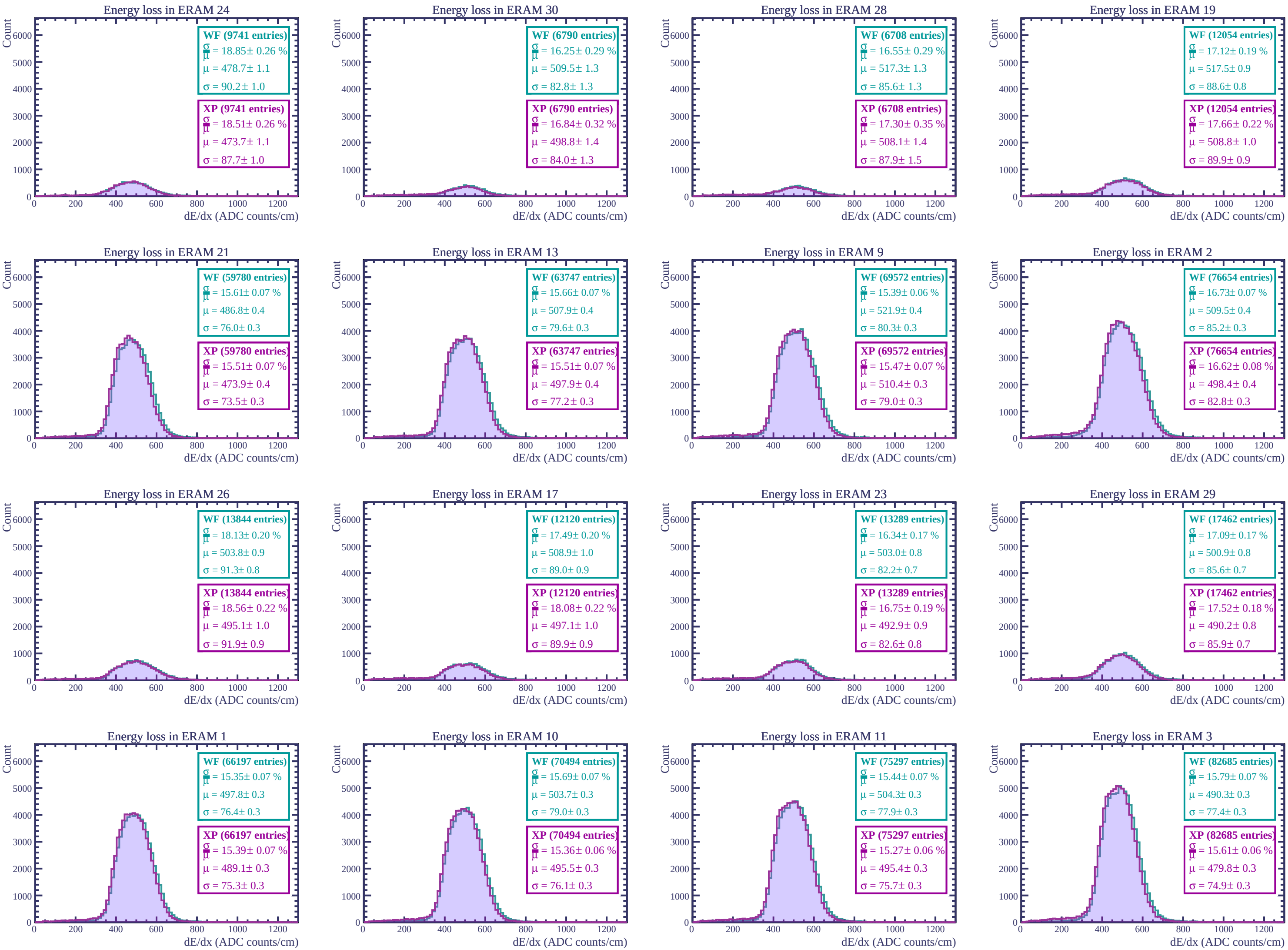
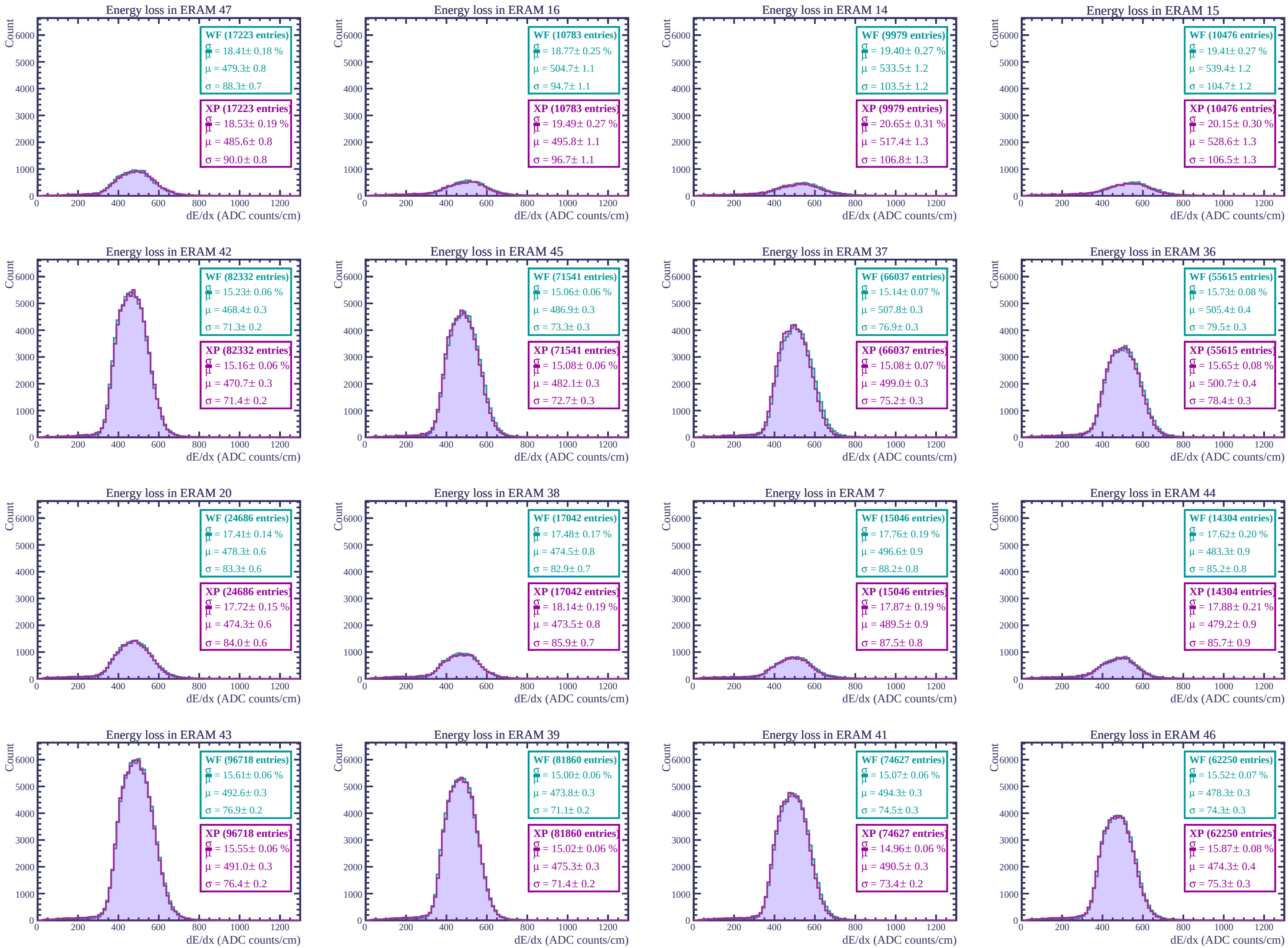
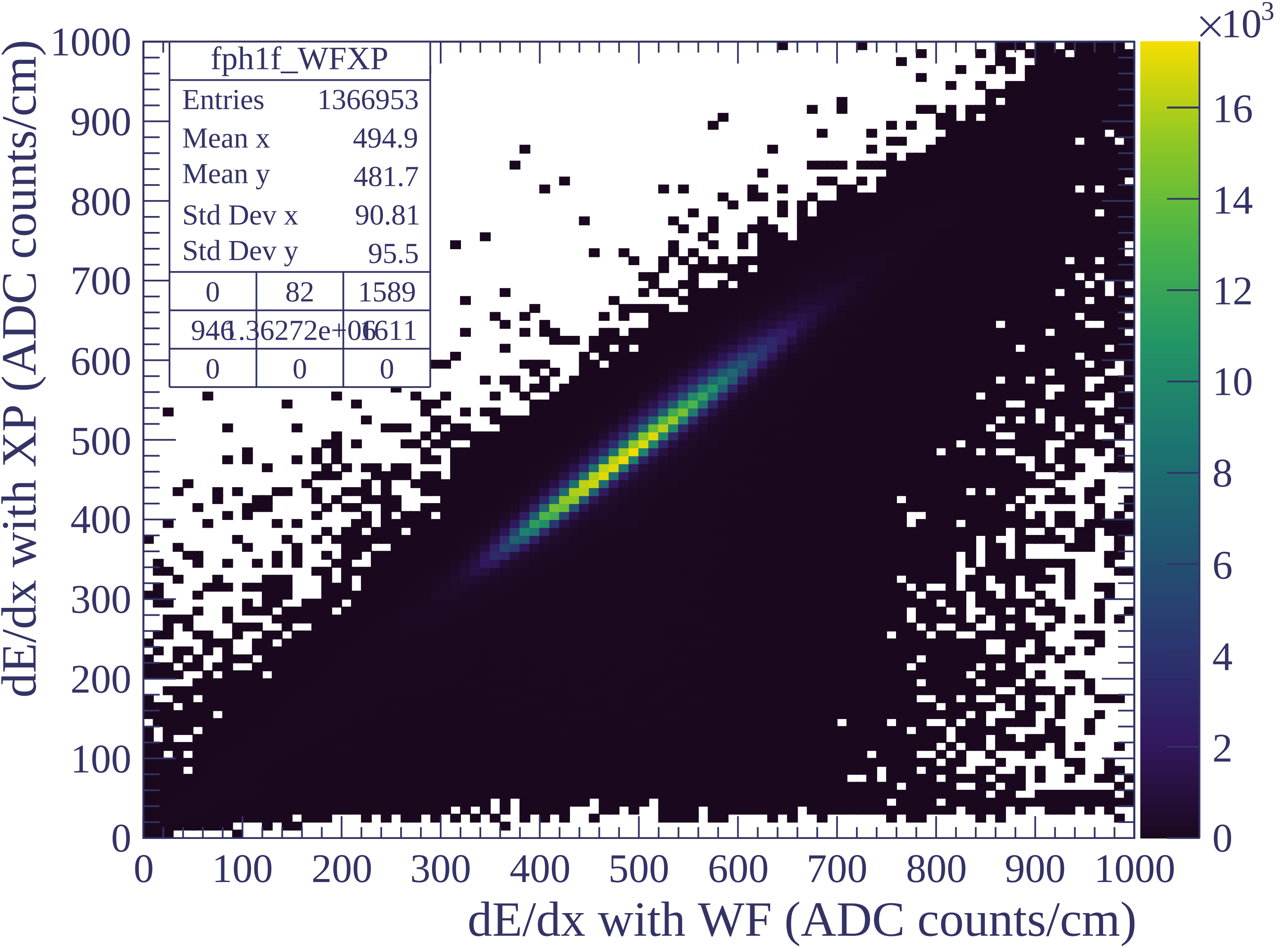


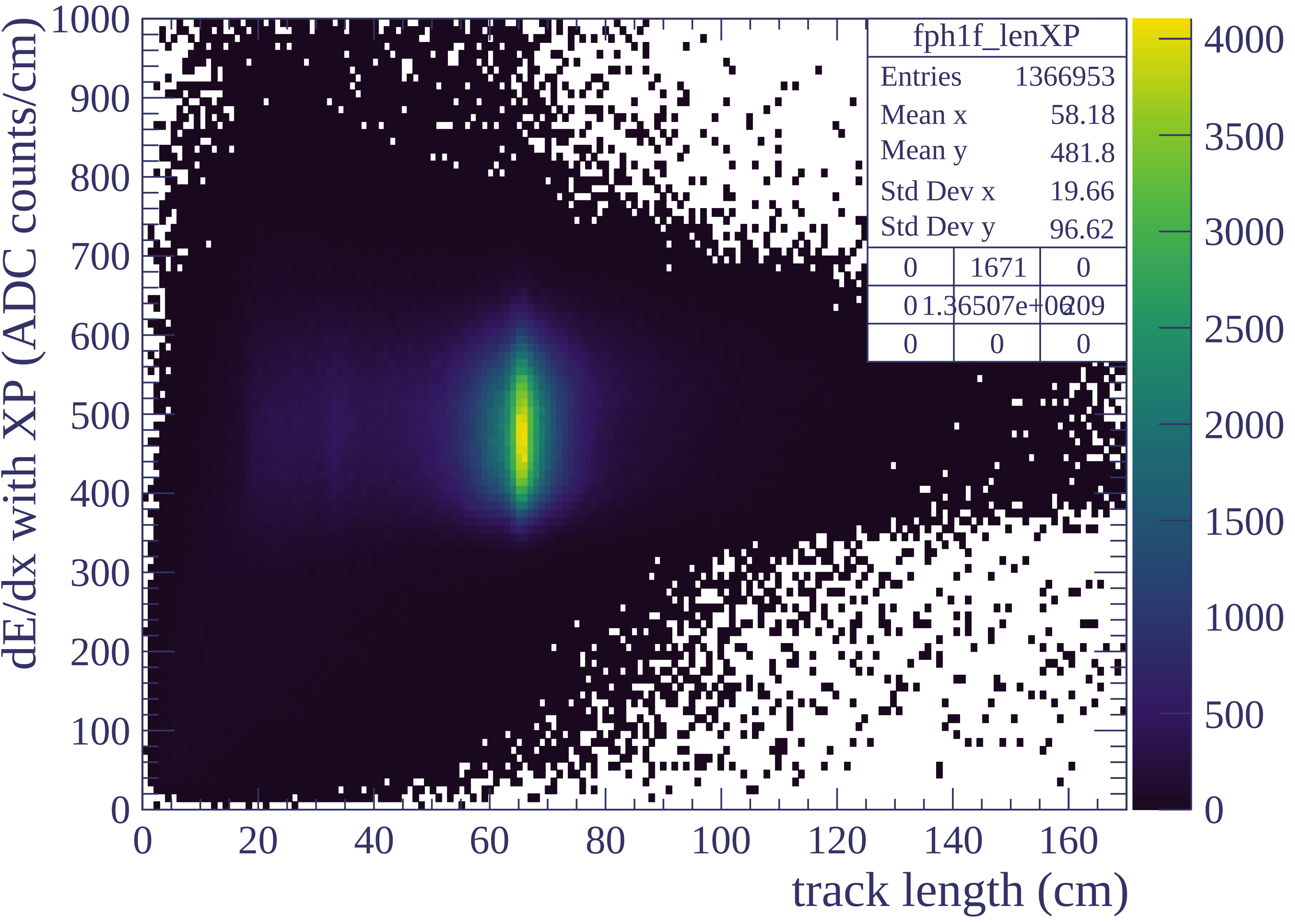
×10^3
Count
XP (1366953 entries)
σ/μ = 16.06± 0.02 %
μ = 488.2± 0.1
σ = 78.4± 0.1
WF (1366953 entries)
σ/μ = 16.13± 0.02 %
μ = 494.5± 0.1
σ = 79.8± 0.1
dE/dx(ADC counts/cm)
0
10
20
30
40
50
60
70
80
200
400
600
800
1000
1200


Energy loss in ERAM 24
WF (9741 entries)
σ/μ = 18.85± 0.26 %
μ = 478.7± 1.1
σ = 90.2± 1.0
XP (9741 entries)
σ/μ = 18.51± 0.26 %
μ = 473.7± 1.1
σ = 87.7± 1.0
Energy loss in ERAM 30
WF (6790 entries)
σ/μ = 16.25± 0.29 %
μ = 509.5± 1.3
σ = 82.8± 1.3
XP (6790 entries)
σ/μ = 16.84± 0.32 %
μ = 498.8± 1.4
σ = 84.0± 1.3
Energy loss in ERAM 28
WF (6708 entries)
σ/μ = 16.55± 0.29 %
μ = 517.3± 1.3
σ = 85.6± 1.3
XP (6708 entries)
σ/μ = 17.30± 0.35 %
μ = 508.1± 1.4
σ = 87.9± 1.5
Energy loss in ERAM 19
WF (12054 entries)
σ/μ = 17.12± 0.19 %
μ = 517.5± 0.9
σ = 88.6± 0.8
XP (12054 entries)
σ/μ = 17.66± 0.22 %
μ = 508.8± 1.0
σ = 89.9± 0.9
Energy loss in ERAM 21
WF (59780 entries)
σ/μ = 15.61± 0.07 %
μ = 486.8± 0.4
σ = 76.0± 0.3
XP (59780 entries)
σ/μ = 15.51± 0.07 %
μ = 473.9± 0.4
σ = 73.5± 0.3
Energy loss in ERAM 13
WF (63747 entries)
σ/μ = 15.66± 0.07 %
μ = 507.9± 0.4
σ = 79.6± 0.3
XP (63747 entries)
σ/μ = 15.51± 0.07 %
μ = 497.9± 0.4
σ = 77.2± 0.3
Energy loss in ERAM 9
WF (69572 entries)
σ/μ = 15.39± 0.06 %
μ = 521.9± 0.4
σ = 80.3± 0.3
XP (69572 entries)
σ/μ = 15.47± 0.07 %
μ = 510.4± 0.3
σ = 79.0± 0.3
Energy loss in ERAM 2
WF (76654 entries)
σ/μ = 16.73± 0.07 %
μ = 509.5± 0.4
σ = 85.2± 0.3
XP (76654 entries)
σ/μ = 16.62± 0.08 %
μ = 498.4± 0.4
σ = 82.8± 0.3
Energy loss in ERAM 26
WF (13844 entries)
σ/μ = 18.13± 0.20 %
μ = 503.8± 0.9
σ = 91.3± 0.8
XP (13844 entries)
σ/μ = 18.56± 0.22 %
μ = 495.1± 1.0
σ = 91.9± 0.9
Energy loss in ERAM 17
WF (12120 entries)
σ/μ = 17.49± 0.20 %
μ = 508.9± 1.0
σ = 89.0± 0.9
XP (12120 entries)
σ/μ = 18.08± 0.22 %
μ = 497.1± 1.0
σ = 89.9± 0.9
Energy loss in ERAM 23
WF (13289 entries)
σ/μ = 16.34± 0.17 %
μ = 503.0± 0.8
σ = 82.2± 0.7
XP (13289 entries)
σ/μ = 16.75± 0.19 %
μ = 492.9± 0.9
σ = 82.6± 0.8
Energy loss in ERAM 29
WF (17462 entries)
σ/μ = 17.09± 0.17 %
μ = 500.9± 0.8
σ = 85.6± 0.7
XP (17462 entries)
σ/μ = 17.52± 0.18 %
μ = 490.2± 0.8
σ = 85.9± 0.7
Energy loss in ERAM 1
WF (66197 entries)
σ/μ = 15.35± 0.07 %
μ = 497.8± 0.3
σ = 76.4± 0.3
XP (66197 entries)
σ/μ = 15.39± 0.07 %
μ = 489.1± 0.3
σ = 75.3± 0.3
Energy loss in ERAM 10
WF (70494 entries)
σ/μ = 15.69± 0.07 %
μ = 503.7± 0.3
σ = 79.0± 0.3
XP (70494 entries)
σ/μ = 15.36± 0.06 %
μ = 495.5± 0.3
σ = 76.1± 0.3
Energy loss in ERAM 11
WF (75297 entries)
σ/μ = 15.44± 0.07 %
μ = 504.3± 0.3
σ = 77.9± 0.3
XP (75297 entries)
σ/μ = 15.27± 0.06 %
μ = 495.4± 0.3
σ = 75.7± 0.3
Energy loss in ERAM 3
WF (82685 entries)
σ/μ = 15.79± 0.07 %
μ = 490.3± 0.3
σ = 77.4± 0.3
XP (82685 entries)
σ/μ = 15.61± 0.06 %
μ = 479.8± 0.3
σ = 74.9± 0.3
Count
dE/dx (ADC counts/cm)

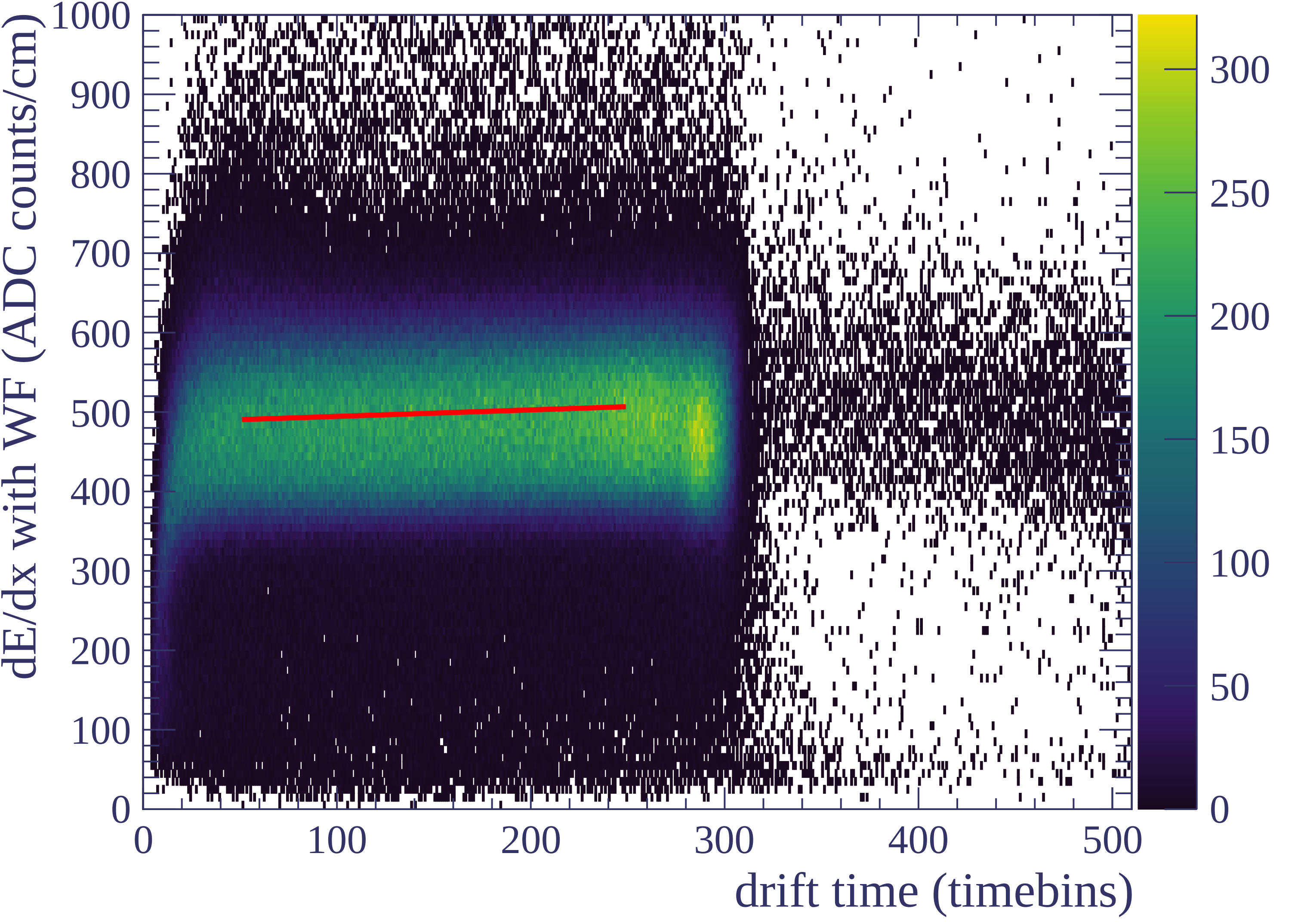


Energy loss in ERAM 47
WF (17223 entries) σ/μ = 18.41± 0.18 % μ = 479.3± 0.8 σ = 88.3± 0.7
XP (17223 entries) σ/μ = 18.53± 0.19 % μ = 485.6± 0.8 σ = 90.0± 0.8
Energy loss in ERAM 16
WF (10783 entries) σ/μ = 18.77± 0.25 % μ = 504.7± 1.1 σ = 94.7± 1.1
XP (10783 entries) σ/μ = 19.49± 0.27 % μ = 495.8± 1.1 σ = 96.7± 1.1
Energy loss in ERAM 14
WF (9979 entries) σ/μ = 19.40± 0.27 % μ = 533.5± 1.2 σ = 103.5± 1.2
XP (9979 entries) σ/μ = 20.65± 0.31 % μ = 517.4± 1.3 σ = 106.8± 1.3
Energy loss in ERAM 15
WF (10476 entries) σ/μ = 19.41± 0.27 % μ = 539.4± 1.2 σ = 104.7± 1.2
XP (10476 entries) σ/μ = 20.15± 0.30 % μ = 528.6± 1.3 σ = 106.5± 1.3
Energy loss in ERAM 42
WF (82332 entries) σ/μ = 15.23± 0.06 % μ = 468.4± 0.3 σ = 71.3± 0.2
XP (82332 entries) σ/μ = 15.16± 0.06 % μ = 470.7± 0.3 σ = 71.4± 0.2
Energy loss in ERAM 45
WF (71541 entries) σ/μ = 15.06± 0.06 % μ = 486.9± 0.3 σ = 73.3± 0.3
XP (71541 entries) σ/μ = 15.08± 0.06 % μ = 482.1± 0.3 σ = 72.7± 0.3
Energy loss in ERAM 37
WF (66037 entries) σ/μ = 15.14± 0.07 % μ = 507.8± 0.3 σ = 76.9± 0.3
XP (66037 entries) σ/μ = 15.08± 0.07 % μ = 499.0± 0.3 σ = 75.2± 0.3
Energy loss in ERAM 36
WF (55615 entries) σ/μ = 15.73± 0.08 % μ = 505.4± 0.4 σ = 79.5± 0.3
XP (55615 entries) σ/μ = 15.65± 0.08 % μ = 500.7± 0.4 σ = 78.4± 0.3
Energy loss in ERAM 20
WF (24686 entries) σ/μ = 17.41± 0.14 % μ = 478.3± 0.6 σ = 83.3± 0.6
XP (24686 entries) σ/μ = 17.72± 0.15 % μ = 474.3± 0.6 σ = 84.0± 0.6
Energy loss in ERAM 38
WF (17042 entries) σ/μ = 17.48± 0.17 % μ = 474.5± 0.8 σ = 82.9± 0.7
XP (17042 entries) σ/μ = 18.14± 0.19 % μ = 473.5± 0.8 σ = 85.9± 0.7
Energy loss in ERAM 7
WF (15046 entries) σ/μ = 17.76± 0.19 % μ = 496.6± 0.9 σ = 88.2± 0.8
XP (15046 entries) σ/μ = 17.87± 0.19 % μ = 489.5± 0.9 σ = 87.5± 0.8
Energy loss in ERAM 44
WF (14304 entries) σ/μ = 17.62± 0.20 % μ = 483.3± 0.9 σ = 85.2± 0.8
XP (14304 entries) σ/μ = 17.88± 0.21 % μ = 479.2± 0.9 σ = 85.7± 0.9
Energy loss in ERAM 43
WF (96718 entries) σ/μ = 15.61± 0.06 % μ = 492.6± 0.3 σ = 76.9± 0.2
XP (96718 entries) σ/μ = 15.55± 0.06 % μ = 491.0± 0.3 σ = 76.4± 0.2
Energy loss in ERAM 39
WF (81860 entries) σ/μ = 15.00± 0.06 % μ = 473.8± 0.3 σ = 71.1± 0.2
XP (81860 entries) σ/μ = 15.02± 0.06 % μ = 475.3± 0.3 σ = 71.4± 0.2
Energy loss in ERAM 41
WF (74627 entries) σ/μ = 15.07± 0.06 % μ = 494.3± 0.3 σ = 74.5± 0.3
XP (74627 entries) σ/μ = 14.96± 0.06 % μ = 490.5± 0.3 σ = 73.4± 0.2
Energy loss in ERAM 46
WF (62250 entries) σ/μ = 15.52± 0.07 % μ = 478.3± 0.3 σ = 74.3± 0.3
XP (62250 entries) σ/μ = 15.87± 0.08 % μ = 474.3± 0.4 σ = 75.3± 0.3
Count
dE/dx (ADC counts/cm)

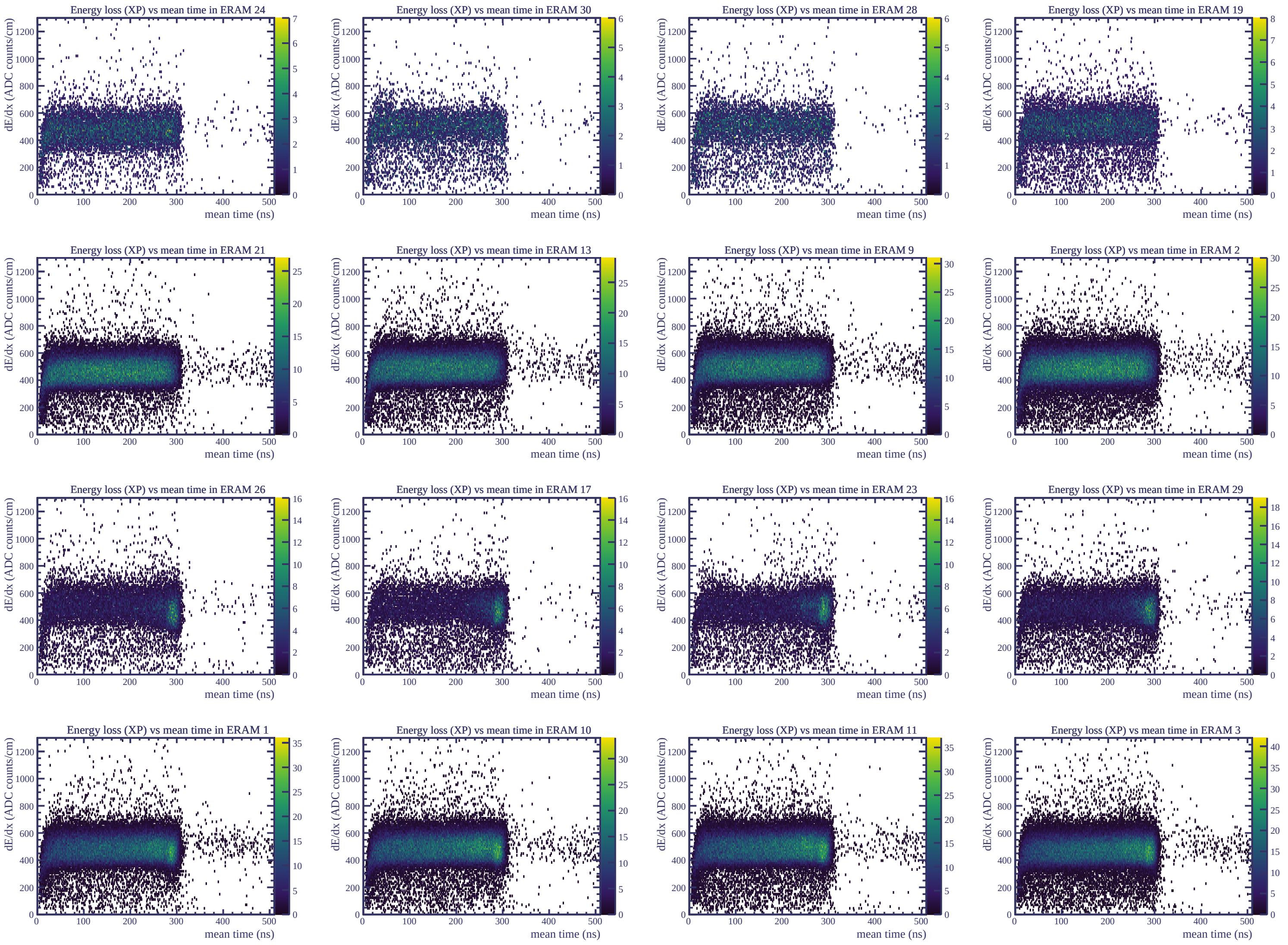

fph1f_WFXP
Entries 1366953
Mean x 494.9
Mean y 481.7
Std Dev x 90.81
Std Dev y 95.5
0 82 1589
946 1.36272e+06 611
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0
100
200
300
400
500
600
700
800
900
1000
2
4
6
8
10
12
14
16



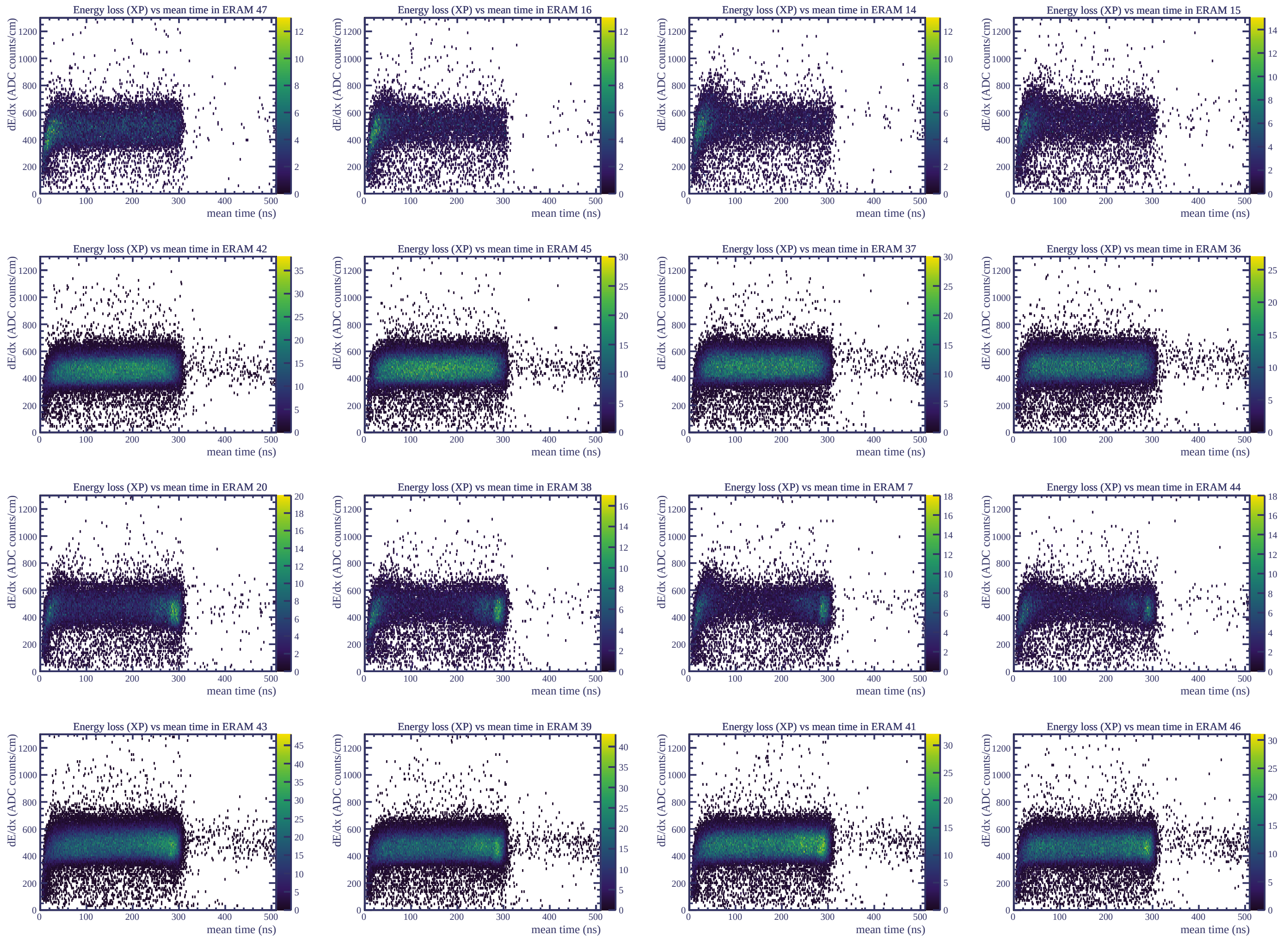
fph1f_lenXP
Entries 1366953
Mean x 58.18
Mean y 481.8
Std Dev x 19.66
Std Dev y 96.62
0 1671 0
0 1.36507e+06 209
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
500
1000
1500
2000
2500
3000
3500
4000

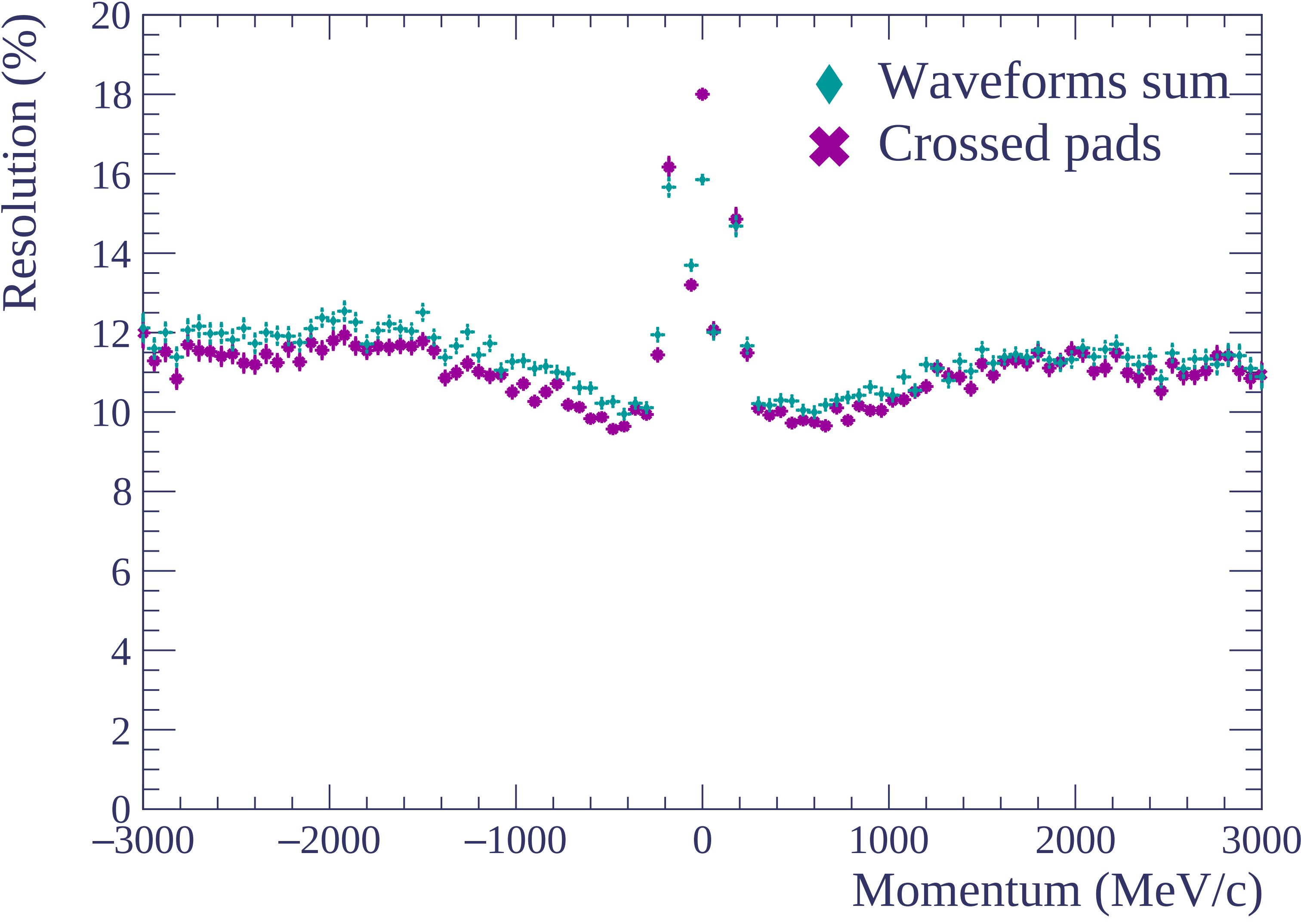


dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
50
100
150
200
250
300

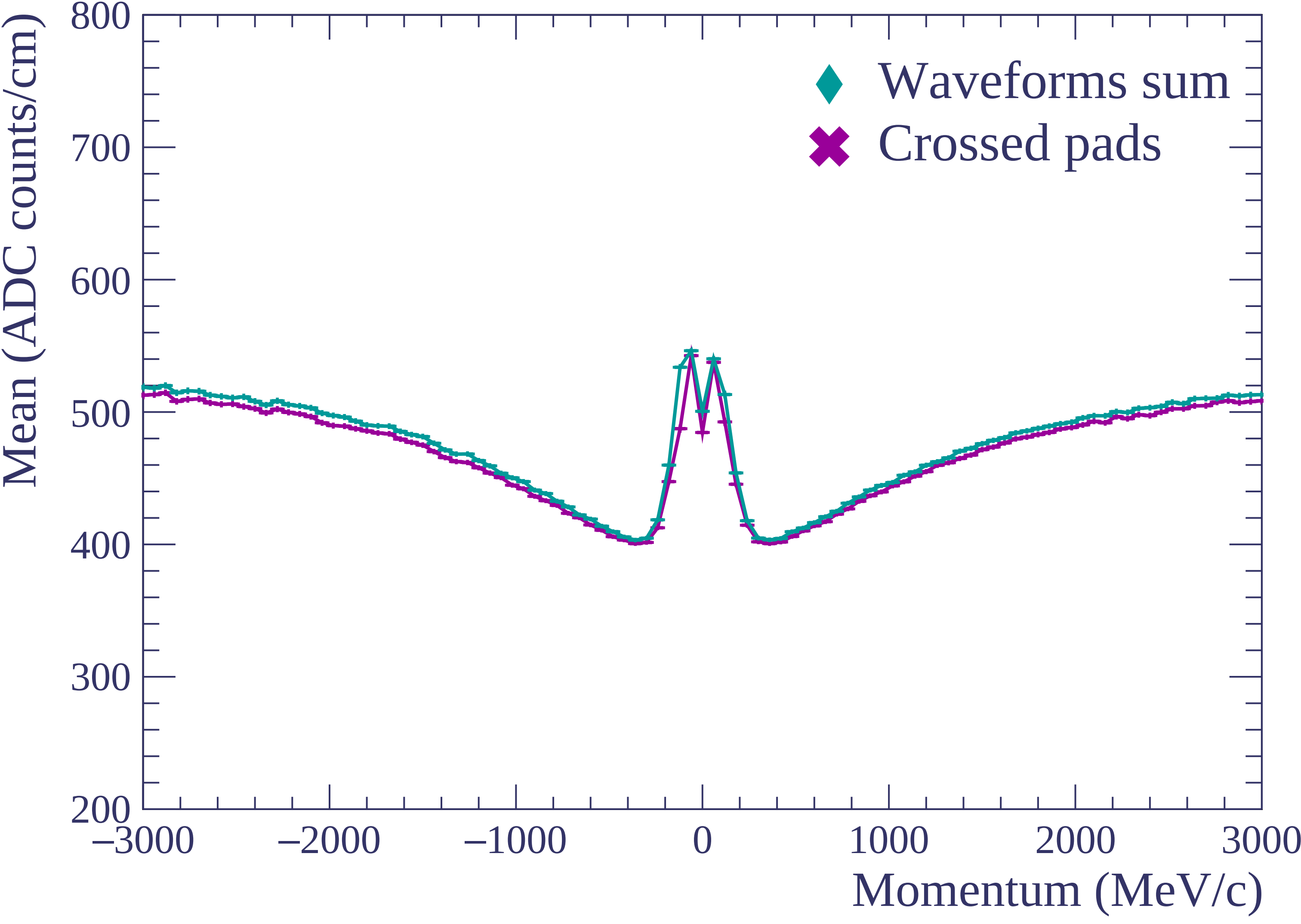


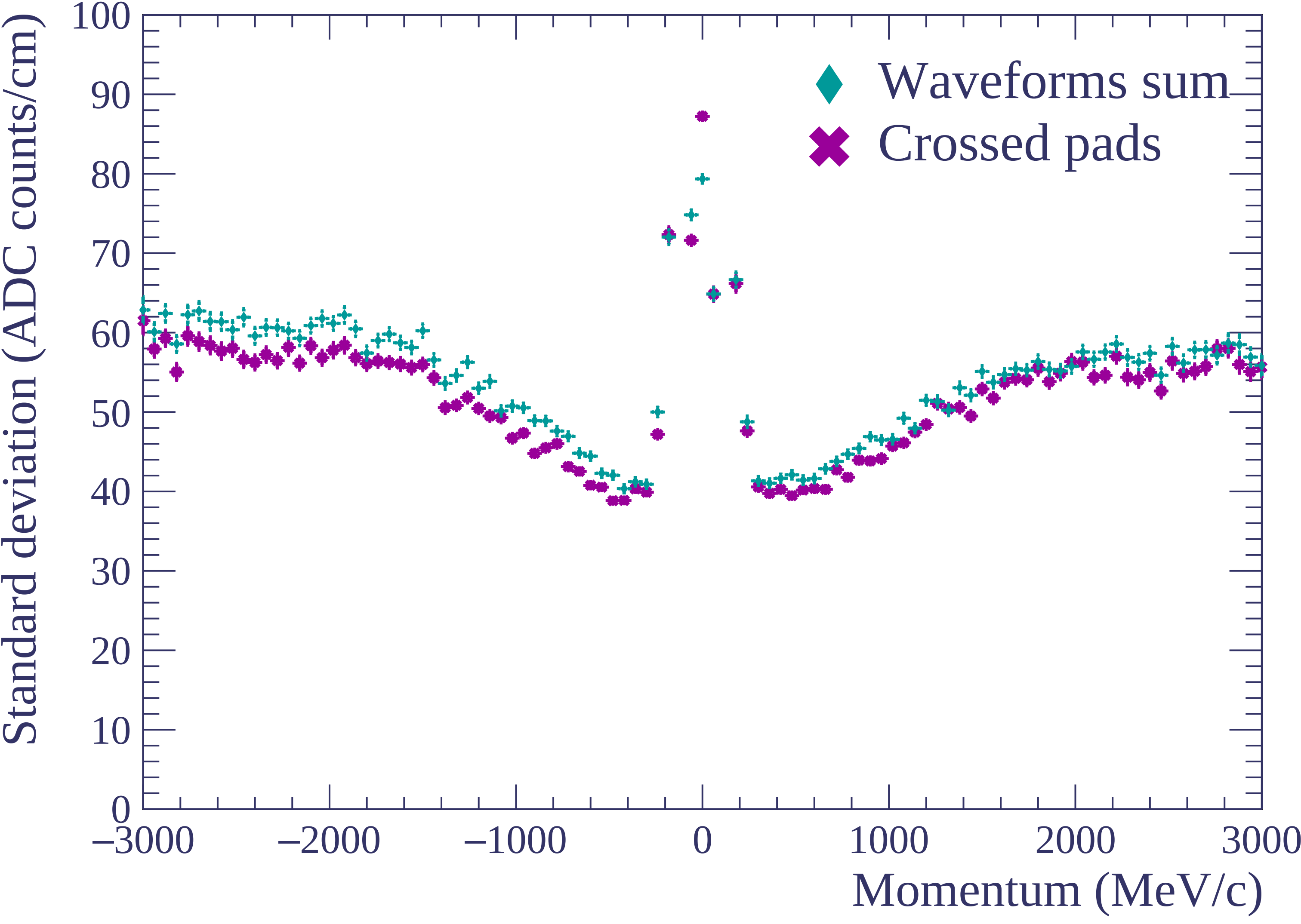
Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)


Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

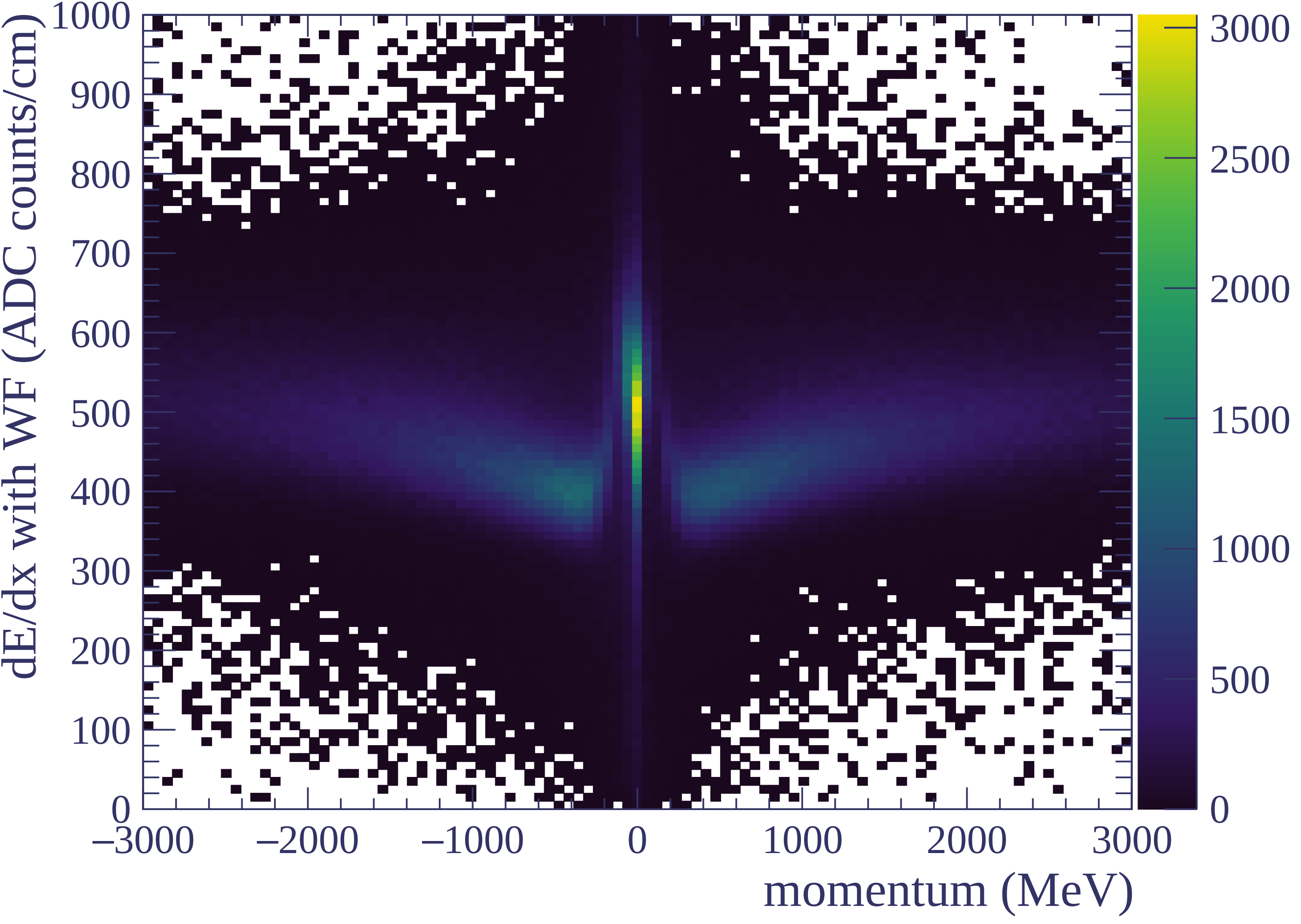


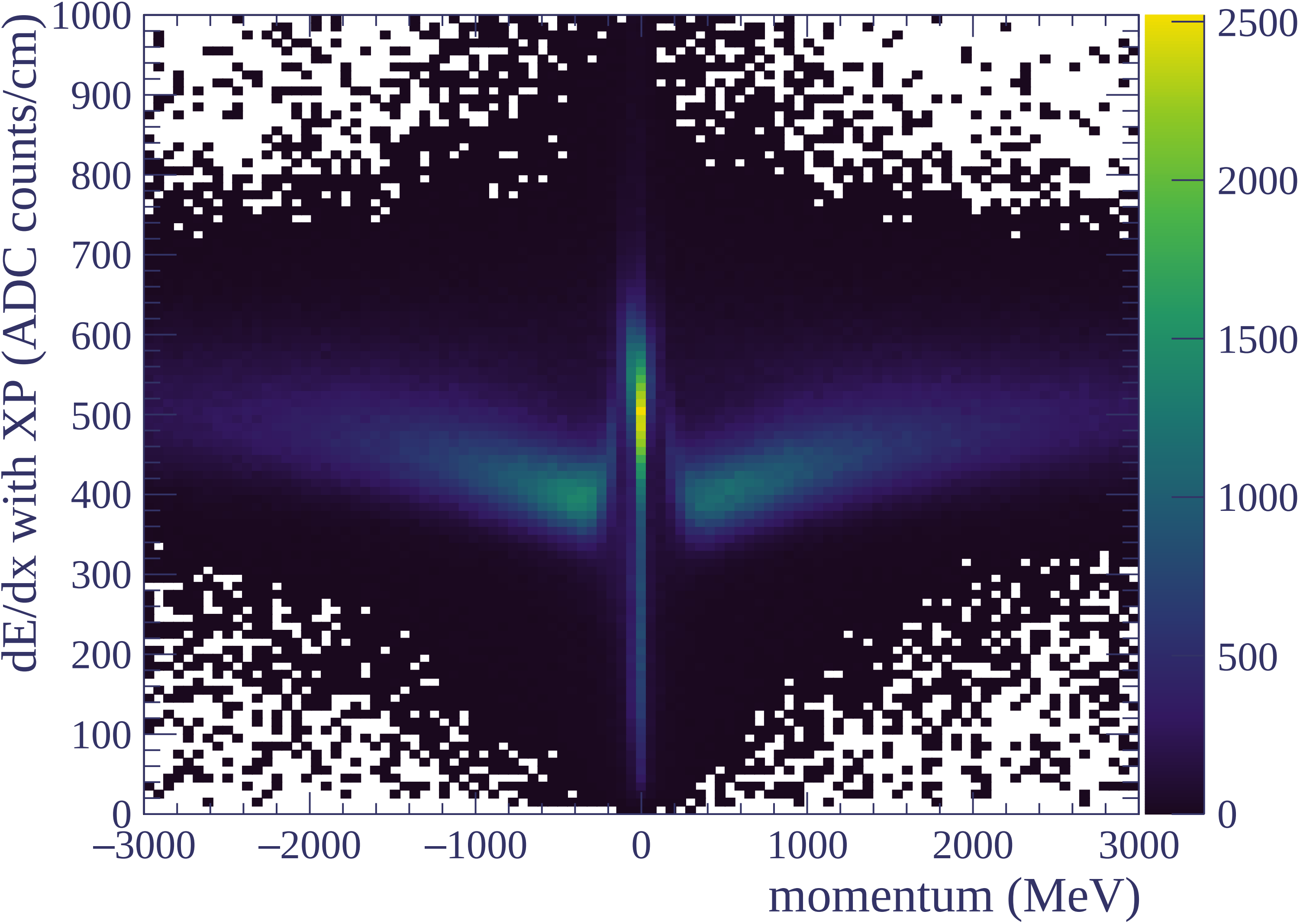
Resolution (%)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000


Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Momentum (MeV/c)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000

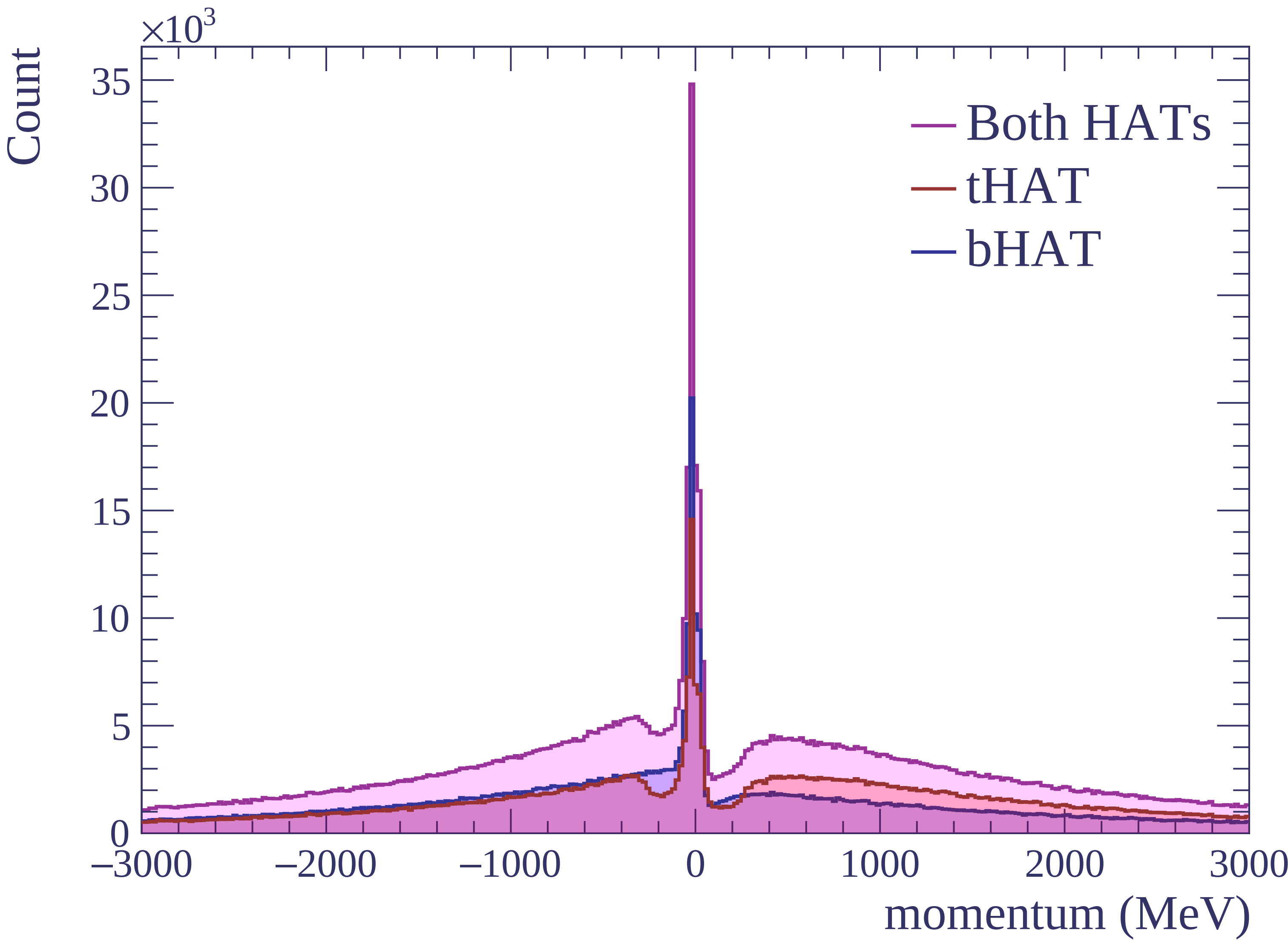


Waveforms sum
Crossed pads
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
0
10
20
30
40
50
60
70
80
90
100
−3000
−2000
−1000
0
1000
2000
3000

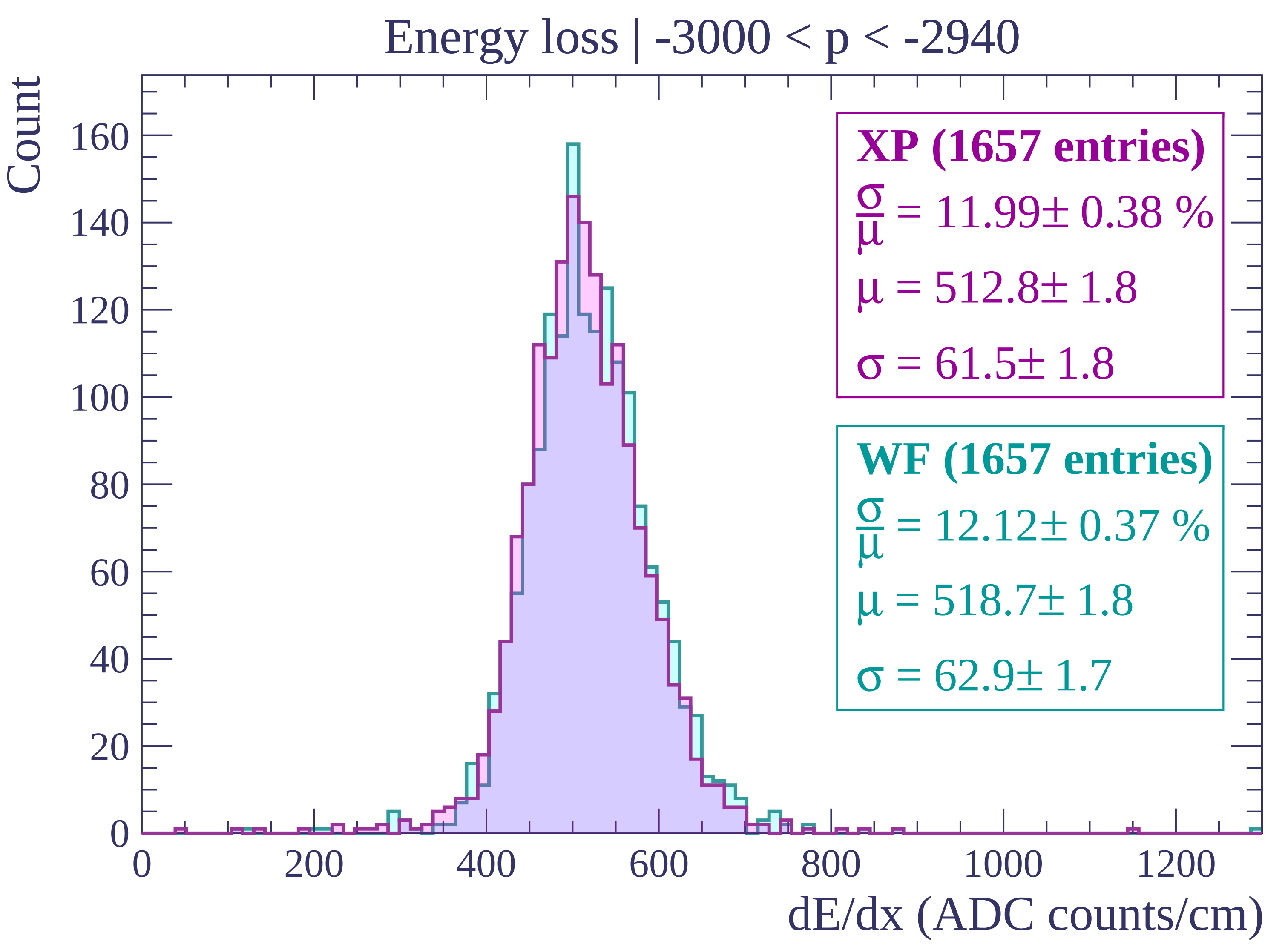

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
3000
2500
2000
1500
1000
500
0



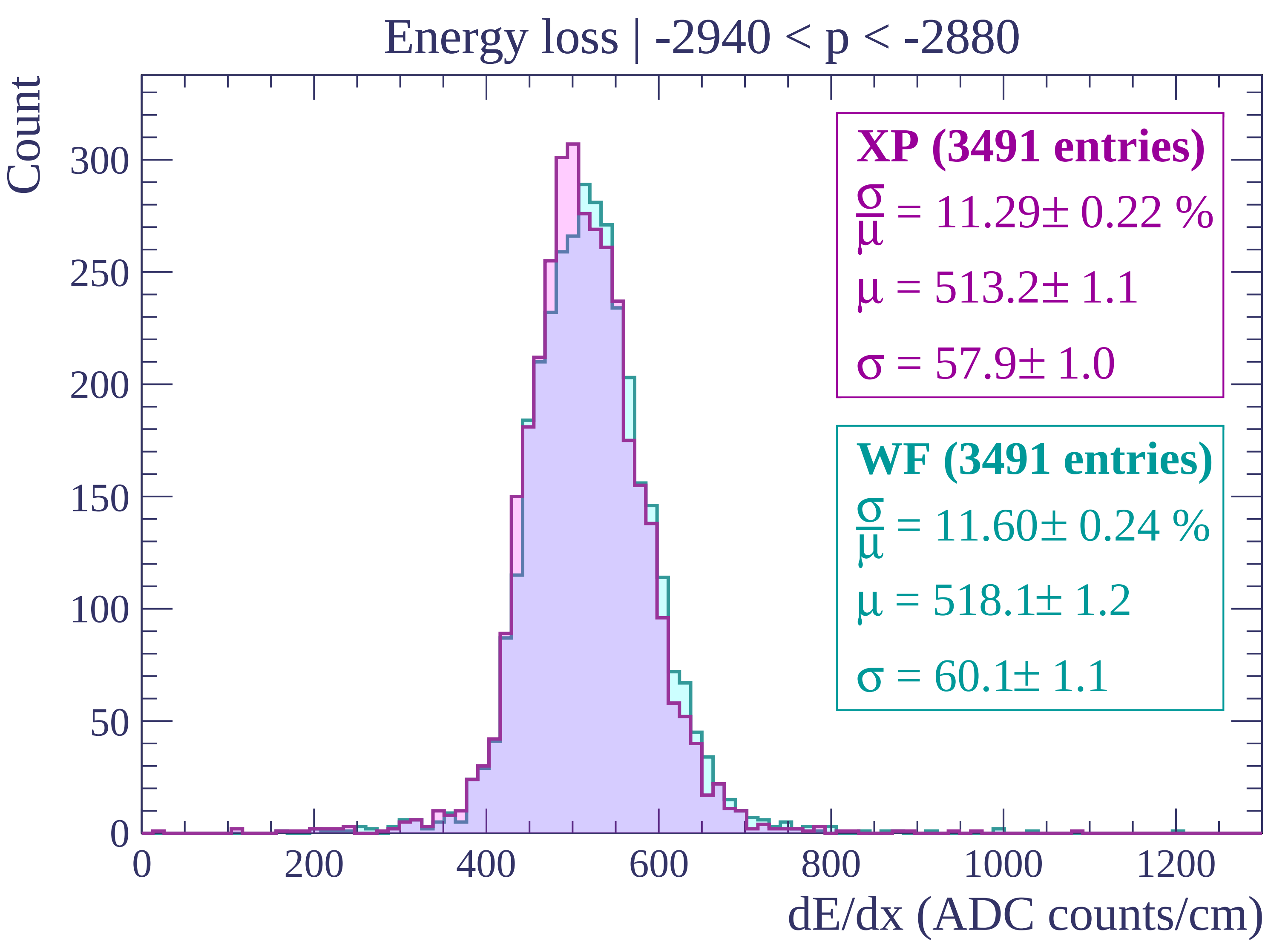
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
2500
2000
1500
1000
500
0

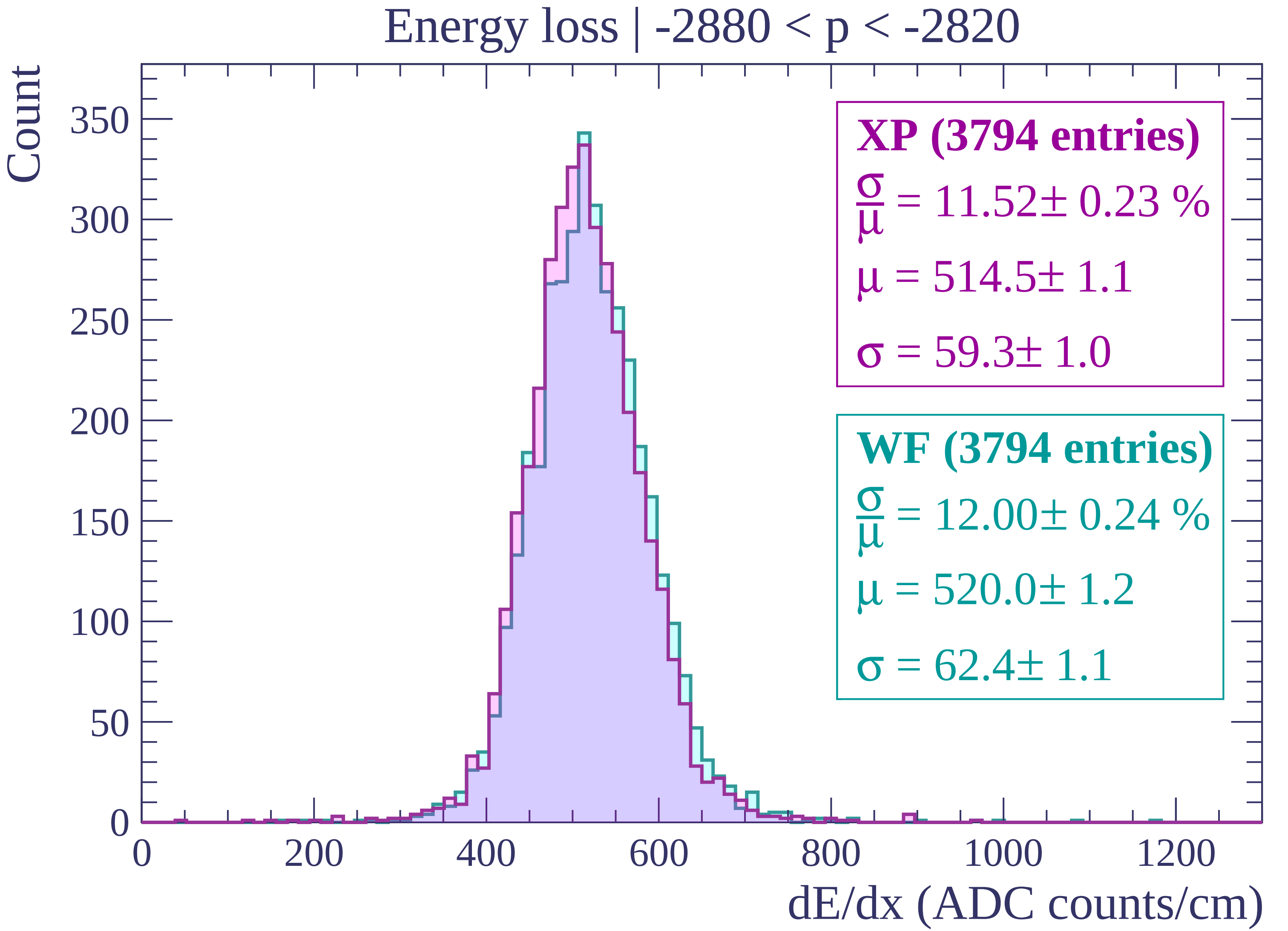


Count
×10^3
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

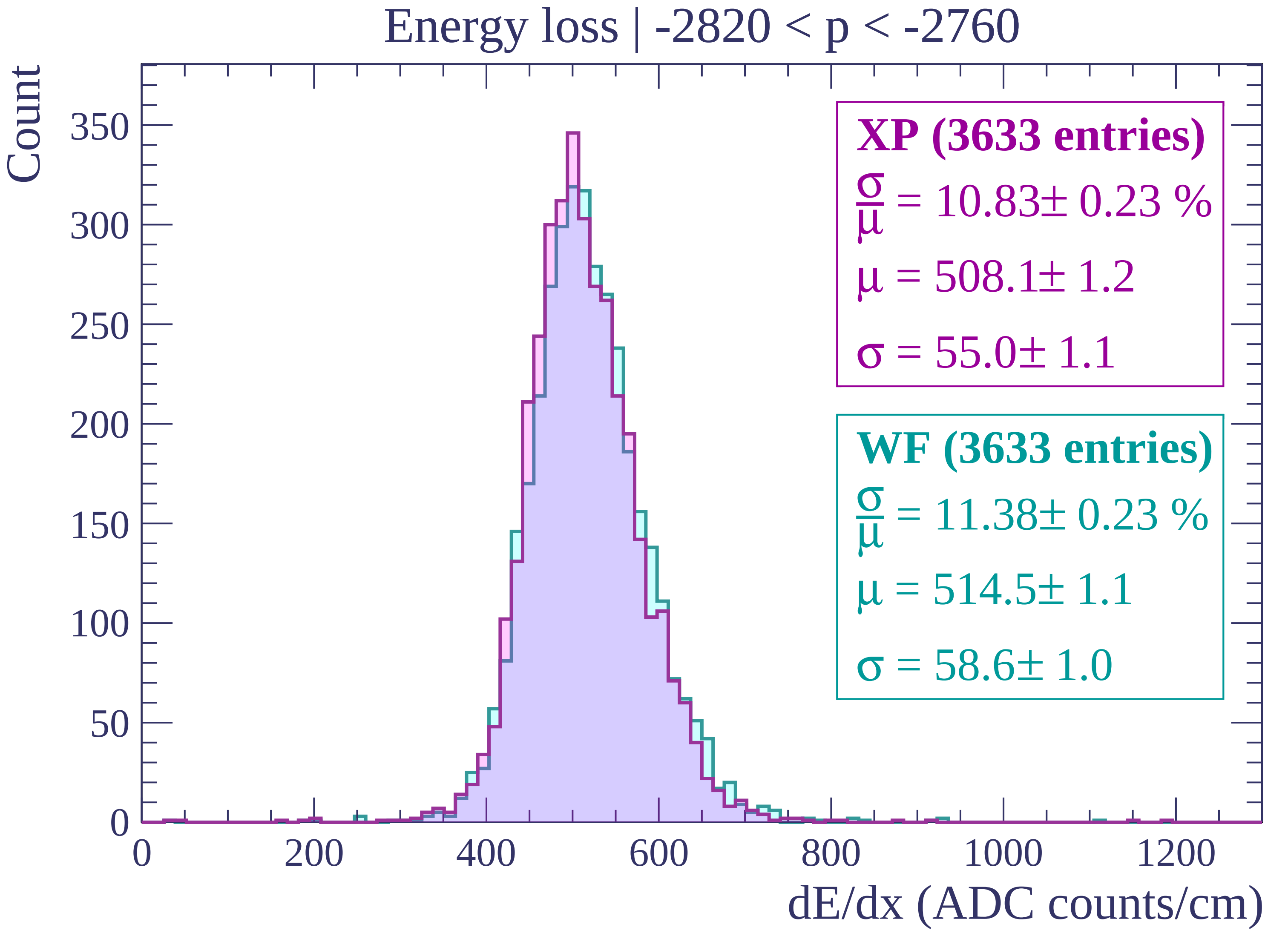


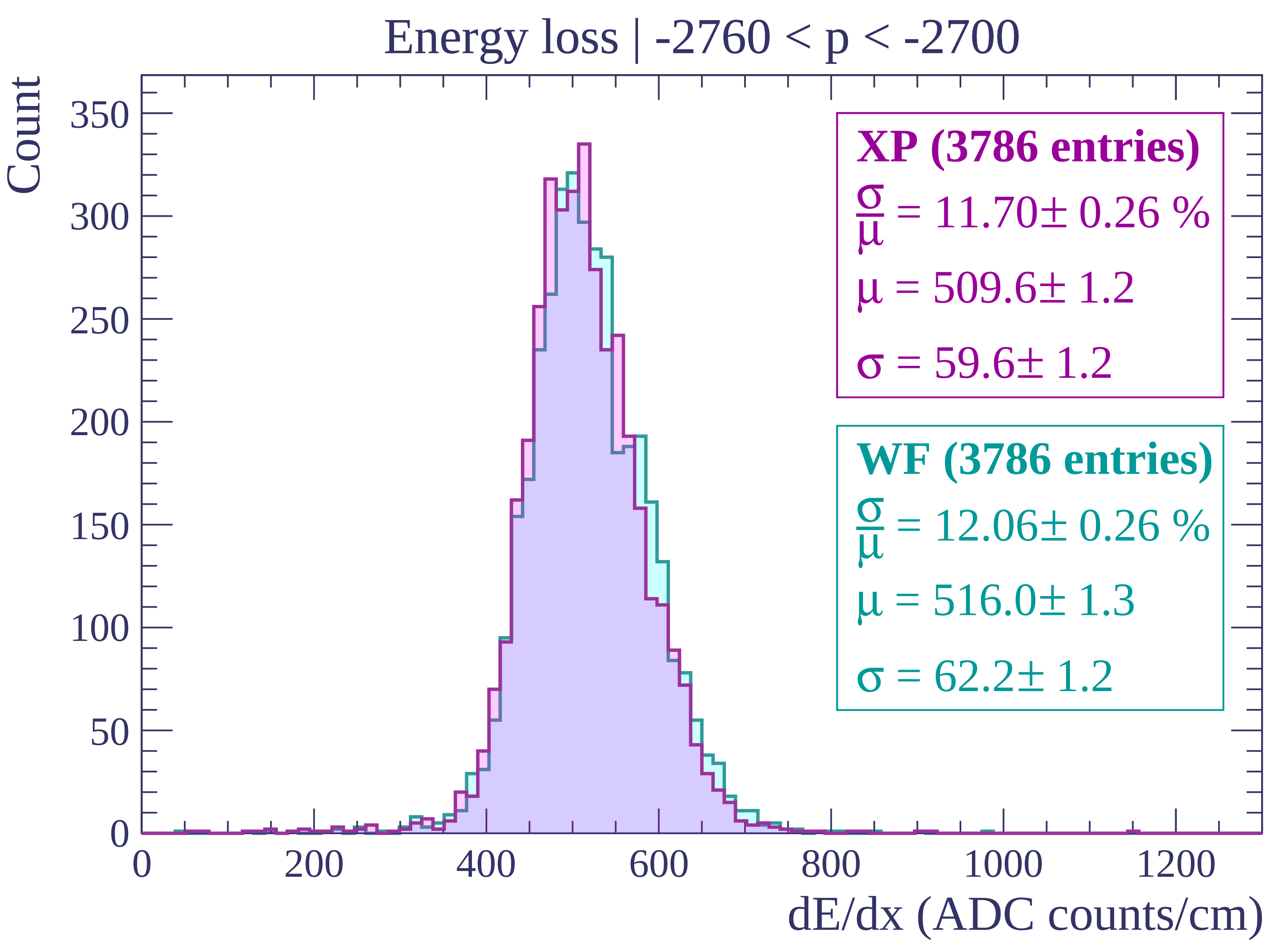
Energy loss | -3000 < p < -2940
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1657 entries)
$\frac{\sigma}{\mu}$ = 11.99± 0.38 %
μ = 512.8± 1.8
σ = 61.5± 1.8
WF (1657 entries)
$\frac{\sigma}{\mu}$ = 12.12± 0.37 %
μ = 518.7± 1.8
σ = 62.9± 1.7


Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3491 entries)
σ/μ = 11.29± 0.22 %
μ = 513.2± 1.1
σ = 57.9± 1.0
WF (3491 entries)
σ/μ = 11.60± 0.24 %
μ = 518.1± 1.2
σ = 60.1± 1.1

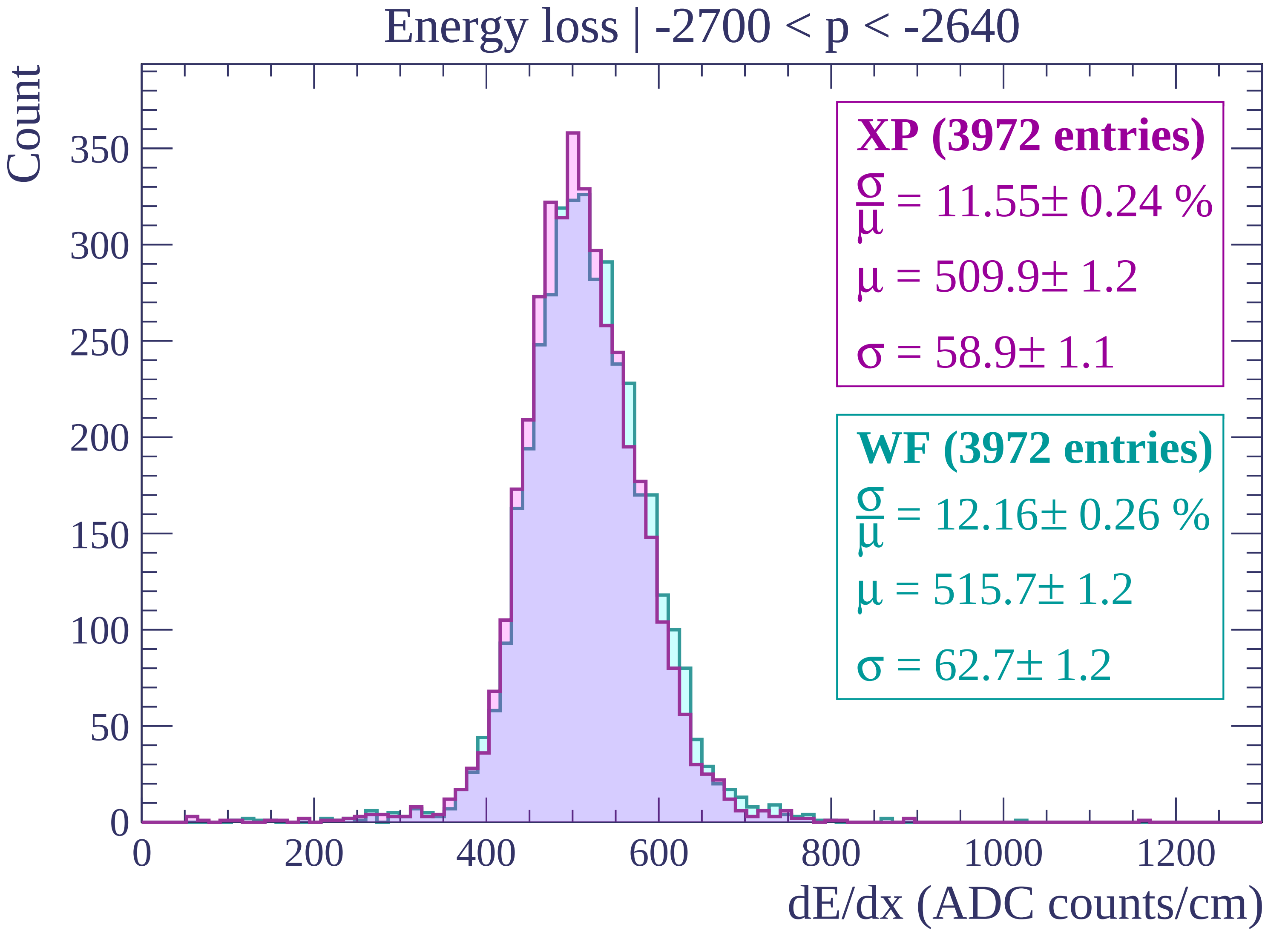


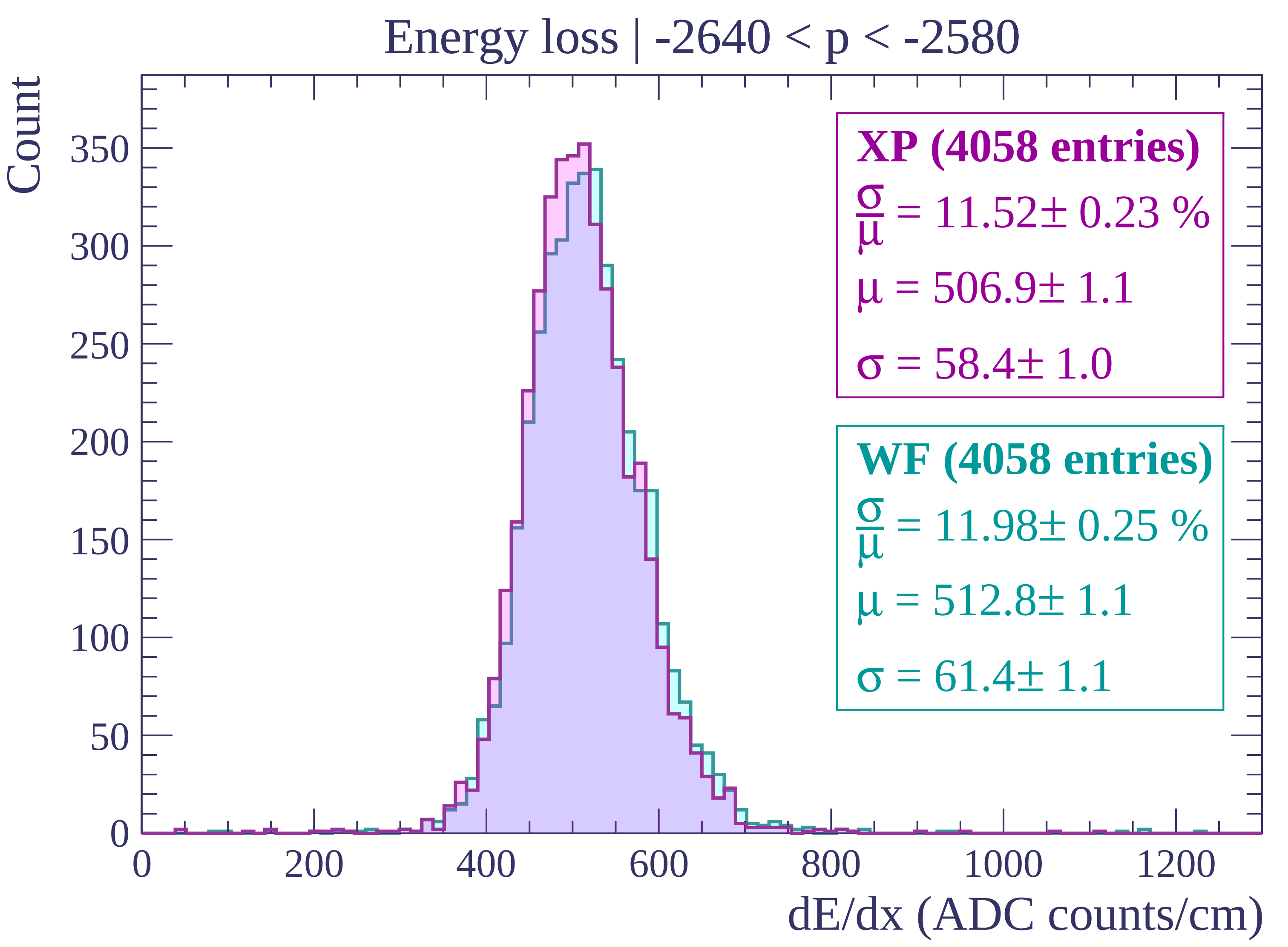
Energy loss | -2880 < p < -2820
Count
XP (3794 entries)
σ/μ = 11.52± 0.23 %
μ = 514.5± 1.1
σ = 59.3± 1.0
WF (3794 entries)
σ/μ = 12.00± 0.24 %
μ = 520.0± 1.2
σ = 62.4± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)


Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
XP (3633 entries)
$\frac{\sigma}{\mu}$ = 10.83± 0.23 %
μ = 508.1± 1.2
σ = 55.0± 1.1
WF (3633 entries)
$\frac{\sigma}{\mu}$ = 11.38± 0.23 %
μ = 514.5± 1.1
σ = 58.6± 1.0
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

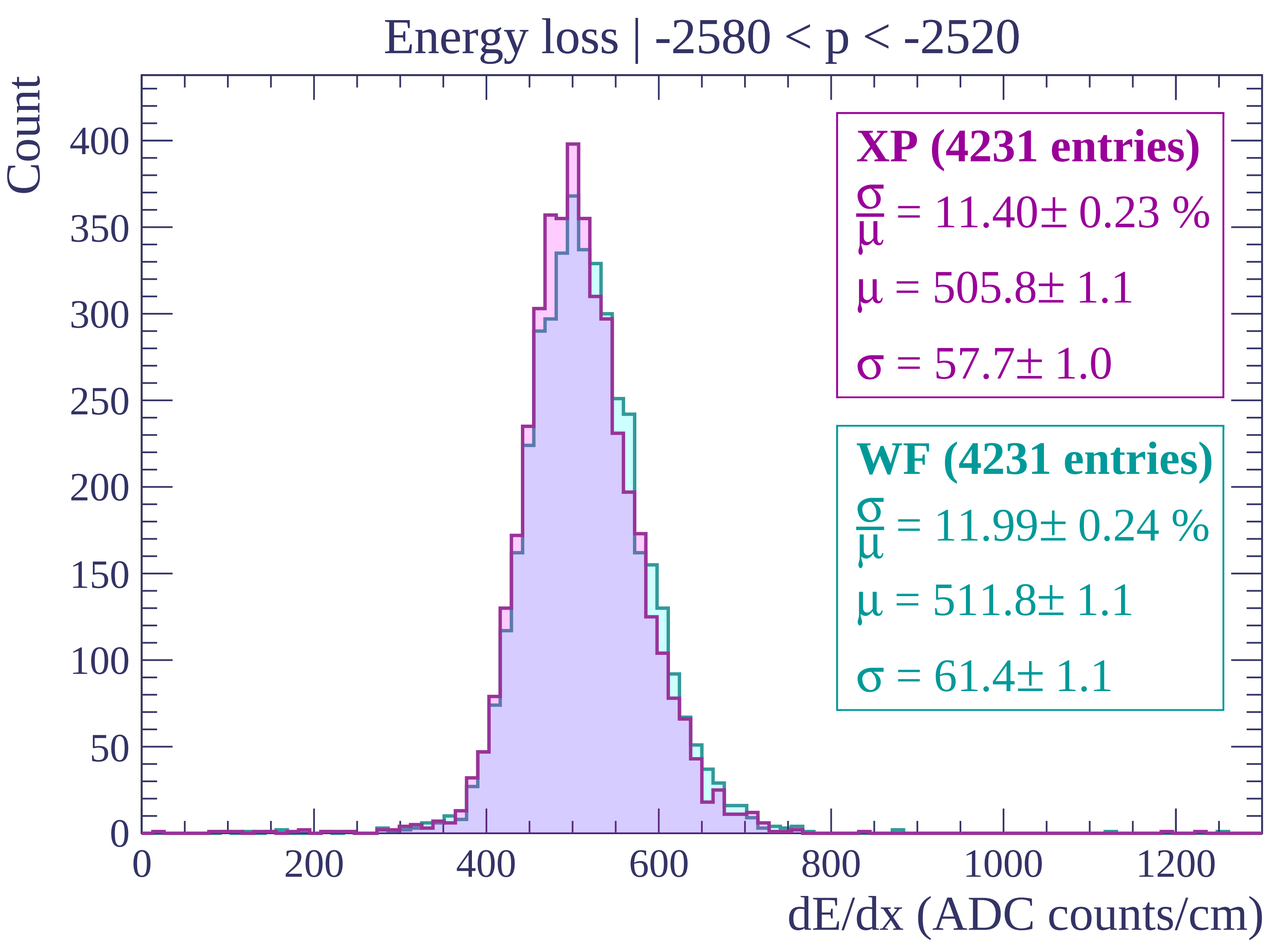


Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
XP (3786 entries)
σ/μ = 11.70± 0.26 %
μ = 509.6± 1.2
σ = 59.6± 1.2
WF (3786 entries)
σ/μ = 12.06± 0.26 %
μ = 516.0± 1.3
σ = 62.2± 1.2
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350

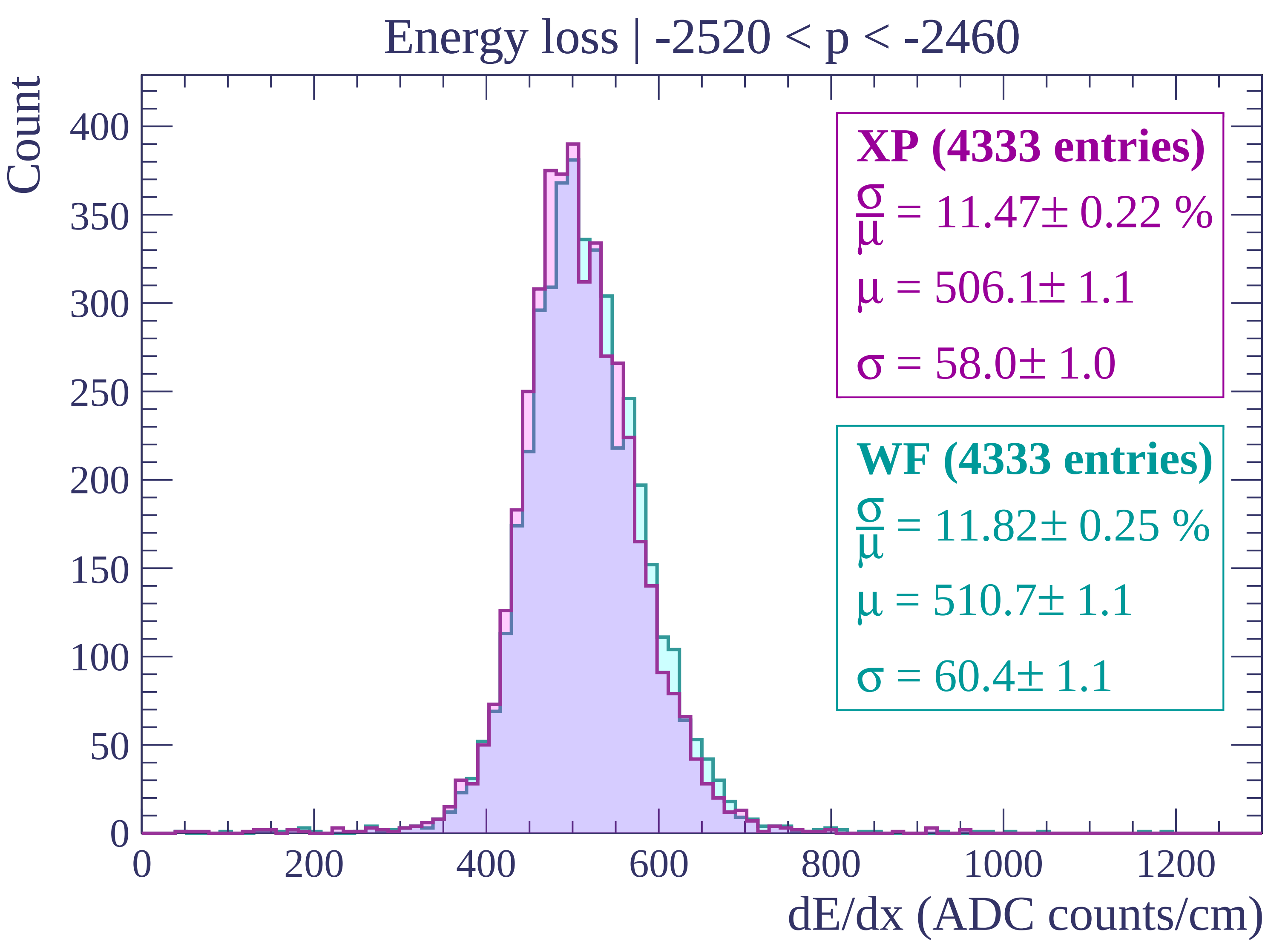

Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3972 entries)
σ/μ = 11.55± 0.24 %
μ = 509.9± 1.2
σ = 58.9± 1.1
WF (3972 entries)
σ/μ = 12.16± 0.26 %
μ = 515.7± 1.2
σ = 62.7± 1.2



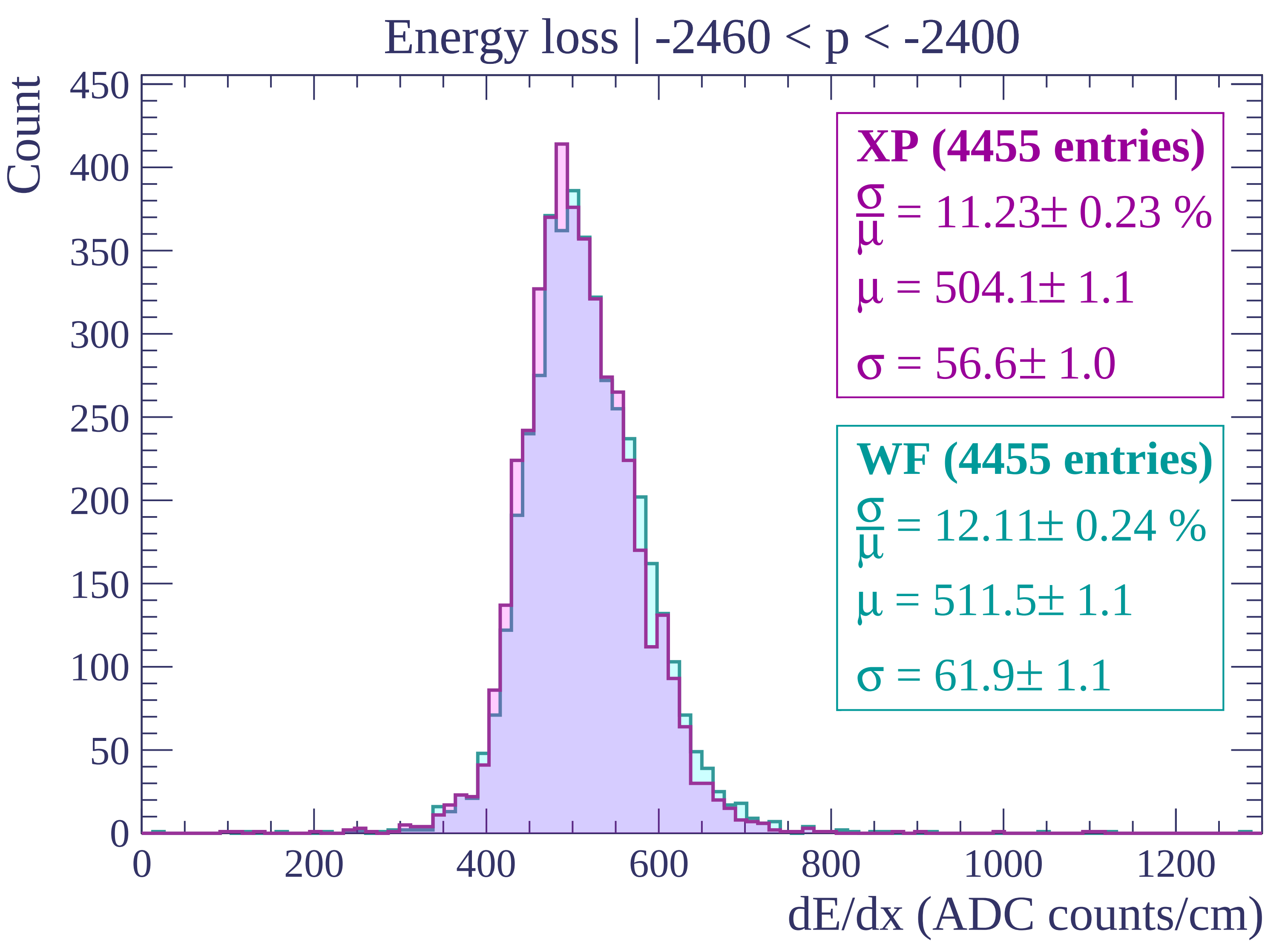
Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP (4058 entries)
σ/μ = 11.52± 0.23 %
μ = 506.9± 1.1
σ = 58.4± 1.0
WF (4058 entries)
σ/μ = 11.98± 0.25 %
μ = 512.8± 1.1
σ = 61.4± 1.1



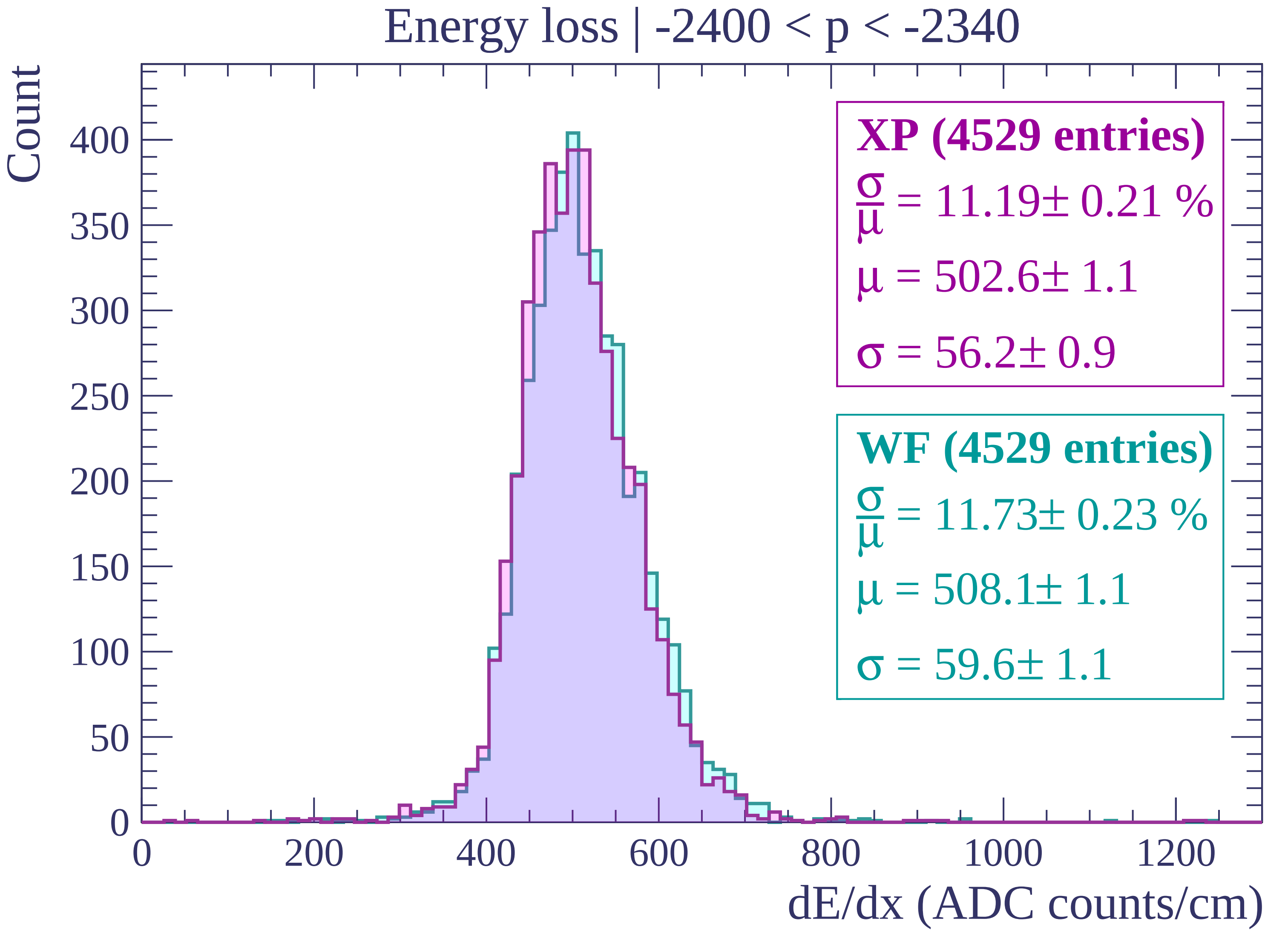
Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
XP (4231 entries)
σ/μ = 11.40± 0.23 %
μ = 505.8± 1.1
σ = 57.7± 1.0
WF (4231 entries)
σ/μ = 11.99± 0.24 %
μ = 511.8± 1.1
σ = 61.4± 1.1

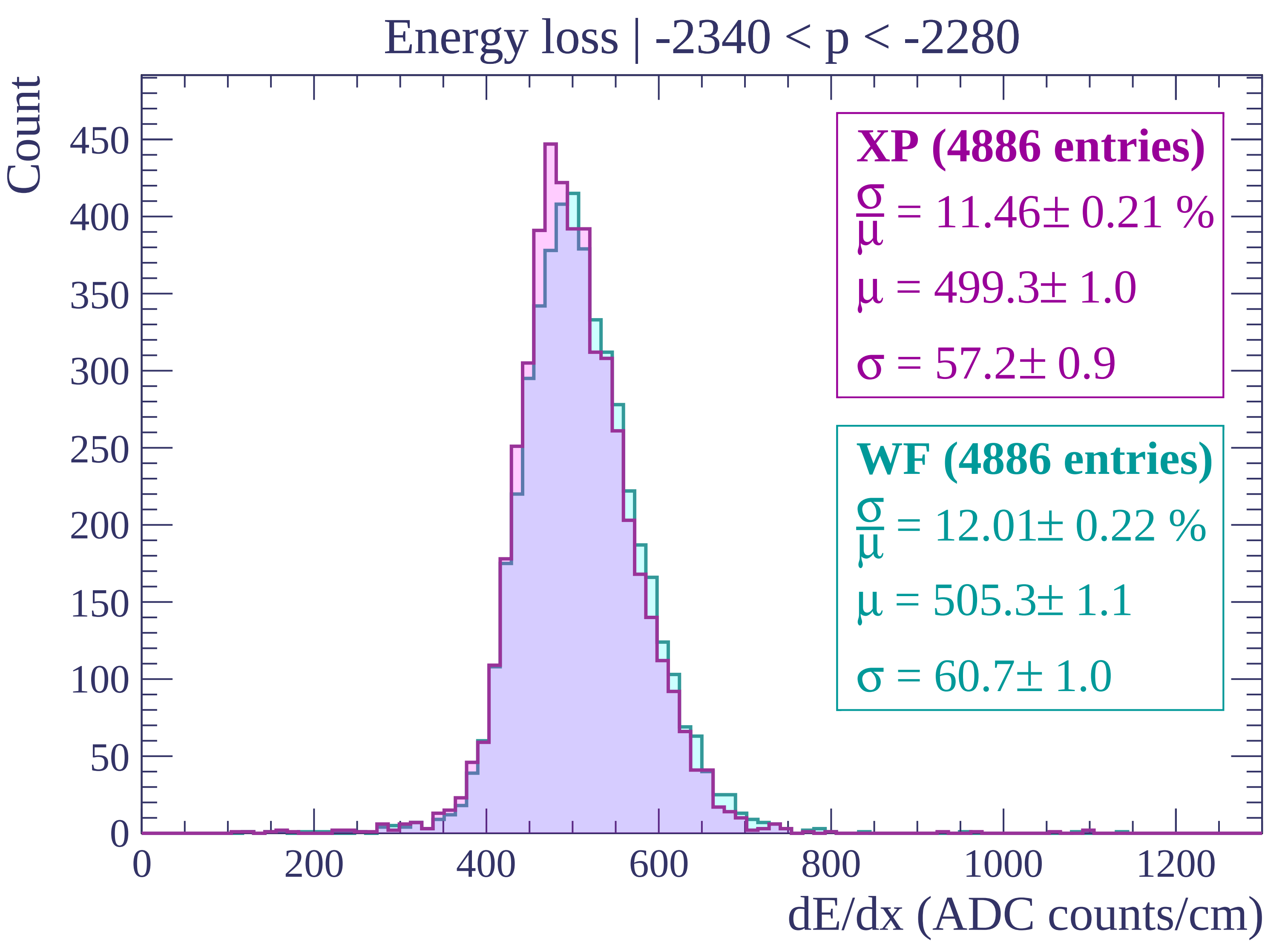


Energy loss | -2520 < p < -2460
Count
XP (4333 entries)
σ/μ = 11.47± 0.22 %
μ = 506.1± 1.1
σ = 58.0± 1.0
WF (4333 entries)
σ/μ = 11.82± 0.25 %
μ = 510.7± 1.1
σ = 60.4± 1.1
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

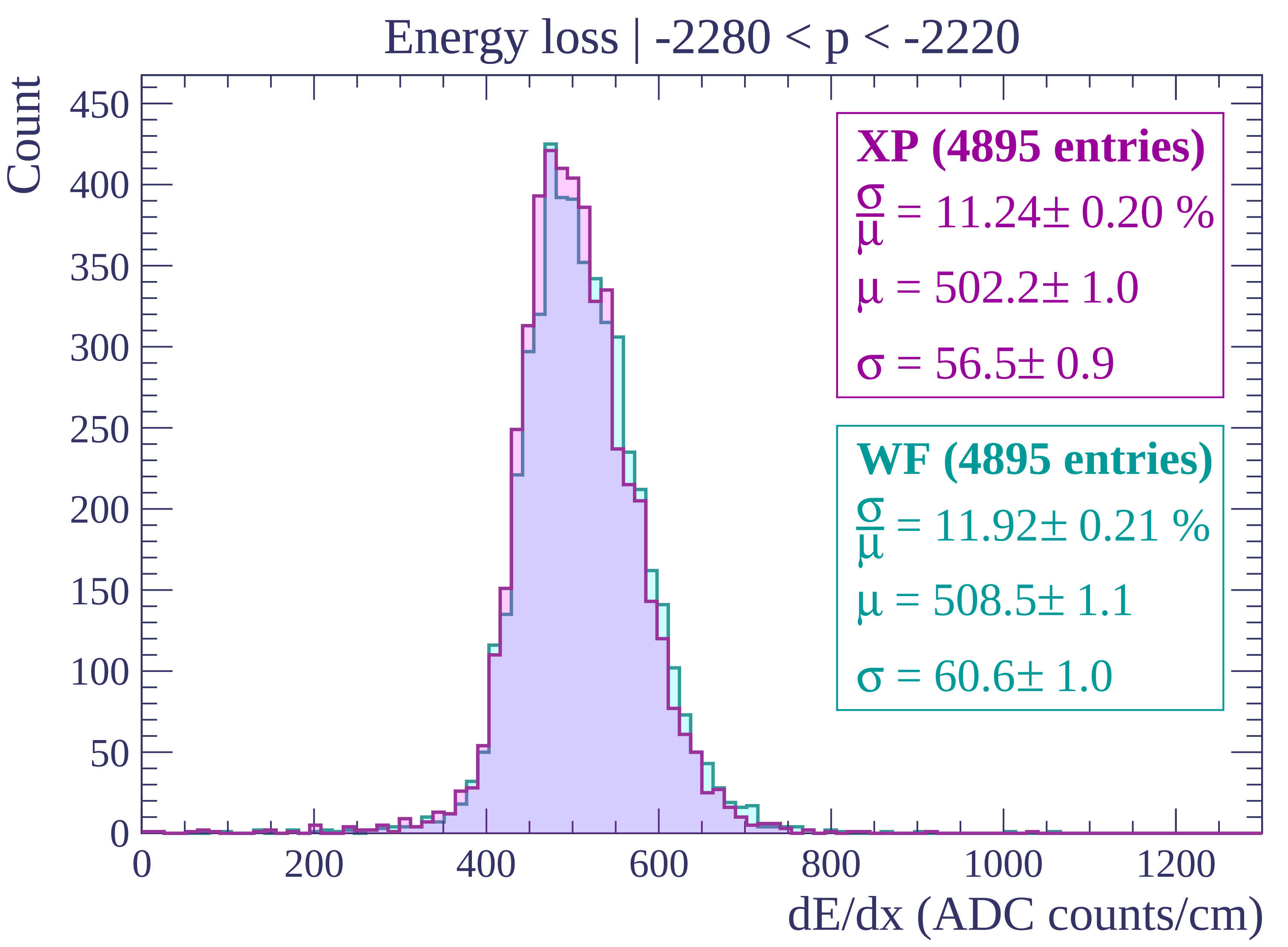


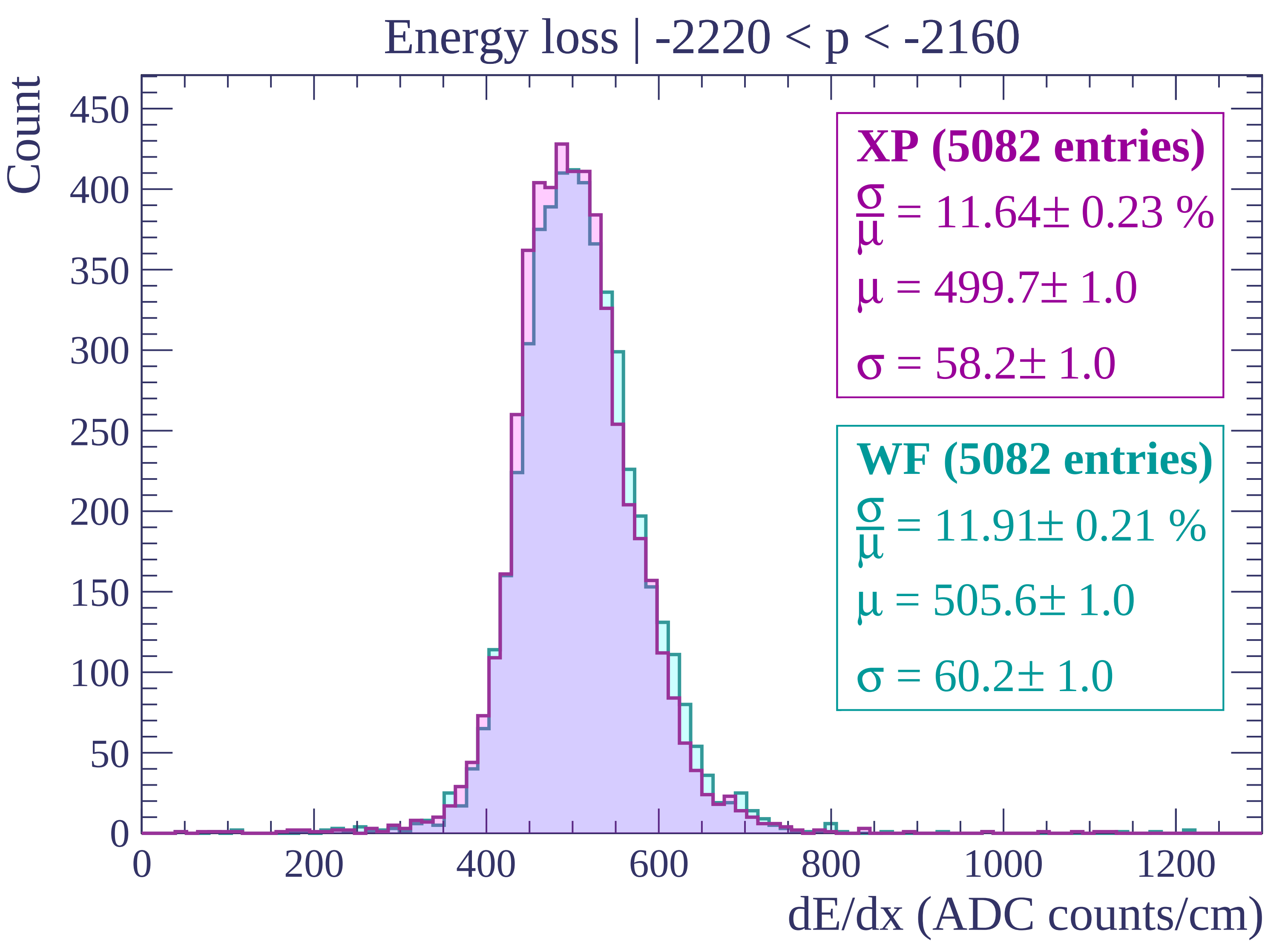
Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4455 entries)
σ/μ = 11.23± 0.23 %
μ = 504.1± 1.1
σ = 56.6± 1.0
WF (4455 entries)
σ/μ = 12.11± 0.24 %
μ = 511.5± 1.1
σ = 61.9± 1.1

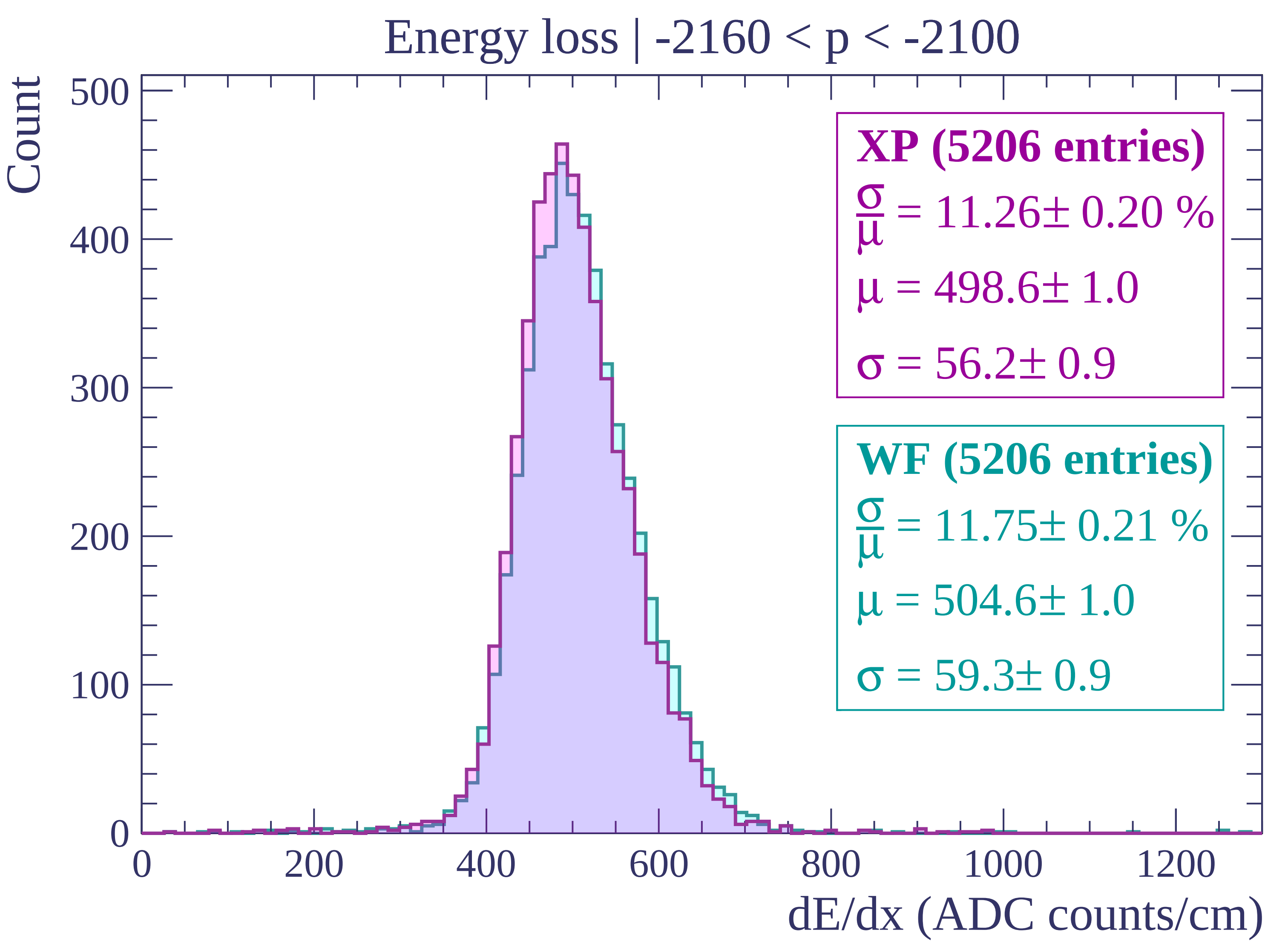

Energy loss | -2400 < p < -2340
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4529 entries)
σ/μ = 11.19± 0.21 %
μ = 502.6± 1.1
σ = 56.2± 0.9
WF (4529 entries)
σ/μ = 11.73± 0.23 %
μ = 508.1± 1.1
σ = 59.6± 1.1



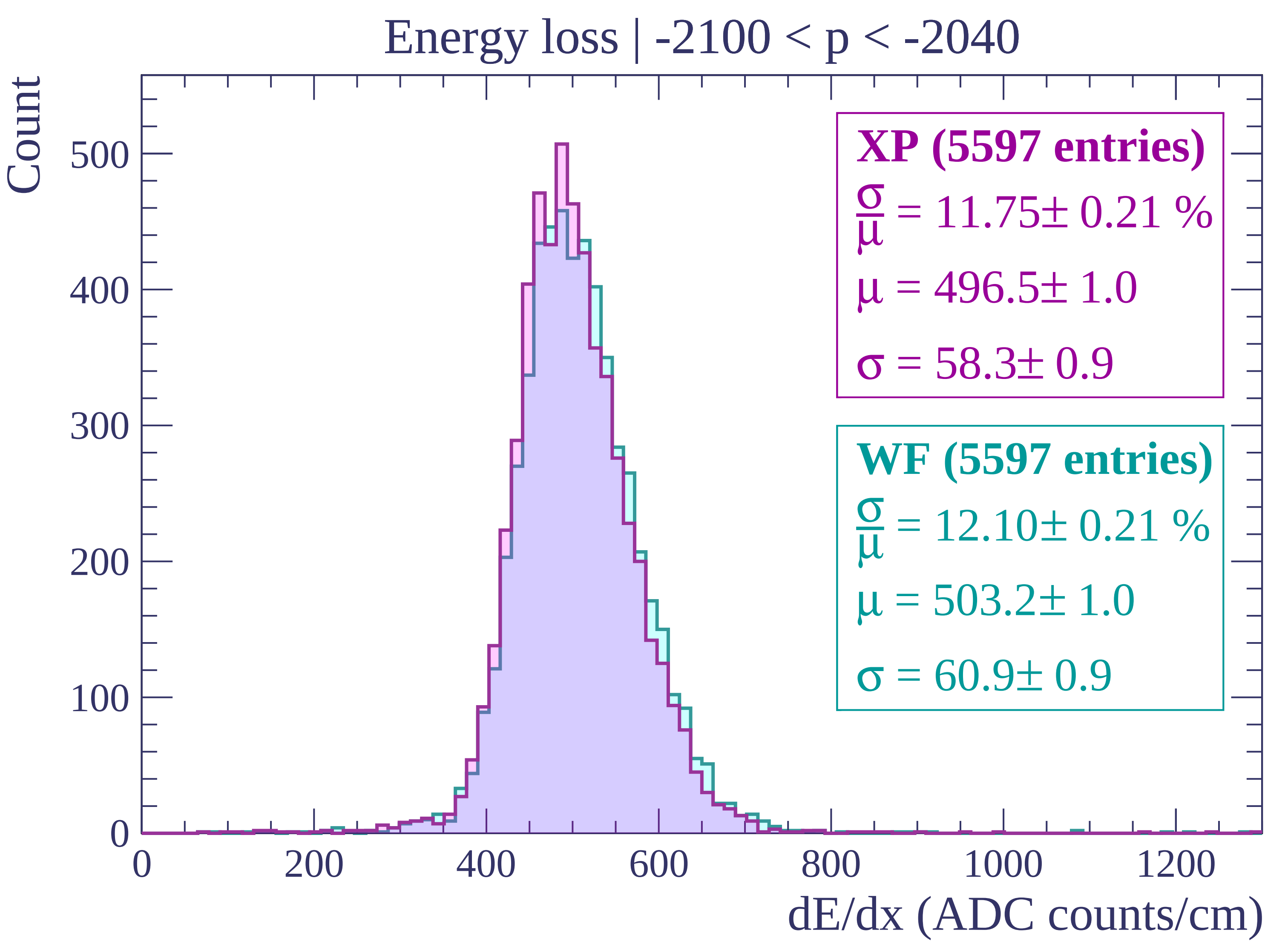
Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4886 entries)
σ/μ = 11.46± 0.21 %
μ = 499.3± 1.0
σ = 57.2± 0.9
WF (4886 entries)
σ/μ = 12.01± 0.22 %
μ = 505.3± 1.1
σ = 60.7± 1.0

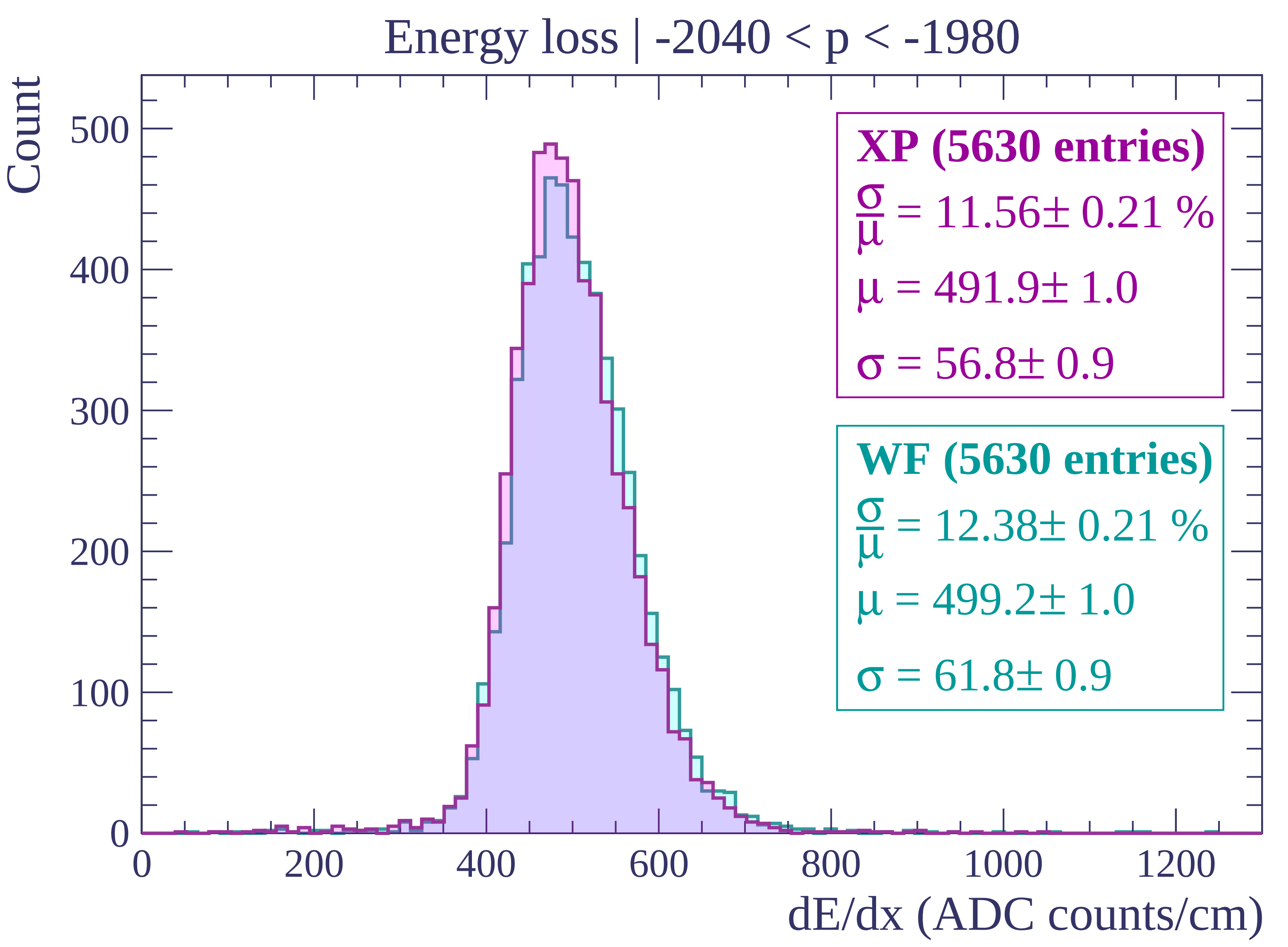

Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4895 entries)
σ/μ = 11.24± 0.20 %
μ = 502.2± 1.0
σ = 56.5± 0.9
WF (4895 entries)
σ/μ = 11.92± 0.21 %
μ = 508.5± 1.1
σ = 60.6± 1.0


Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
XP (5082 entries)
σ/μ = 11.64± 0.23 %
μ = 499.7± 1.0
σ = 58.2± 1.0
WF (5082 entries)
σ/μ = 11.91± 0.21 %
μ = 505.6± 1.0
σ = 60.2± 1.0

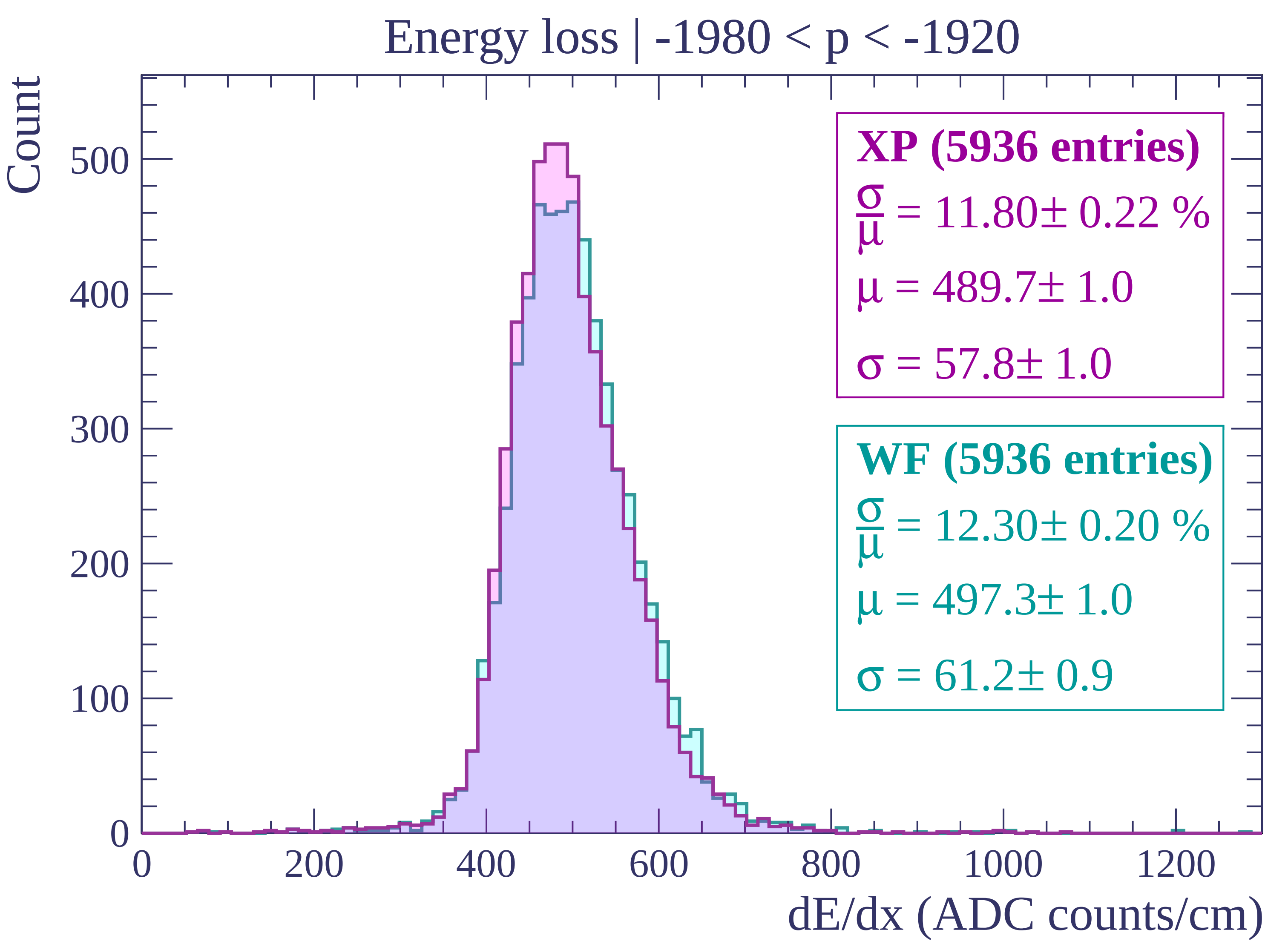


Energy loss | -2160 < p < -2100
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5206 entries)
$\frac{\sigma}{\mu}$ = 11.26± 0.20 %
μ = 498.6± 1.0
σ = 56.2± 0.9
WF (5206 entries)
$\frac{\sigma}{\mu}$ = 11.75± 0.21 %
μ = 504.6± 1.0
σ = 59.3± 0.9

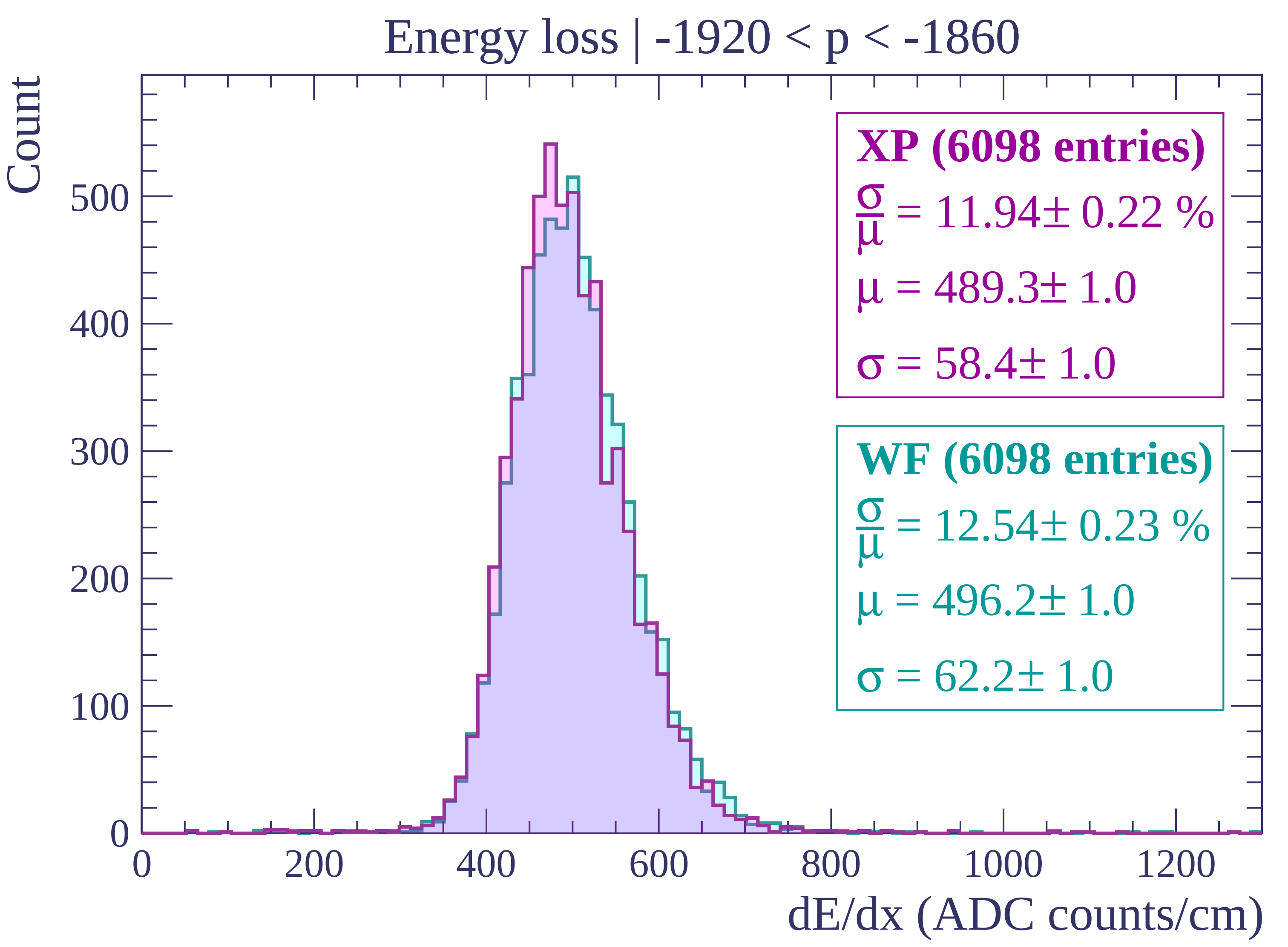

Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5597 entries)
$\frac{\sigma}{\mu}$ = 11.75± 0.21 %
μ = 496.5± 1.0
σ = 58.3± 0.9
WF (5597 entries)
$\frac{\sigma}{\mu}$ = 12.10± 0.21 %
μ = 503.2± 1.0
σ = 60.9± 0.9



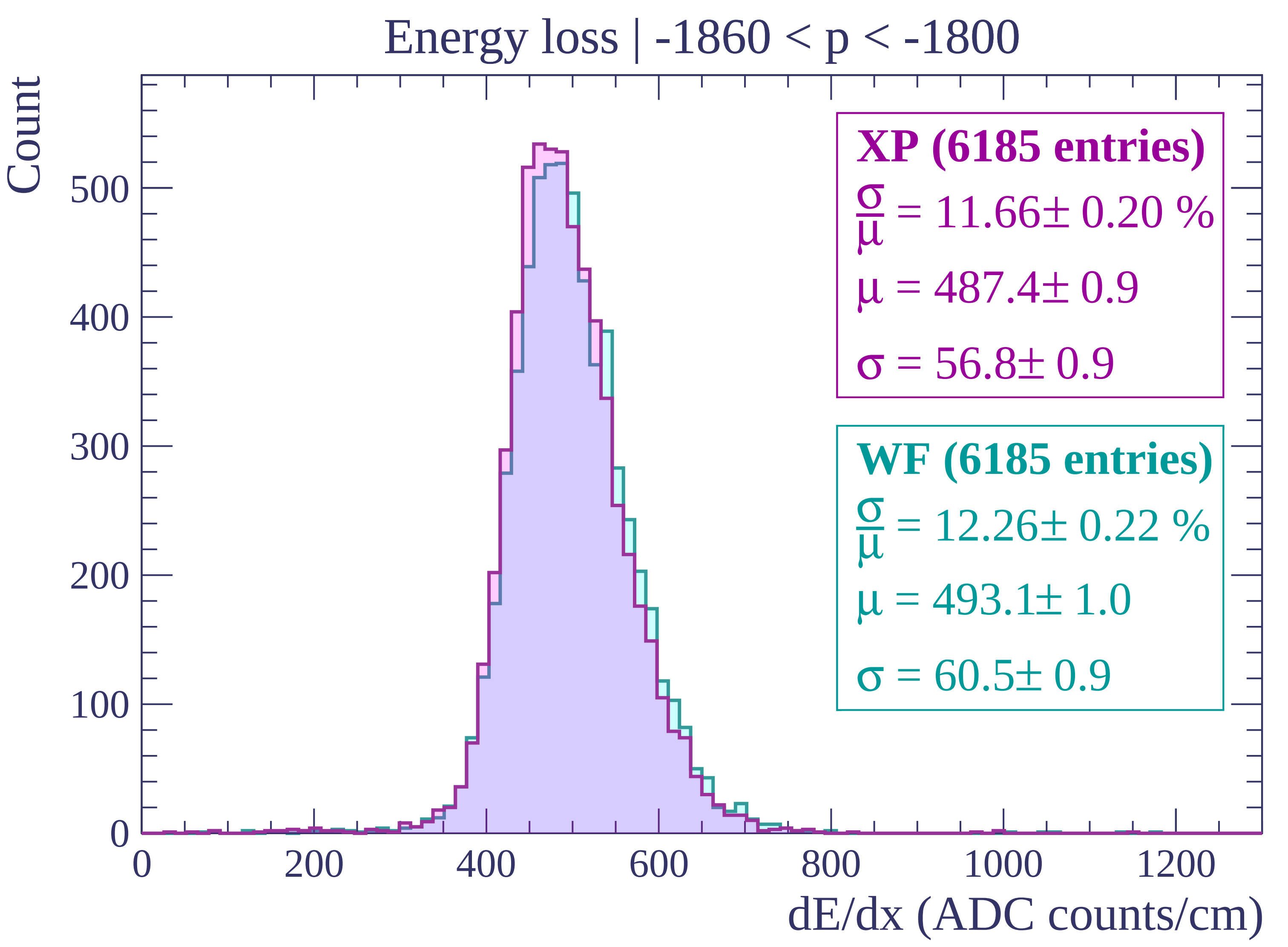
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5630 entries)
σ/μ = 11.56± 0.21 %
μ = 491.9± 1.0
σ = 56.8± 0.9
WF (5630 entries)
σ/μ = 12.38± 0.21 %
μ = 499.2± 1.0
σ = 61.8± 0.9



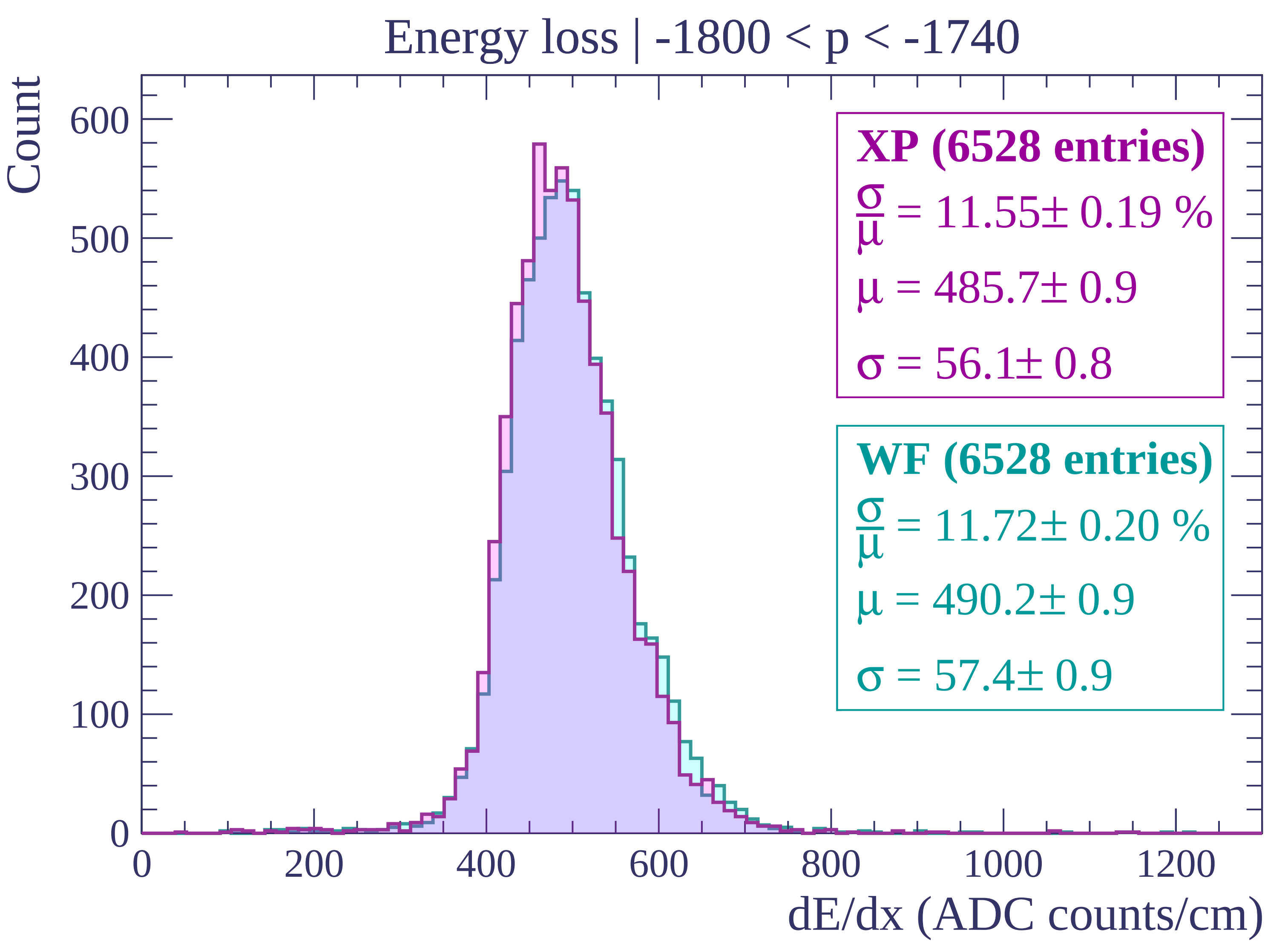
Energy loss | -1980 < p < -1920
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5936 entries)
σ/μ = 11.80± 0.22 %
μ = 489.7± 1.0
σ = 57.8± 1.0
WF (5936 entries)
σ/μ = 12.30± 0.20 %
μ = 497.3± 1.0
σ = 61.2± 0.9



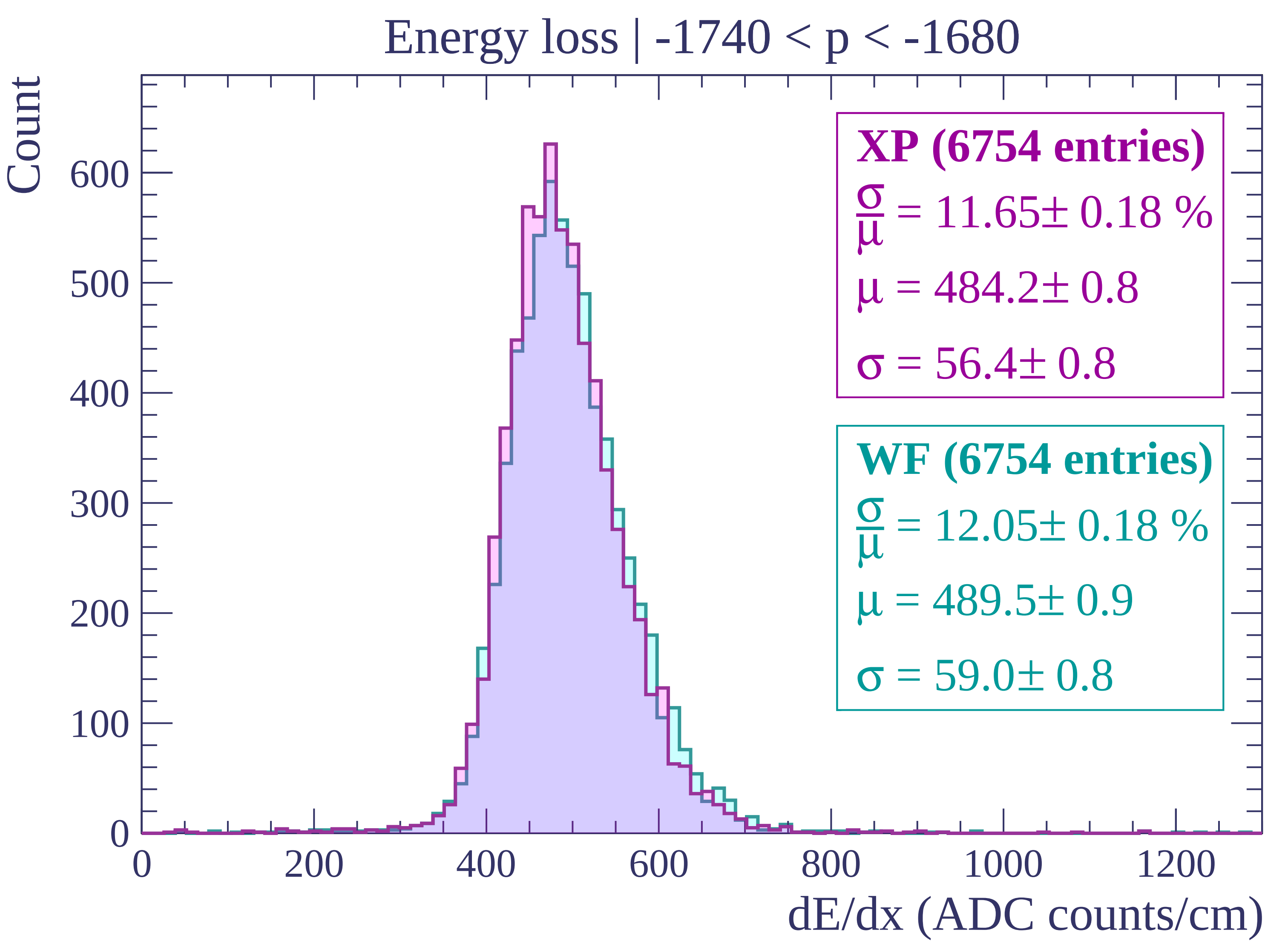
Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (6098 entries)
σ/μ = 11.94± 0.22 %
μ = 489.3± 1.0
σ = 58.4± 1.0
WF (6098 entries)
σ/μ = 12.54± 0.23 %
μ = 496.2± 1.0
σ = 62.2± 1.0



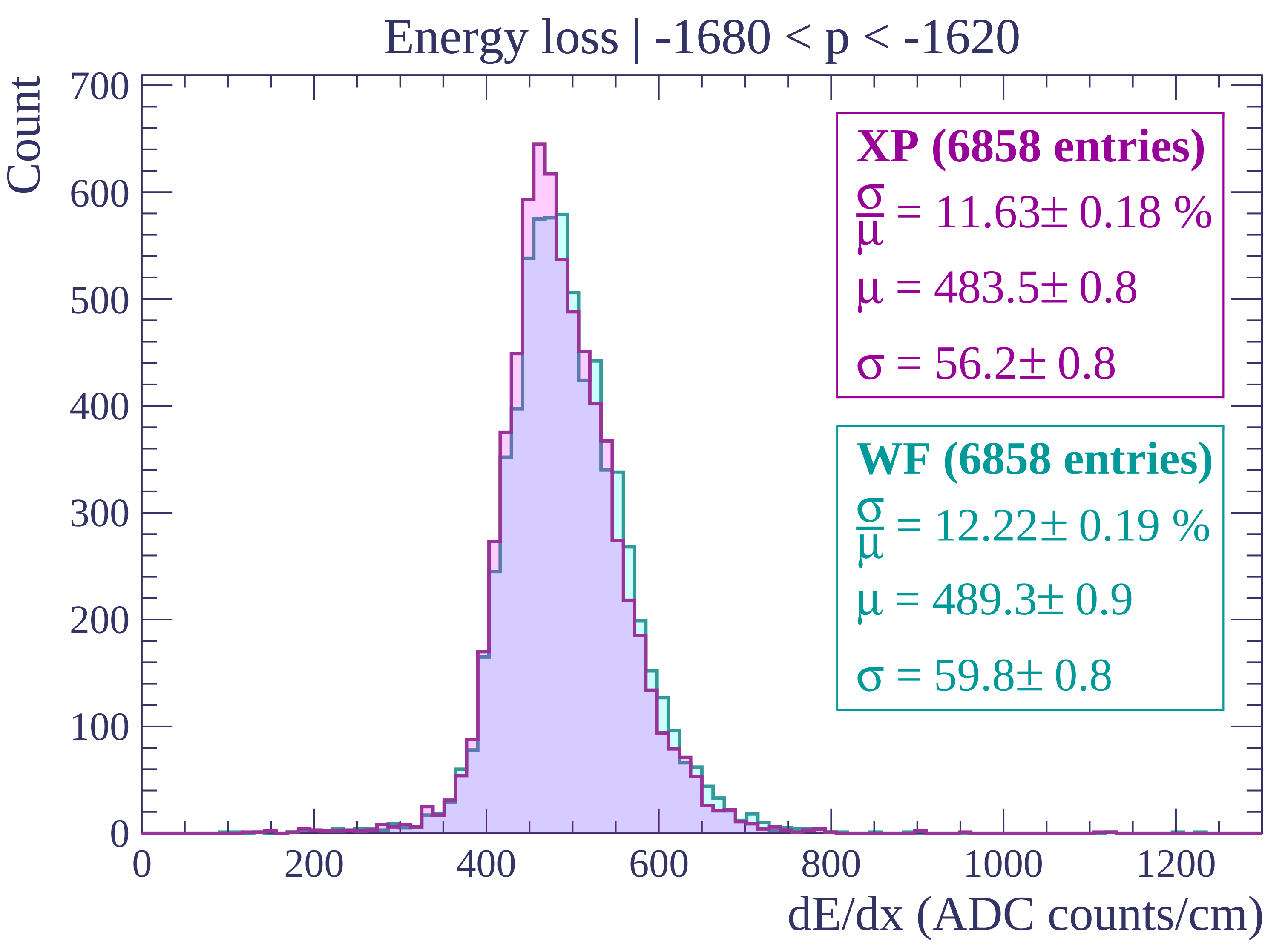
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (6185 entries)
σ/μ = 11.66± 0.20 %
μ = 487.4± 0.9
σ = 56.8± 0.9
WF (6185 entries)
σ/μ = 12.26± 0.22 %
μ = 493.1± 1.0
σ = 60.5± 0.9

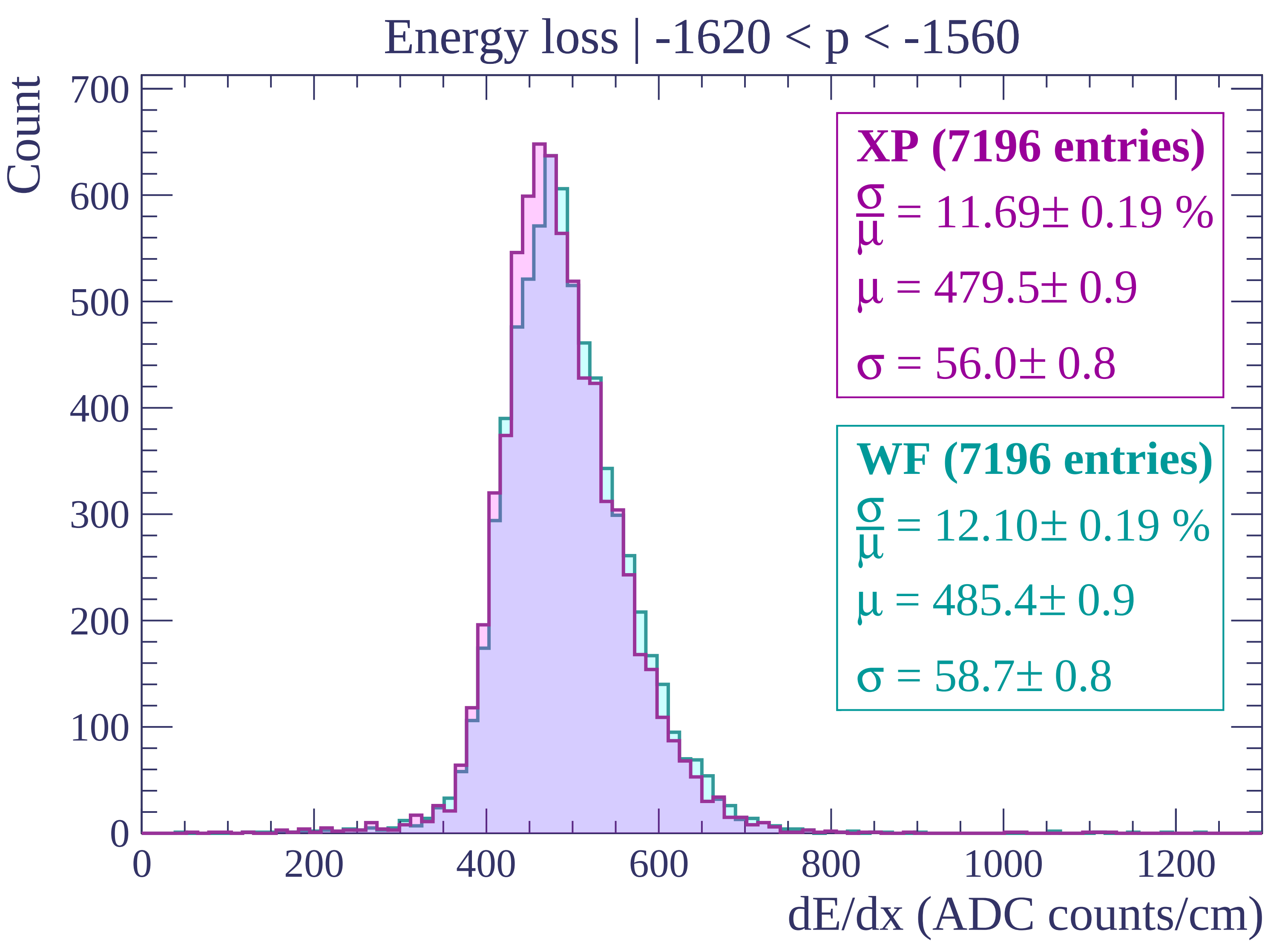


Energy loss | -1800 < p < -1740
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6528 entries)
$\frac{\sigma}{\mu}$ = 11.55± 0.19 %
μ = 485.7± 0.9
σ = 56.1± 0.8
WF (6528 entries)
$\frac{\sigma}{\mu}$ = 11.72± 0.20 %
μ = 490.2± 0.9
σ = 57.4± 0.9

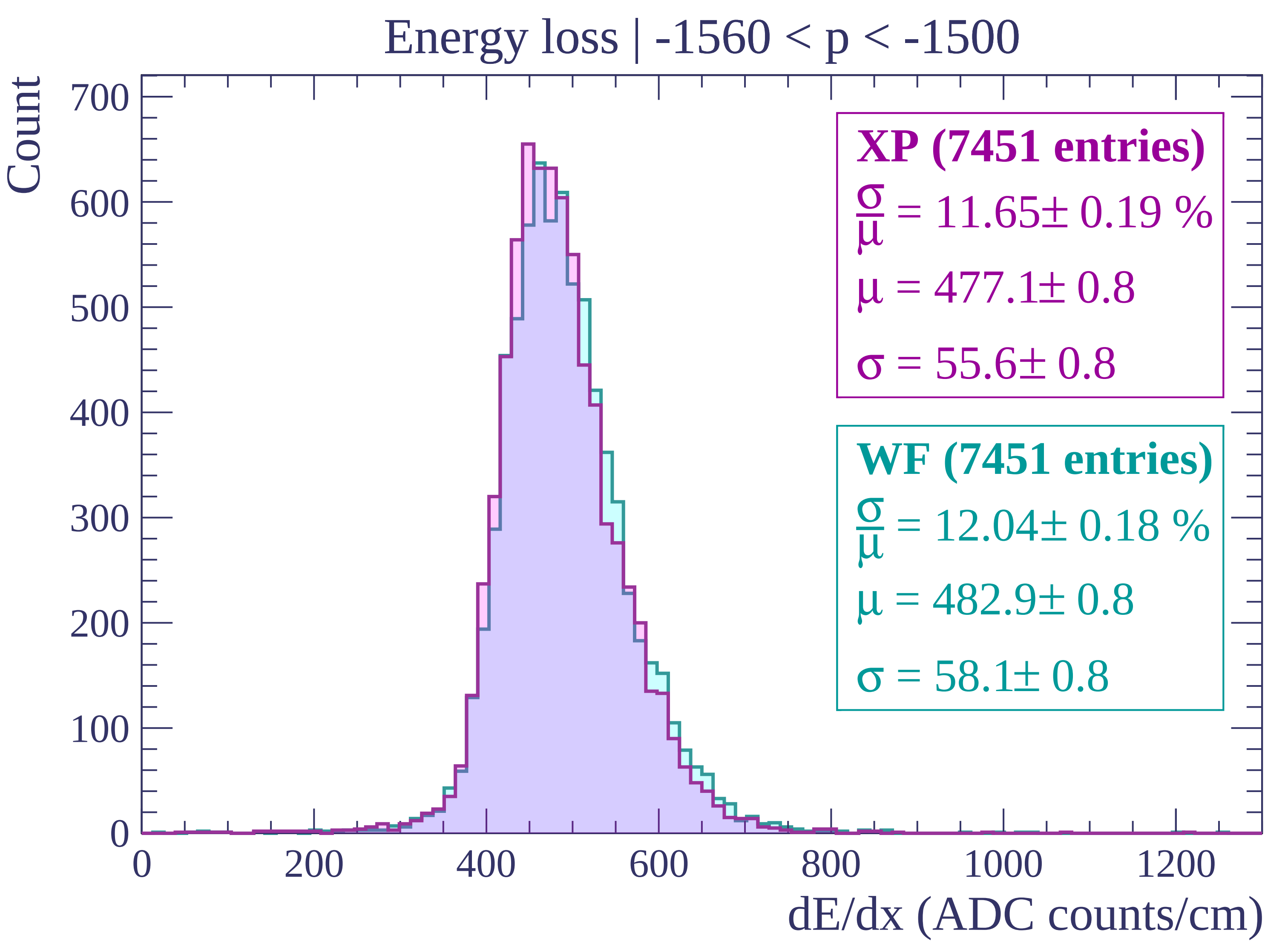


Energy loss | -1740 < p < -1680
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6754 entries)
σ/μ = 11.65± 0.18 %
μ = 484.2± 0.8
σ = 56.4± 0.8
WF (6754 entries)
σ/μ = 12.05± 0.18 %
μ = 489.5± 0.9
σ = 59.0± 0.8

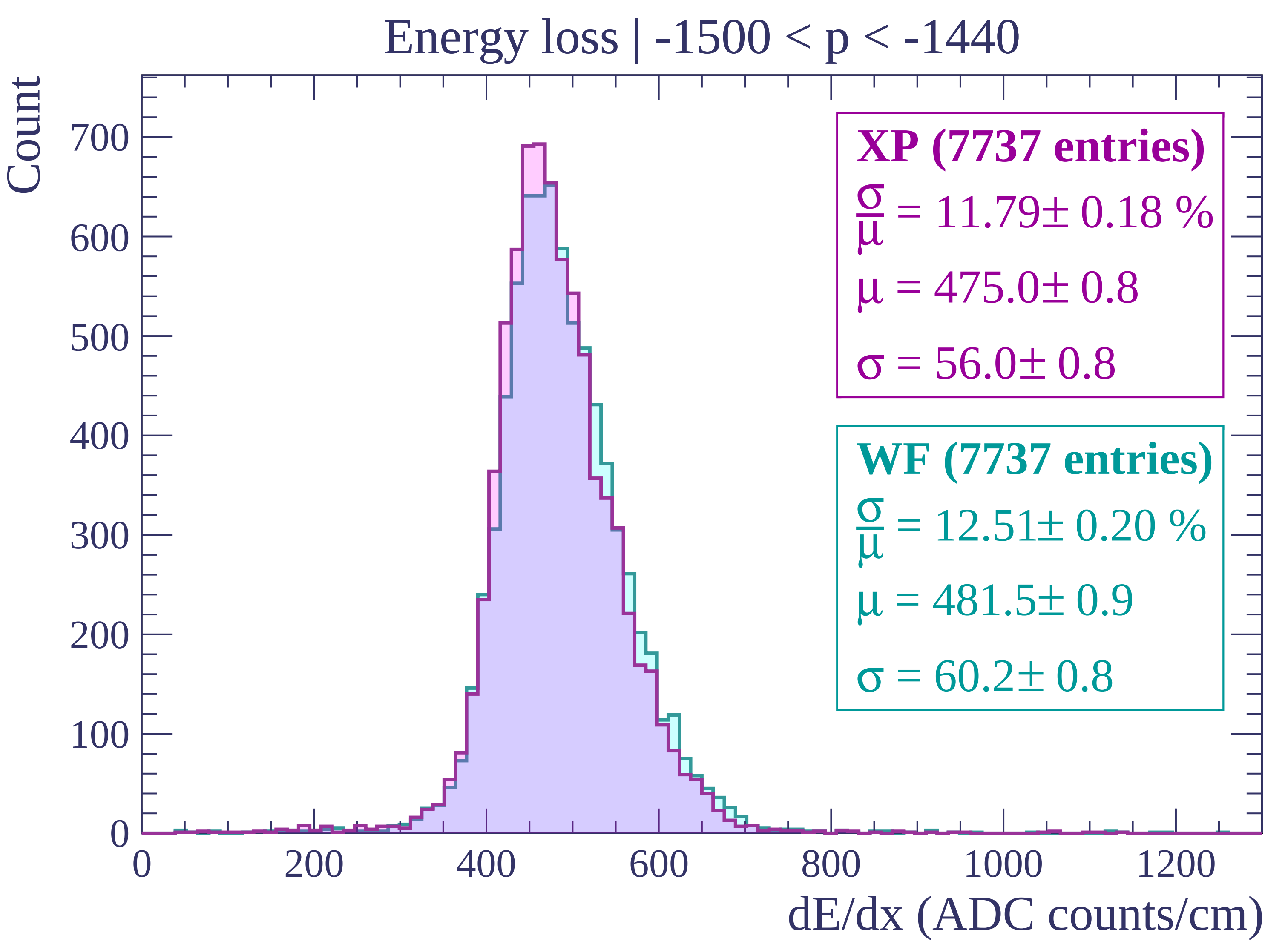

Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (6858 entries)
σ/μ = 11.63± 0.18 %
μ = 483.5± 0.8
σ = 56.2± 0.8
WF (6858 entries)
σ/μ = 12.22± 0.19 %
μ = 489.3± 0.9
σ = 59.8± 0.8



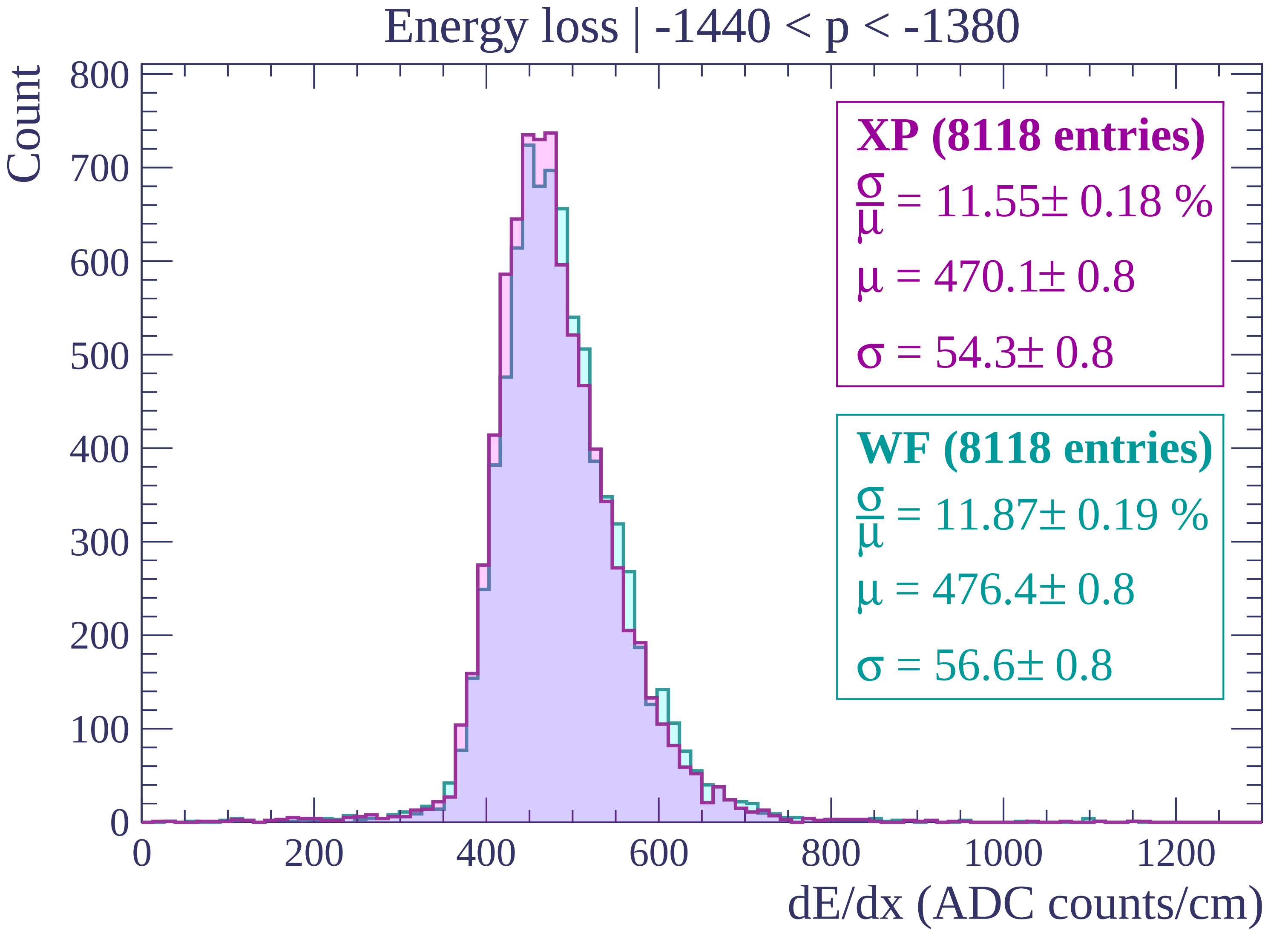
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (7196 entries)
σ/μ = 11.69± 0.19 %
μ = 479.5± 0.9
σ = 56.0± 0.8
WF (7196 entries)
σ/μ = 12.10± 0.19 %
μ = 485.4± 0.9
σ = 58.7± 0.8



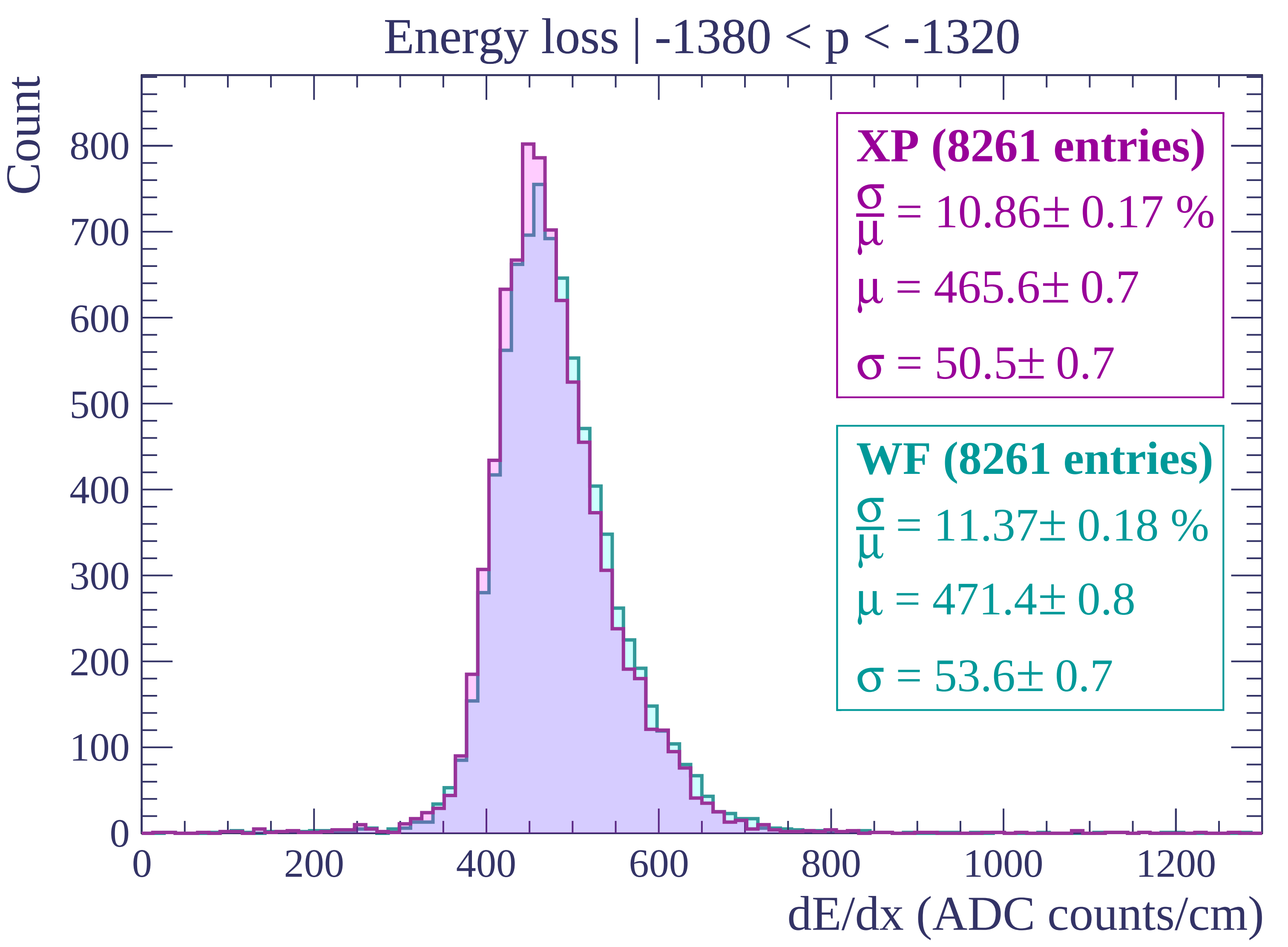
Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (7451 entries)
$\frac{\sigma}{\mu}$ = 11.65± 0.19 %
μ = 477.1± 0.8
σ = 55.6± 0.8
WF (7451 entries)
$\frac{\sigma}{\mu}$ = 12.04± 0.18 %
μ = 482.9± 0.8
σ = 58.1± 0.8

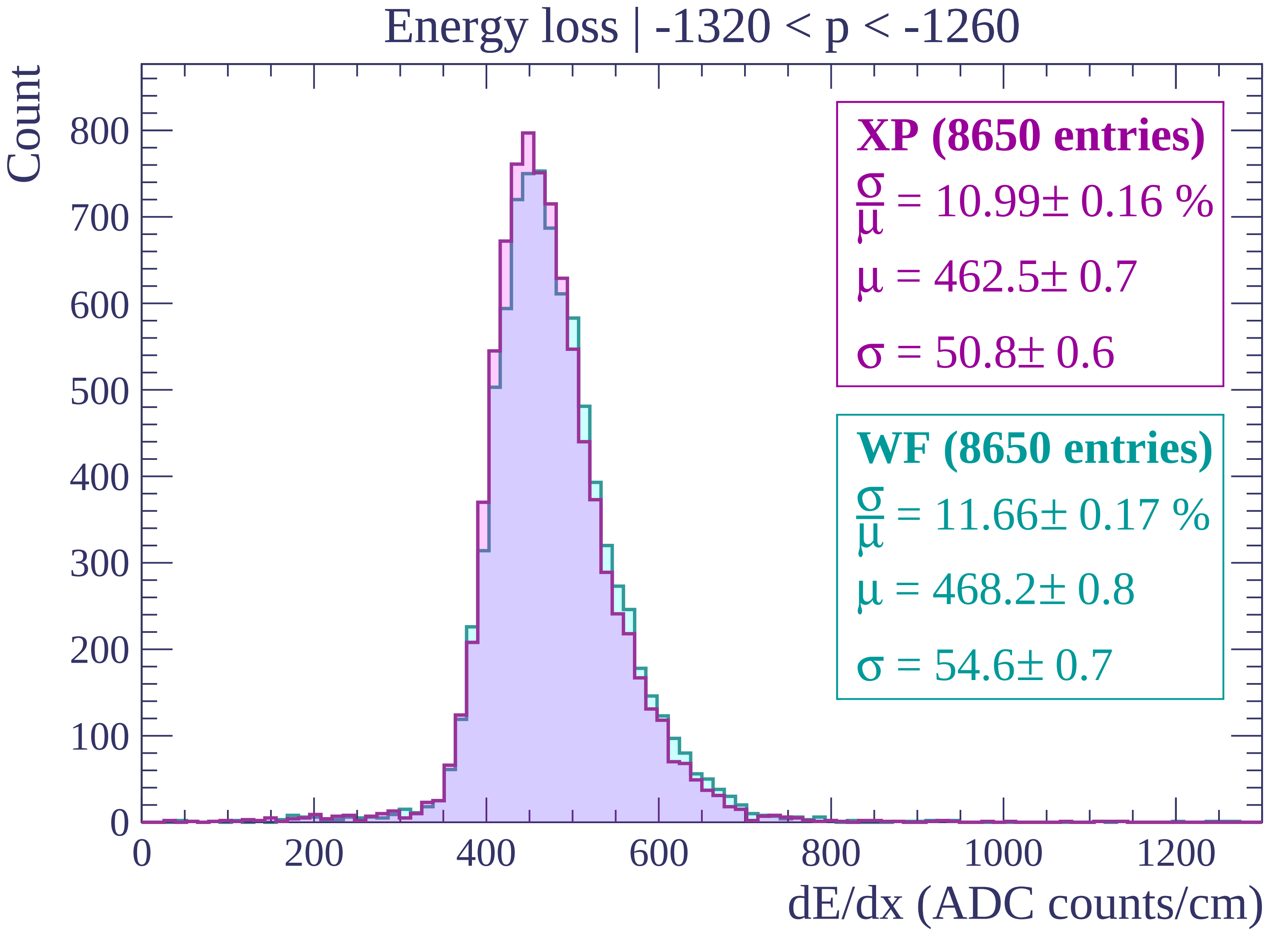


Energy loss | -1500 < p < -1440
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7737 entries)
$\frac{\sigma}{\mu}$ = 11.79± 0.18 %
μ = 475.0± 0.8
σ = 56.0± 0.8
WF (7737 entries)
$\frac{\sigma}{\mu}$ = 12.51± 0.20 %
μ = 481.5± 0.9
σ = 60.2± 0.8

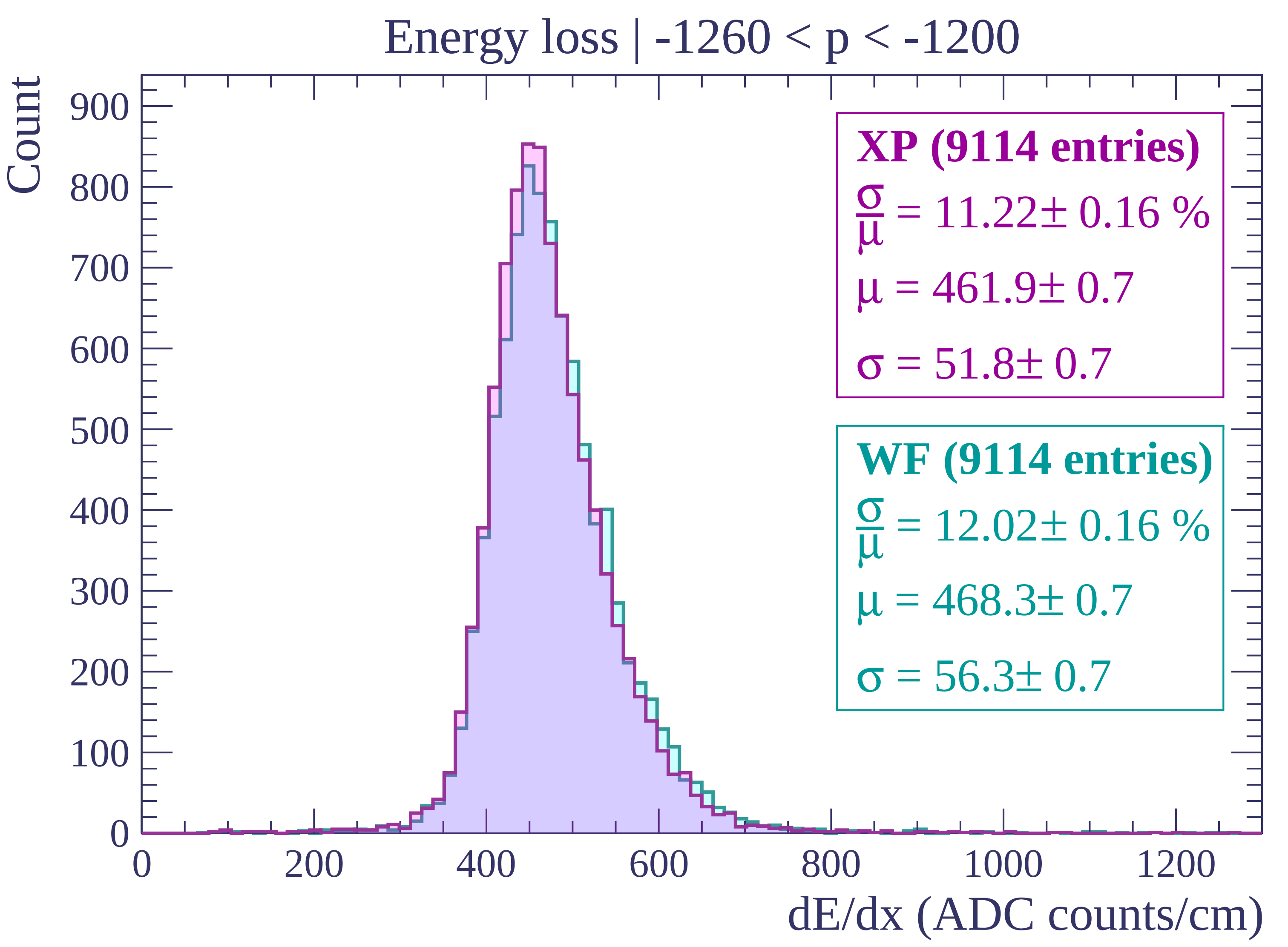


Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
XP (8118 entries)
$\frac{\sigma}{\mu}$ = 11.55± 0.18 %
μ = 470.1± 0.8
σ = 54.3± 0.8
WF (8118 entries)
$\frac{\sigma}{\mu}$ = 11.87± 0.19 %
μ = 476.4± 0.8
σ = 56.6± 0.8

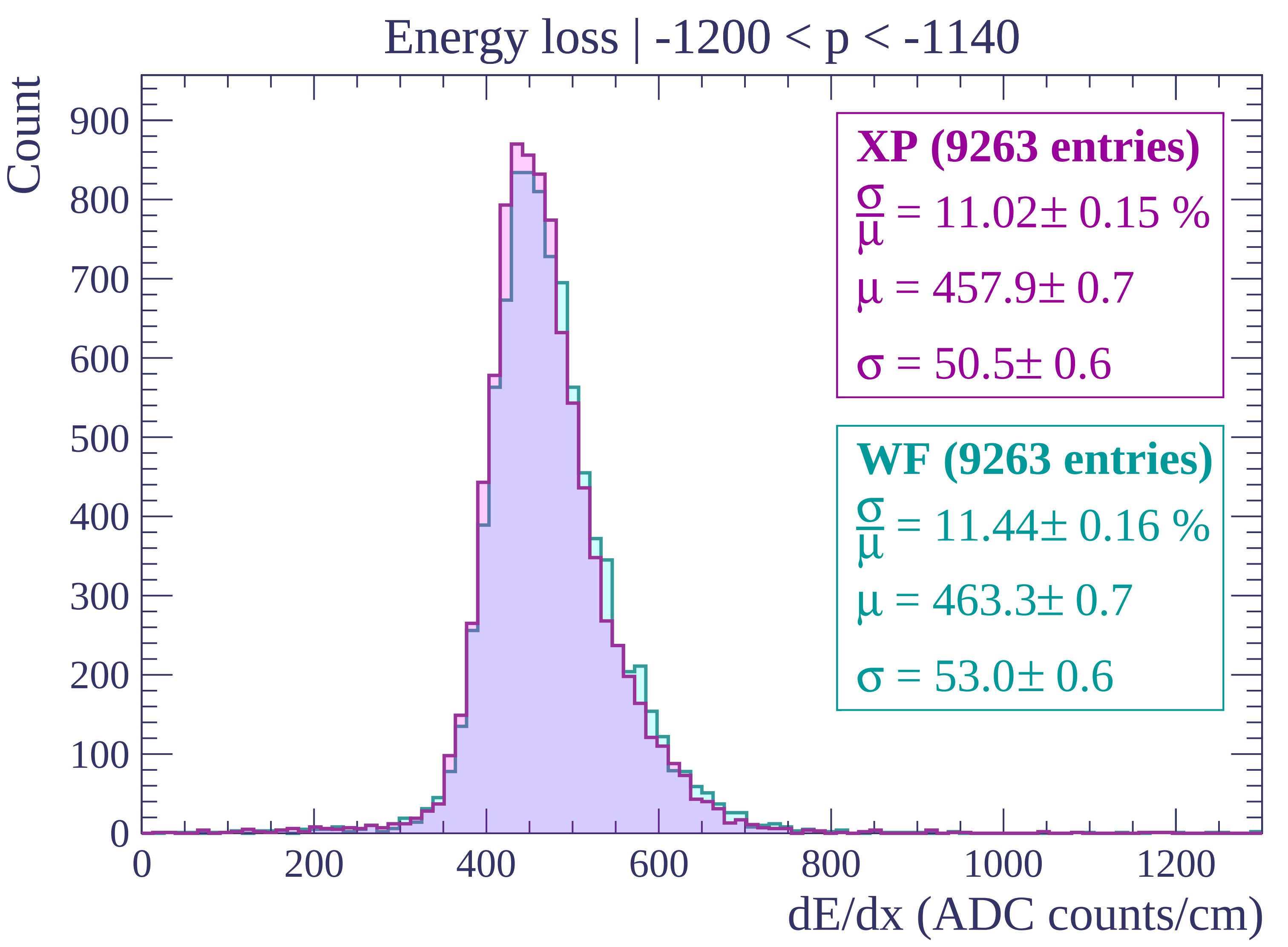

Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (8261 entries)
$\frac{\sigma}{\mu}$ = 10.86± 0.17 %
μ = 465.6± 0.7
σ = 50.5± 0.7
WF (8261 entries)
$\frac{\sigma}{\mu}$ = 11.37± 0.18 %
μ = 471.4± 0.8
σ = 53.6± 0.7



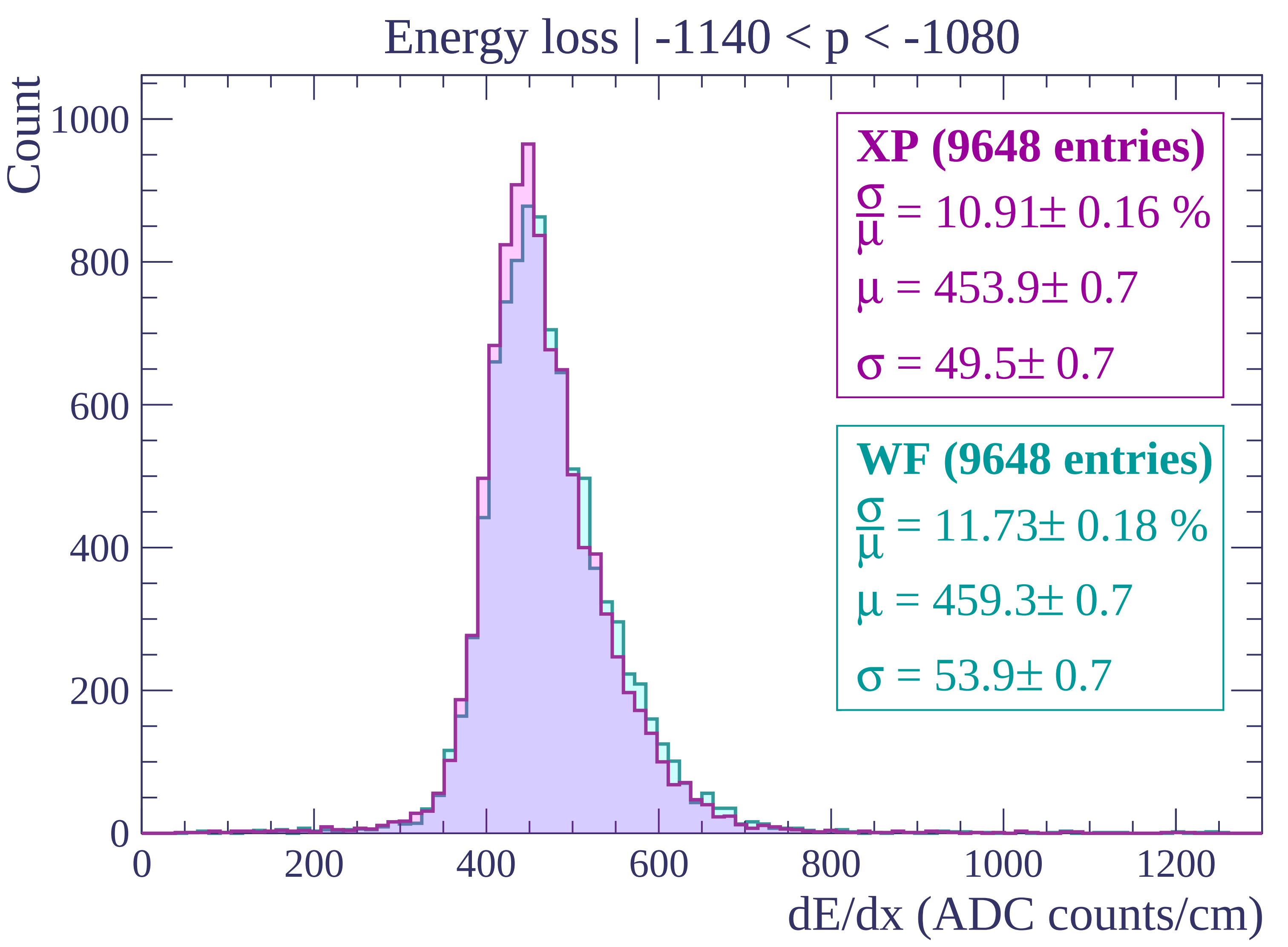
Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (8650 entries)
σ/μ = 10.99± 0.16 %
μ = 462.5± 0.7
σ = 50.8± 0.6
WF (8650 entries)
σ/μ = 11.66± 0.17 %
μ = 468.2± 0.8
σ = 54.6± 0.7

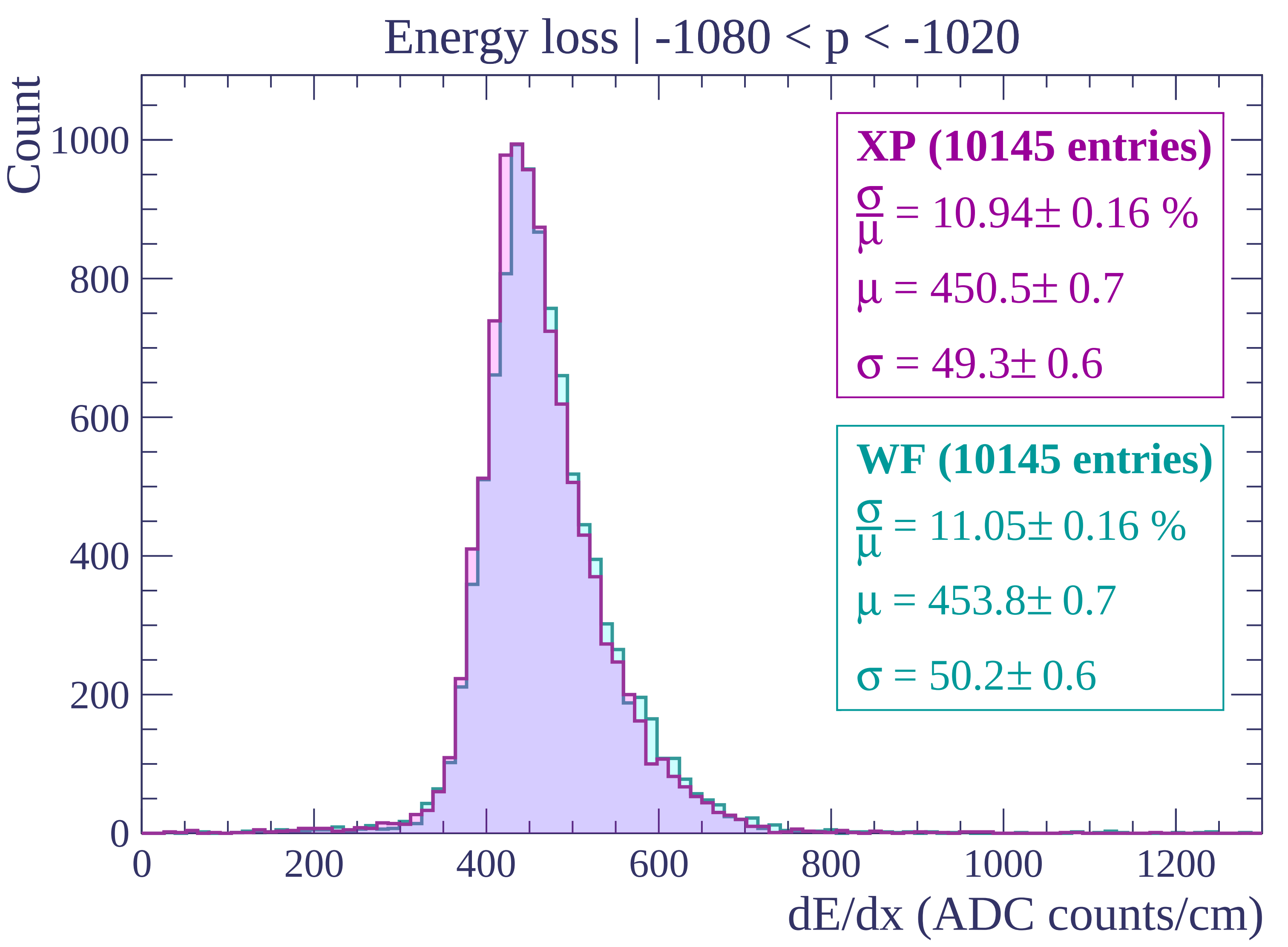


Energy loss | -1260 < p < -1200
Count
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9114 entries)
σ/μ = 11.22± 0.16 %
μ = 461.9± 0.7
σ = 51.8± 0.7
WF (9114 entries)
σ/μ = 12.02± 0.16 %
μ = 468.3± 0.7
σ = 56.3± 0.7

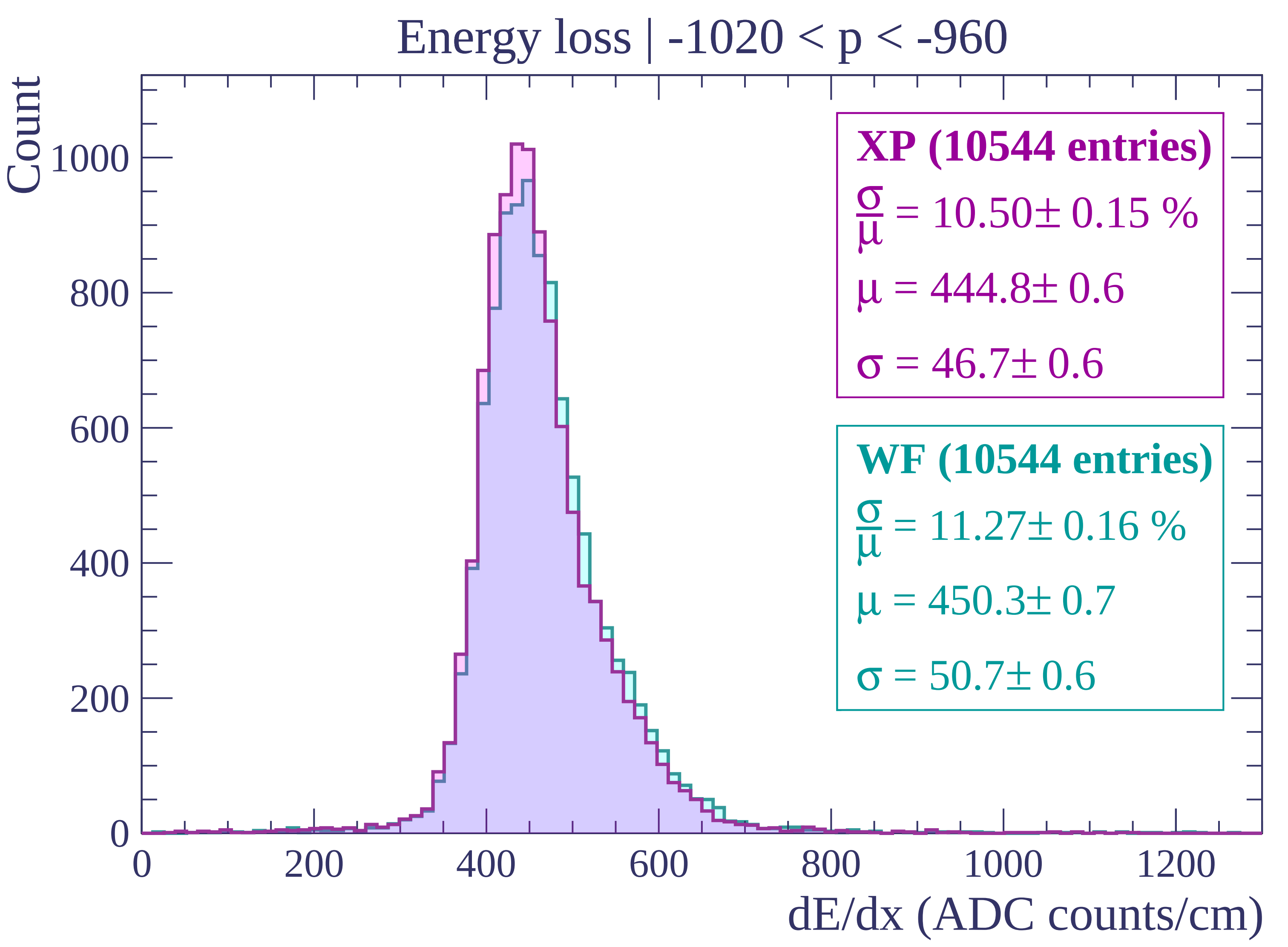


Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
XP (9263 entries)
$\frac{\sigma}{\mu}$ = 11.02± 0.15 %
μ = 457.9± 0.7
σ = 50.5± 0.6
WF (9263 entries)
$\frac{\sigma}{\mu}$ = 11.44± 0.16 %
μ = 463.3± 0.7
σ = 53.0± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

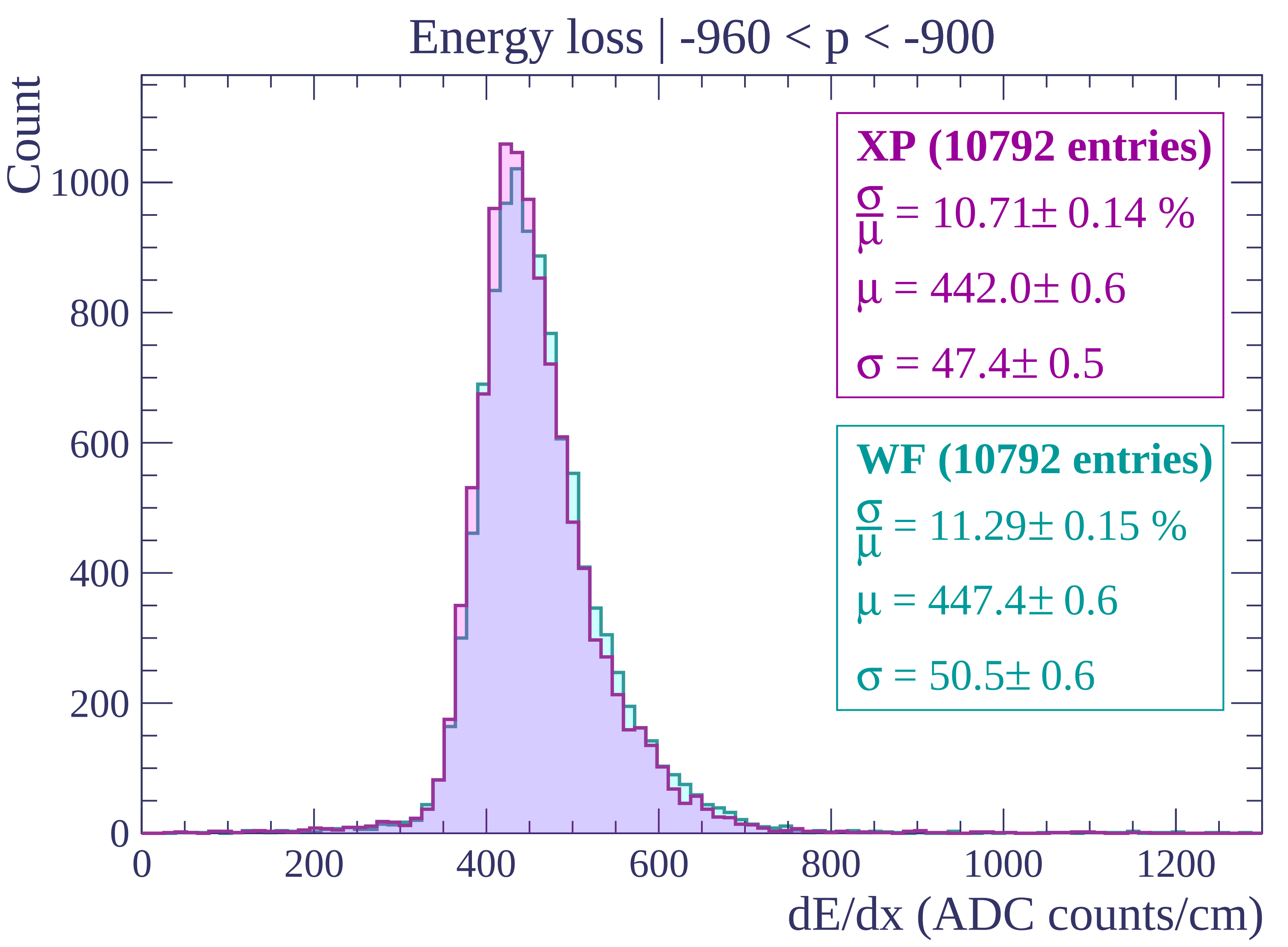

Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
XP (9648 entries)
σ/μ = 10.91± 0.16 %
μ = 453.9± 0.7
σ = 49.5± 0.7
WF (9648 entries)
σ/μ = 11.73± 0.18 %
μ = 459.3± 0.7
σ = 53.9± 0.7



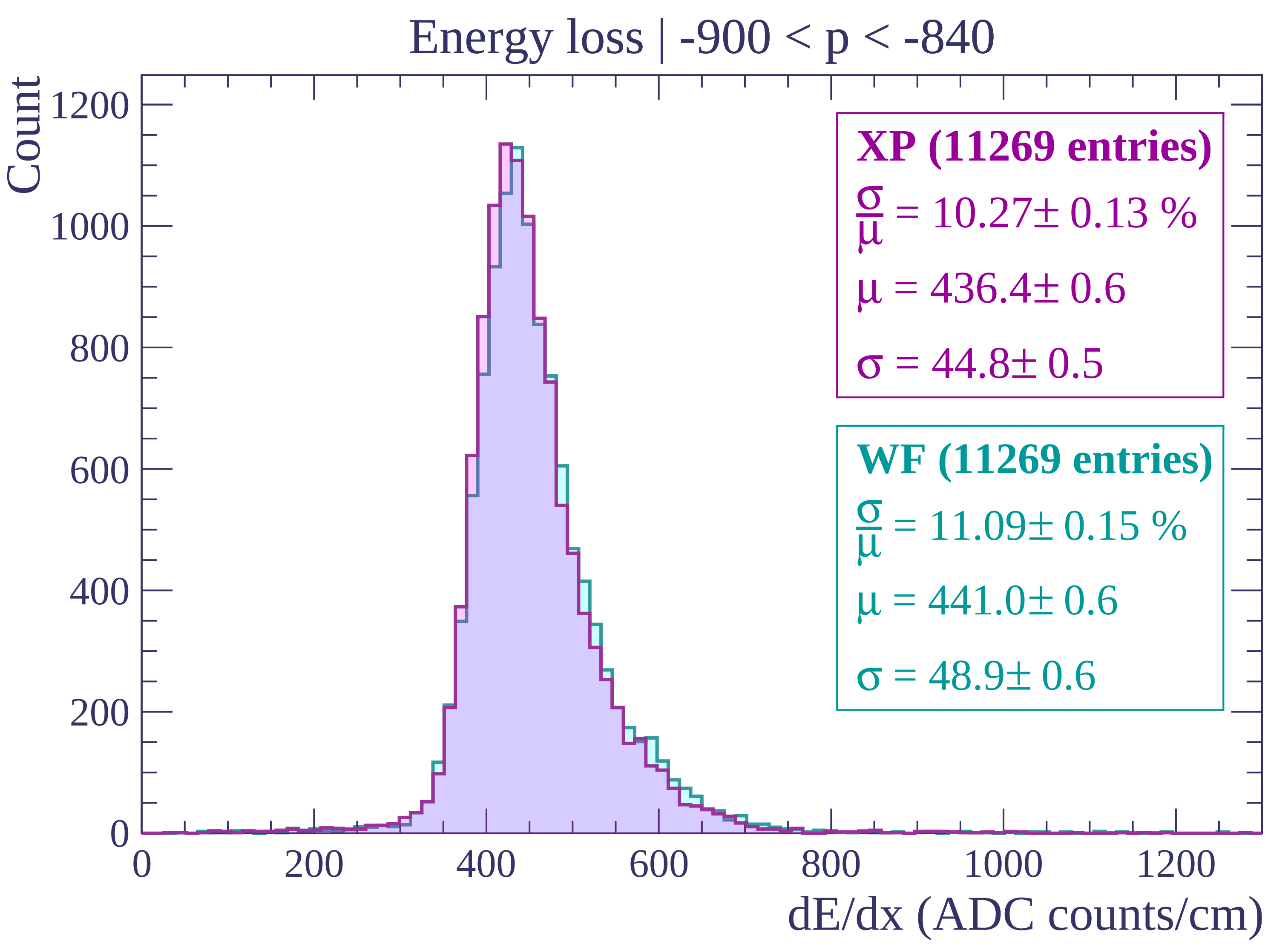
Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10145 entries)
σ/μ = 10.94± 0.16 %
μ = 450.5± 0.7
σ = 49.3± 0.6
WF (10145 entries)
σ/μ = 11.05± 0.16 %
μ = 453.8± 0.7
σ = 50.2± 0.6



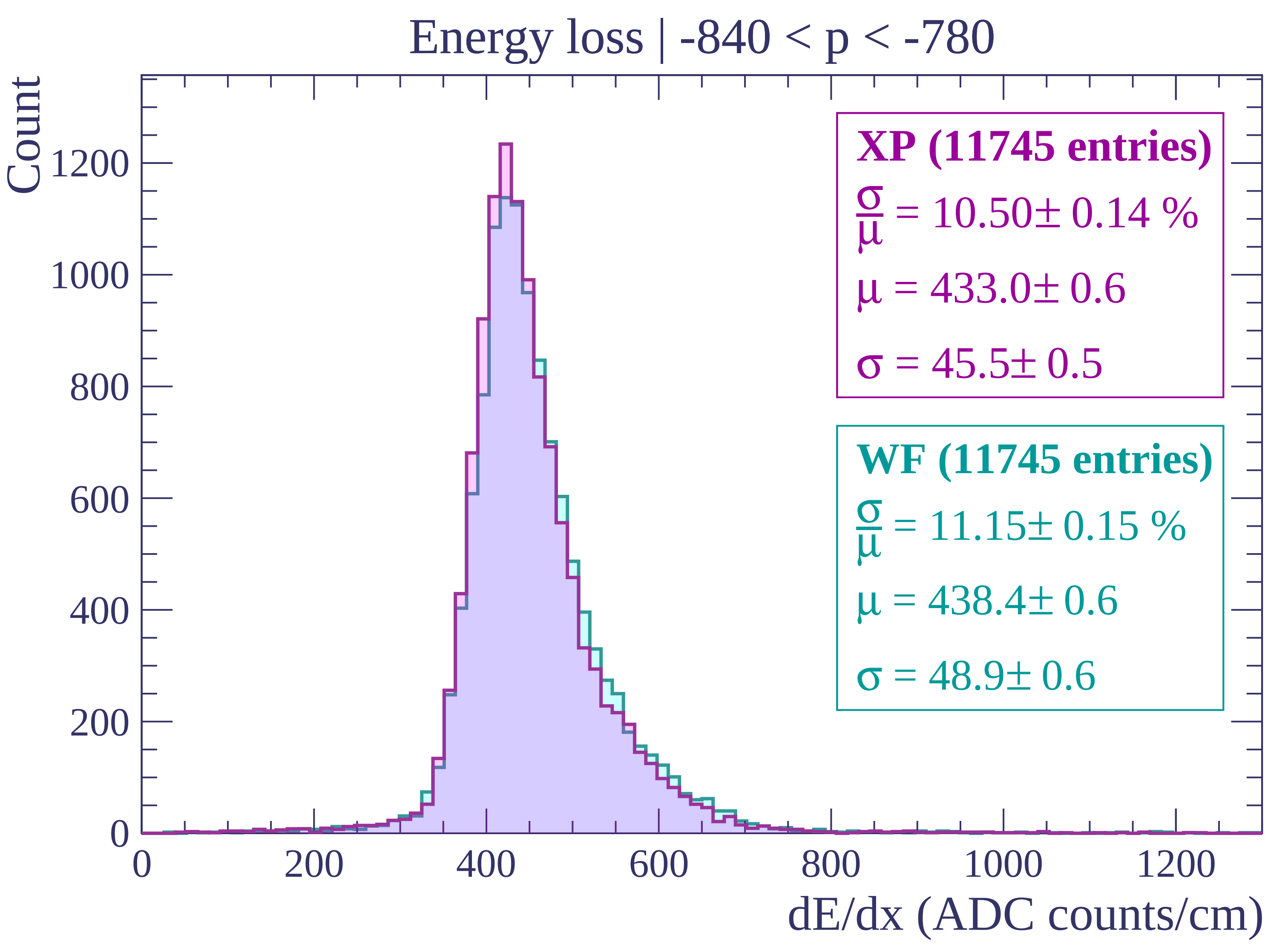
Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10544 entries)
σ/μ = 10.50± 0.15 %
μ = 444.8± 0.6
σ = 46.7± 0.6
WF (10544 entries)
σ/μ = 11.27± 0.16 %
μ = 450.3± 0.7
σ = 50.7± 0.6



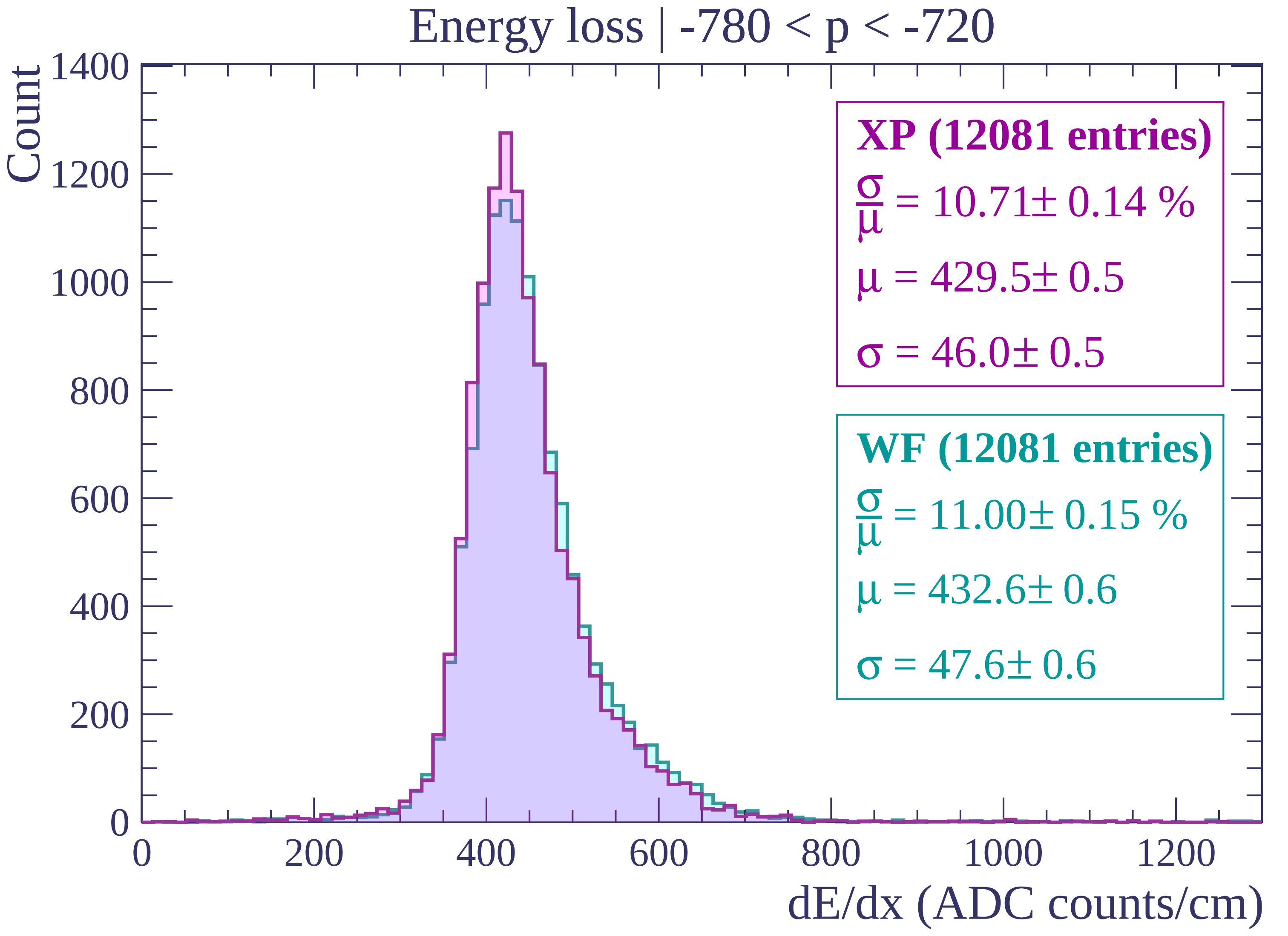
Energy loss | -960 < p < -900
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10792 entries)
σ/μ = 10.71± 0.14 %
μ = 442.0± 0.6
σ = 47.4± 0.5
WF (10792 entries)
σ/μ = 11.29± 0.15 %
μ = 447.4± 0.6
σ = 50.5± 0.6



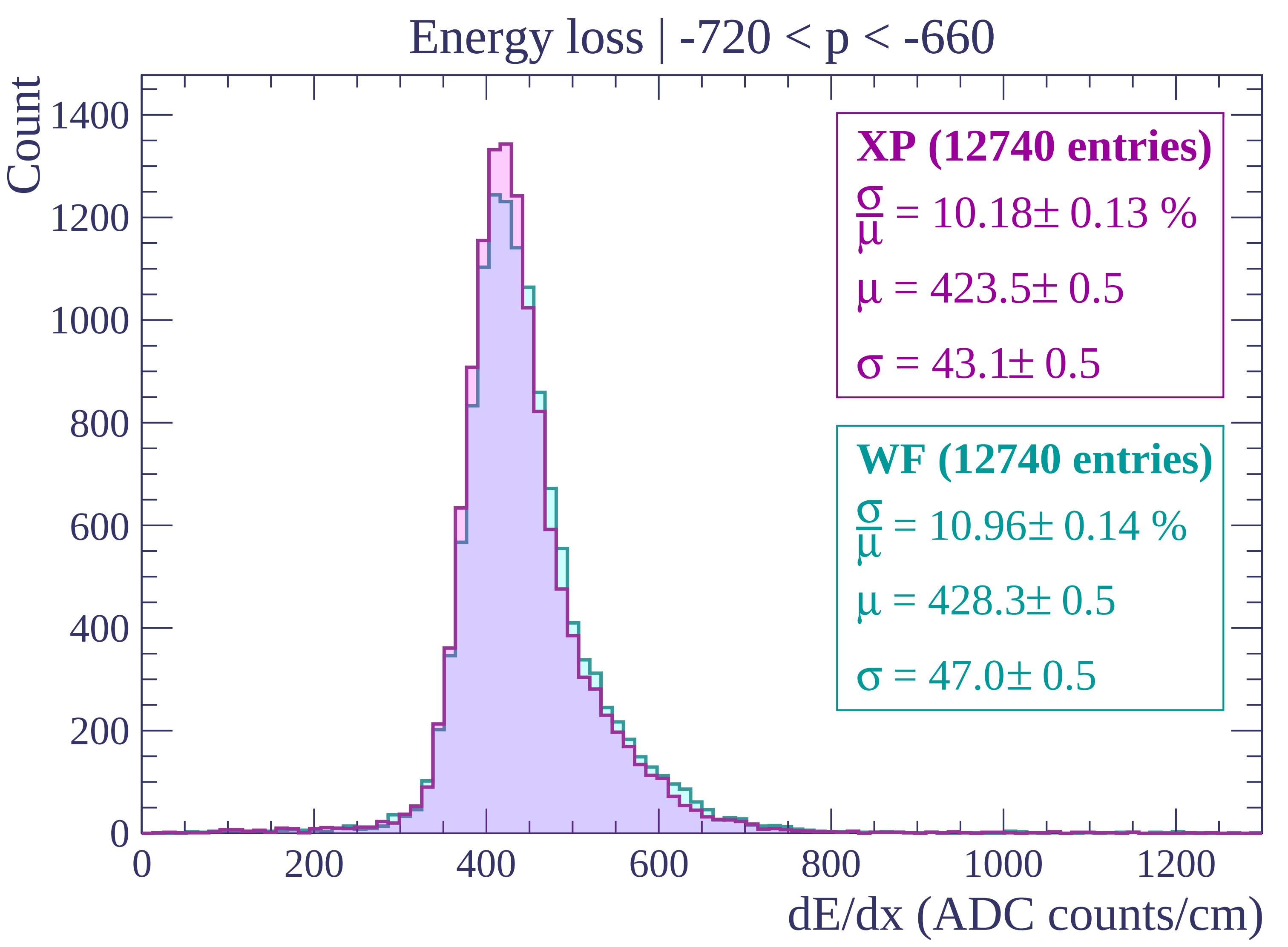
Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11269 entries)
σ/μ = 10.27± 0.13 %
μ = 436.4± 0.6
σ = 44.8± 0.5
WF (11269 entries)
σ/μ = 11.09± 0.15 %
μ = 441.0± 0.6
σ = 48.9± 0.6



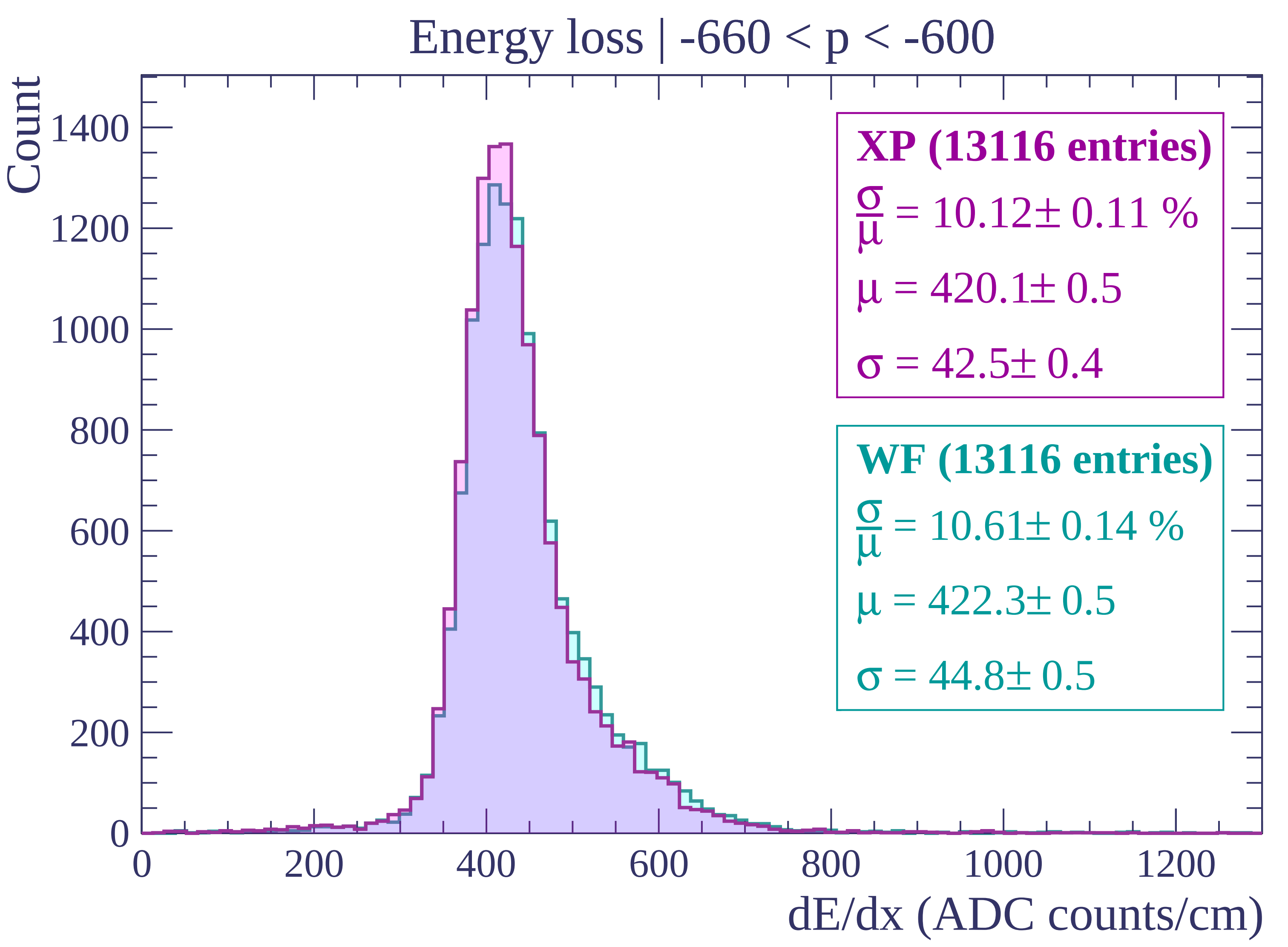
Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11745 entries)
σ/μ = 10.50± 0.14 %
μ = 433.0± 0.6
σ = 45.5± 0.5
WF (11745 entries)
σ/μ = 11.15± 0.15 %
μ = 438.4± 0.6
σ = 48.9± 0.6



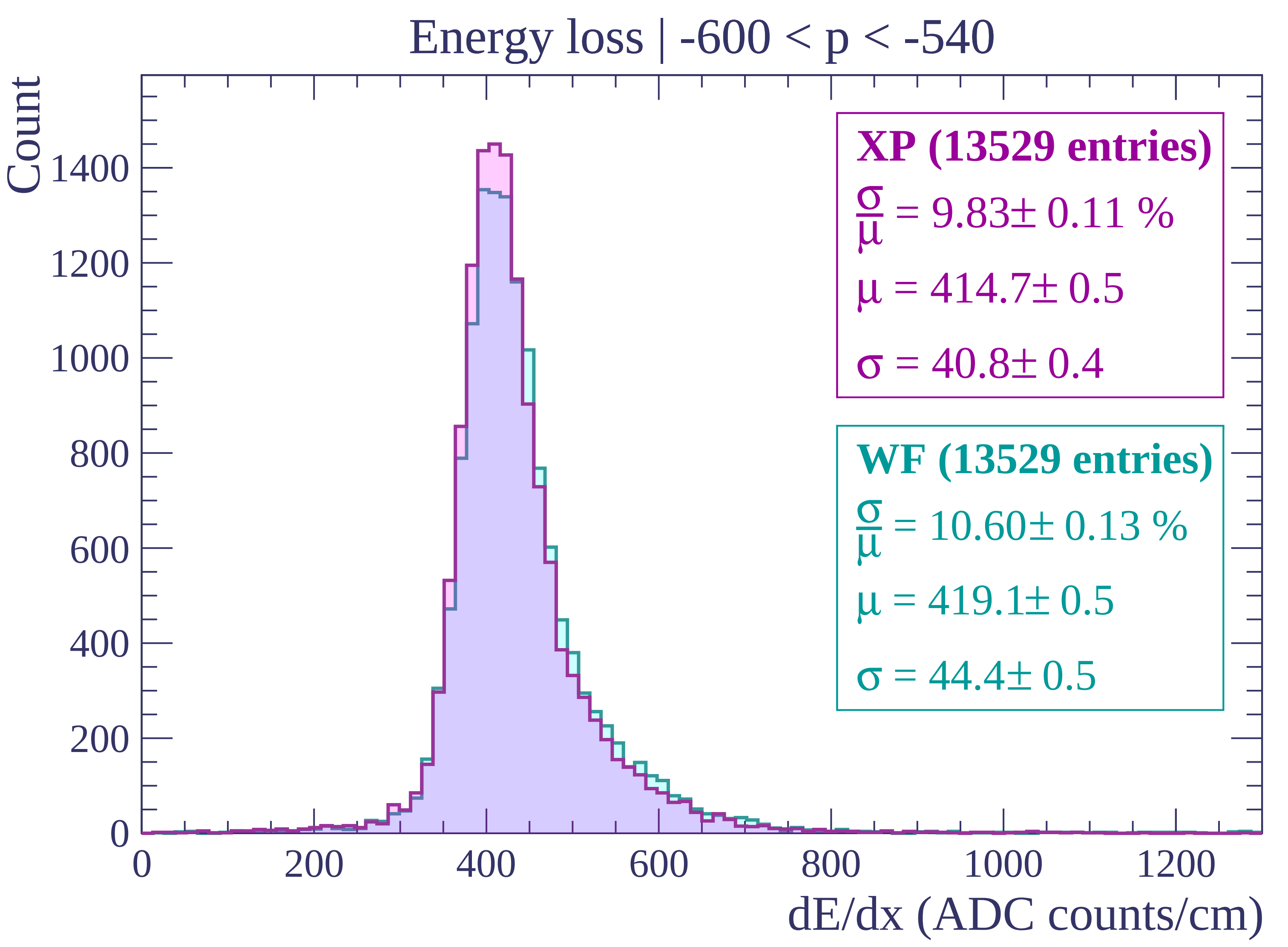
Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
XP (12081 entries)
σ/μ = 10.71± 0.14 %
μ = 429.5± 0.5
σ = 46.0± 0.5
WF (12081 entries)
σ/μ = 11.00± 0.15 %
μ = 432.6± 0.6
σ = 47.6± 0.6



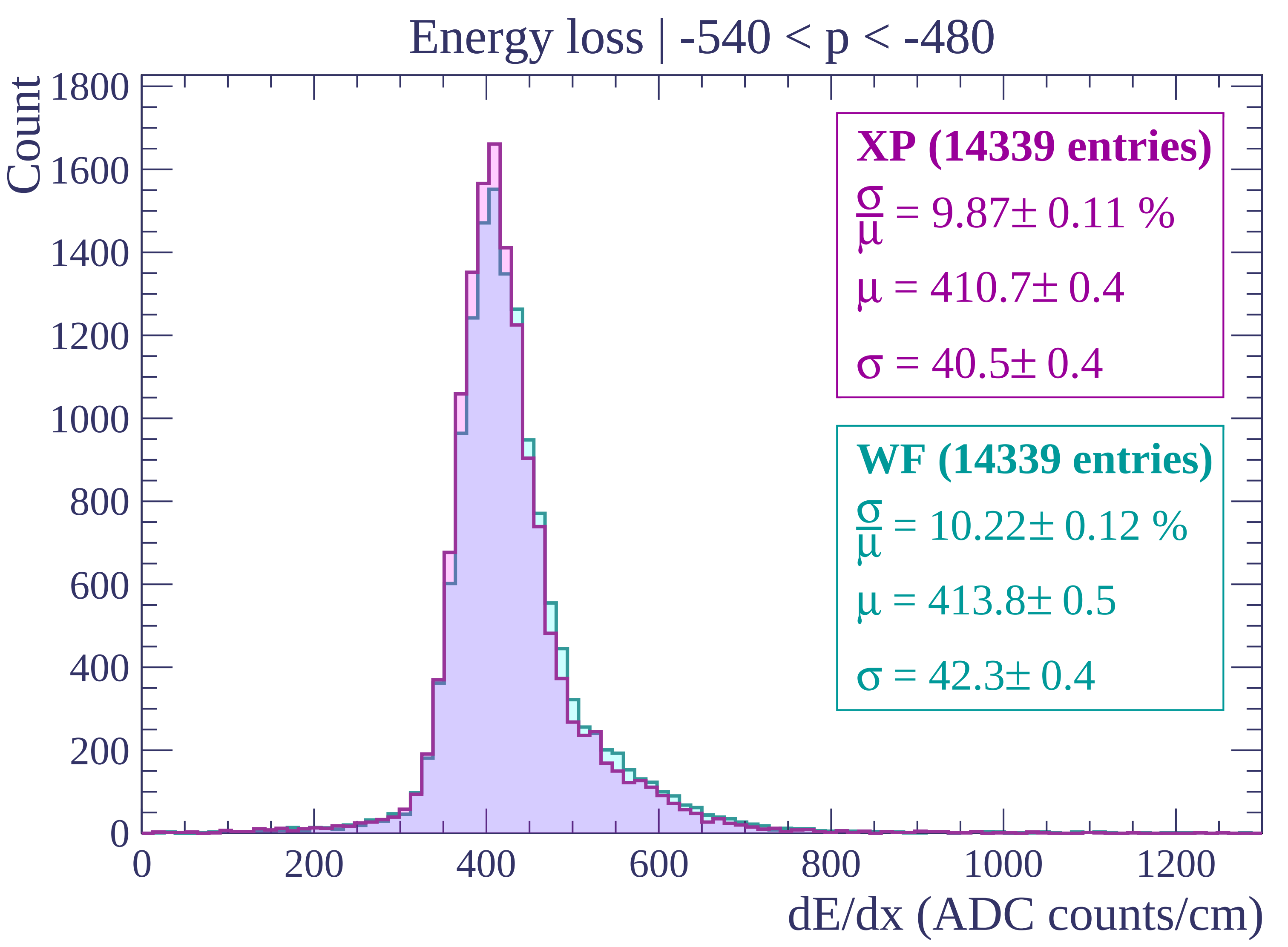
Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
XP (12740 entries)
σ/μ = 10.18± 0.13 %
μ = 423.5± 0.5
σ = 43.1± 0.5
WF (12740 entries)
σ/μ = 10.96± 0.14 %
μ = 428.3± 0.5
σ = 47.0± 0.5
0
200
400
600
800
1000
1200
1400



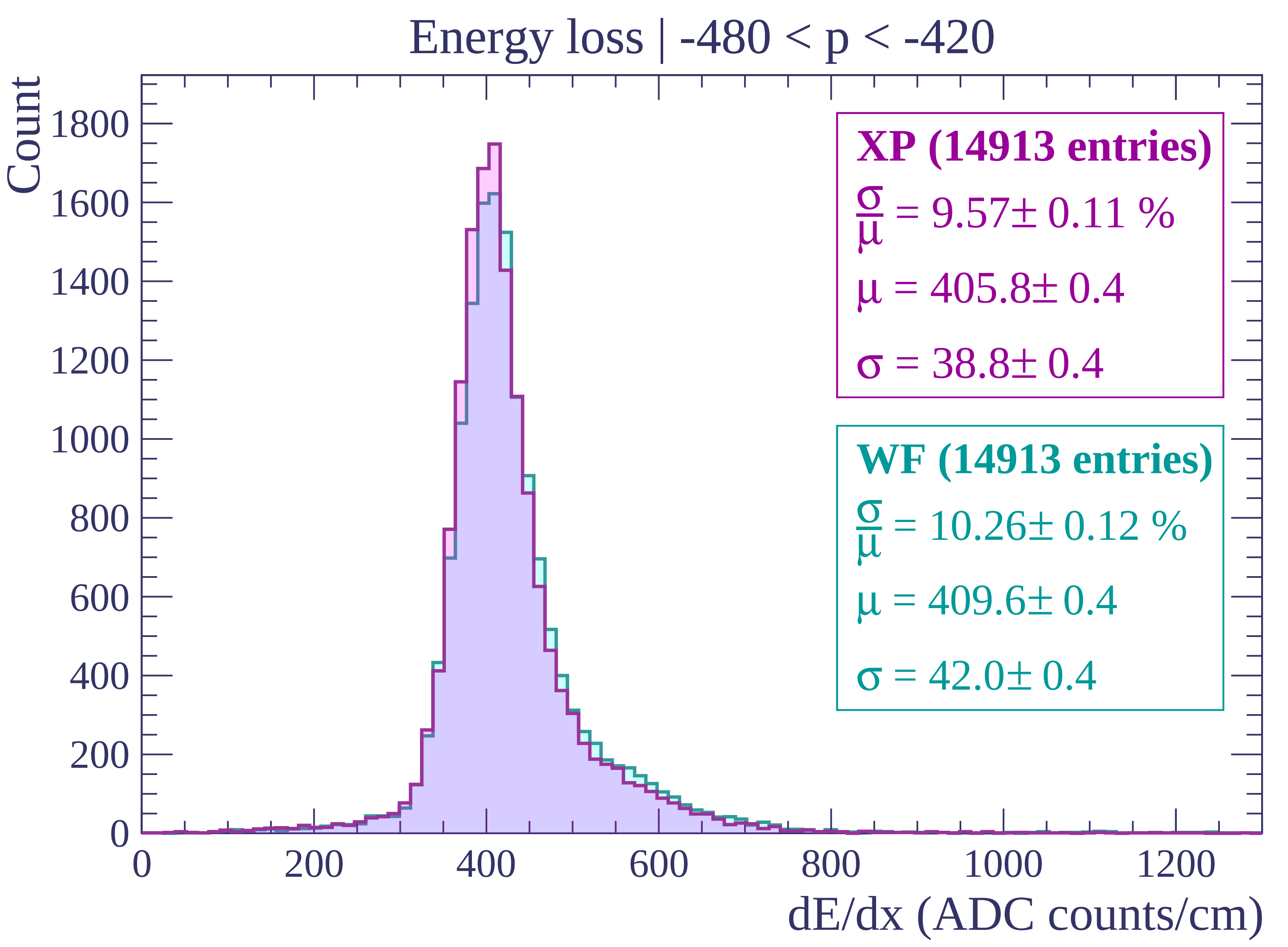
Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
XP (13116 entries)
$\frac{\sigma}{\mu}$ = 10.12± 0.11 %
μ = 420.1± 0.5
σ = 42.5± 0.4
WF (13116 entries)
$\frac{\sigma}{\mu}$ = 10.61± 0.14 %
μ = 422.3± 0.5
σ = 44.8± 0.5
0
200
400
600
800
1000
1200
1400



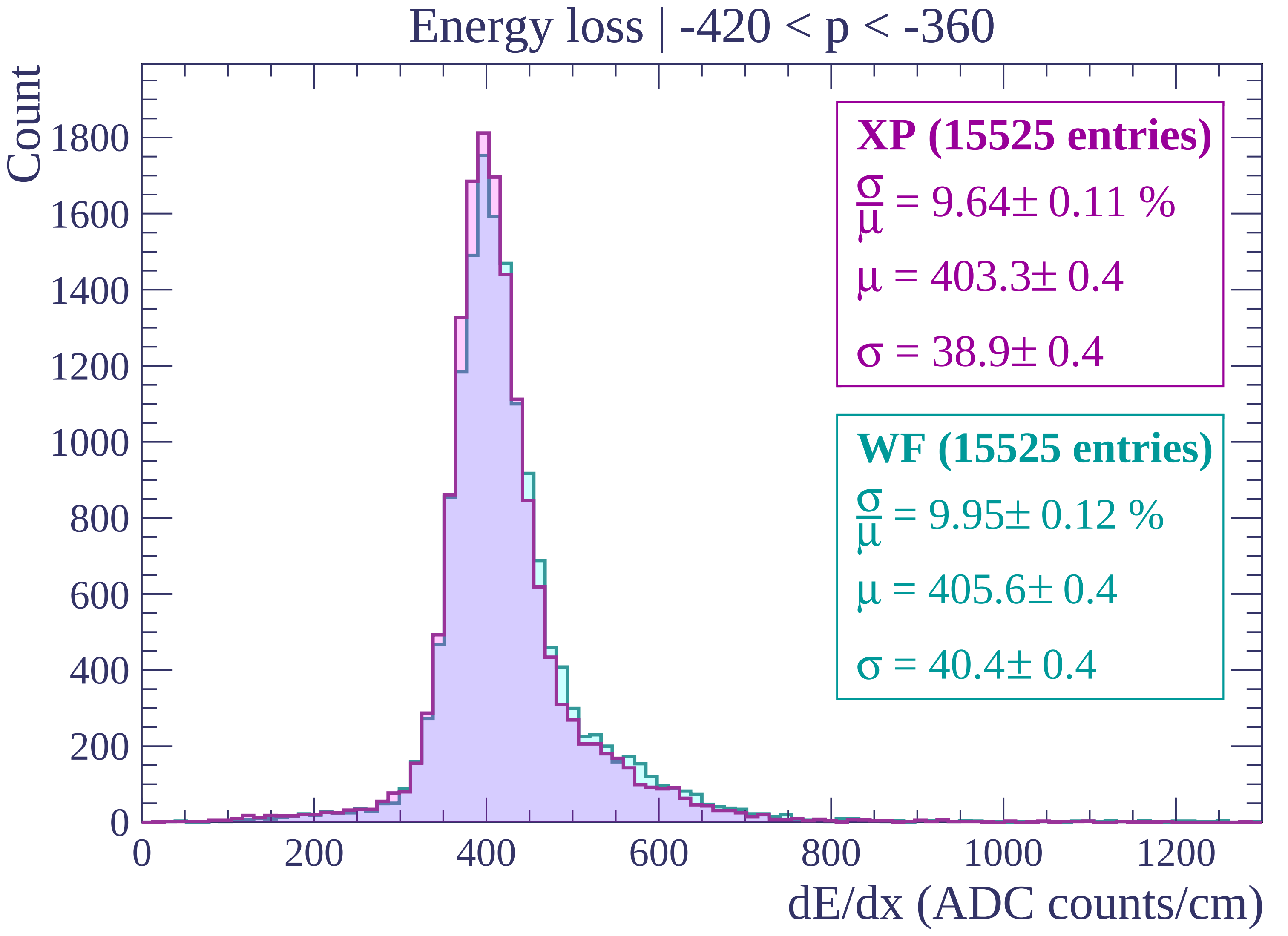
Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
0
200
400
600
800
1000
1200
XP (13529 entries)
σ/μ = 9.83± 0.11 %
μ = 414.7± 0.5
σ = 40.8± 0.4
WF (13529 entries)
σ/μ = 10.60± 0.13 %
μ = 419.1± 0.5
σ = 44.4± 0.5



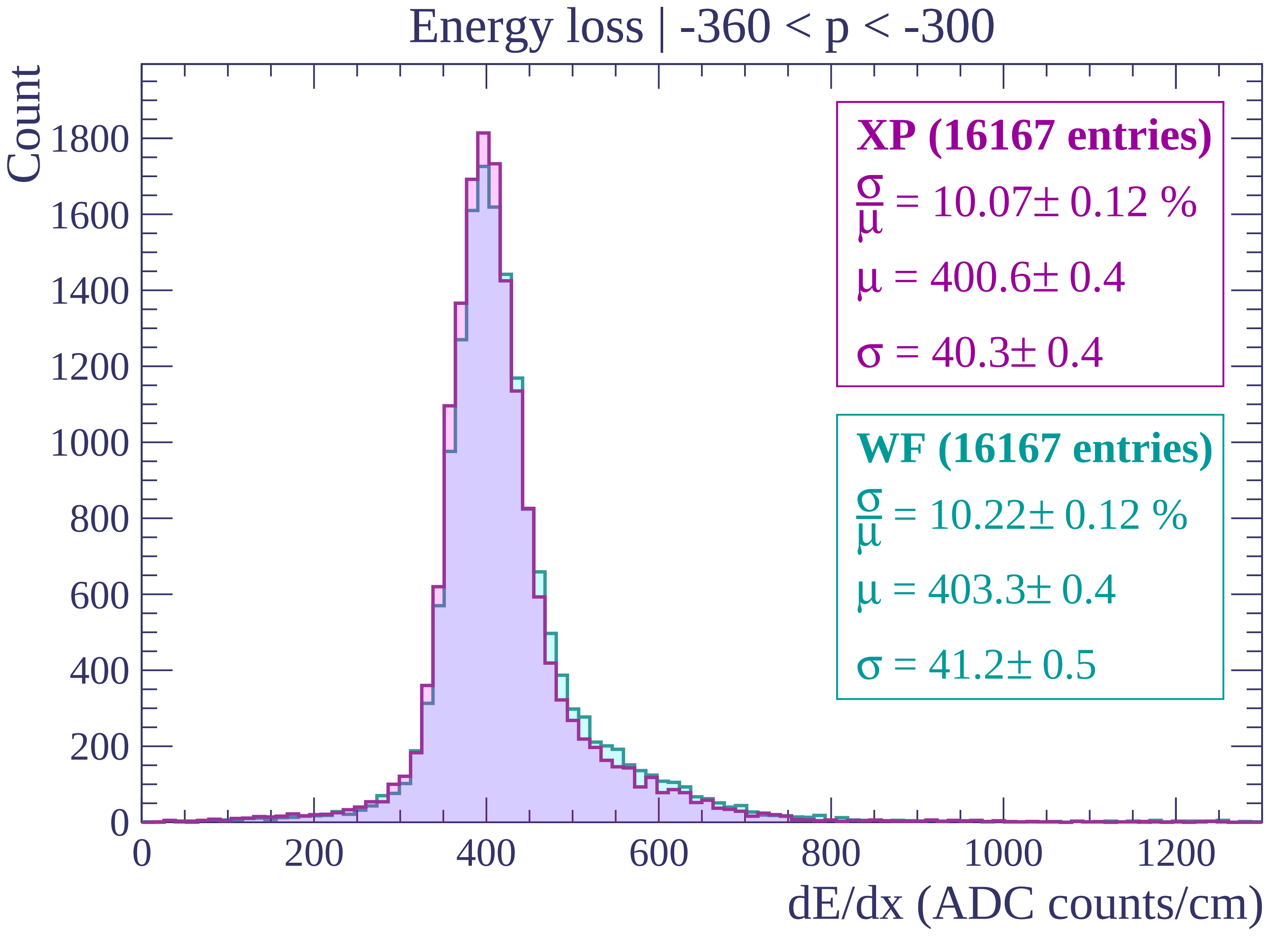
Energy loss | -540 < p < -480
Count
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (14339 entries)
$\frac{\sigma}{\mu}$ = 9.87± 0.11 %
μ = 410.7± 0.4
σ = 40.5± 0.4
WF (14339 entries)
$\frac{\sigma}{\mu}$ = 10.22± 0.12 %
μ = 413.8± 0.5
σ = 42.3± 0.4



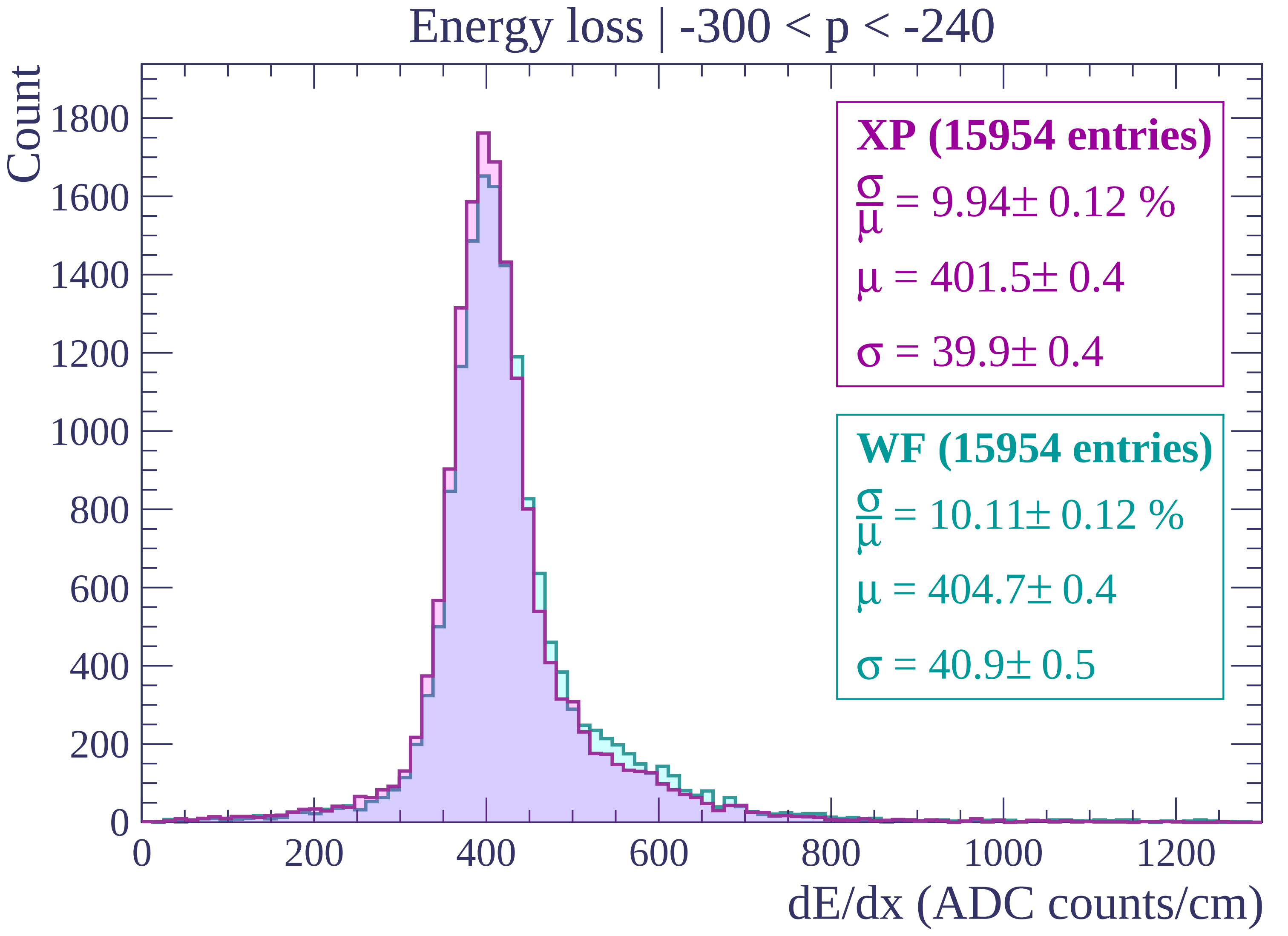
Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
XP (14913 entries)
$\frac{\sigma}{\mu}$ = 9.57± 0.11 %
μ = 405.8± 0.4
σ = 38.8± 0.4
WF (14913 entries)
$\frac{\sigma}{\mu}$ = 10.26± 0.12 %
μ = 409.6± 0.4
σ = 42.0± 0.4

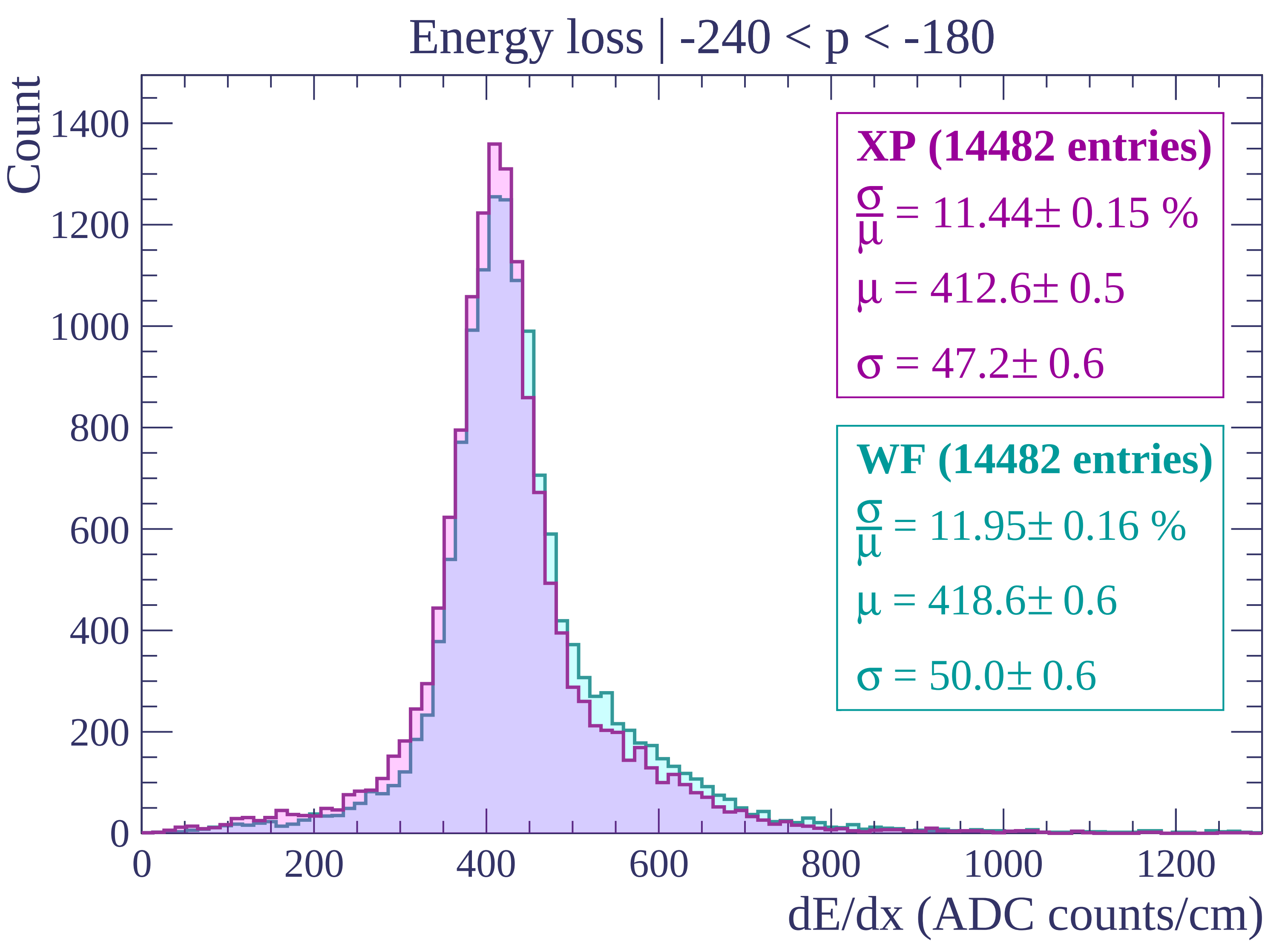


Energy loss | -420 < p < -360
Count
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (15525 entries)
σ/μ = 9.64± 0.11 %
μ = 403.3± 0.4
σ = 38.9± 0.4
WF (15525 entries)
σ/μ = 9.95± 0.12 %
μ = 405.6± 0.4
σ = 40.4± 0.4

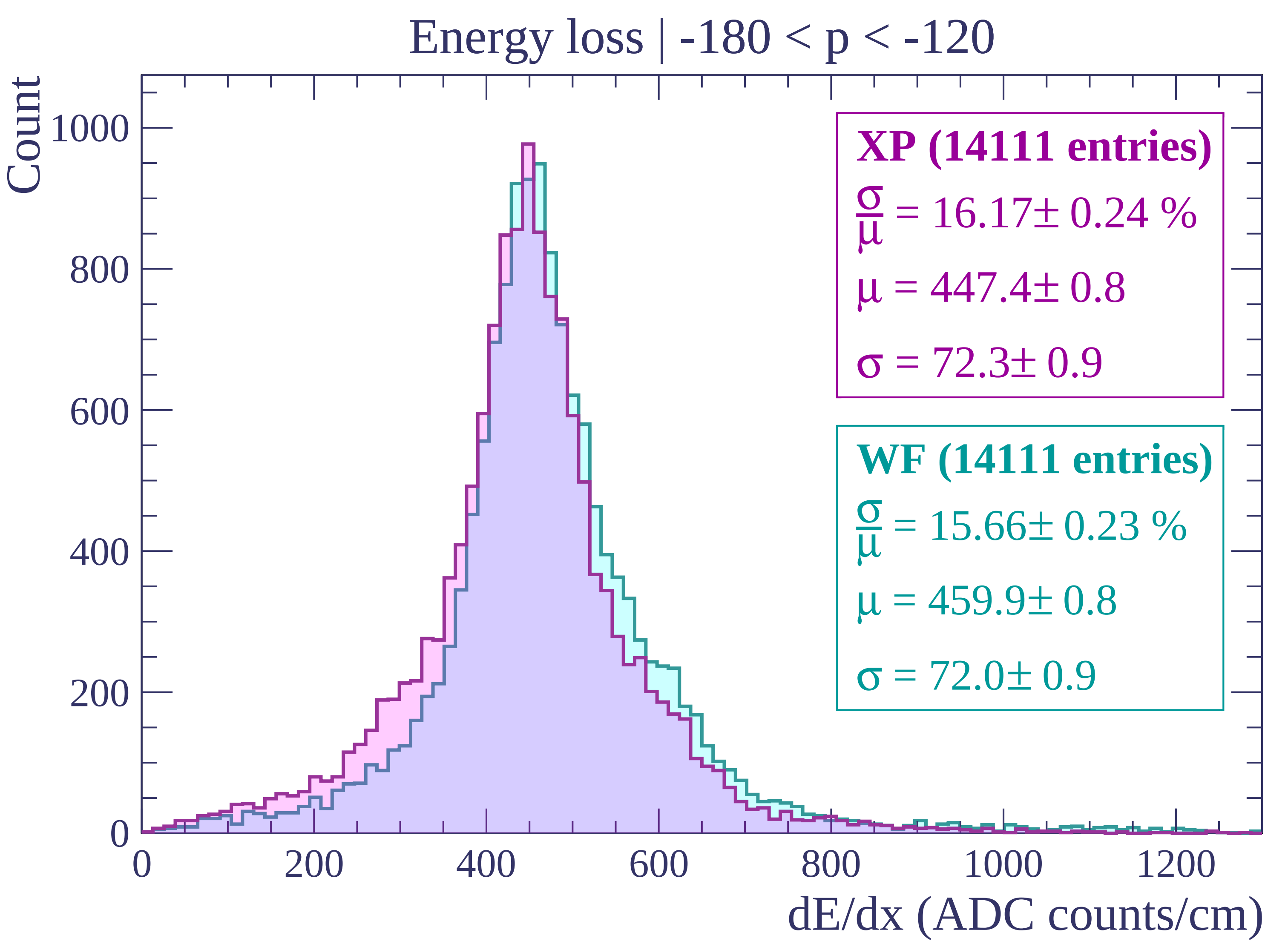


Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
XP (16167 entries)
σ/μ = 10.07± 0.12 %
μ = 400.6± 0.4
σ = 40.3± 0.4
WF (16167 entries)
σ/μ = 10.22± 0.12 %
μ = 403.3± 0.4
σ = 41.2± 0.5

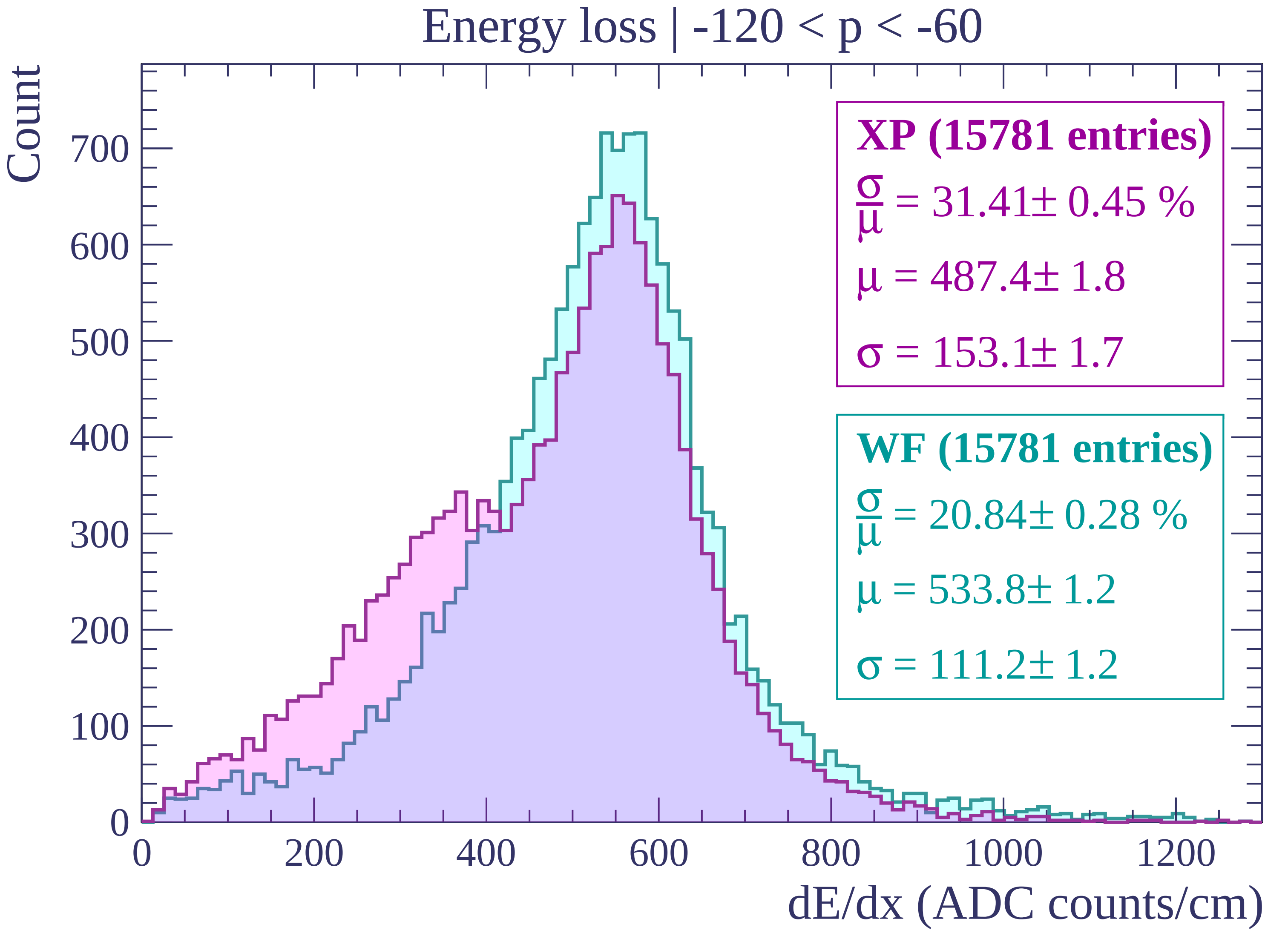

Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
XP (15954 entries)
σ/μ = 9.94± 0.12 %
μ = 401.5± 0.4
σ = 39.9± 0.4
WF (15954 entries)
σ/μ = 10.11± 0.12 %
μ = 404.7± 0.4
σ = 40.9± 0.5
0
200
400
600
800
1000
1200
1400
1600
1800



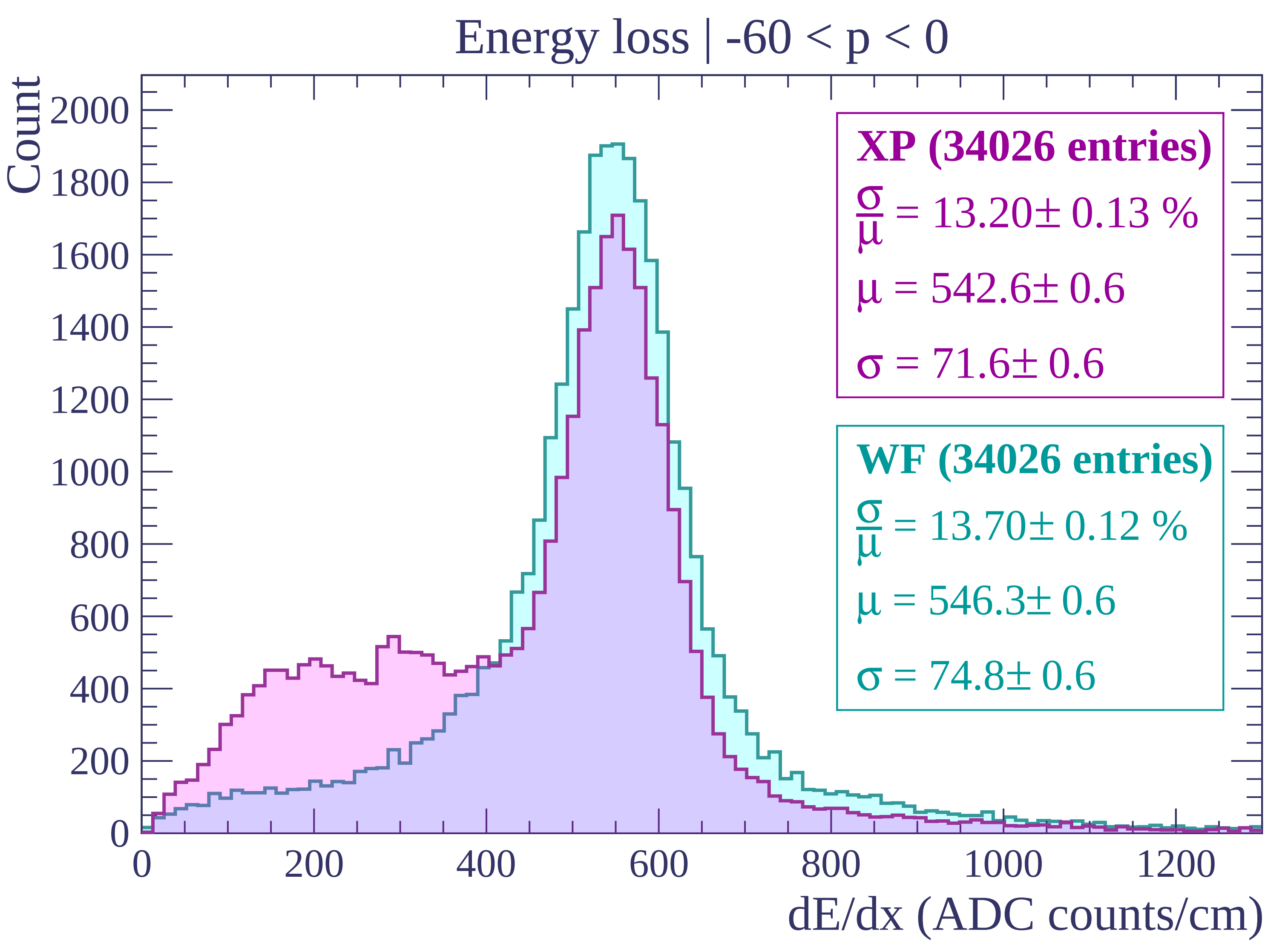
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (14482 entries)
σ/μ = 11.44± 0.15 %
μ = 412.6± 0.5
σ = 47.2± 0.6
WF (14482 entries)
σ/μ = 11.95± 0.16 %
μ = 418.6± 0.6
σ = 50.0± 0.6



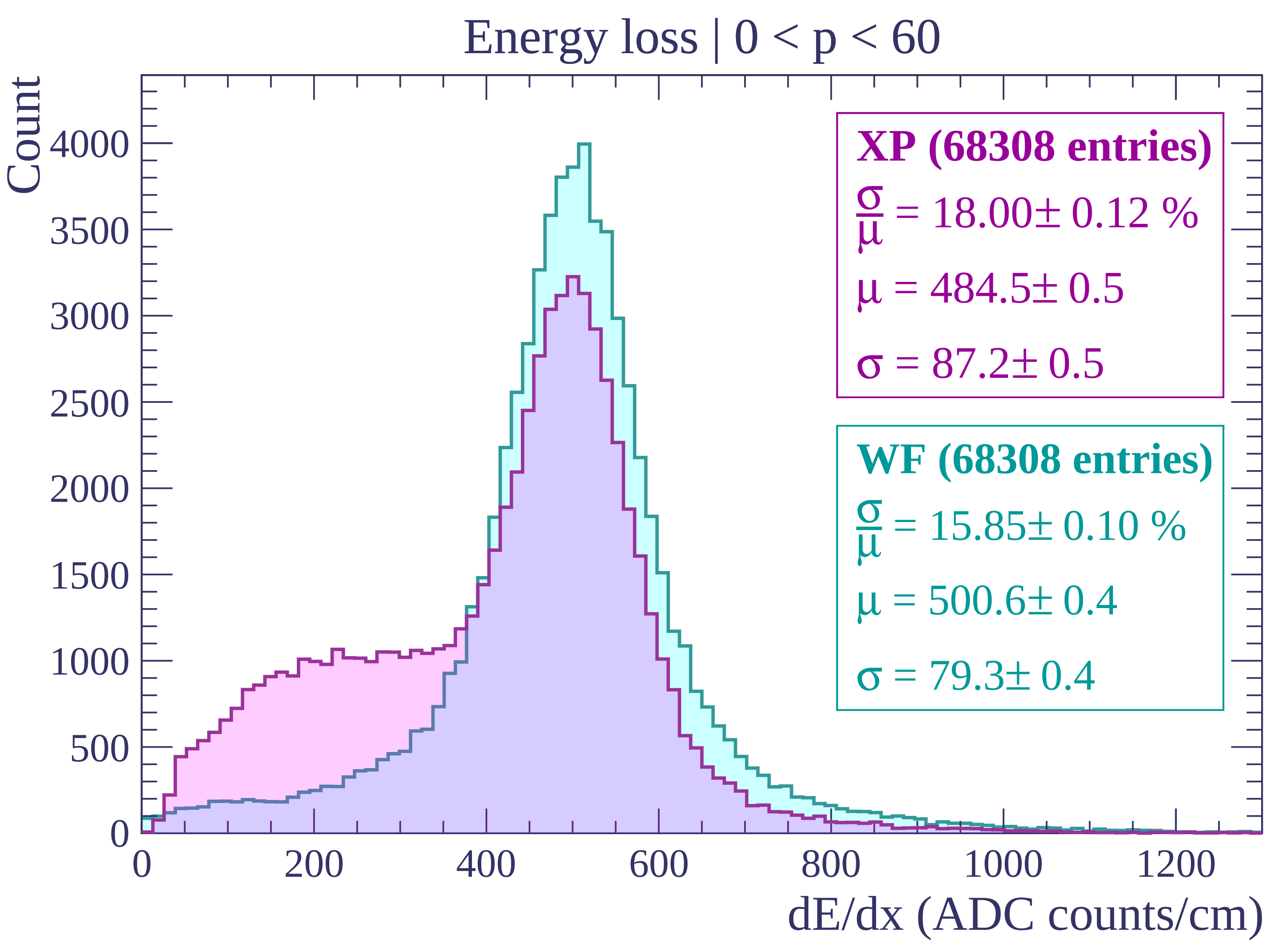
Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (14111 entries)
σ/μ = 16.17± 0.24 %
μ = 447.4± 0.8
σ = 72.3± 0.9
WF (14111 entries)
σ/μ = 15.66± 0.23 %
μ = 459.9± 0.8
σ = 72.0± 0.9



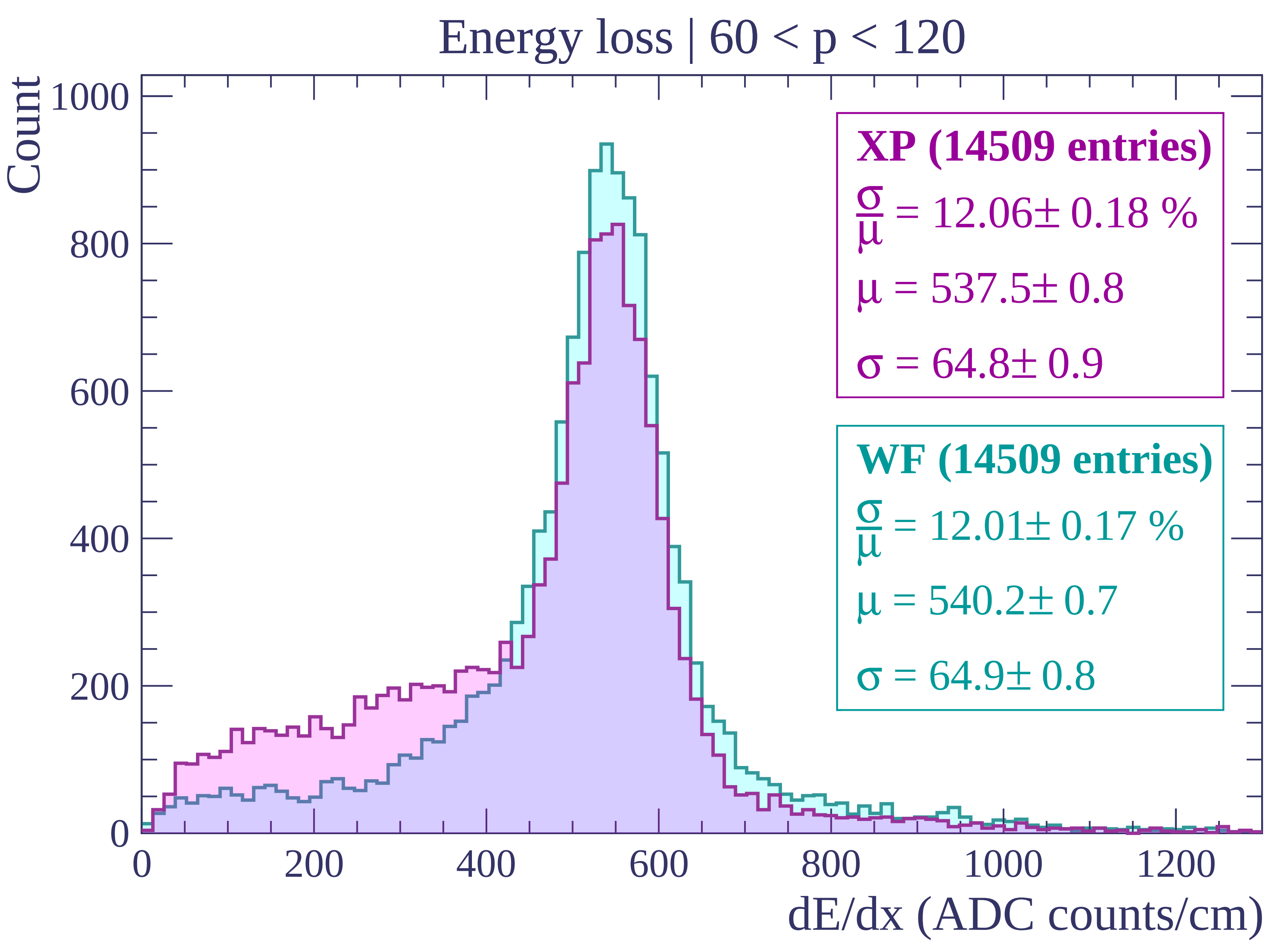
Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
XP (15781 entries)
σ/μ = 31.41± 0.45 %
μ = 487.4± 1.8
σ = 153.1± 1.7
WF (15781 entries)
σ/μ = 20.84± 0.28 %
μ = 533.8± 1.2
σ = 111.2± 1.2
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200



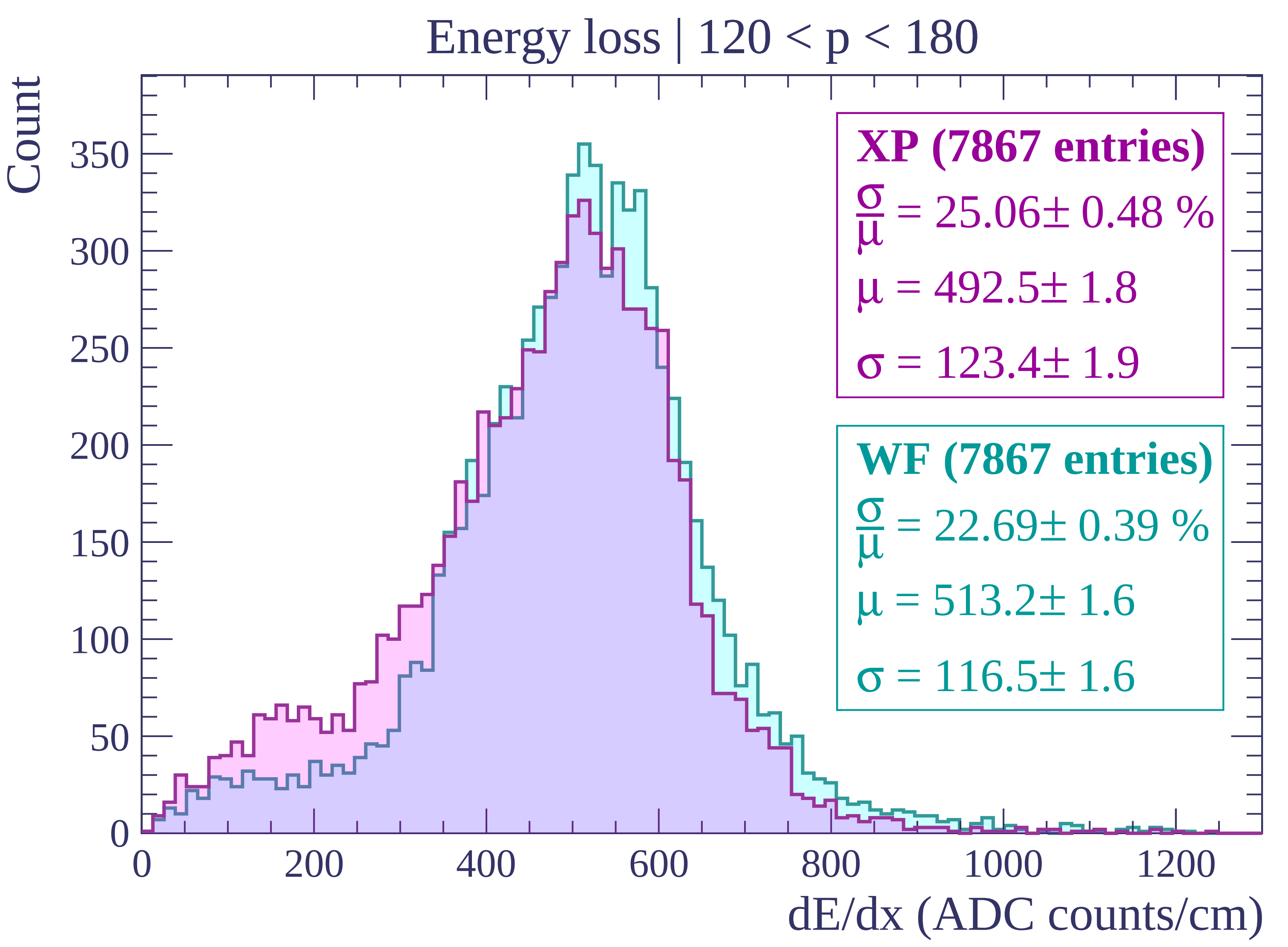
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP (34026 entries)
$\frac{\sigma}{\mu}$ = 13.20± 0.13 %
μ = 542.6± 0.6
σ = 71.6± 0.6
WF (34026 entries)
$\frac{\sigma}{\mu}$ = 13.70± 0.12 %
μ = 546.3± 0.6
σ = 74.8± 0.6



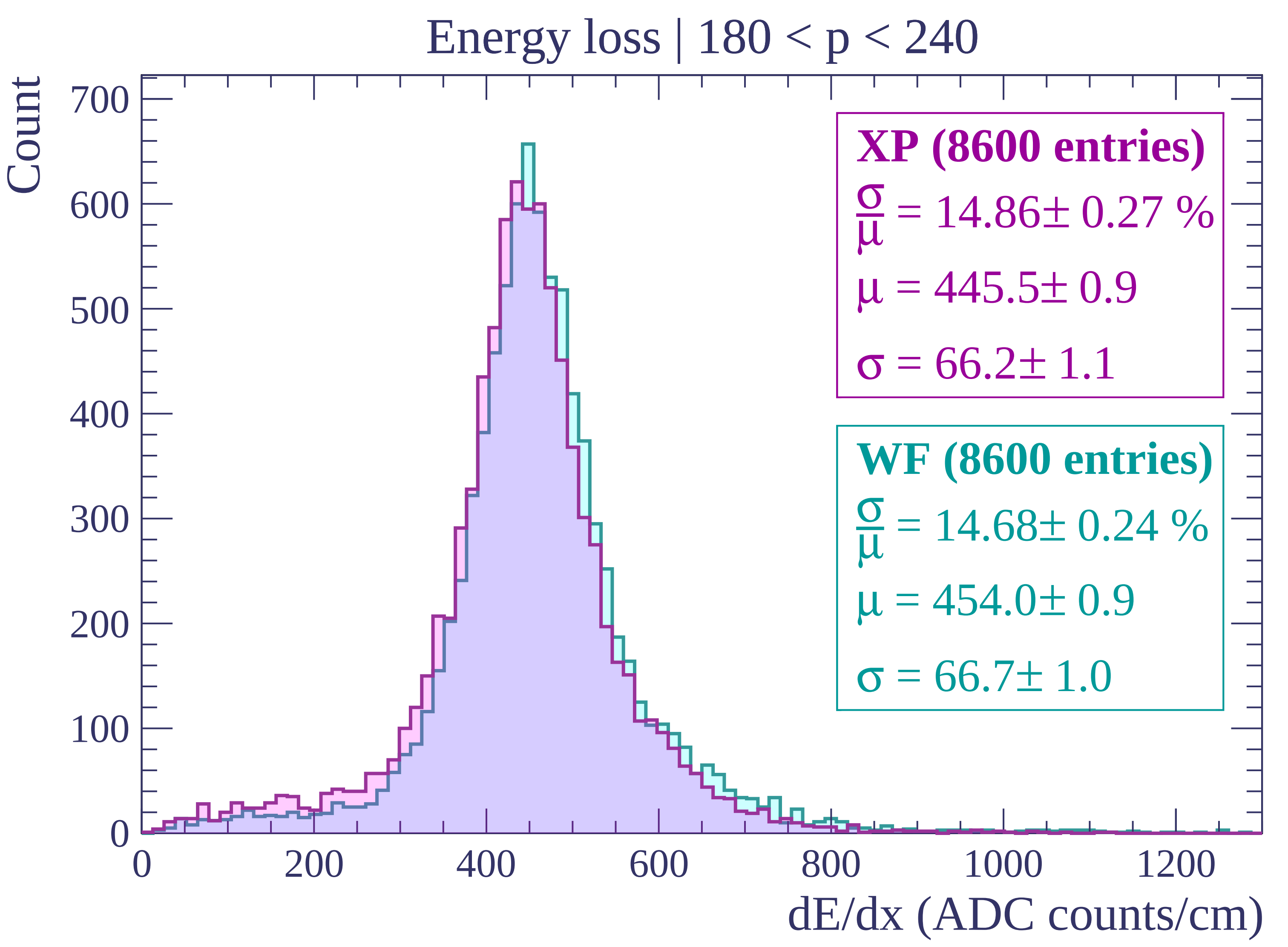
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
4000
XP (68308 entries)
σ/μ = 18.00± 0.12 %
μ = 484.5± 0.5
σ = 87.2± 0.5
WF (68308 entries)
σ/μ = 15.85± 0.10 %
μ = 500.6± 0.4
σ = 79.3± 0.4



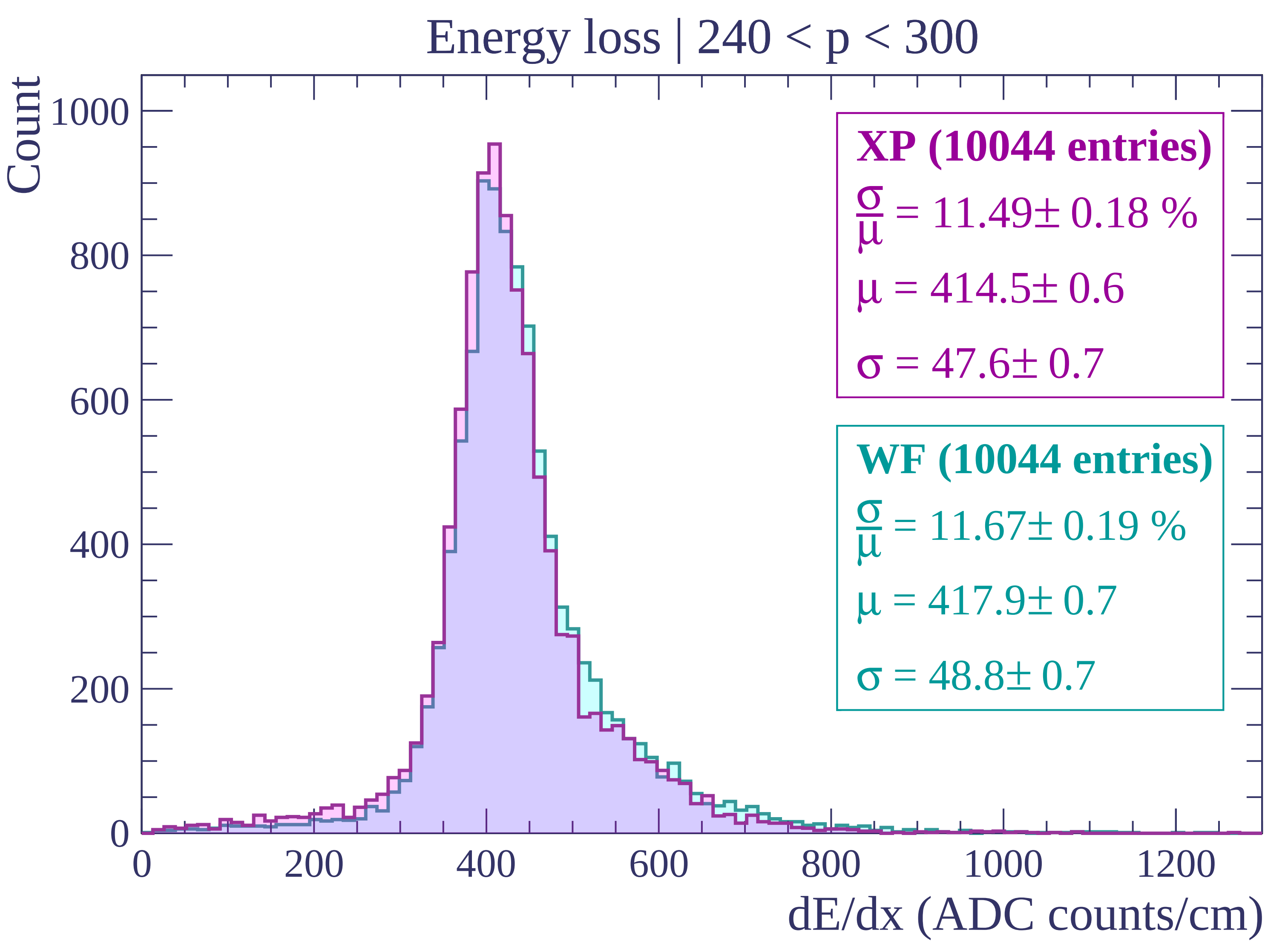
Energy loss | 60 < p < 120
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (14509 entries)
$\frac{\sigma}{\mu}$ = 12.06± 0.18 %
μ = 537.5± 0.8
σ = 64.8± 0.9
WF (14509 entries)
$\frac{\sigma}{\mu}$ = 12.01± 0.17 %
μ = 540.2± 0.7
σ = 64.9± 0.8



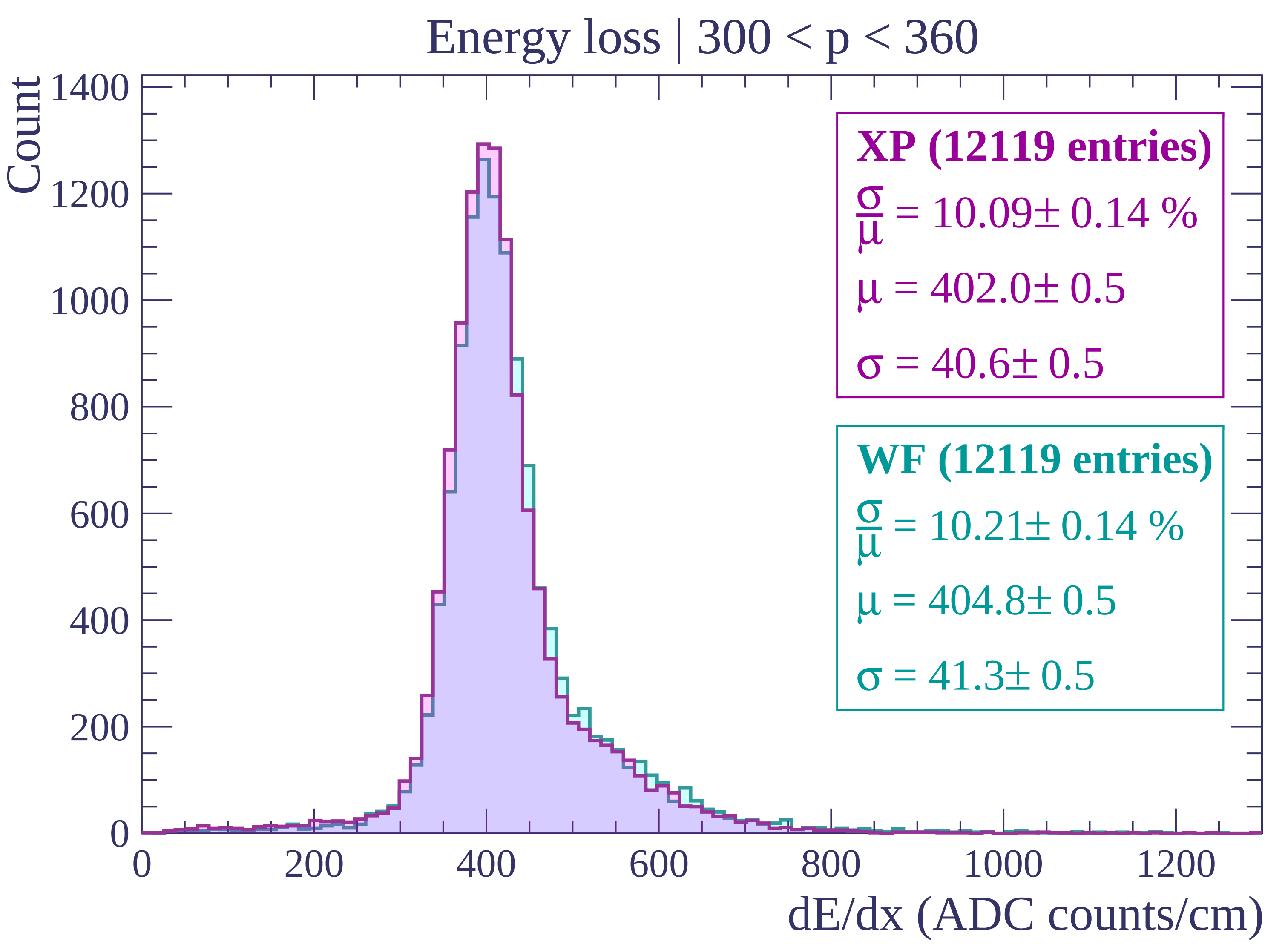
Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (7867 entries)
σ/μ = 25.06± 0.48 %
μ = 492.5± 1.8
σ = 123.4± 1.9
WF (7867 entries)
σ/μ = 22.69± 0.39 %
μ = 513.2± 1.6
σ = 116.5± 1.6



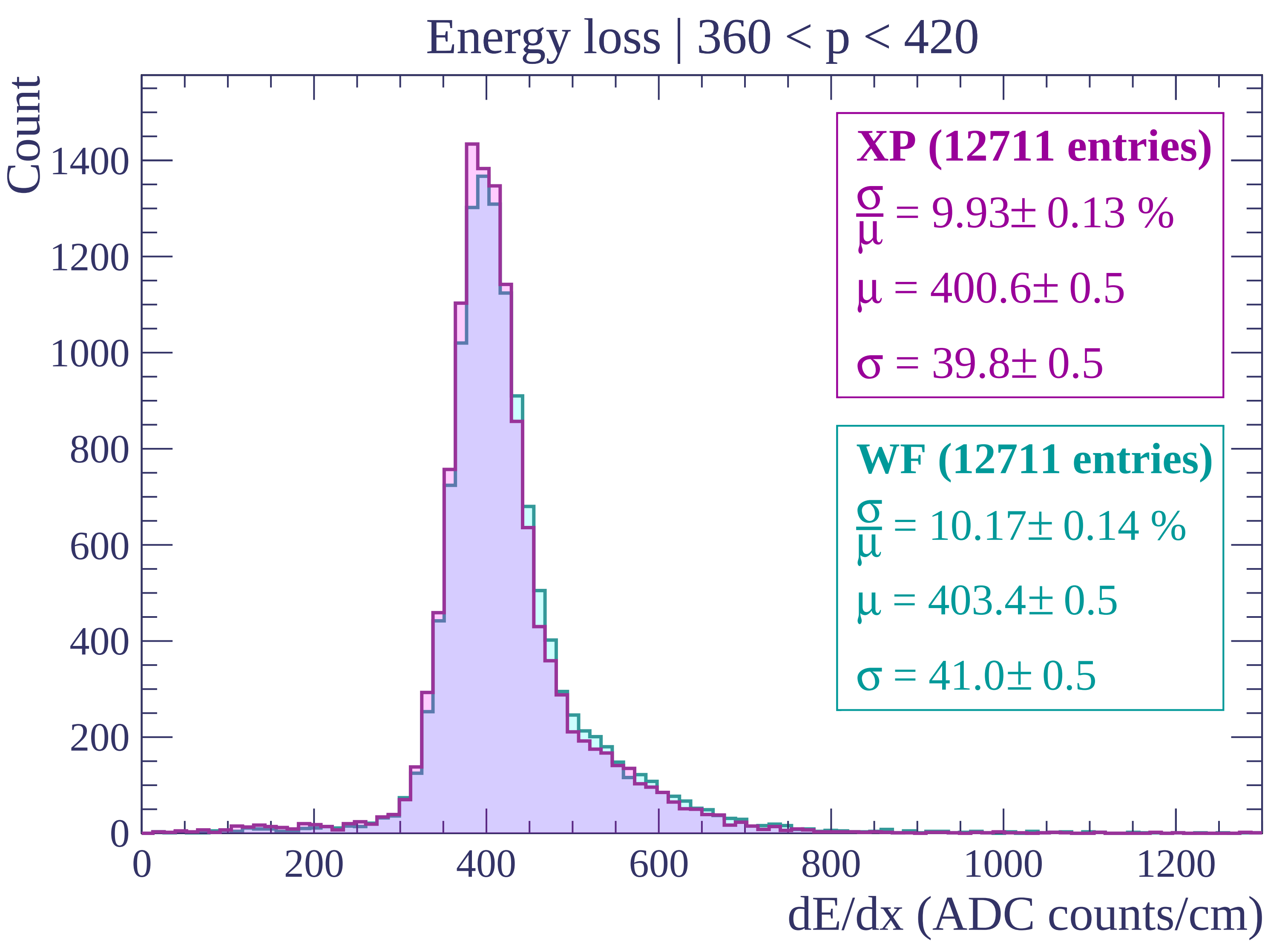
Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (8600 entries)
$\frac{\sigma}{\mu}$ = 14.86± 0.27 %
μ = 445.5± 0.9
σ = 66.2± 1.1
WF (8600 entries)
$\frac{\sigma}{\mu}$ = 14.68± 0.24 %
μ = 454.0± 0.9
σ = 66.7± 1.0

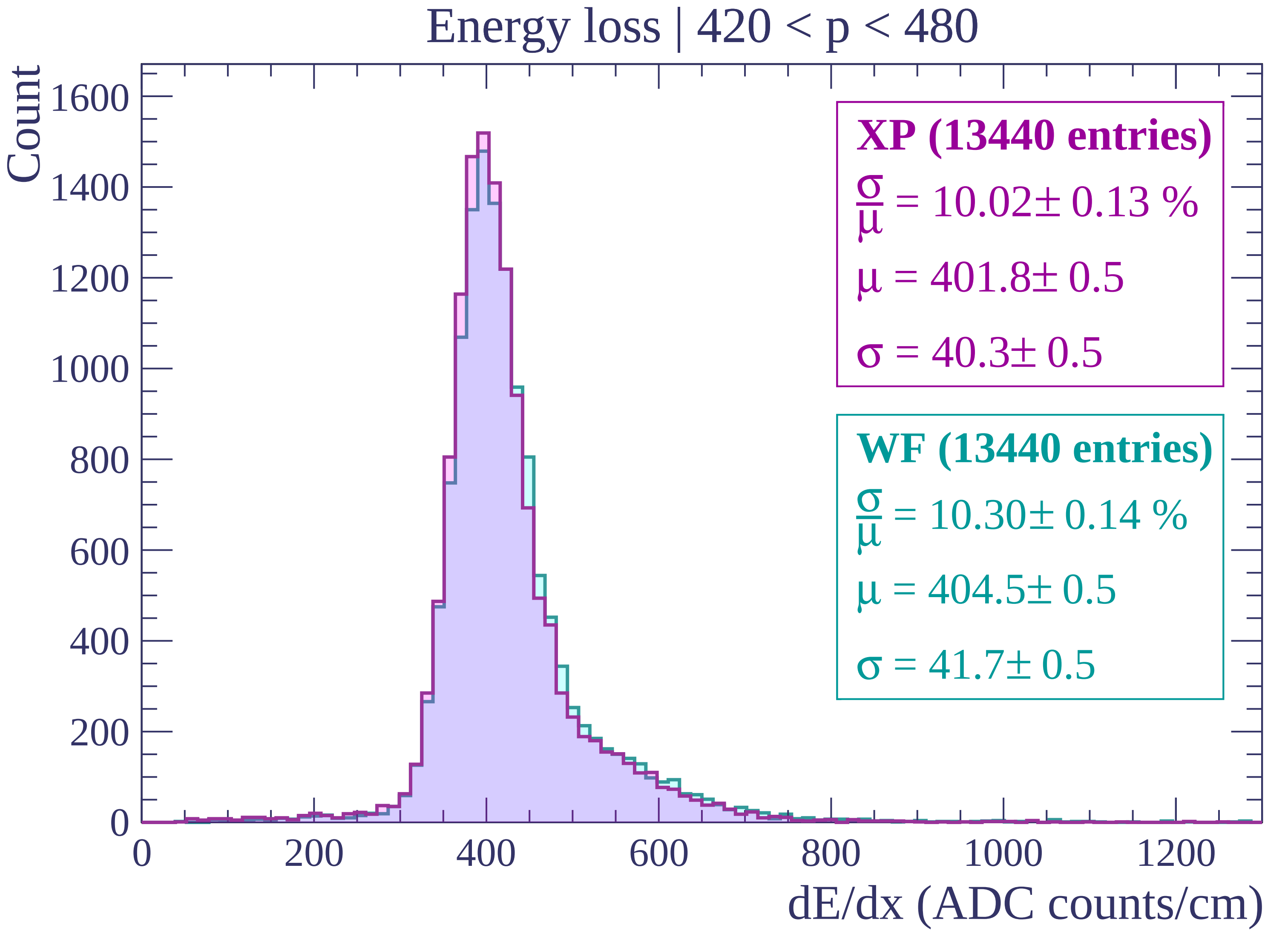


Energy loss | 240 < p < 300
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10044 entries)
σ/μ = 11.49± 0.18 %
μ = 414.5± 0.6
σ = 47.6± 0.7
WF (10044 entries)
σ/μ = 11.67± 0.19 %
μ = 417.9± 0.7
σ = 48.8± 0.7

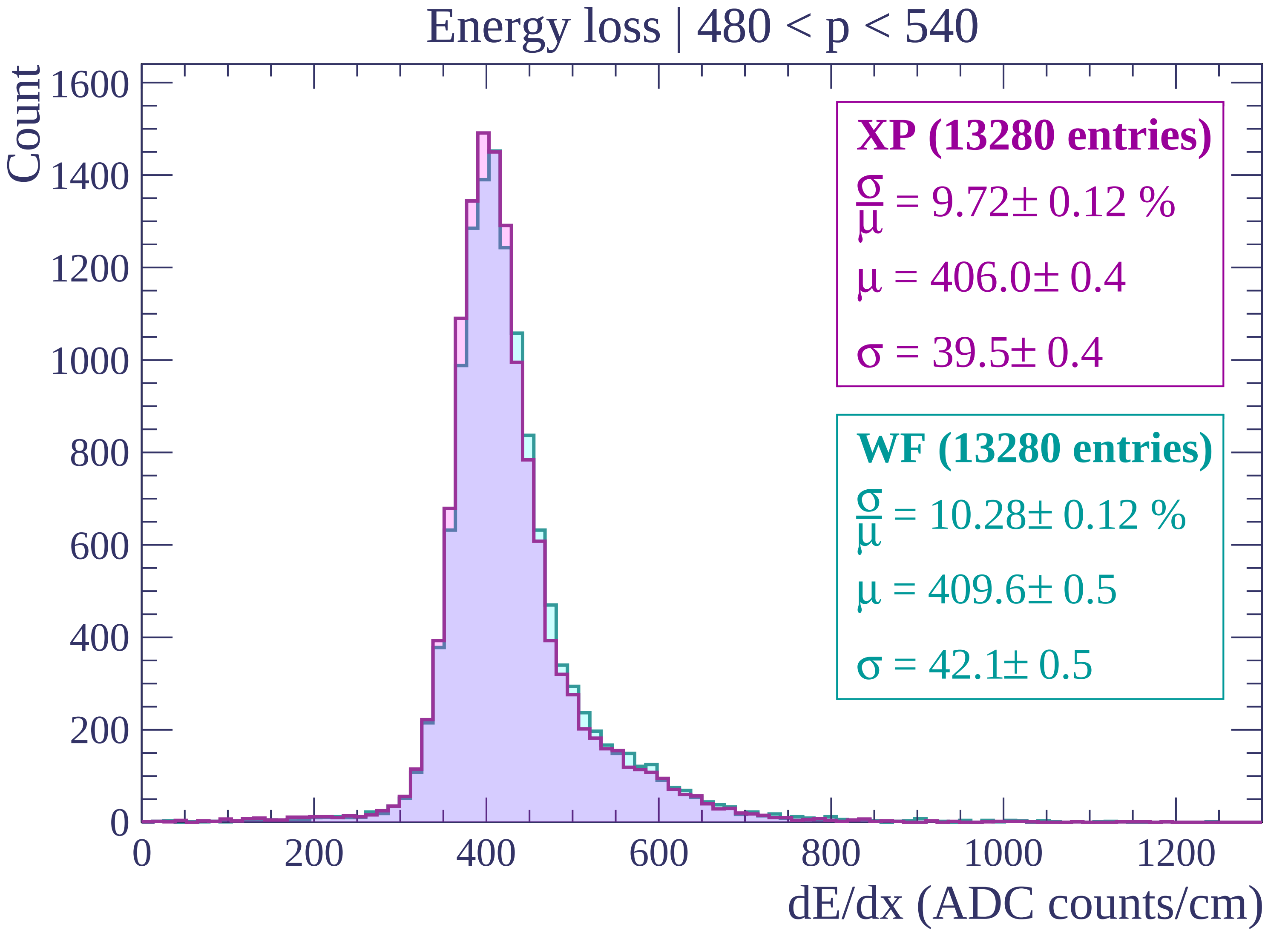


Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
XP (12119 entries)
$\frac{\sigma}{\mu}$ = 10.09± 0.14 %
μ = 402.0± 0.5
σ = 40.6± 0.5
WF (12119 entries)
$\frac{\sigma}{\mu}$ = 10.21± 0.14 %
μ = 404.8± 0.5
σ = 41.3± 0.5
0
200
400
600
800
1000
1200
1400

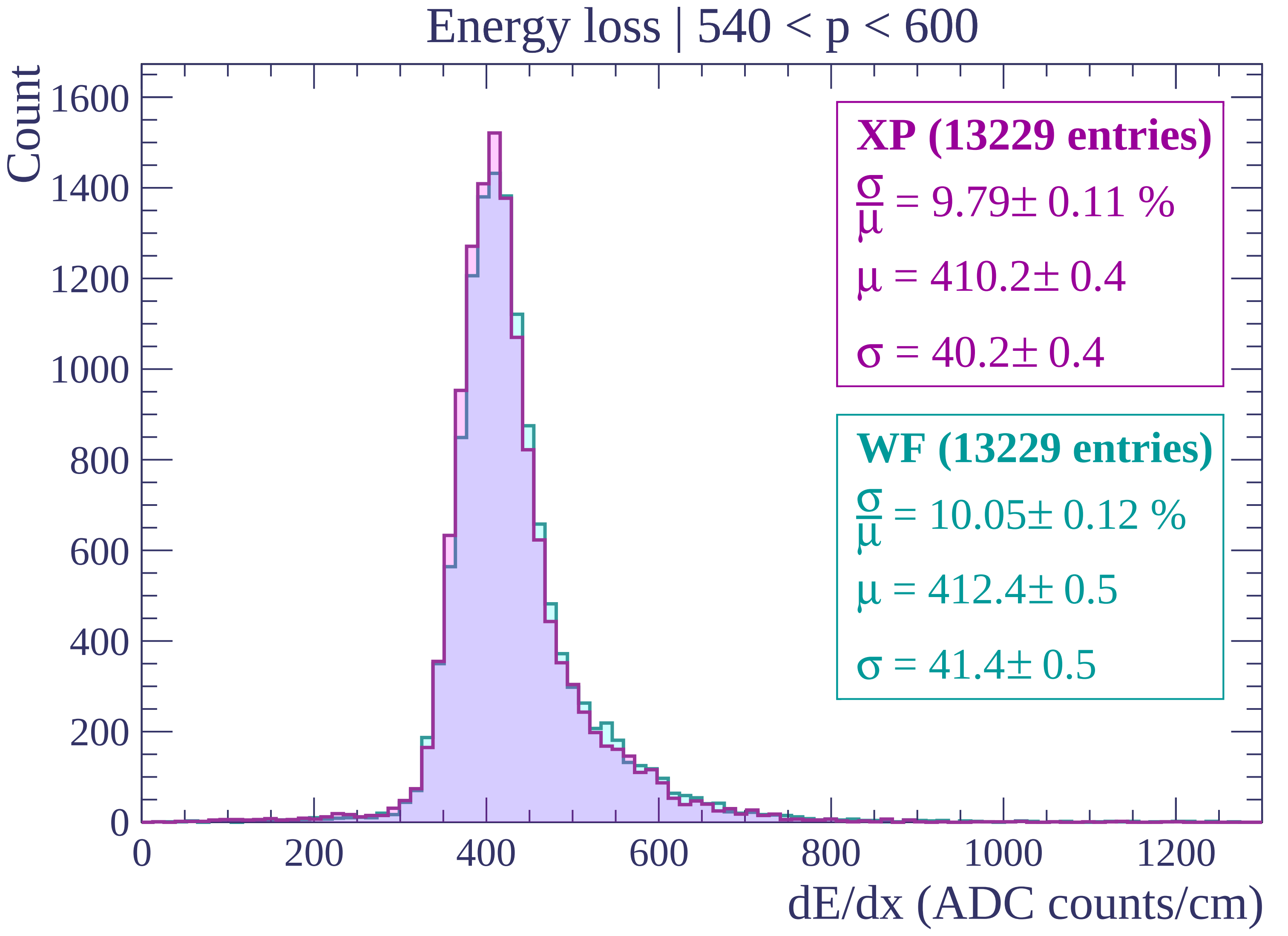

Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
0
200
400
600
800
1000
1200
XP (12711 entries)
σ/μ = 9.93± 0.13 %
μ = 400.6± 0.5
σ = 39.8± 0.5
WF (12711 entries)
σ/μ = 10.17± 0.14 %
μ = 403.4± 0.5
σ = 41.0± 0.5



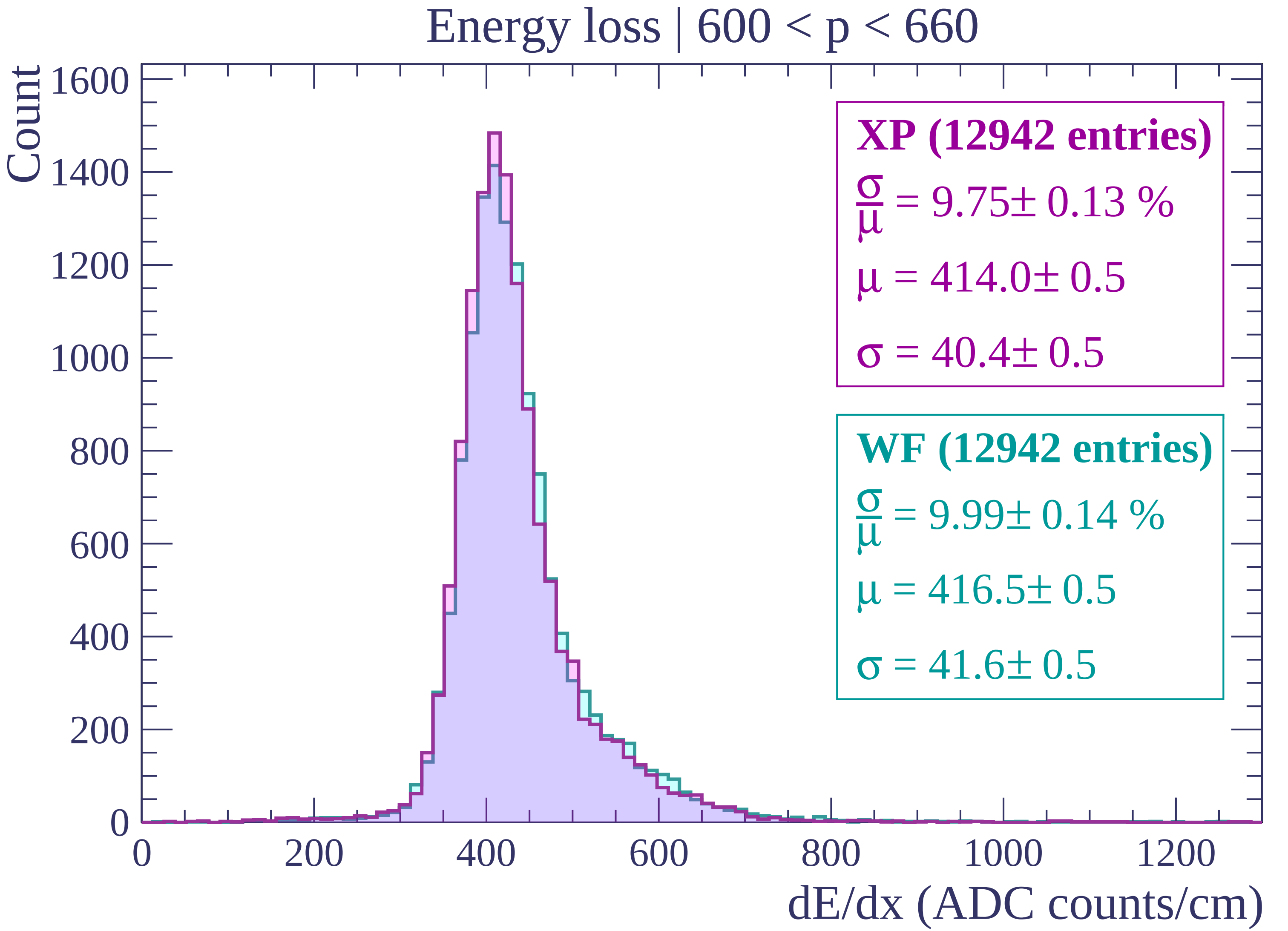
Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP (13440 entries)
σ/μ = 10.02± 0.13 %
μ = 401.8± 0.5
σ = 40.3± 0.5
WF (13440 entries)
σ/μ = 10.30± 0.14 %
μ = 404.5± 0.5
σ = 41.7± 0.5



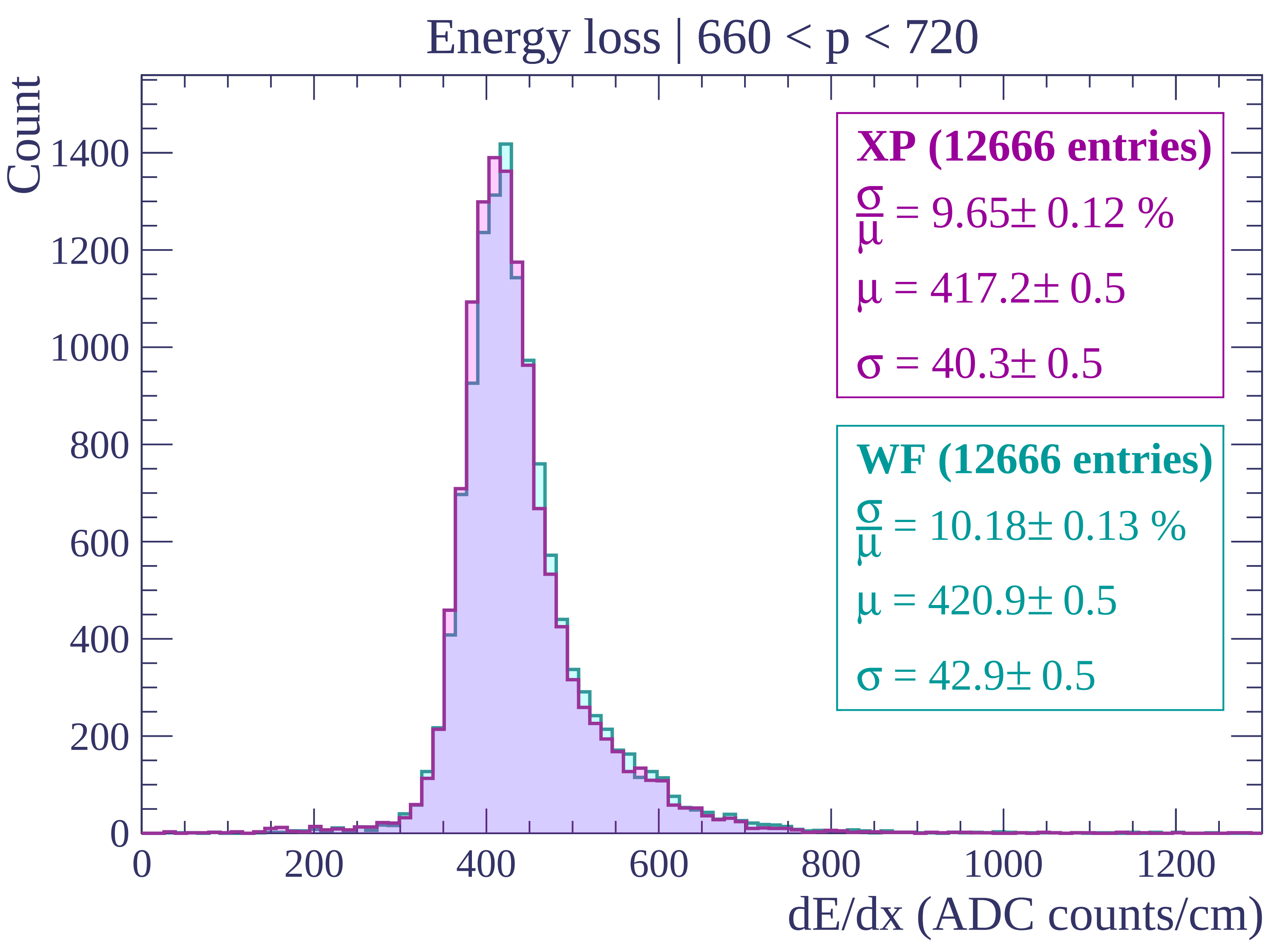
Energy loss | 480 < p < 540
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13280 entries)
$\frac{\sigma}{\mu}$ = 9.72± 0.12 %
μ = 406.0± 0.4
σ = 39.5± 0.4
WF (13280 entries)
$\frac{\sigma}{\mu}$ = 10.28± 0.12 %
μ = 409.6± 0.5
σ = 42.1± 0.5

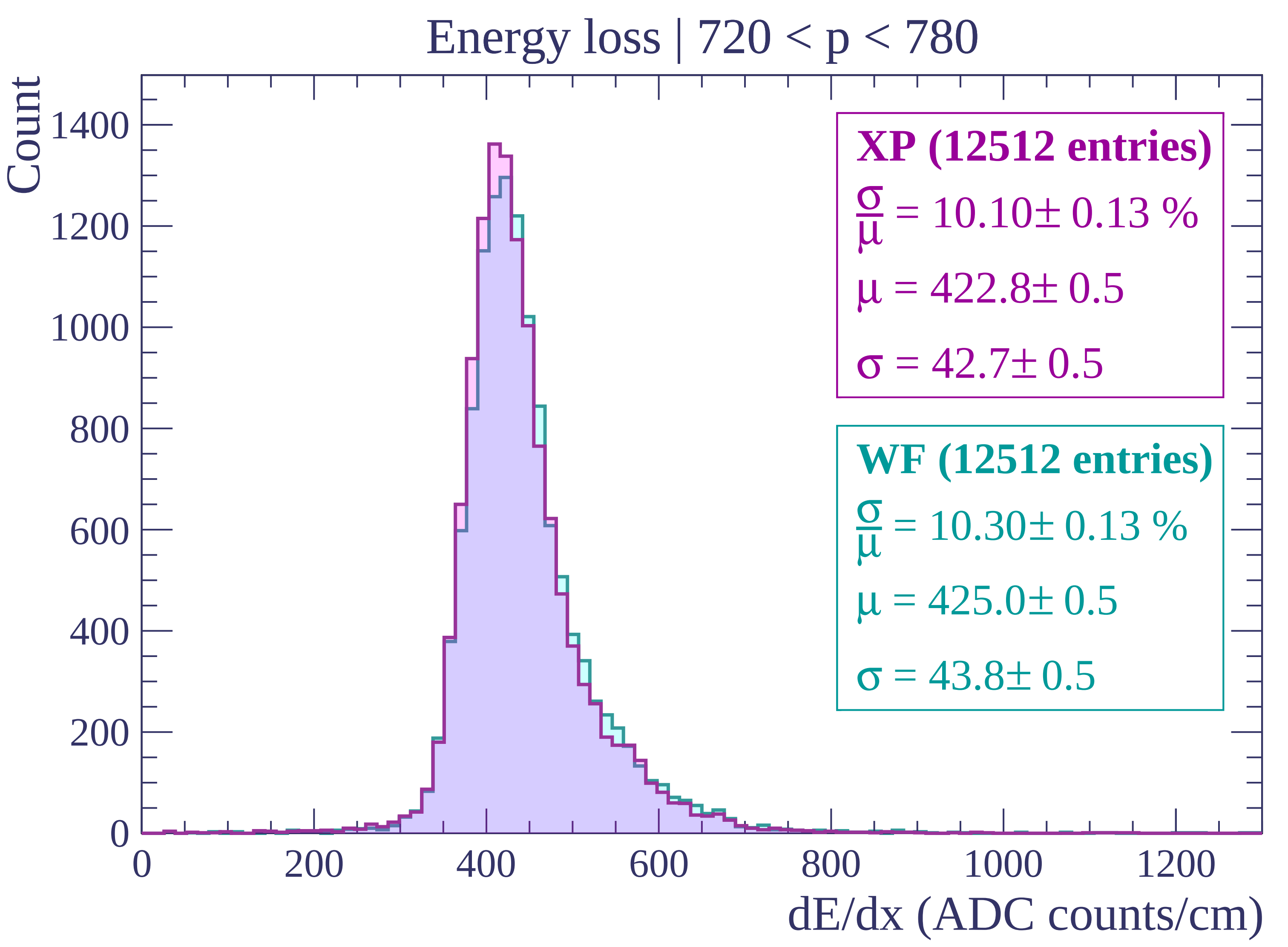


Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
XP (13229 entries)
σ/μ = 9.79± 0.11 %
μ = 410.2± 0.4
σ = 40.2± 0.4
WF (13229 entries)
σ/μ = 10.05± 0.12 %
μ = 412.4± 0.5
σ = 41.4± 0.5
0
200
400
600
800
1000
1200
1400
1600

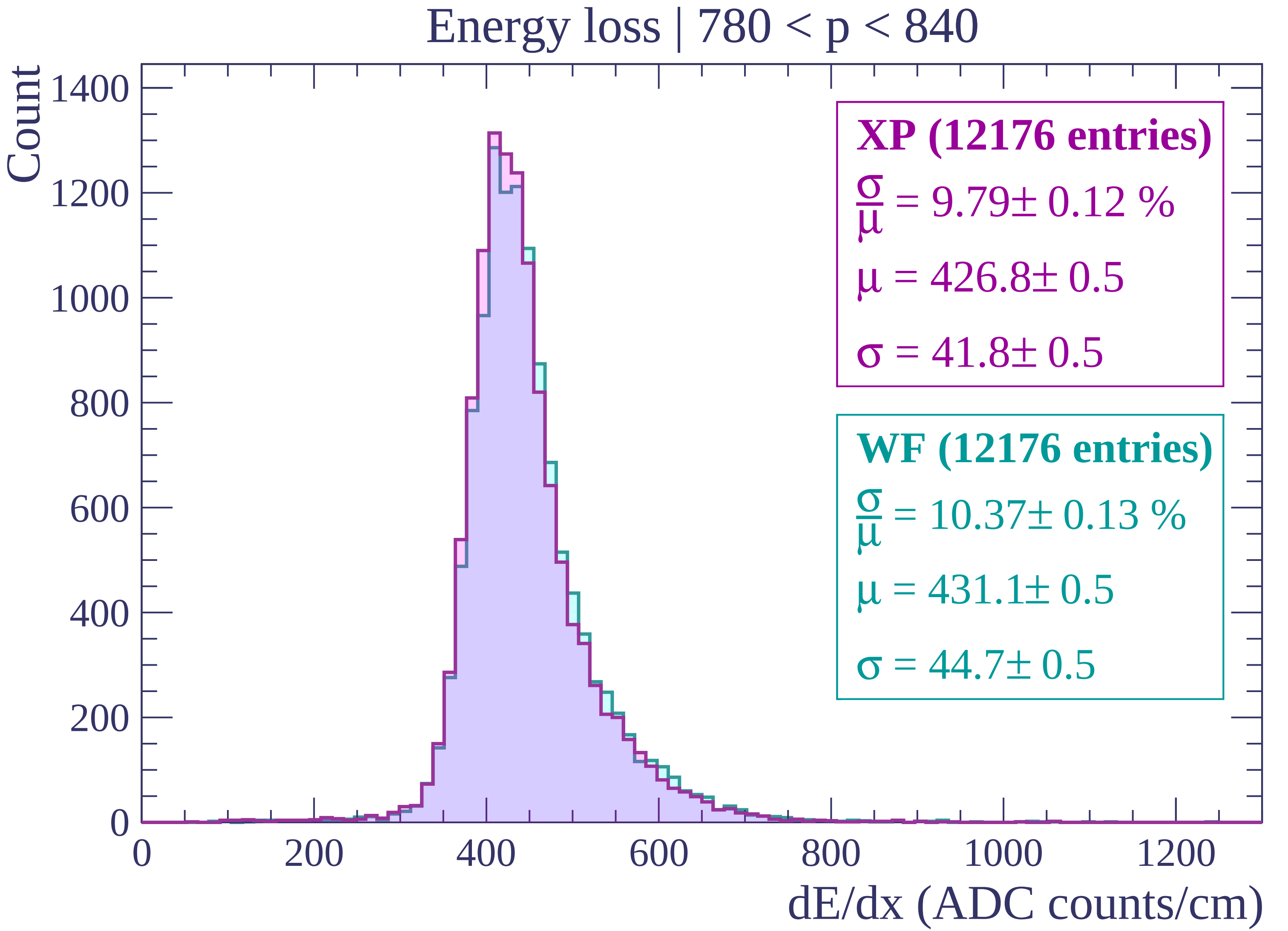


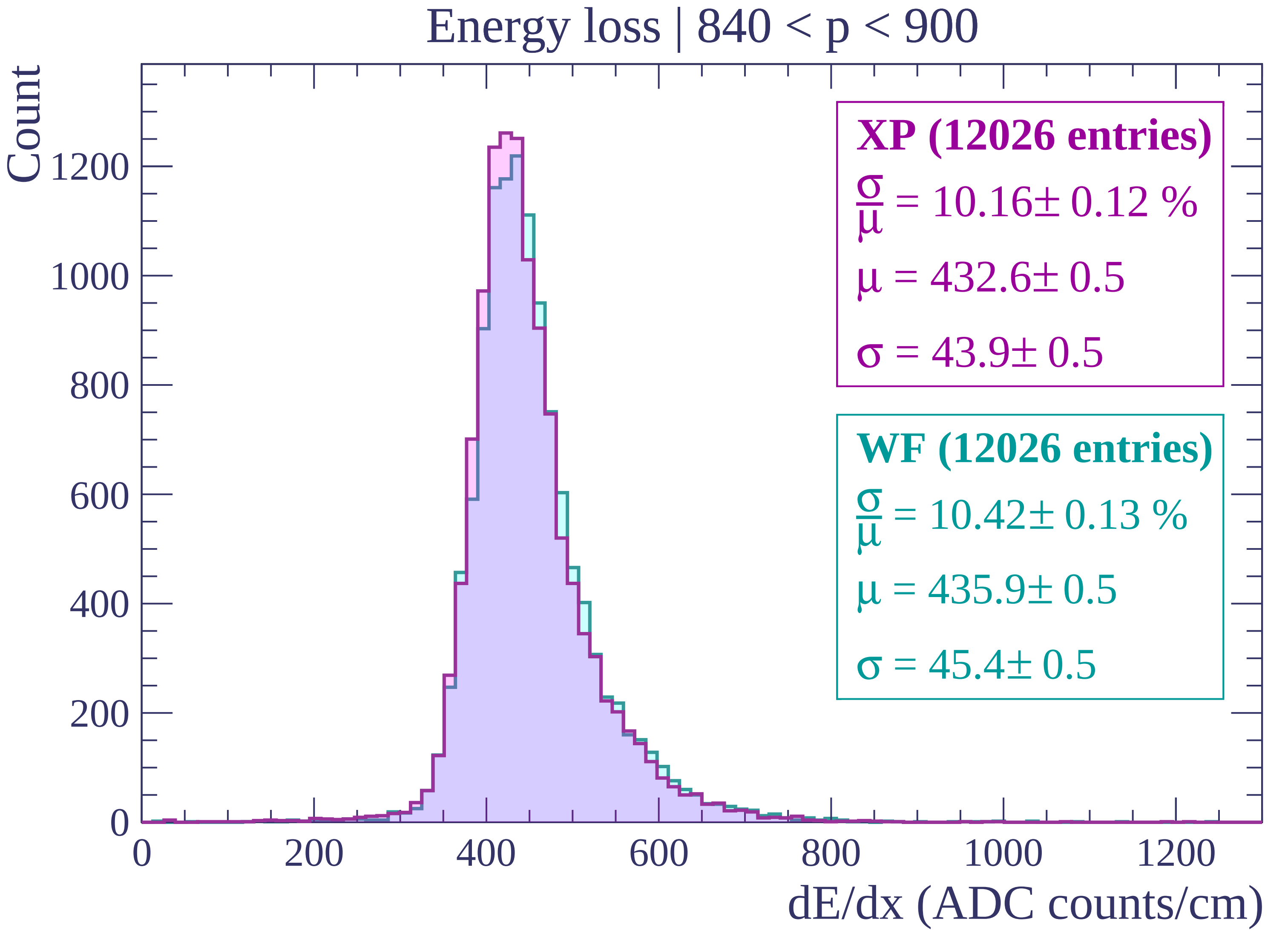
Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
XP (12942 entries)
σ/μ = 9.75± 0.13 %
μ = 414.0± 0.5
σ = 40.4± 0.5
WF (12942 entries)
σ/μ = 9.99± 0.14 %
μ = 416.5± 0.5
σ = 41.6± 0.5
0
200
400
600
800
1000
1200
1400
1600


Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
0
200
400
600
800
1000
1200
XP (12666 entries)
$\frac{\sigma}{\mu}$ = 9.65± 0.12 %
μ = 417.2± 0.5
σ = 40.3± 0.5
WF (12666 entries)
$\frac{\sigma}{\mu}$ = 10.18± 0.13 %
μ = 420.9± 0.5
σ = 42.9± 0.5

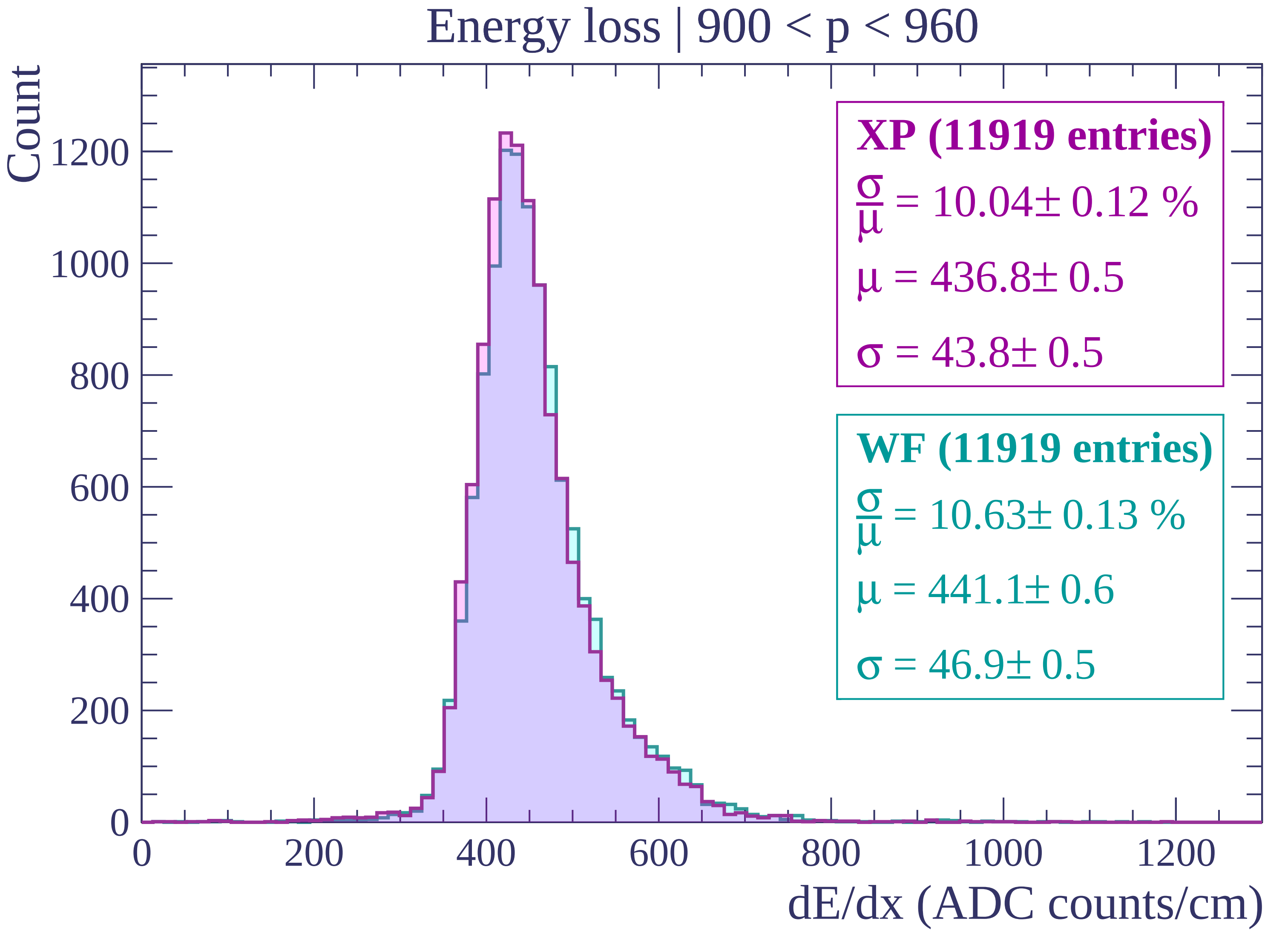


Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (12512 entries)
$\frac{\sigma}{\mu}$ = 10.10± 0.13 %
μ = 422.8± 0.5
σ = 42.7± 0.5
WF (12512 entries)
$\frac{\sigma}{\mu}$ = 10.30± 0.13 %
μ = 425.0± 0.5
σ = 43.8± 0.5

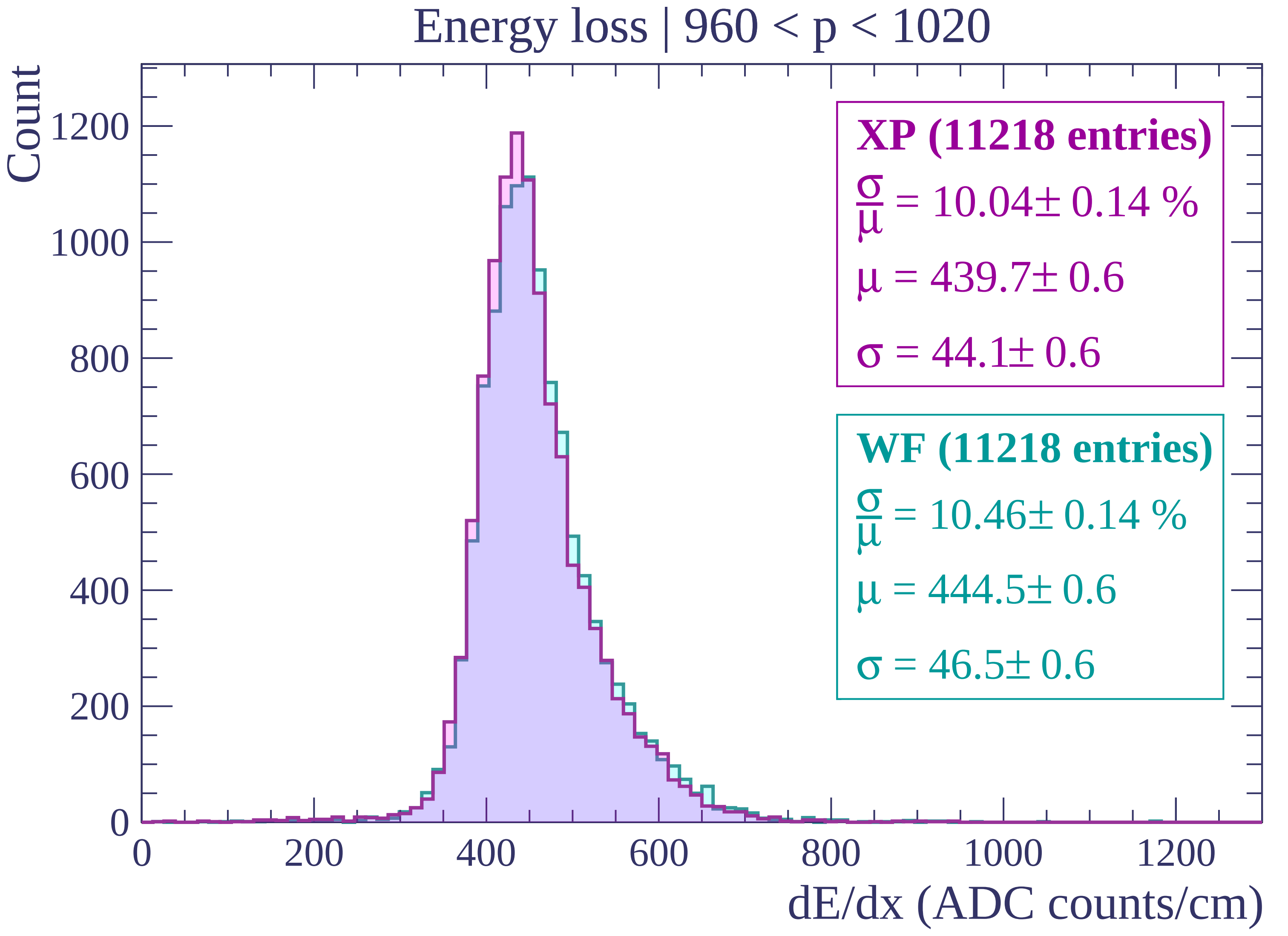

Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
XP (12176 entries)
$\frac{\sigma}{\mu}$ = 9.79± 0.12 %
μ = 426.8± 0.5
σ = 41.8± 0.5
WF (12176 entries)
$\frac{\sigma}{\mu}$ = 10.37± 0.13 %
μ = 431.1± 0.5
σ = 44.7± 0.5
0
200
400
600
800
1000
1200
1400



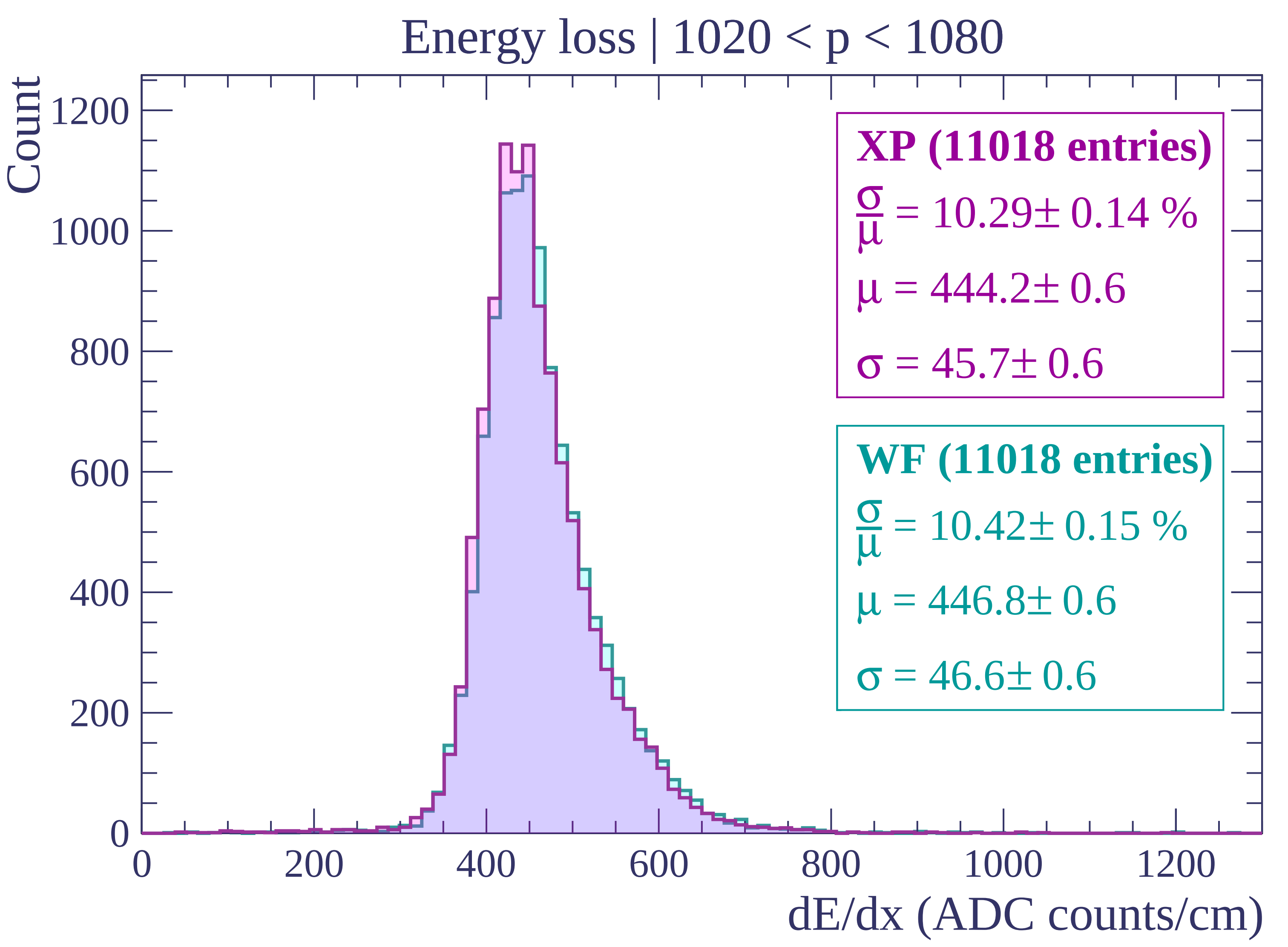
Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (12026 entries)
σ/μ = 10.16± 0.12 %
μ = 432.6± 0.5
σ = 43.9± 0.5
WF (12026 entries)
σ/μ = 10.42± 0.13 %
μ = 435.9± 0.5
σ = 45.4± 0.5



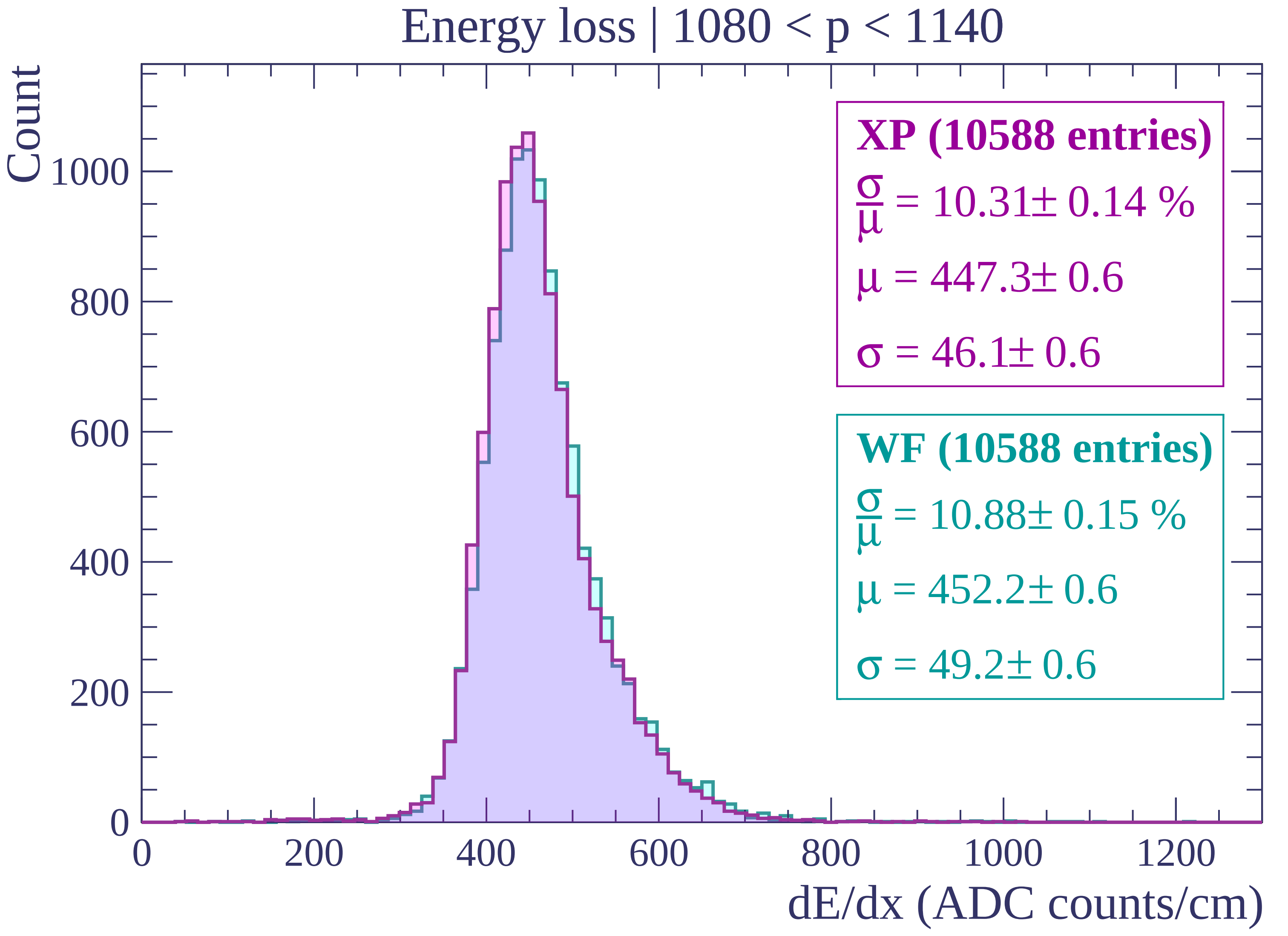
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
XP (11919 entries)
$\frac{\sigma}{\mu}$ = 10.04± 0.12 %
μ = 436.8± 0.5
σ = 43.8± 0.5
WF (11919 entries)
$\frac{\sigma}{\mu}$ = 10.63± 0.13 %
μ = 441.1± 0.6
σ = 46.9± 0.5
0
200
400
600
800
1000
1200



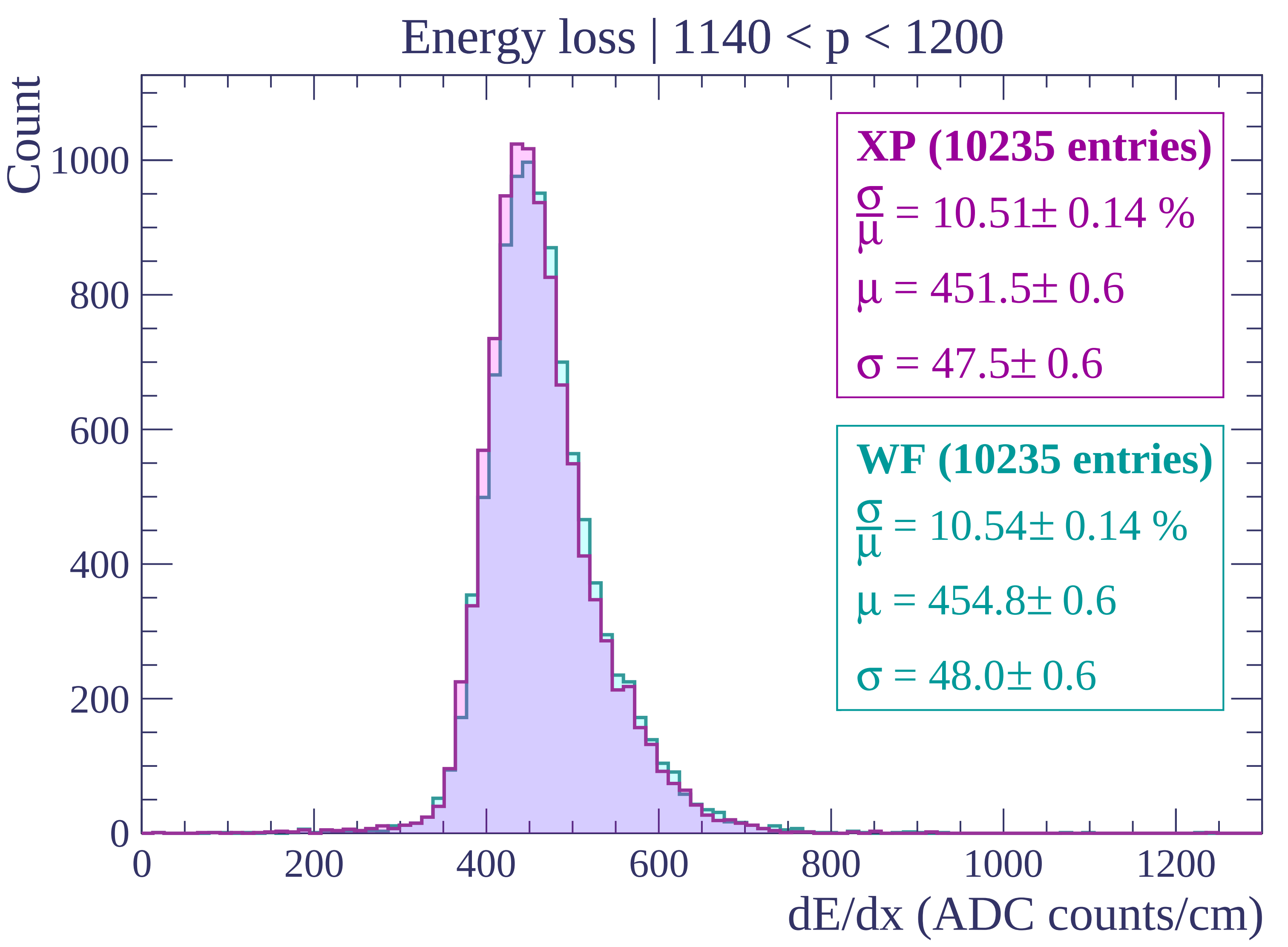
Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11218 entries)
σ/μ = 10.04± 0.14 %
μ = 439.7± 0.6
σ = 44.1± 0.6
WF (11218 entries)
σ/μ = 10.46± 0.14 %
μ = 444.5± 0.6
σ = 46.5± 0.6



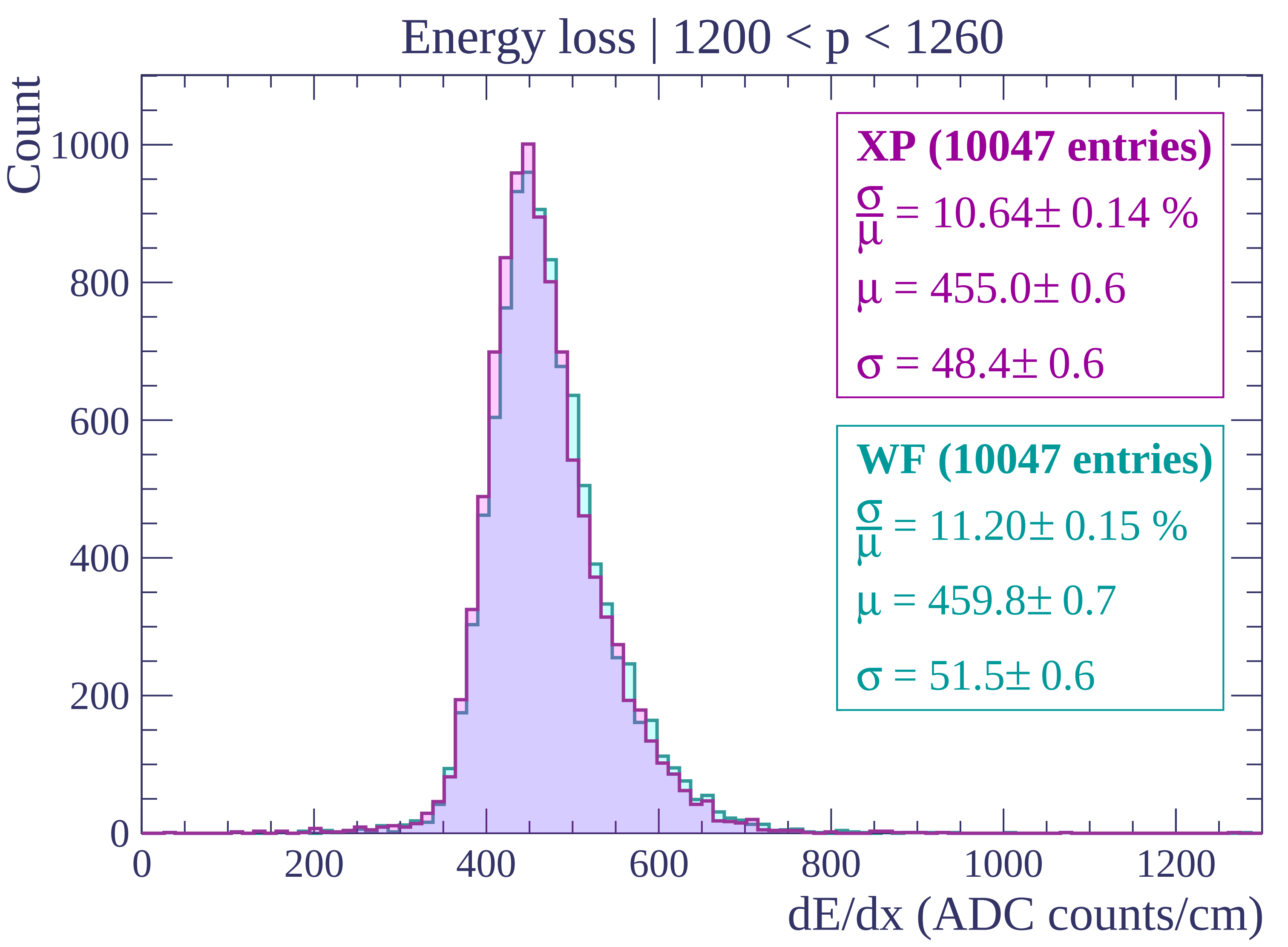
Energy loss | 1020 < p < 1080
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (11018 entries)
$\frac{\sigma}{\mu}$ = 10.29± 0.14 %
μ = 444.2± 0.6
σ = 45.7± 0.6
WF (11018 entries)
$\frac{\sigma}{\mu}$ = 10.42± 0.15 %
μ = 446.8± 0.6
σ = 46.6± 0.6



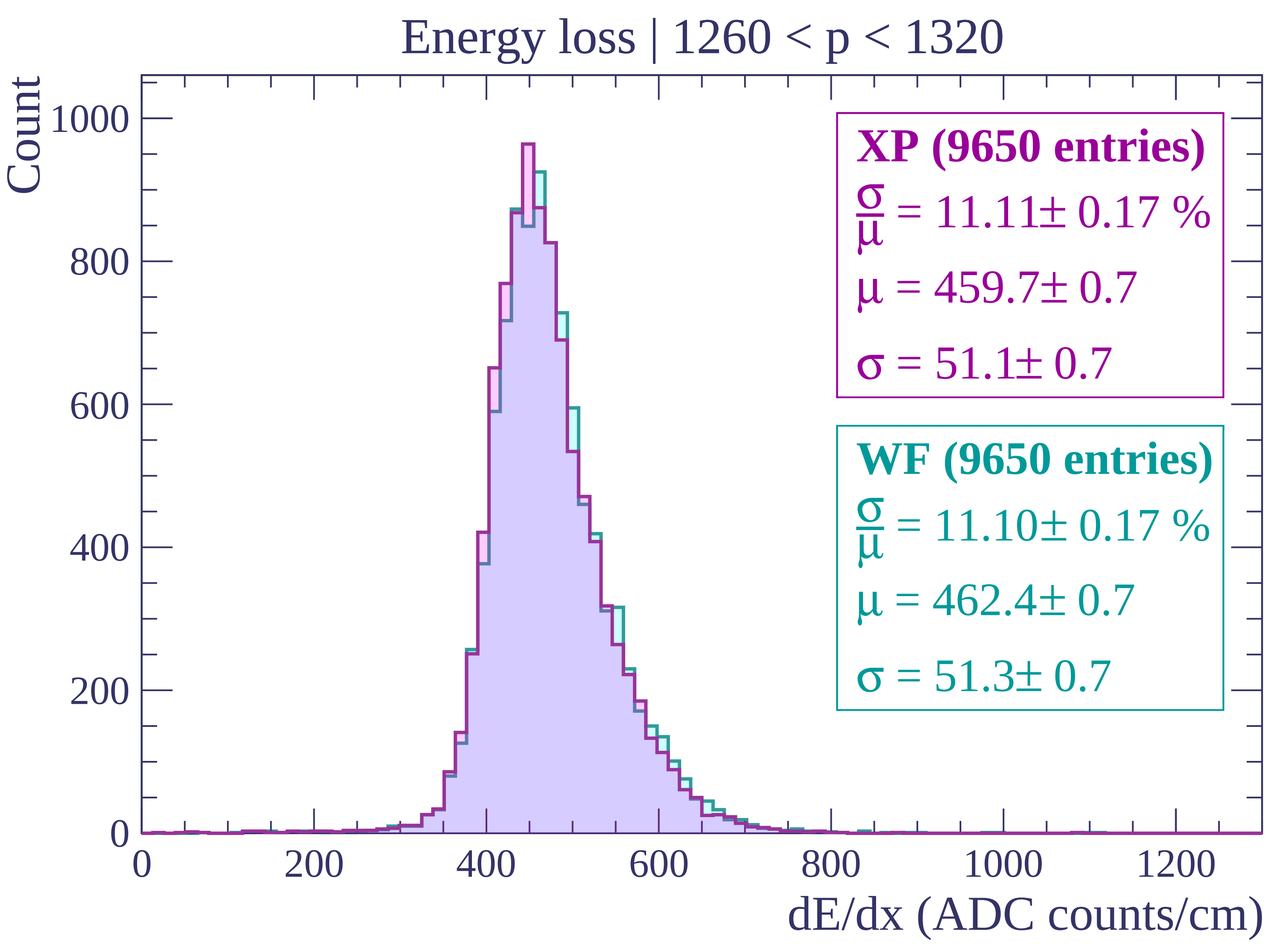
Energy loss | 1080 < p < 1140
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10588 entries)
σ/μ = 10.31± 0.14 %
μ = 447.3± 0.6
σ = 46.1± 0.6
WF (10588 entries)
σ/μ = 10.88± 0.15 %
μ = 452.2± 0.6
σ = 49.2± 0.6



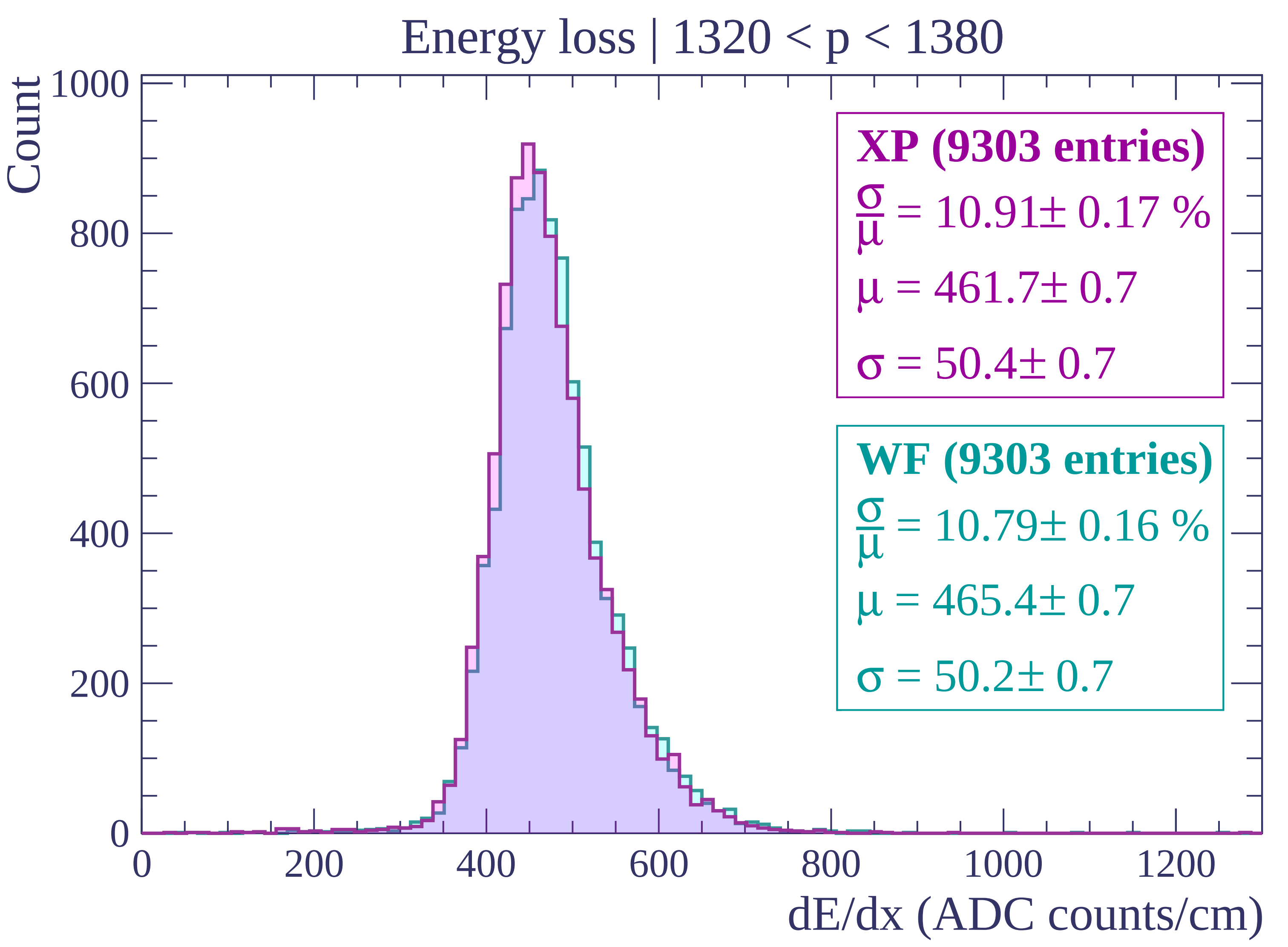
Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10235 entries)
σ/μ = 10.51± 0.14 %
μ = 451.5± 0.6
σ = 47.5± 0.6
WF (10235 entries)
σ/μ = 10.54± 0.14 %
μ = 454.8± 0.6
σ = 48.0± 0.6



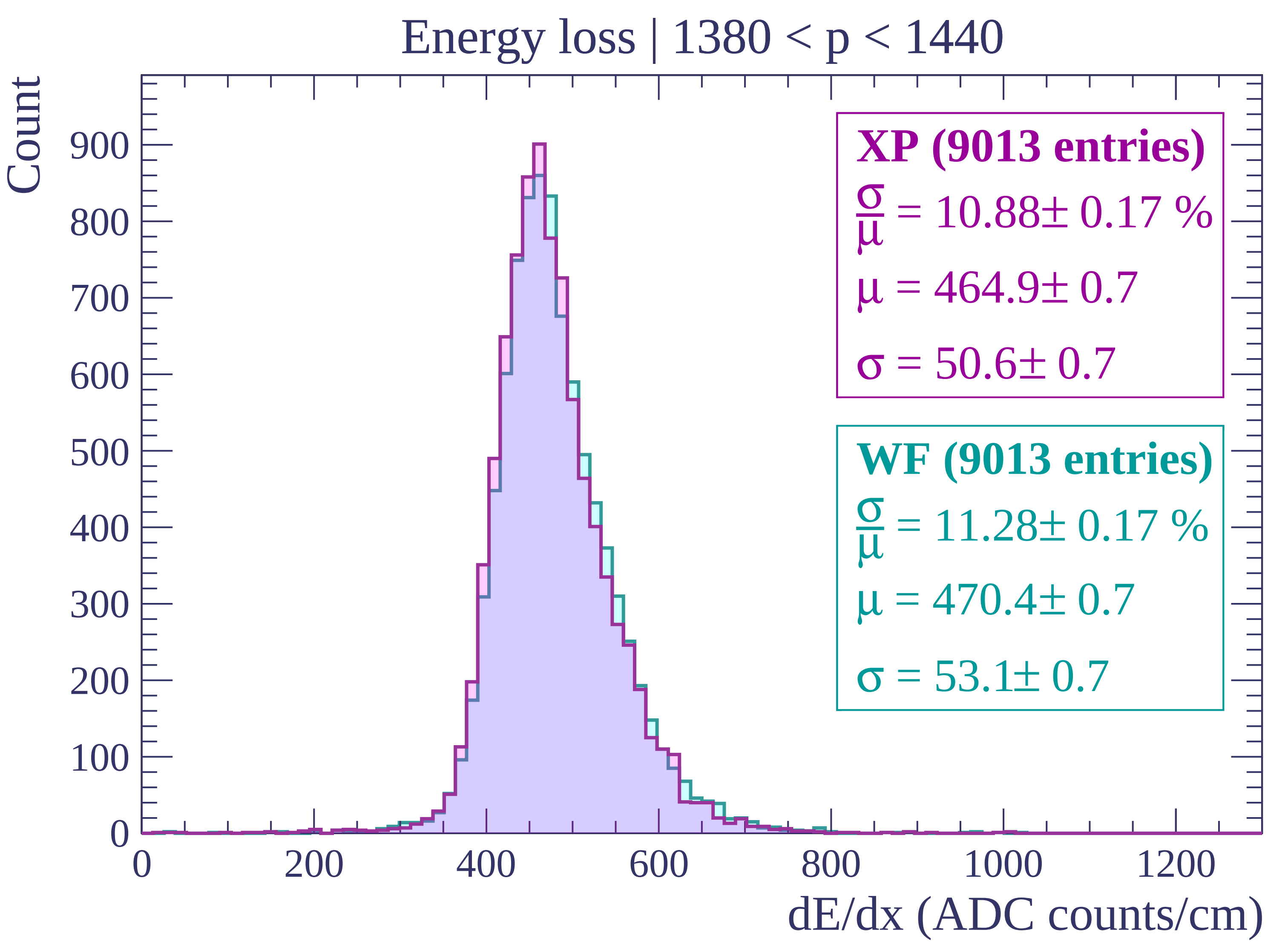
Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10047 entries)
$\frac{\sigma}{\mu}$ = 10.64± 0.14 %
μ = 455.0± 0.6
σ = 48.4± 0.6
WF (10047 entries)
$\frac{\sigma}{\mu}$ = 11.20± 0.15 %
μ = 459.8± 0.7
σ = 51.5± 0.6



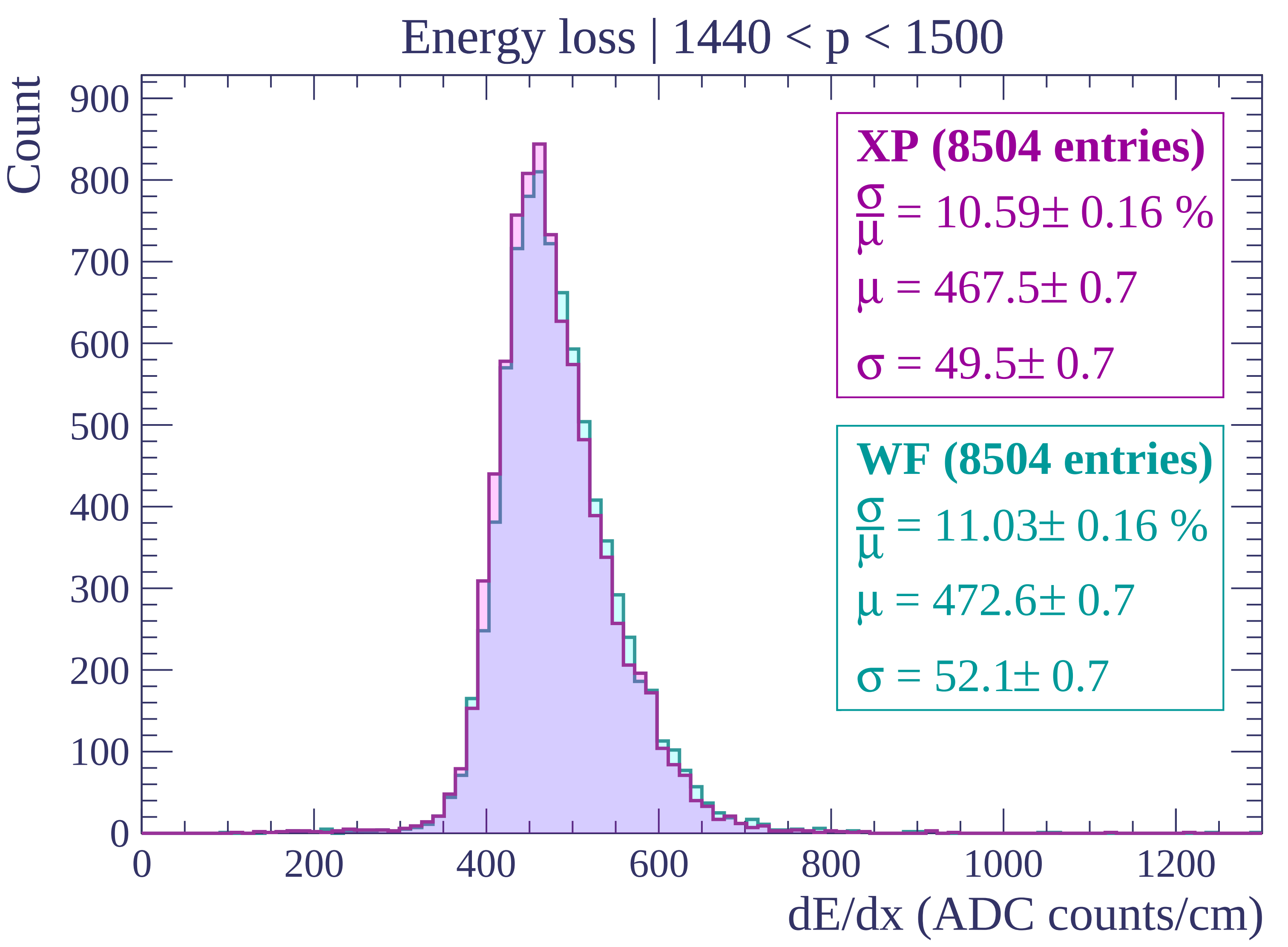
Energy loss | 1260 < p < 1320
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9650 entries)
σ/μ = 11.11± 0.17 %
μ = 459.7± 0.7
σ = 51.1± 0.7
WF (9650 entries)
σ/μ = 11.10± 0.17 %
μ = 462.4± 0.7
σ = 51.3± 0.7



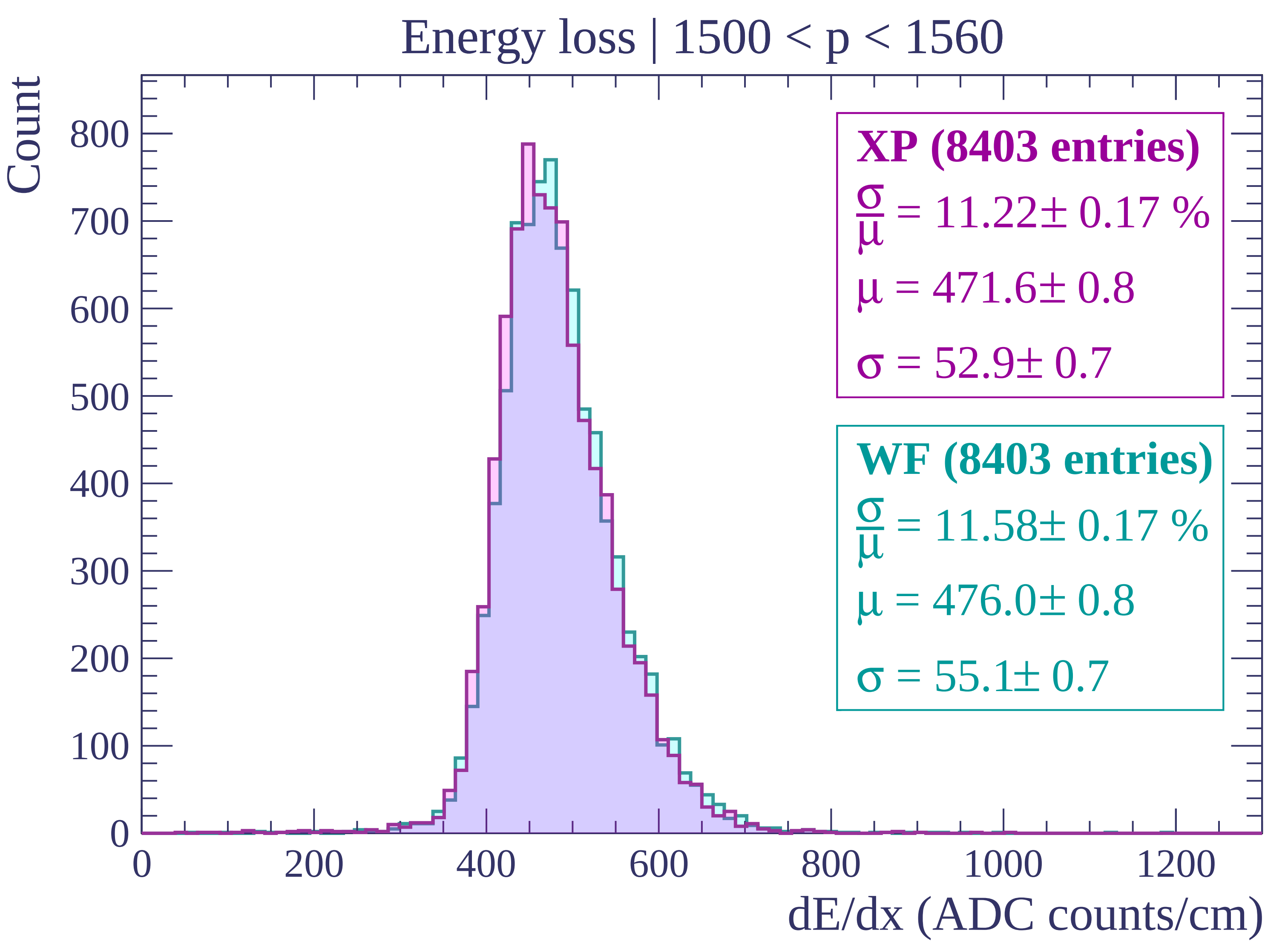
Energy loss | 1320 < p < 1380
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9303 entries)
$\frac{\sigma}{\mu}$ = 10.91± 0.17 %
μ = 461.7± 0.7
σ = 50.4± 0.7
WF (9303 entries)
$\frac{\sigma}{\mu}$ = 10.79± 0.16 %
μ = 465.4± 0.7
σ = 50.2± 0.7



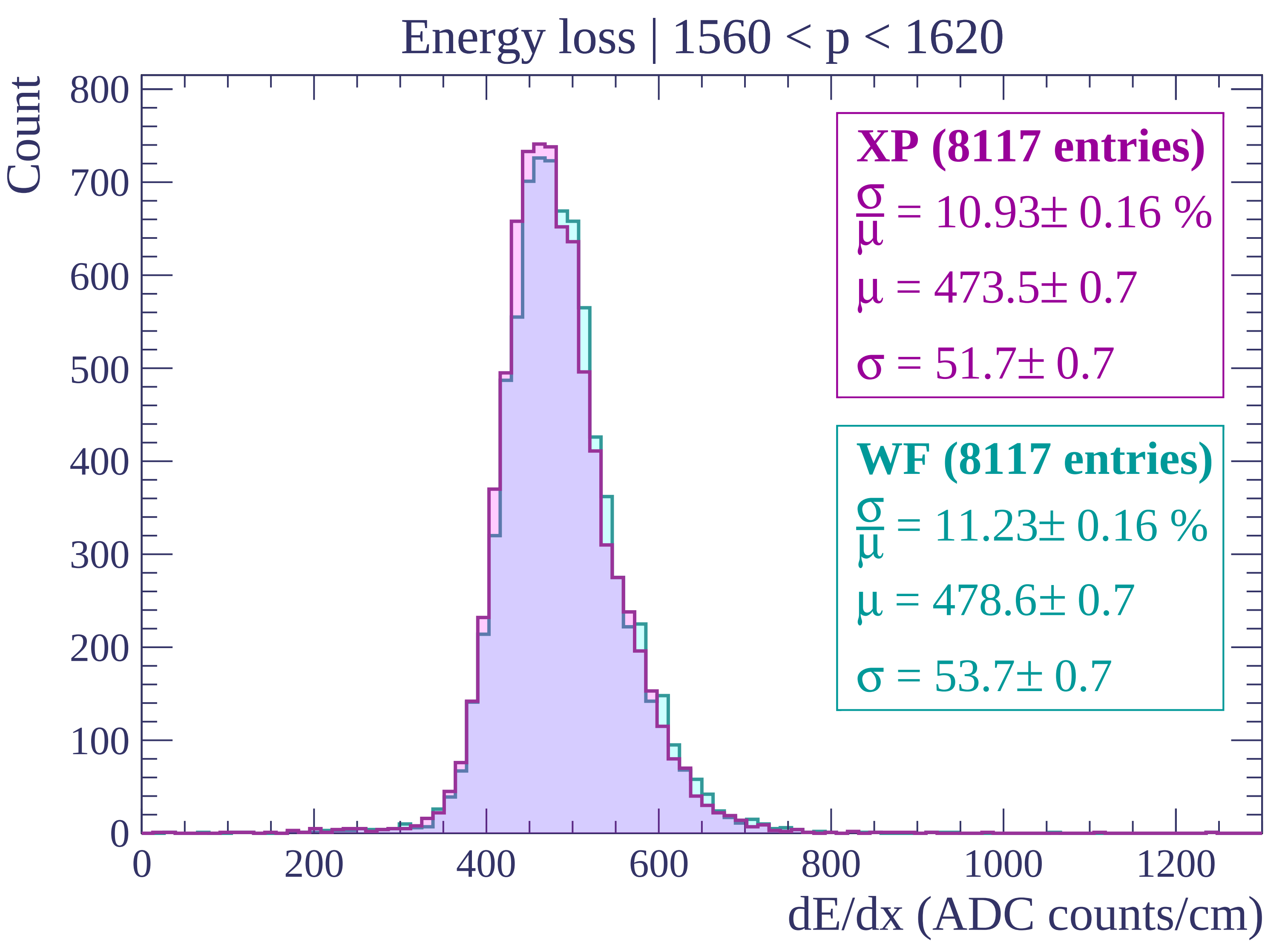
Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (9013 entries)
σ/μ = 10.88± 0.17 %
μ = 464.9± 0.7
σ = 50.6± 0.7
WF (9013 entries)
σ/μ = 11.28± 0.17 %
μ = 470.4± 0.7
σ = 53.1± 0.7



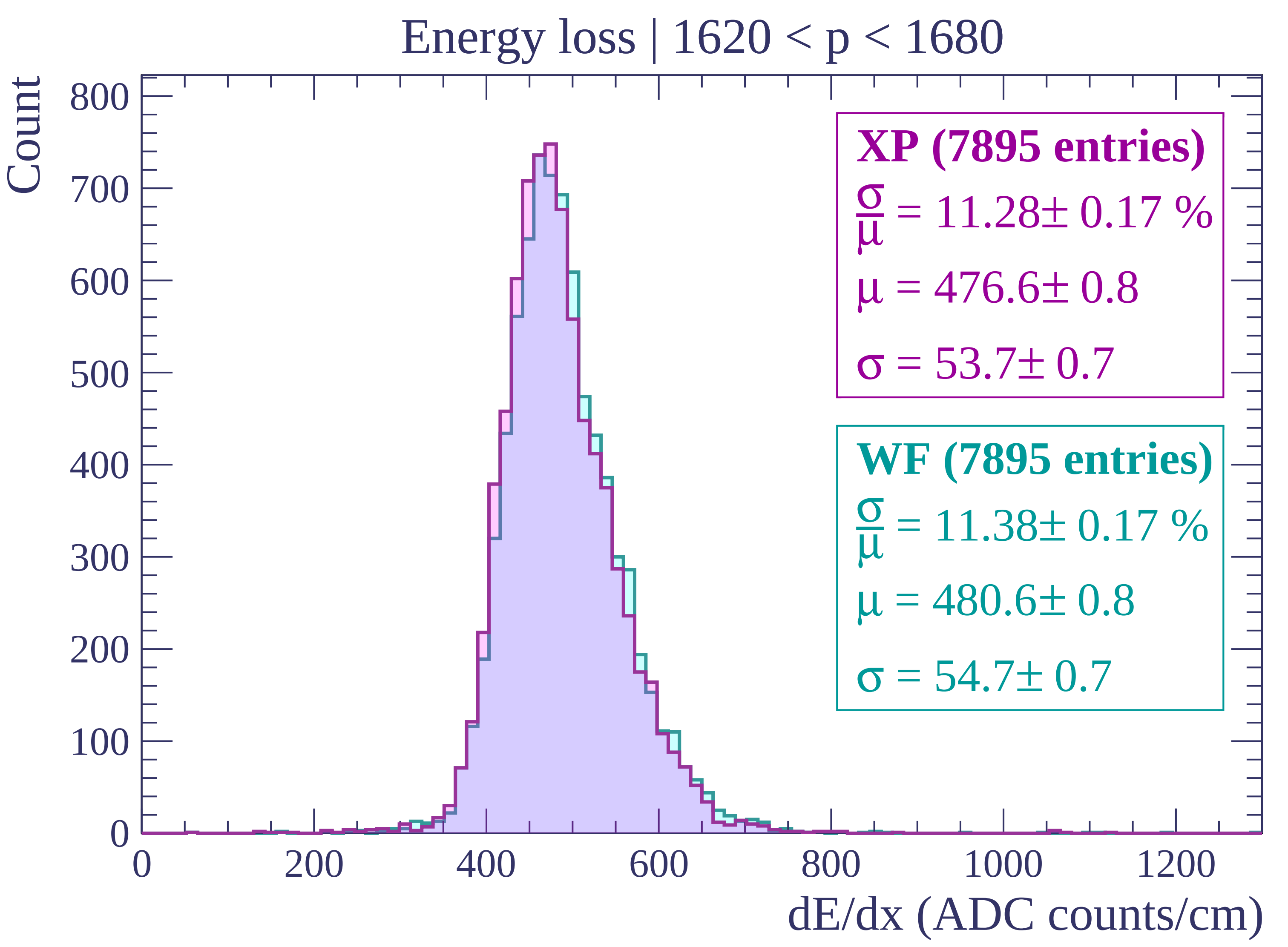
Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (8504 entries)
$\frac{\sigma}{\mu}$ = 10.59± 0.16 %
μ = 467.5± 0.7
σ = 49.5± 0.7
WF (8504 entries)
$\frac{\sigma}{\mu}$ = 11.03± 0.16 %
μ = 472.6± 0.7
σ = 52.1± 0.7

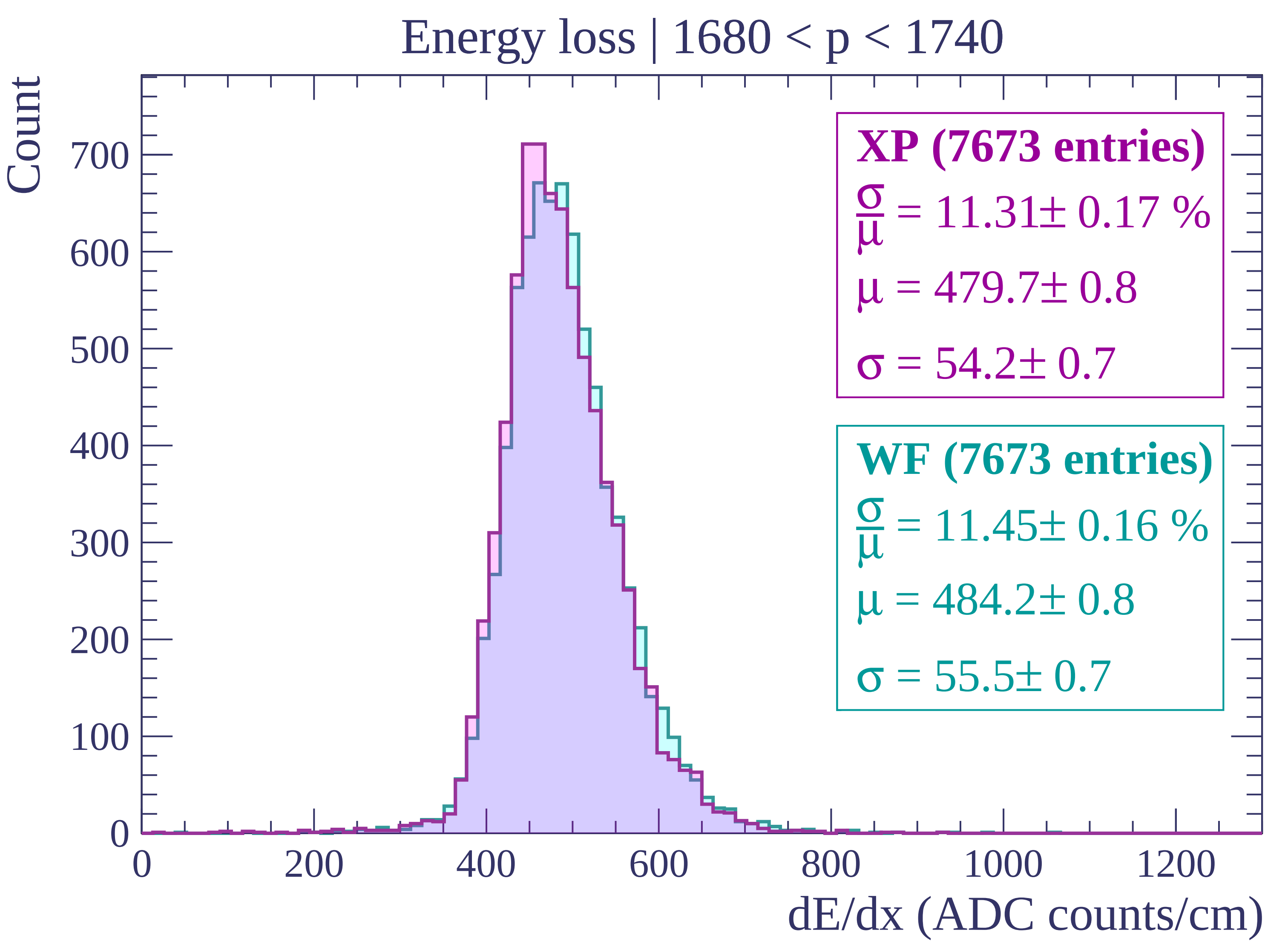


Energy loss | 1500 < p < 1560
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8403 entries)
σ/μ = 11.22± 0.17 %
μ = 471.6± 0.8
σ = 52.9± 0.7
WF (8403 entries)
σ/μ = 11.58± 0.17 %
μ = 476.0± 0.8
σ = 55.1± 0.7

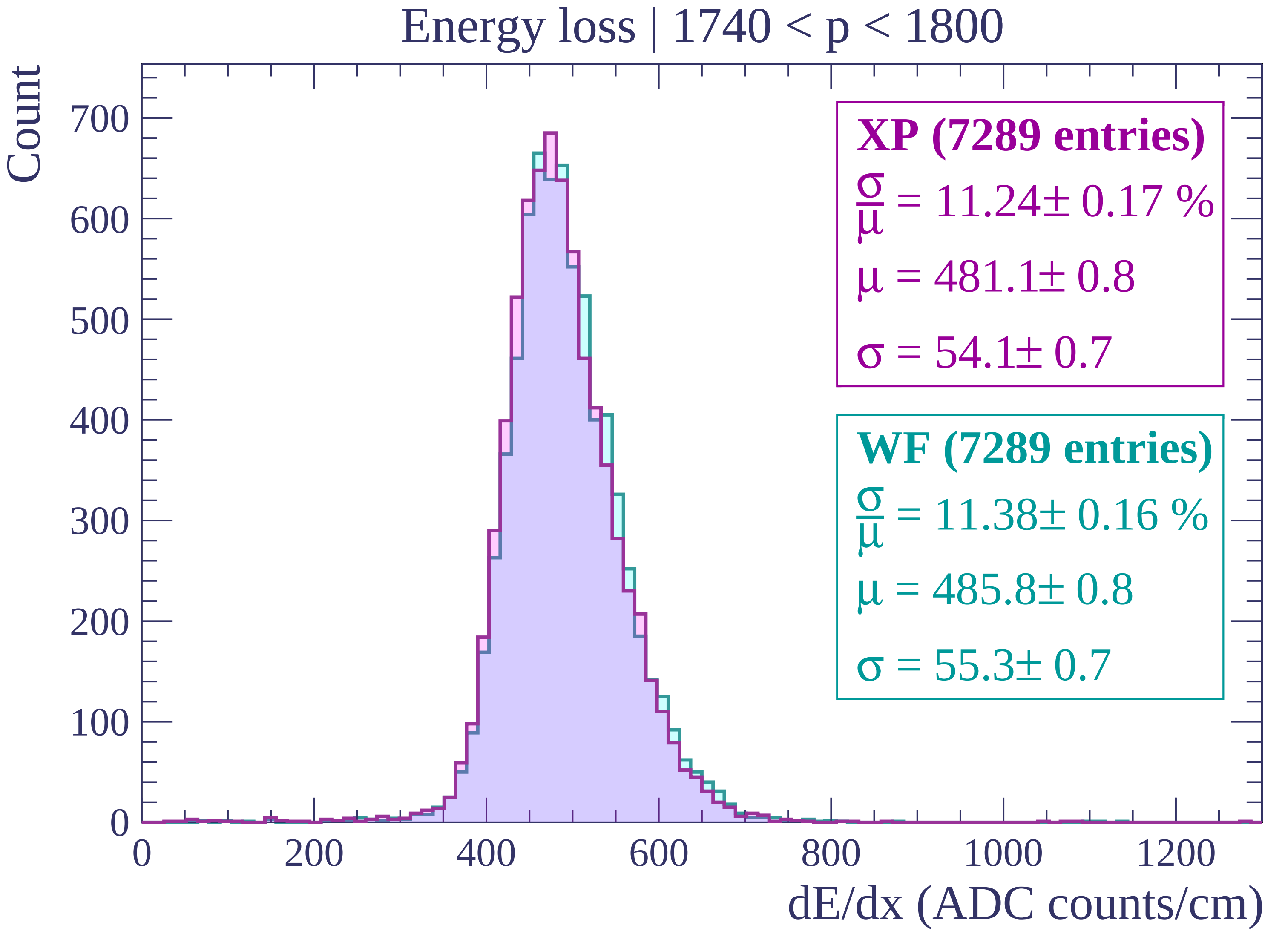


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (8117 entries)
$\frac{\sigma}{\mu}$ = 10.93± 0.16 %
μ = 473.5± 0.7
σ = 51.7± 0.7
WF (8117 entries)
$\frac{\sigma}{\mu}$ = 11.23± 0.16 %
μ = 478.6± 0.7
σ = 53.7± 0.7

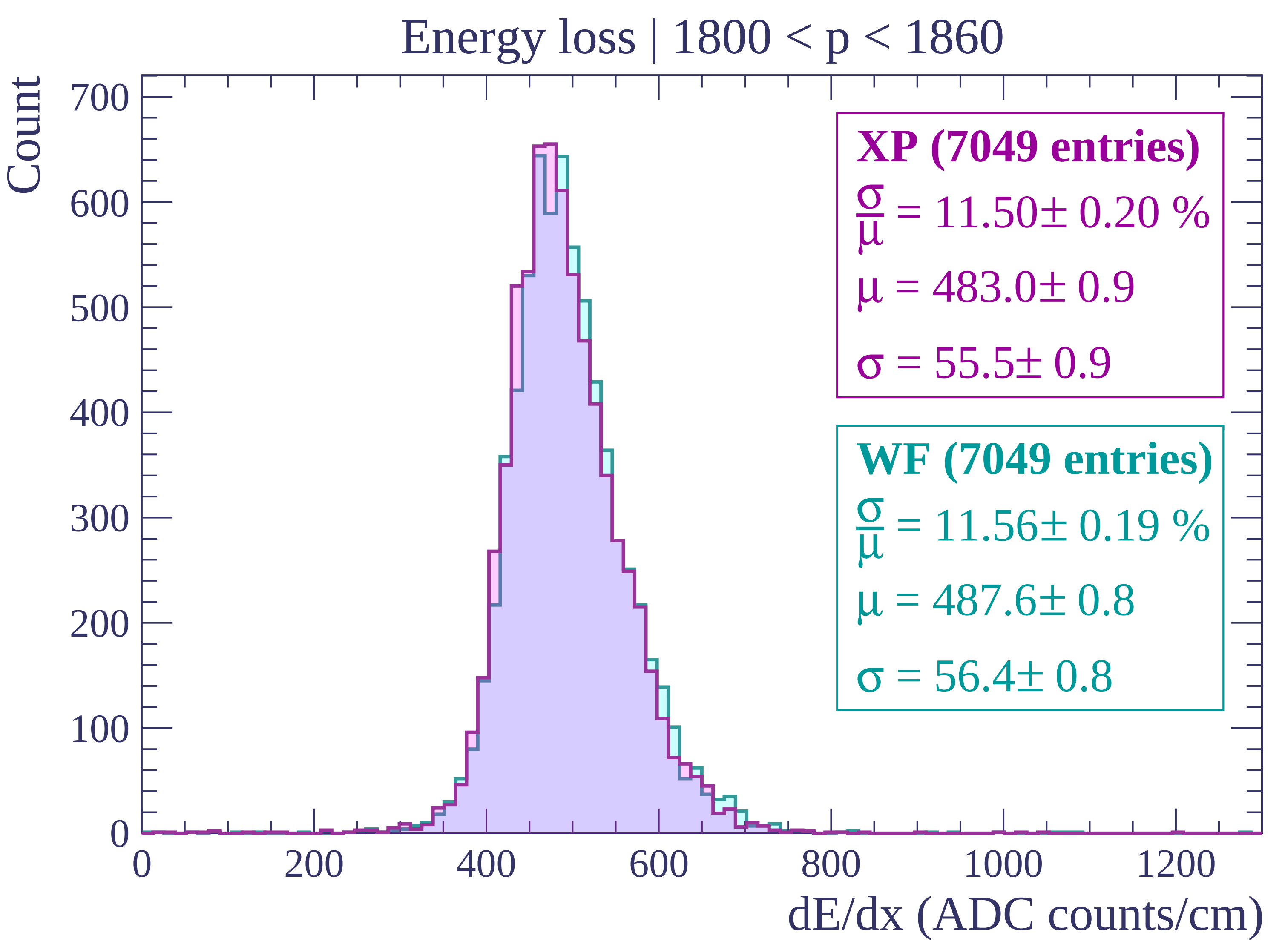

Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP (7895 entries)
σ/μ = 11.28± 0.17 %
μ = 476.6± 0.8
σ = 53.7± 0.7
WF (7895 entries)
σ/μ = 11.38± 0.17 %
μ = 480.6± 0.8
σ = 54.7± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200



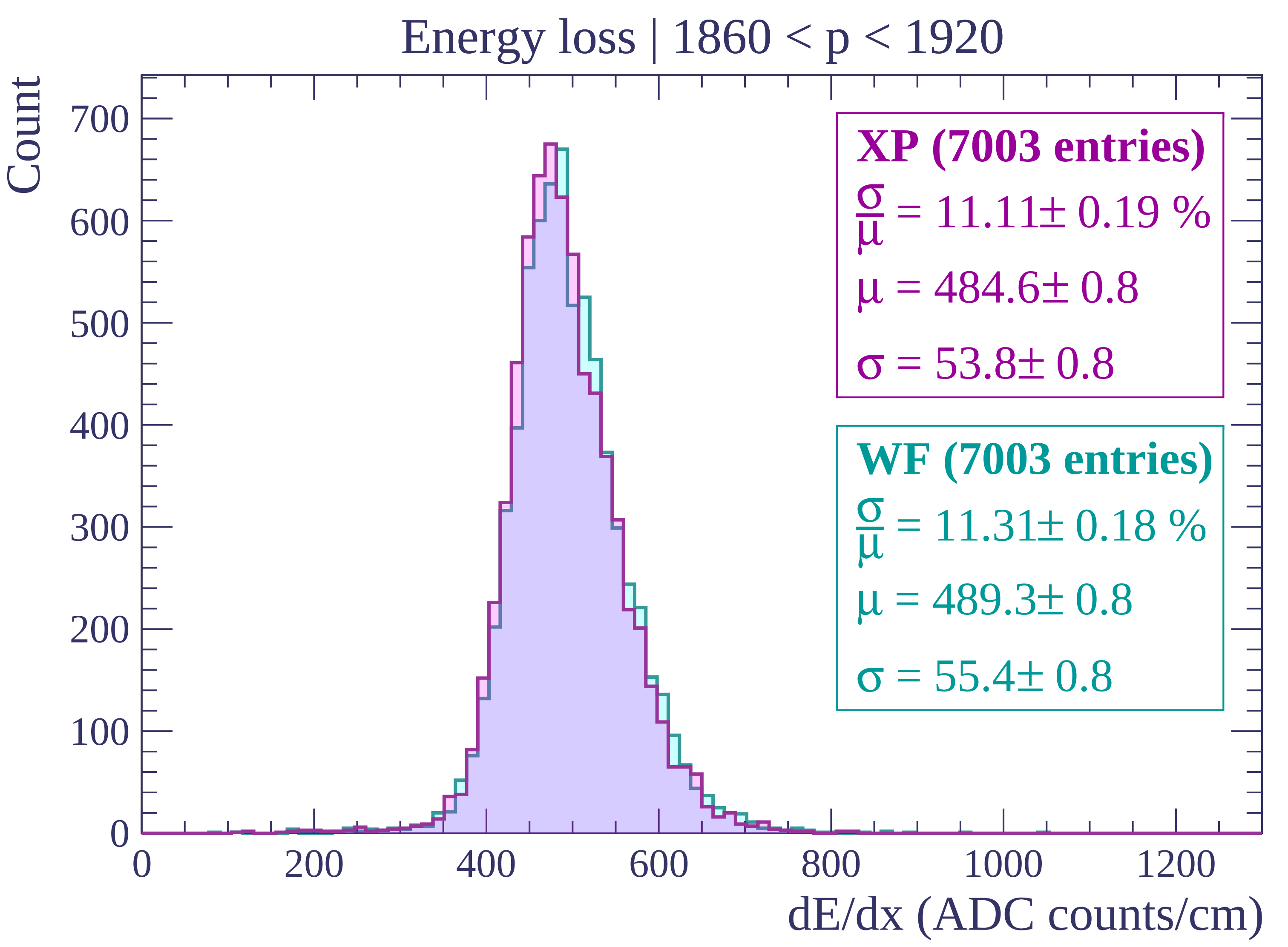
Energy loss | 1680 < p < 1740
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7673 entries)
σ/μ = 11.31± 0.17 %
μ = 479.7± 0.8
σ = 54.2± 0.7
WF (7673 entries)
σ/μ = 11.45± 0.16 %
μ = 484.2± 0.8
σ = 55.5± 0.7



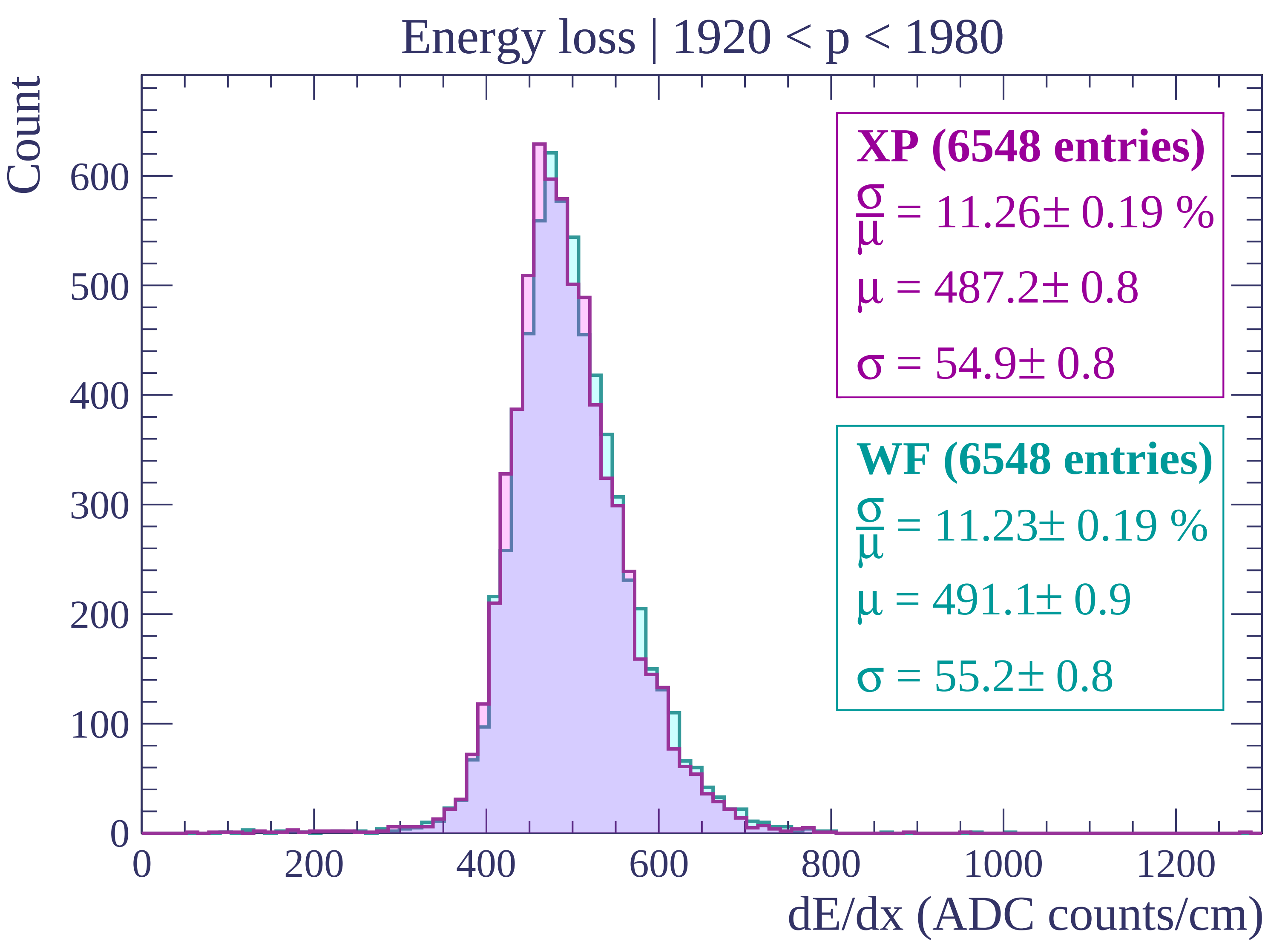
Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
XP (7289 entries)
σ/μ = 11.24± 0.17 %
μ = 481.1± 0.8
σ = 54.1± 0.7
WF (7289 entries)
σ/μ = 11.38± 0.16 %
μ = 485.8± 0.8
σ = 55.3± 0.7
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200



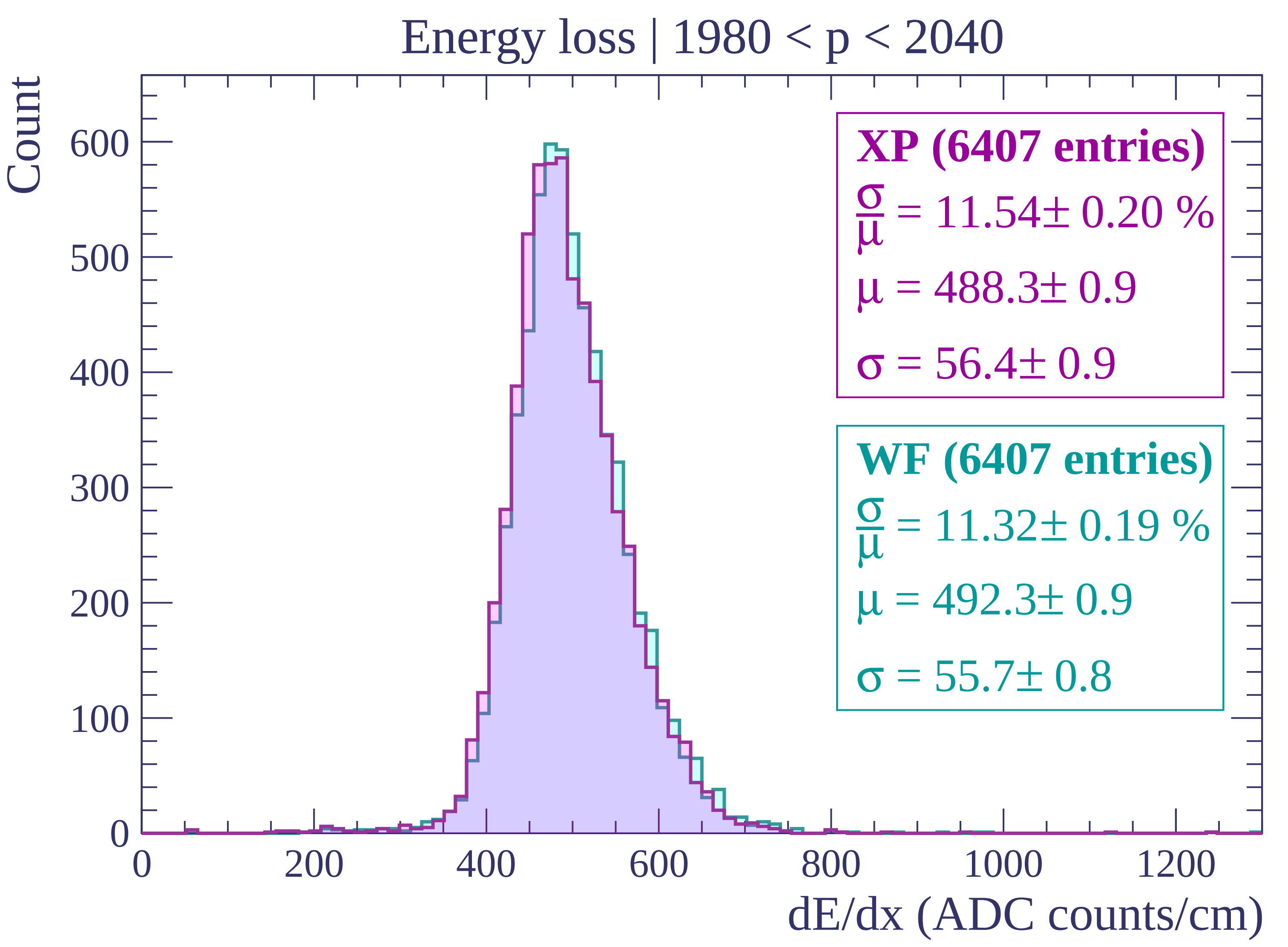
Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (7049 entries)
$\frac{\sigma}{\mu}$ = 11.50± 0.20 %
μ = 483.0± 0.9
σ = 55.5± 0.9
WF (7049 entries)
$\frac{\sigma}{\mu}$ = 11.56± 0.19 %
μ = 487.6± 0.8
σ = 56.4± 0.8



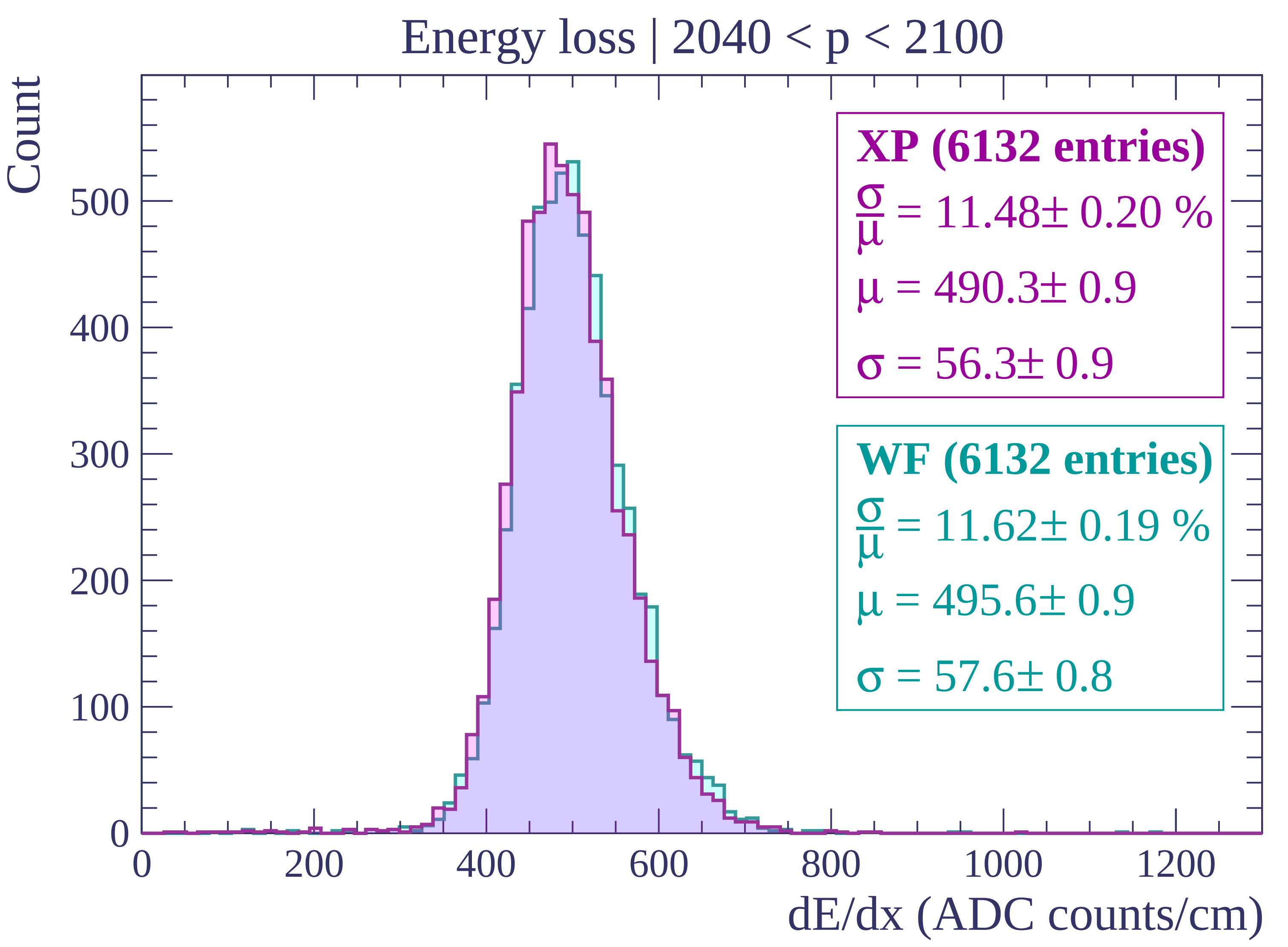
Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (7003 entries)
σ/μ = 11.11± 0.19 %
μ = 484.6± 0.8
σ = 53.8± 0.8
WF (7003 entries)
σ/μ = 11.31± 0.18 %
μ = 489.3± 0.8
σ = 55.4± 0.8

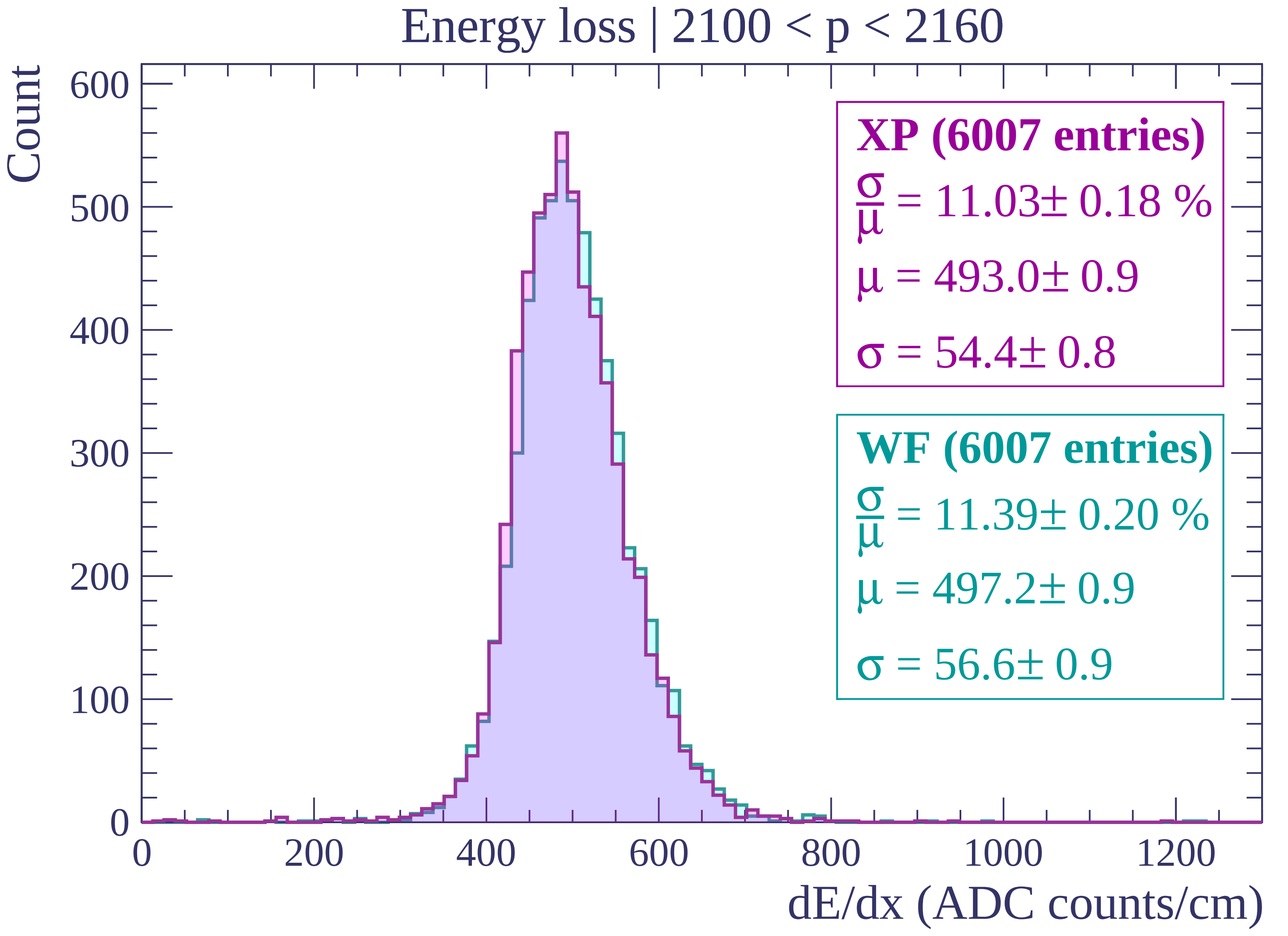


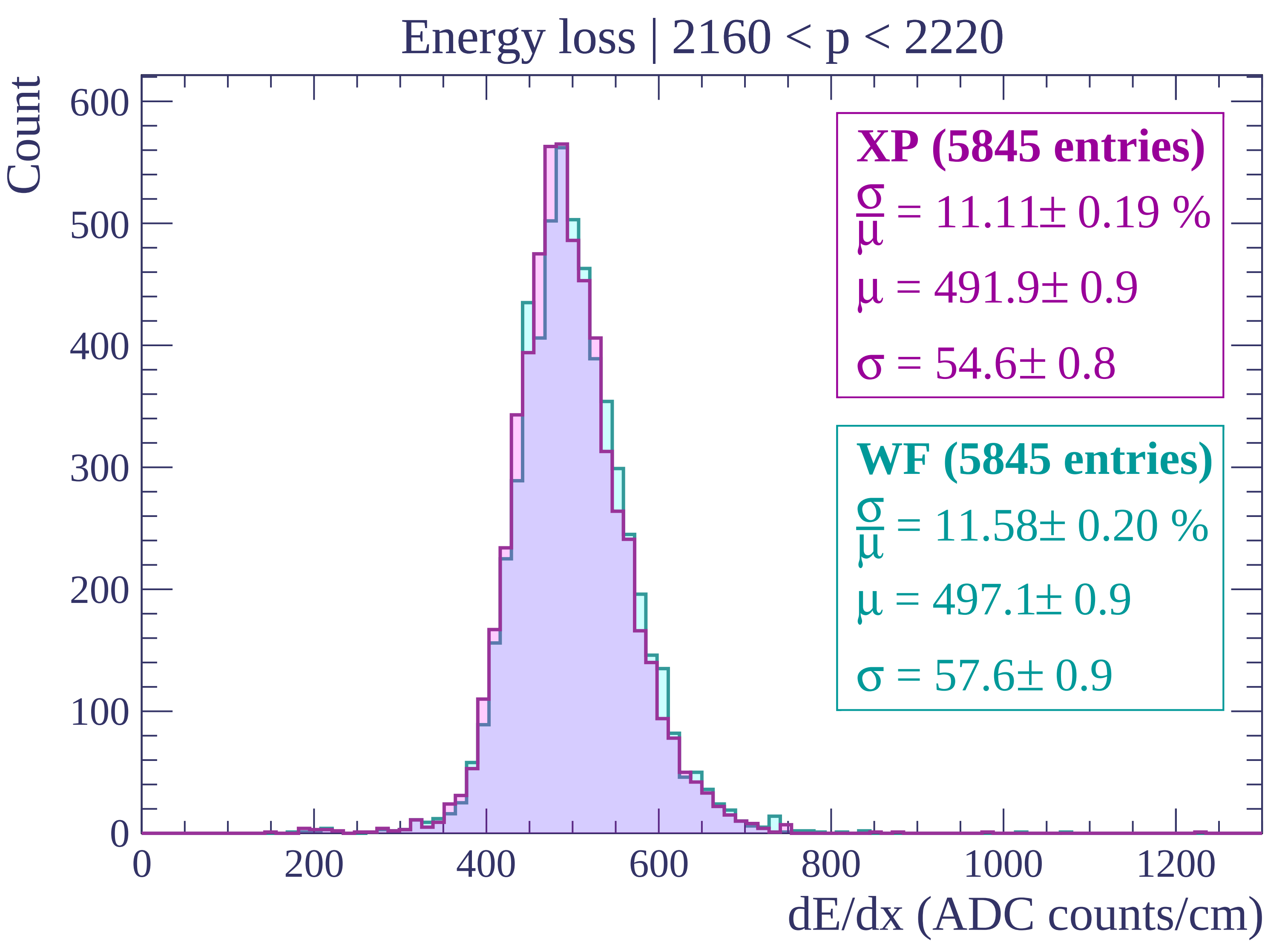
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (6548 entries)
σ/μ = 11.26± 0.19 %
μ = 487.2± 0.8
σ = 54.9± 0.8
WF (6548 entries)
σ/μ = 11.23± 0.19 %
μ = 491.1± 0.9
σ = 55.2± 0.8



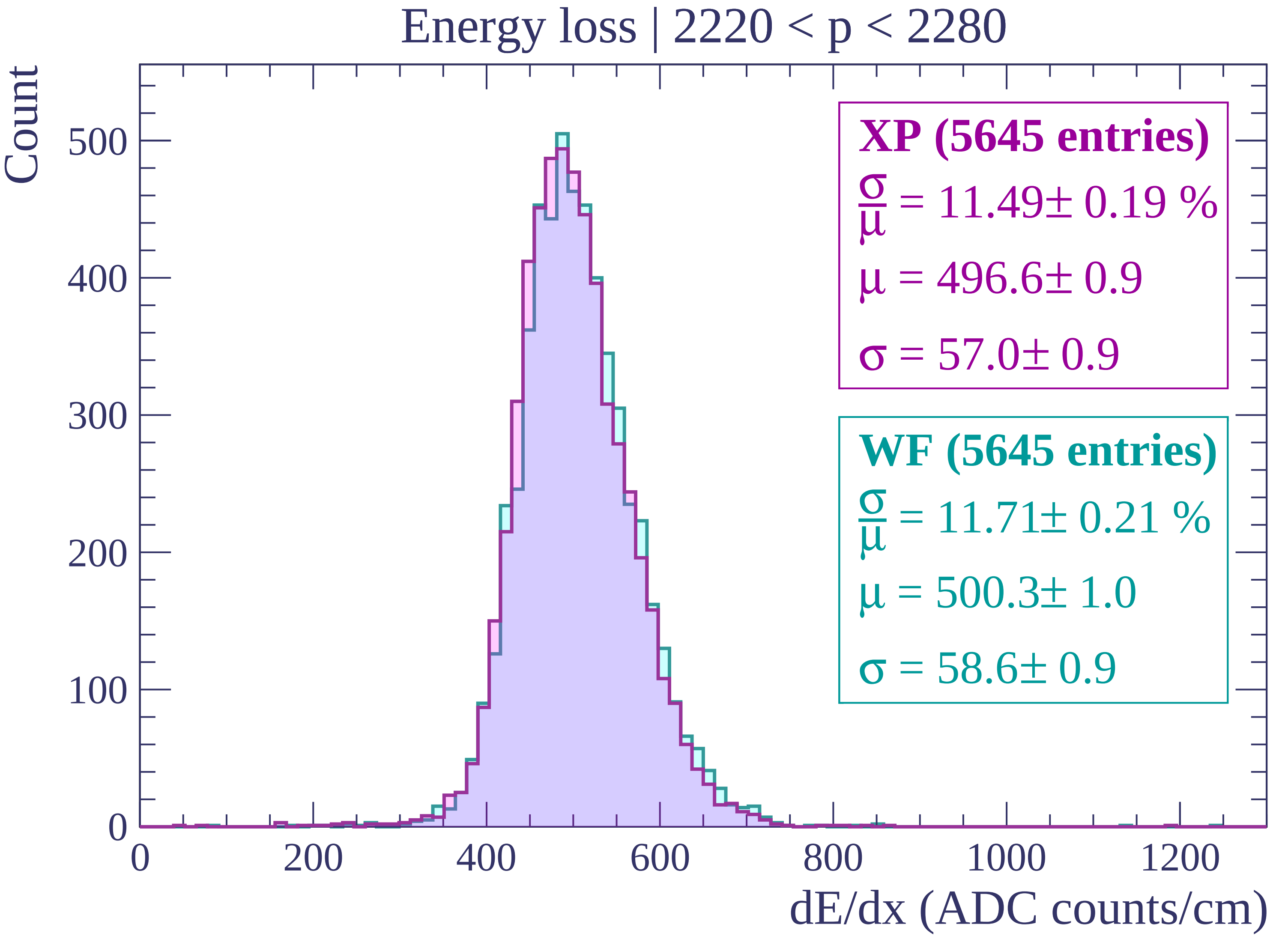
Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
XP (6407 entries)
$\frac{\sigma}{\mu}$ = 11.54± 0.20 %
μ = 488.3± 0.9
σ = 56.4± 0.9
WF (6407 entries)
$\frac{\sigma}{\mu}$ = 11.32± 0.19 %
μ = 492.3± 0.9
σ = 55.7± 0.8


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (6132 entries)
σ/μ = 11.48± 0.20 %
μ = 490.3± 0.9
σ = 56.3± 0.9
WF (6132 entries)
σ/μ = 11.62± 0.19 %
μ = 495.6± 0.9
σ = 57.6± 0.8

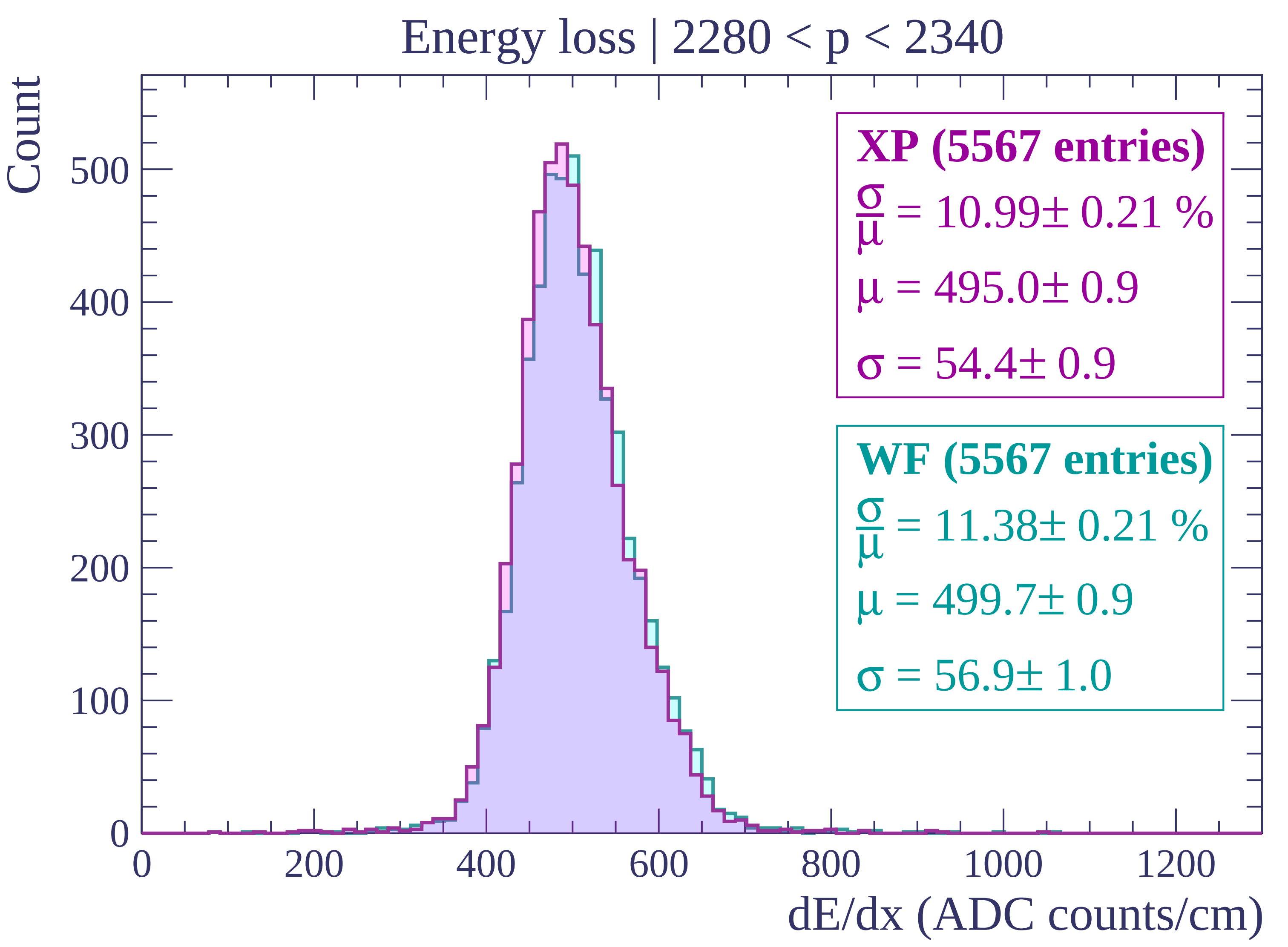

Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (6007 entries)
σ/μ = 11.03± 0.18 %
μ = 493.0± 0.9
σ = 54.4± 0.8
WF (6007 entries)
σ/μ = 11.39± 0.20 %
μ = 497.2± 0.9
σ = 56.6± 0.9



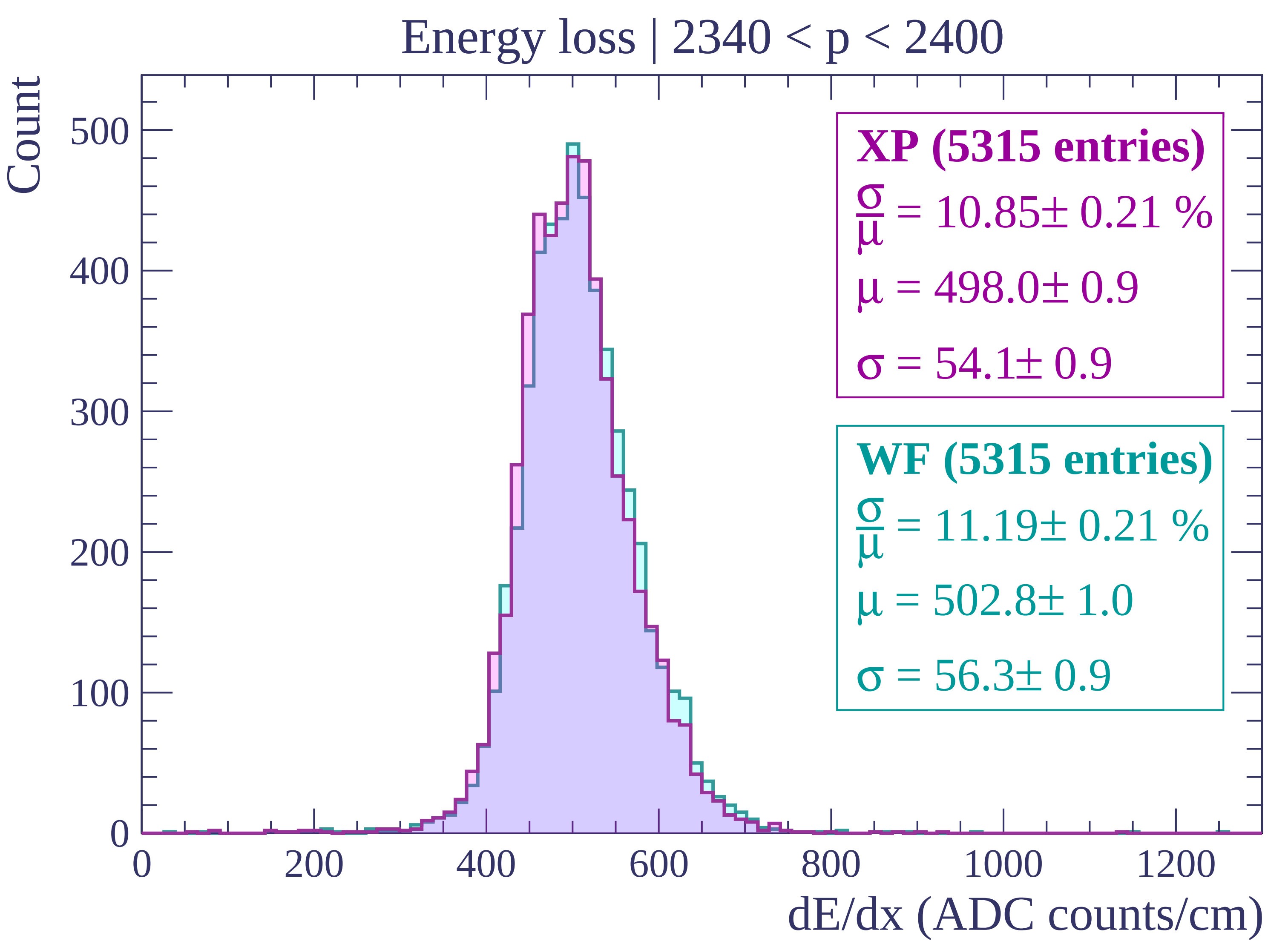
Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5845 entries)
σ/μ = 11.11± 0.19 %
μ = 491.9± 0.9
σ = 54.6± 0.8
WF (5845 entries)
σ/μ = 11.58± 0.20 %
μ = 497.1± 0.9
σ = 57.6± 0.9



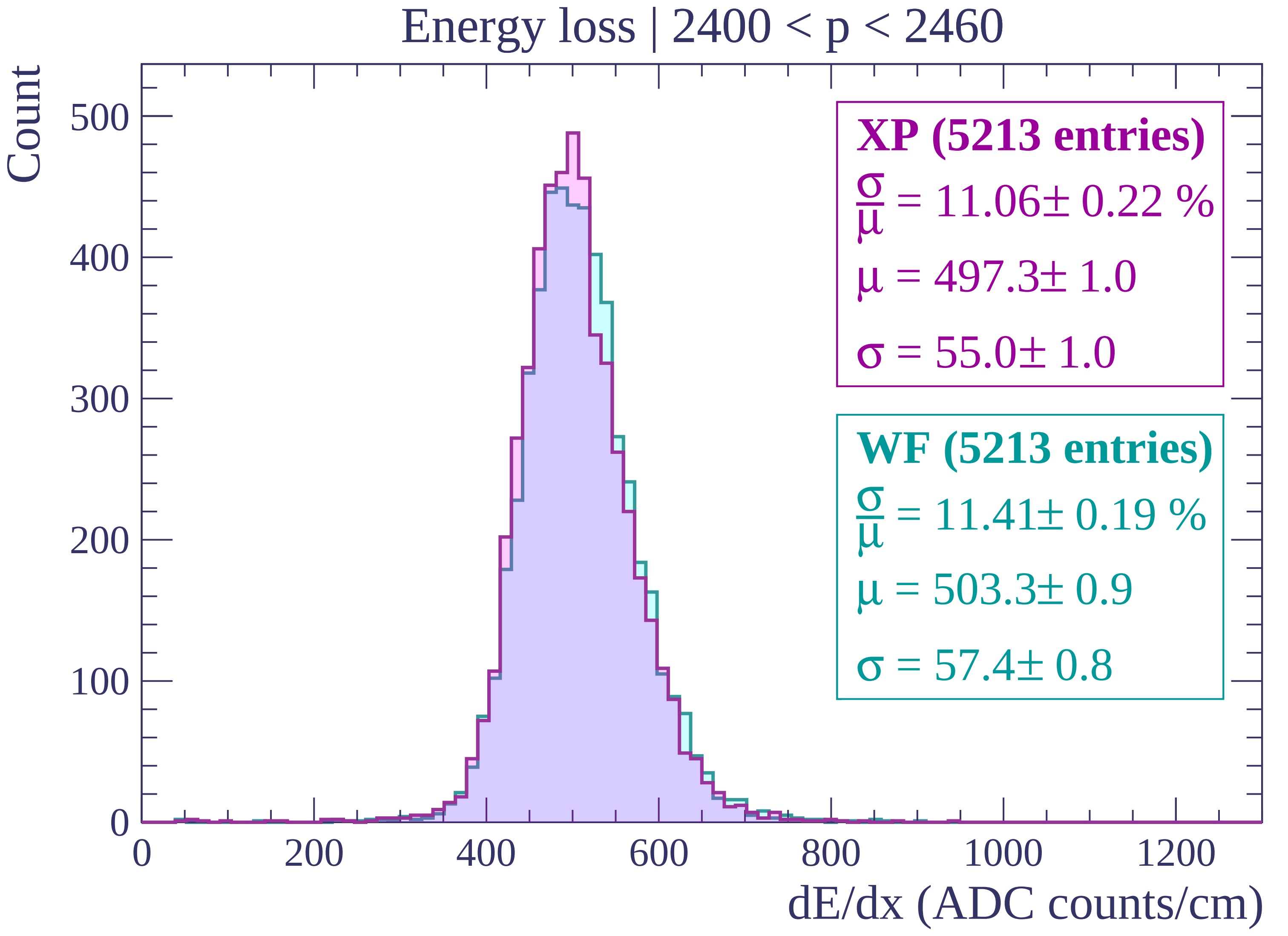
Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5645 entries)
σ/μ = 11.49± 0.19 %
μ = 496.6± 0.9
σ = 57.0± 0.9
WF (5645 entries)
σ/μ = 11.71± 0.21 %
μ = 500.3± 1.0
σ = 58.6± 0.9



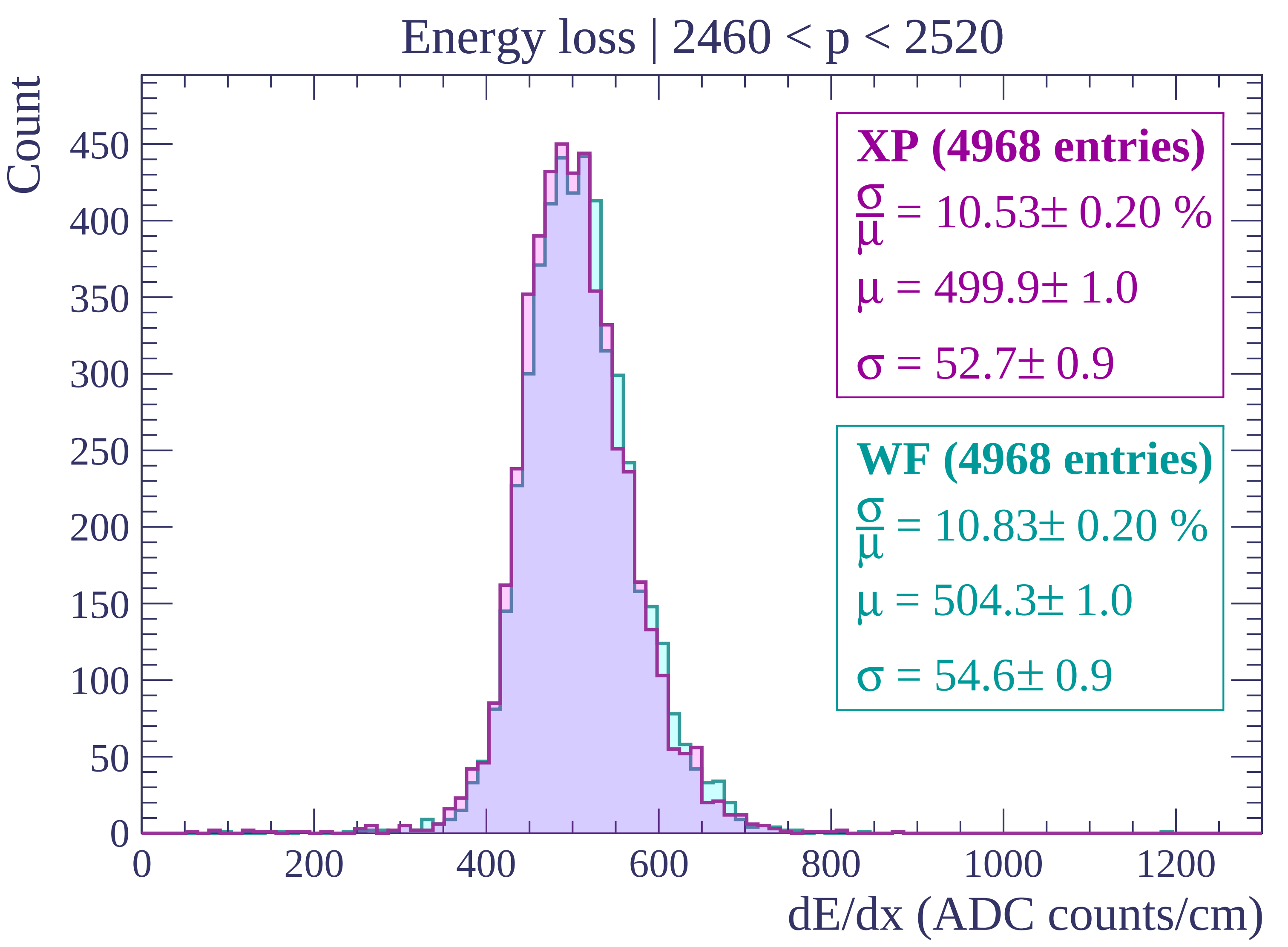
Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5567 entries)
σ/μ = 10.99± 0.21 %
μ = 495.0± 0.9
σ = 54.4± 0.9
WF (5567 entries)
σ/μ = 11.38± 0.21 %
μ = 499.7± 0.9
σ = 56.9± 1.0



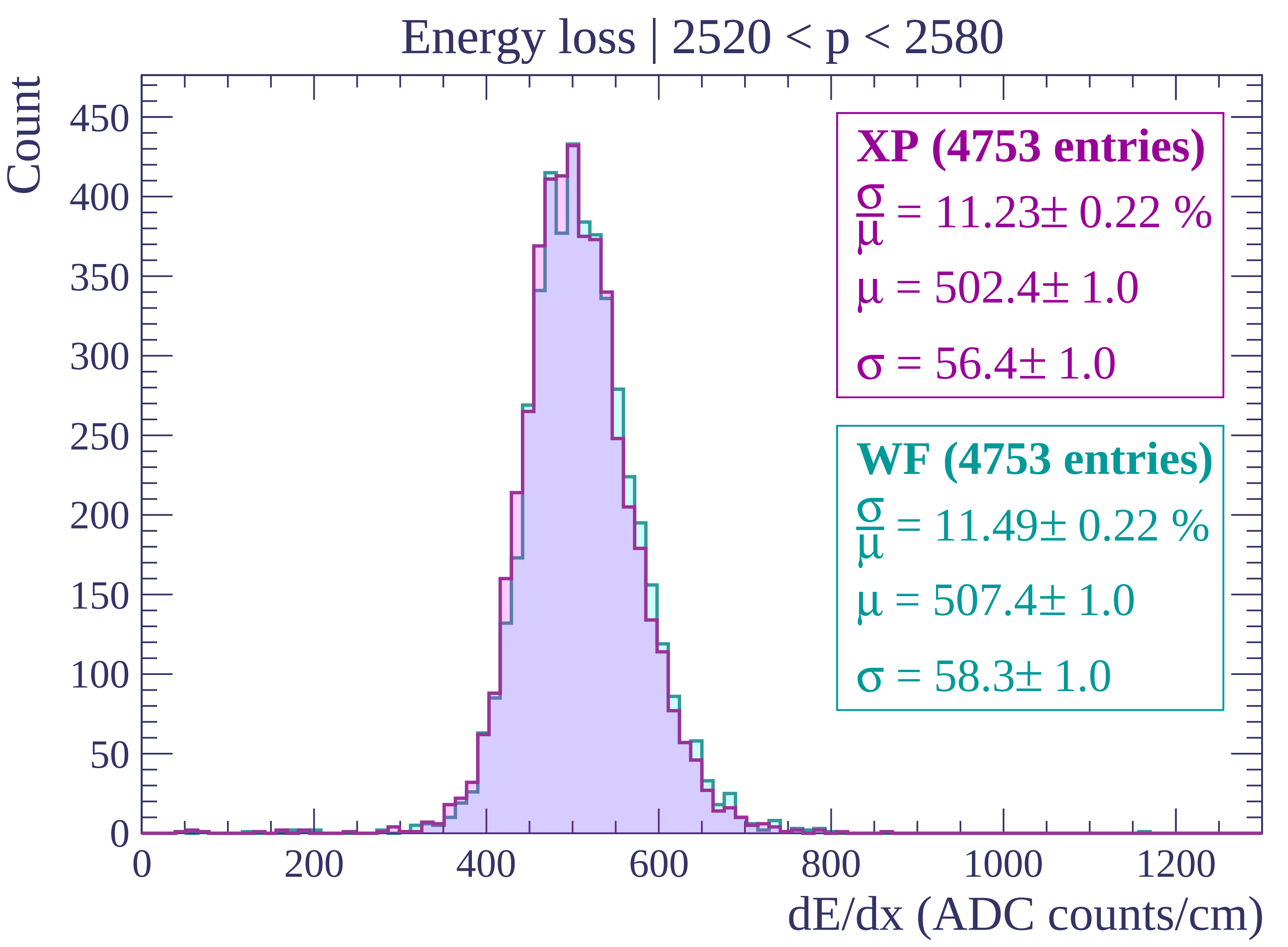
Energy loss | 2340 < p < 2400
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5315 entries)
σ/μ = 10.85± 0.21 %
μ = 498.0± 0.9
σ = 54.1± 0.9
WF (5315 entries)
σ/μ = 11.19± 0.21 %
μ = 502.8± 1.0
σ = 56.3± 0.9

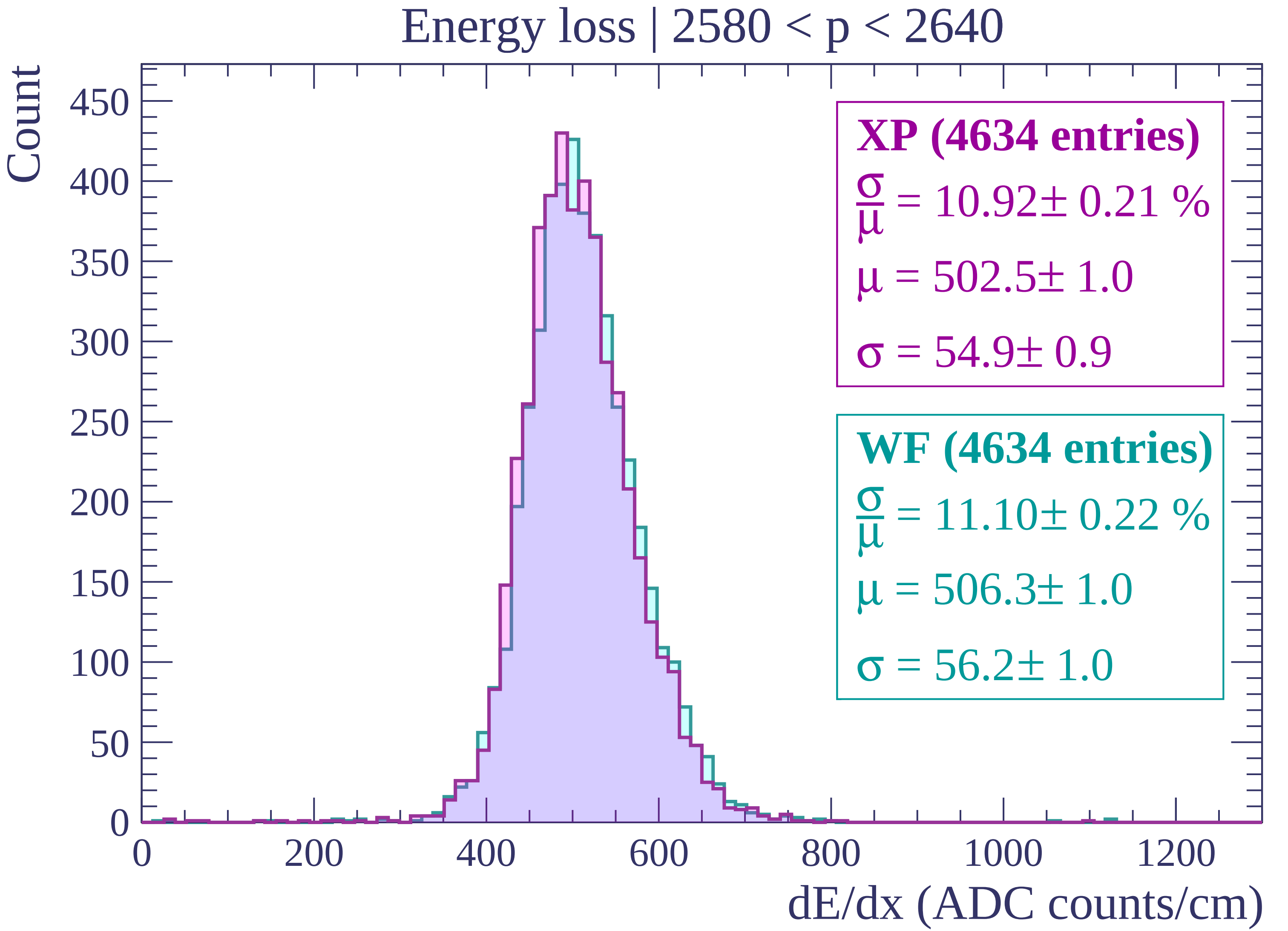


Energy loss | 2400 < p < 2460
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5213 entries)
σ/μ = 11.06± 0.22 %
μ = 497.3± 1.0
σ = 55.0± 1.0
WF (5213 entries)
σ/μ = 11.41± 0.19 %
μ = 503.3± 0.9
σ = 57.4± 0.8

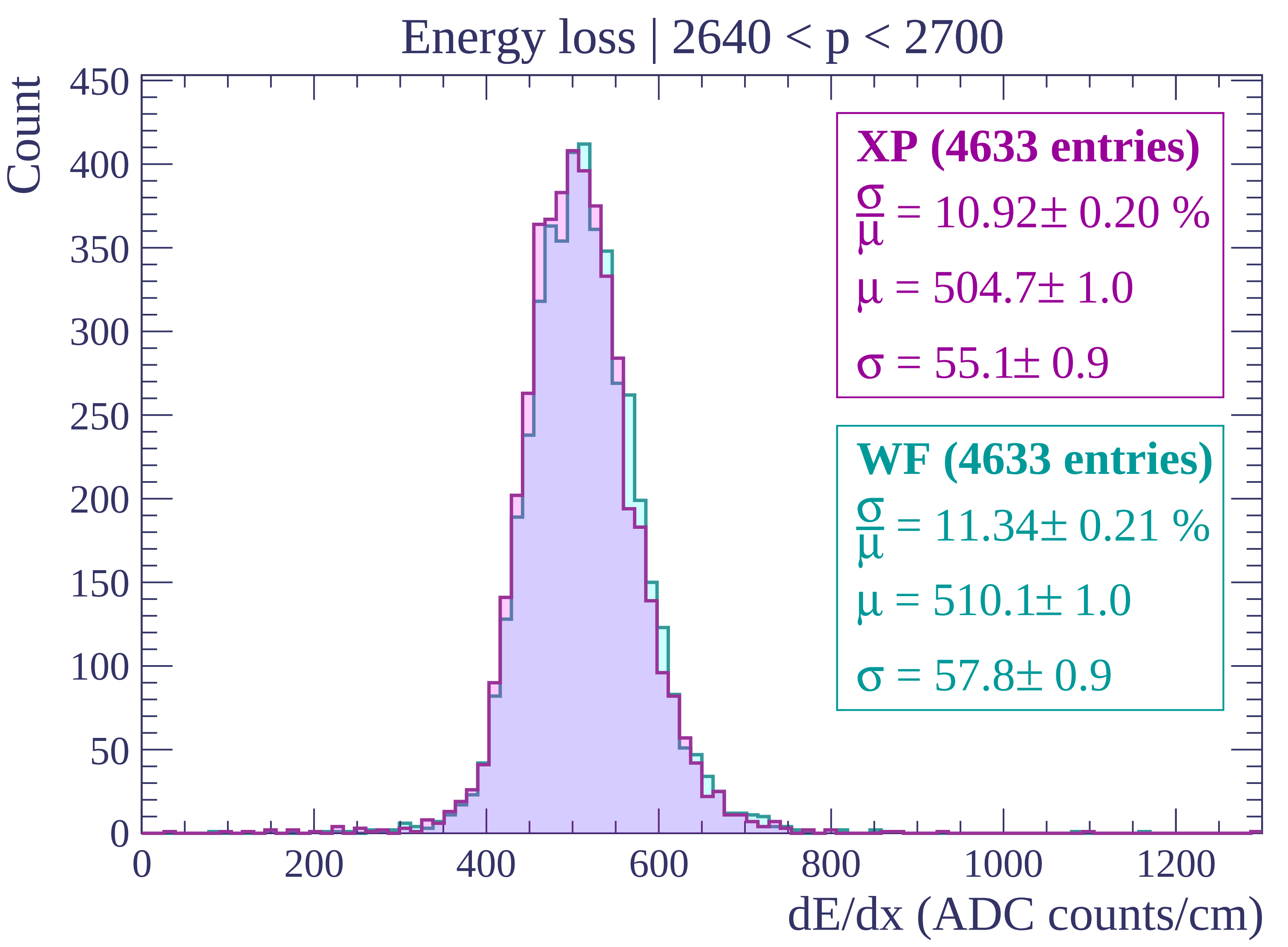


Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
XP (4968 entries)
σ/μ = 10.53± 0.20 %
μ = 499.9± 1.0
σ = 52.7± 0.9
WF (4968 entries)
σ/μ = 10.83± 0.20 %
μ = 504.3± 1.0
σ = 54.6± 0.9
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

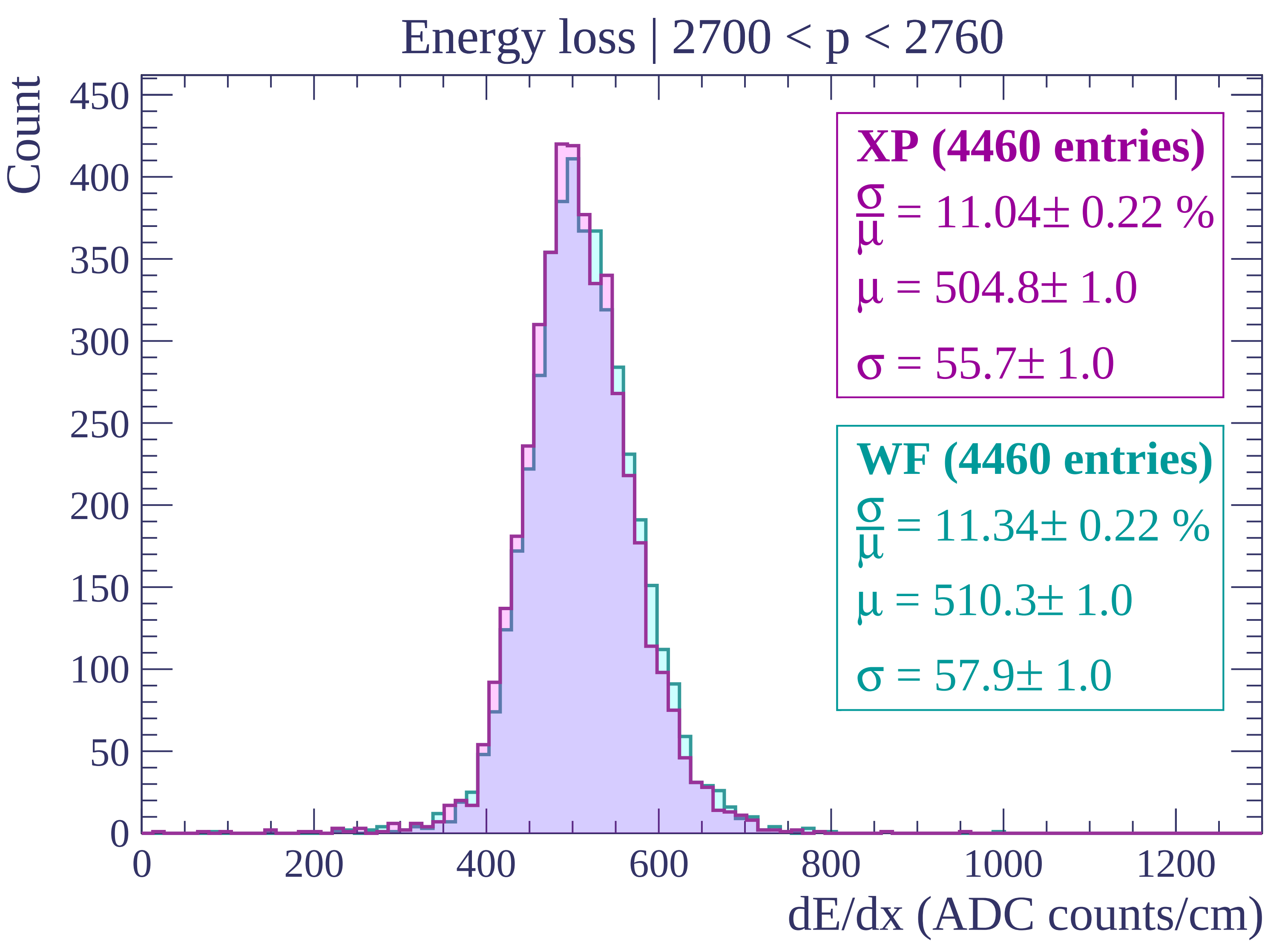

Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4753 entries)
σ/μ = 11.23± 0.22 %
μ = 502.4± 1.0
σ = 56.4± 1.0
WF (4753 entries)
σ/μ = 11.49± 0.22 %
μ = 507.4± 1.0
σ = 58.3± 1.0



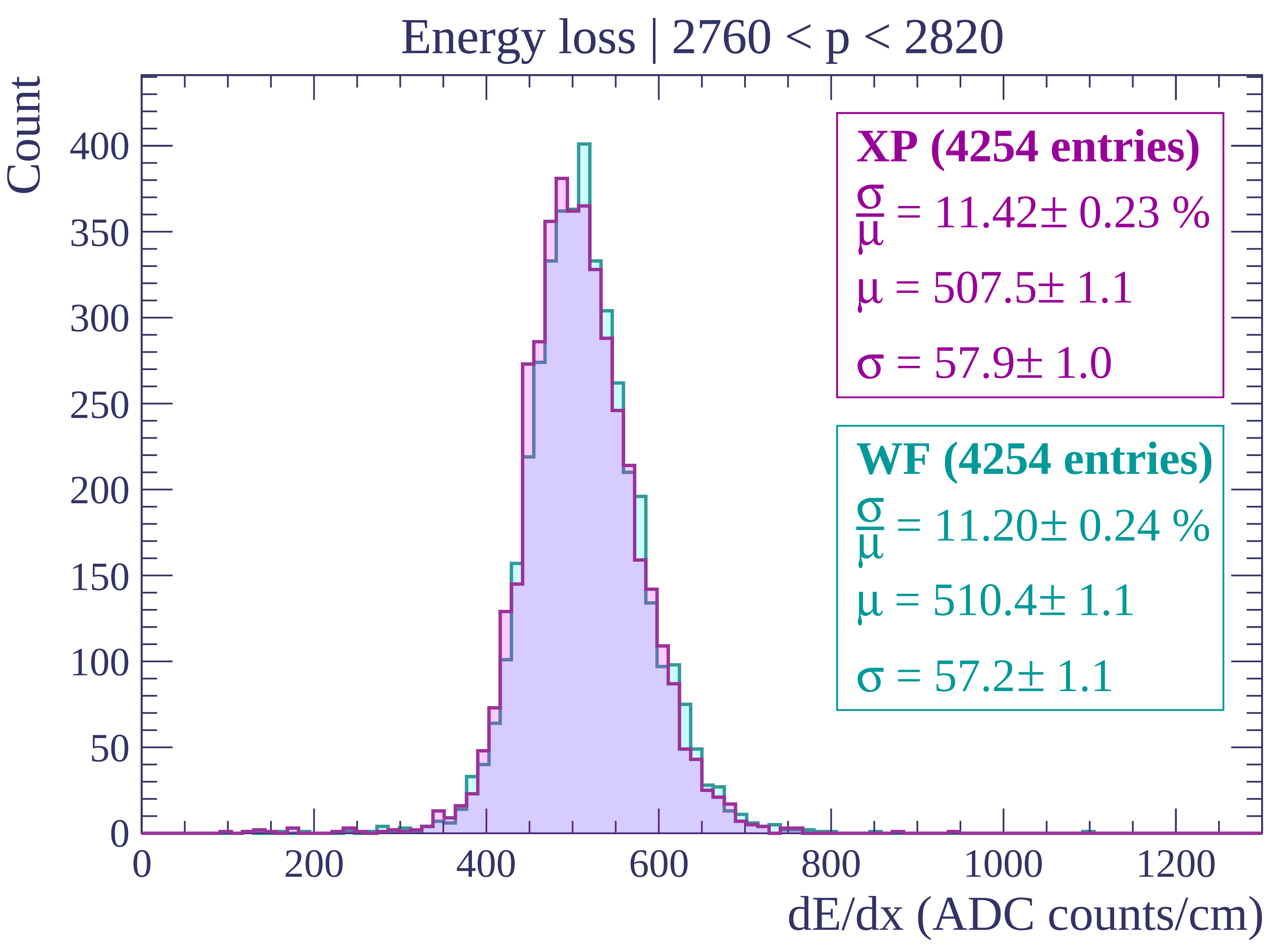
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4634 entries)
$\frac{\sigma}{\mu}$ = 10.92± 0.21 %
μ = 502.5± 1.0
σ = 54.9± 0.9
WF (4634 entries)
$\frac{\sigma}{\mu}$ = 11.10± 0.22 %
μ = 506.3± 1.0
σ = 56.2± 1.0



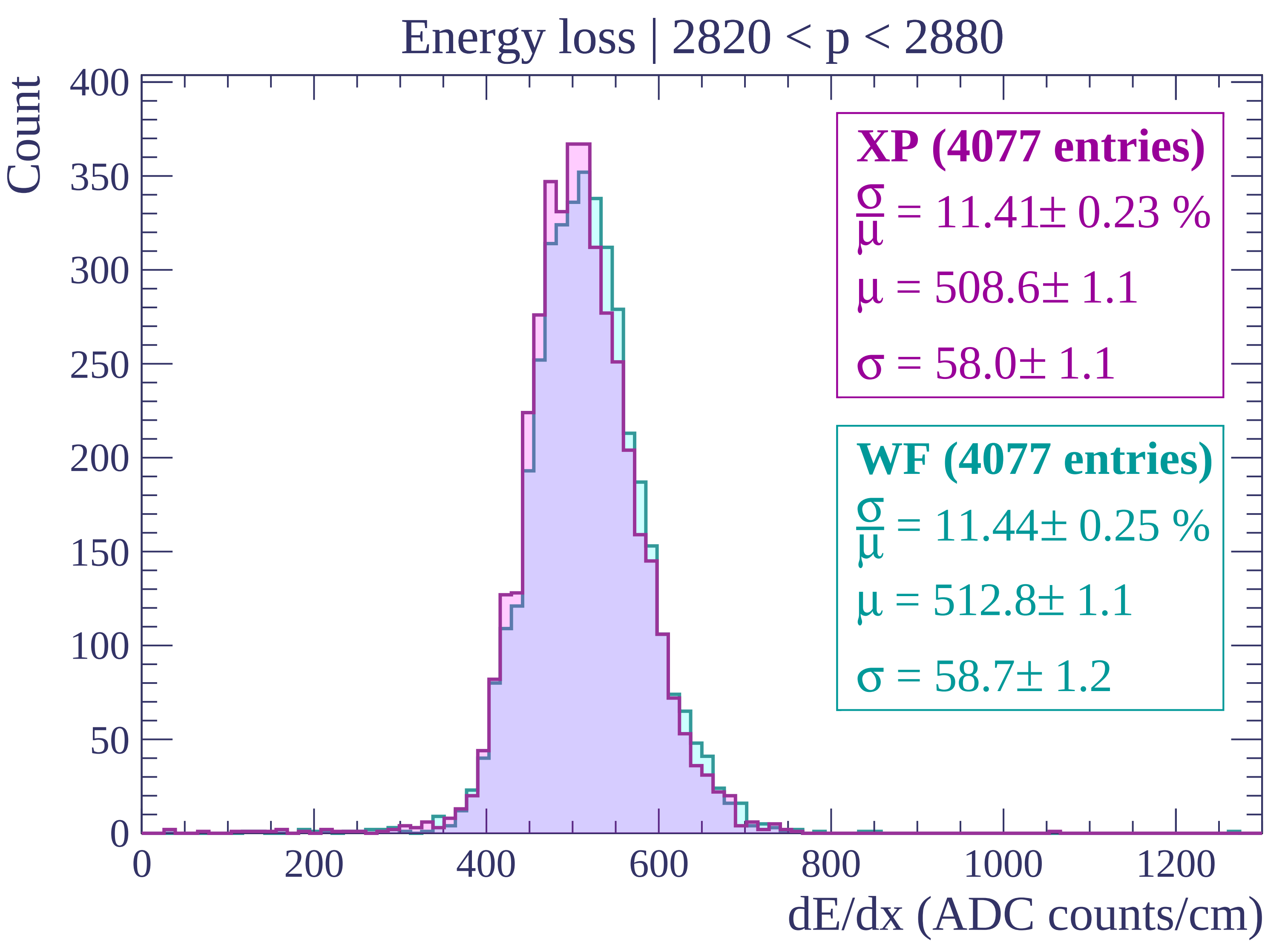
Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP (4633 entries)
σ/μ = 10.92± 0.20 %
μ = 504.7± 1.0
σ = 55.1± 0.9
WF (4633 entries)
σ/μ = 11.34± 0.21 %
μ = 510.1± 1.0
σ = 57.8± 0.9

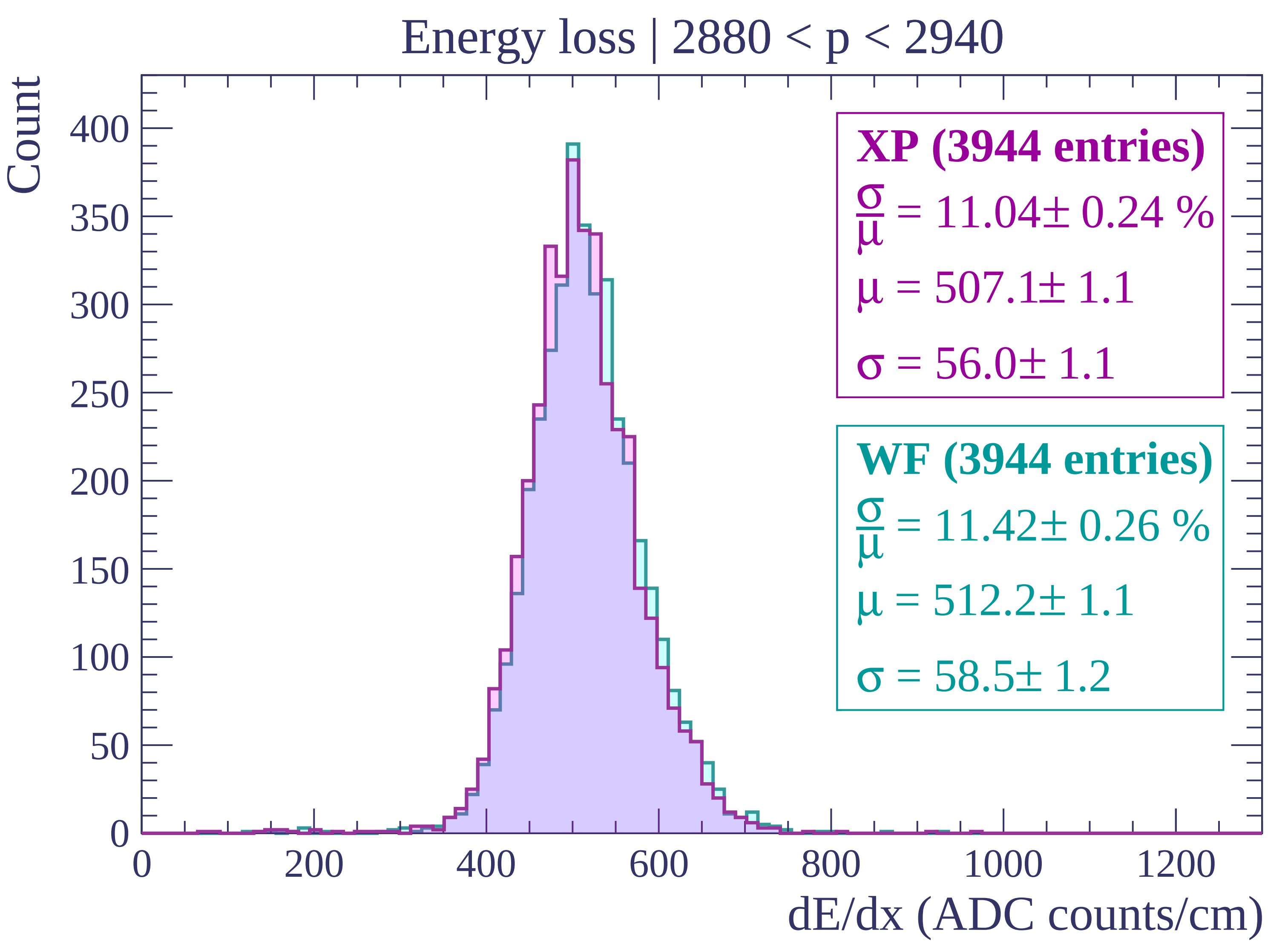


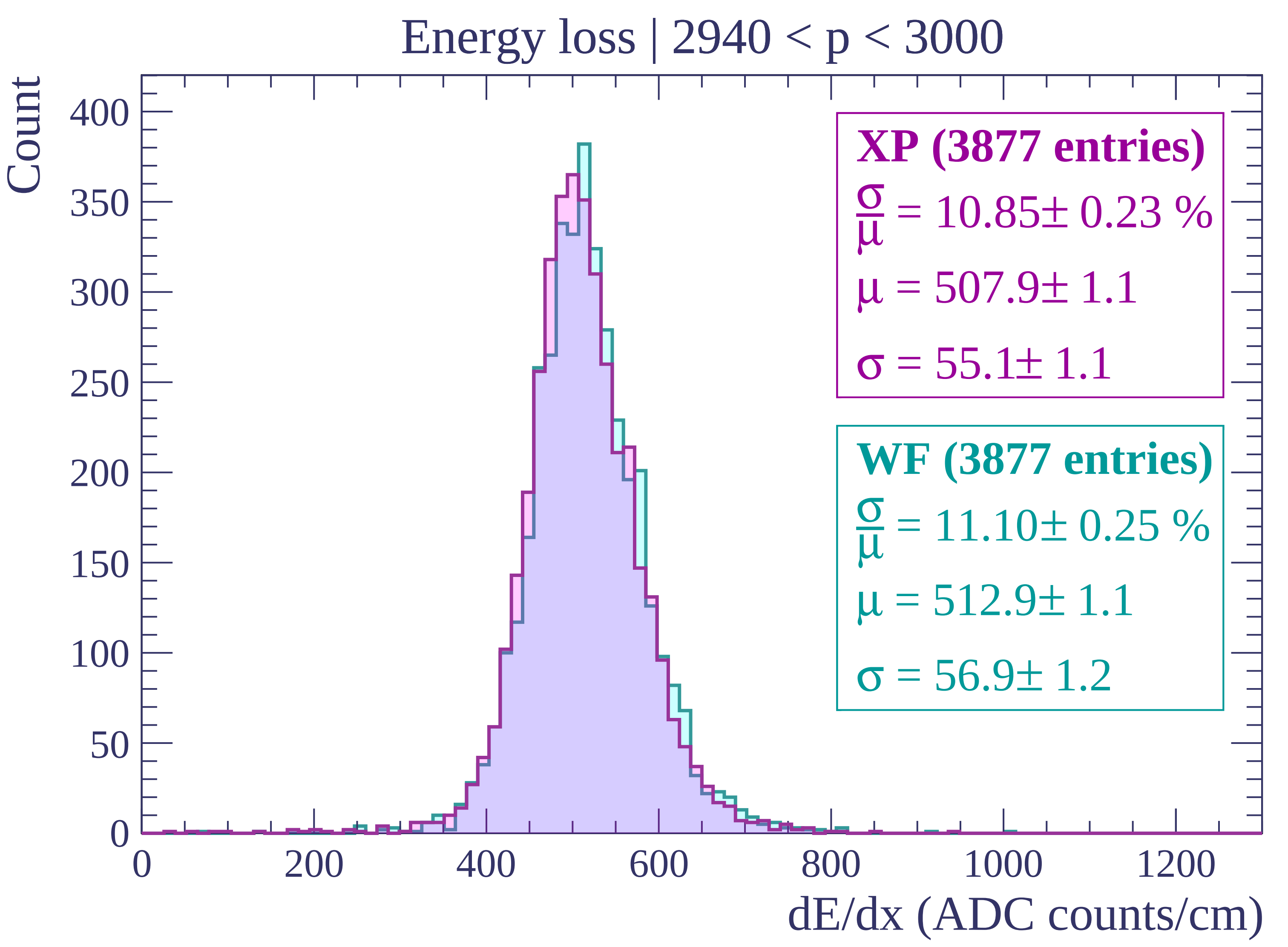
Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
XP (4460 entries)
$\frac{\sigma}{\mu}$ = 11.04± 0.22 %
μ = 504.8± 1.0
σ = 55.7± 1.0
WF (4460 entries)
$\frac{\sigma}{\mu}$ = 11.34± 0.22 %
μ = 510.3± 1.0
σ = 57.9± 1.0
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



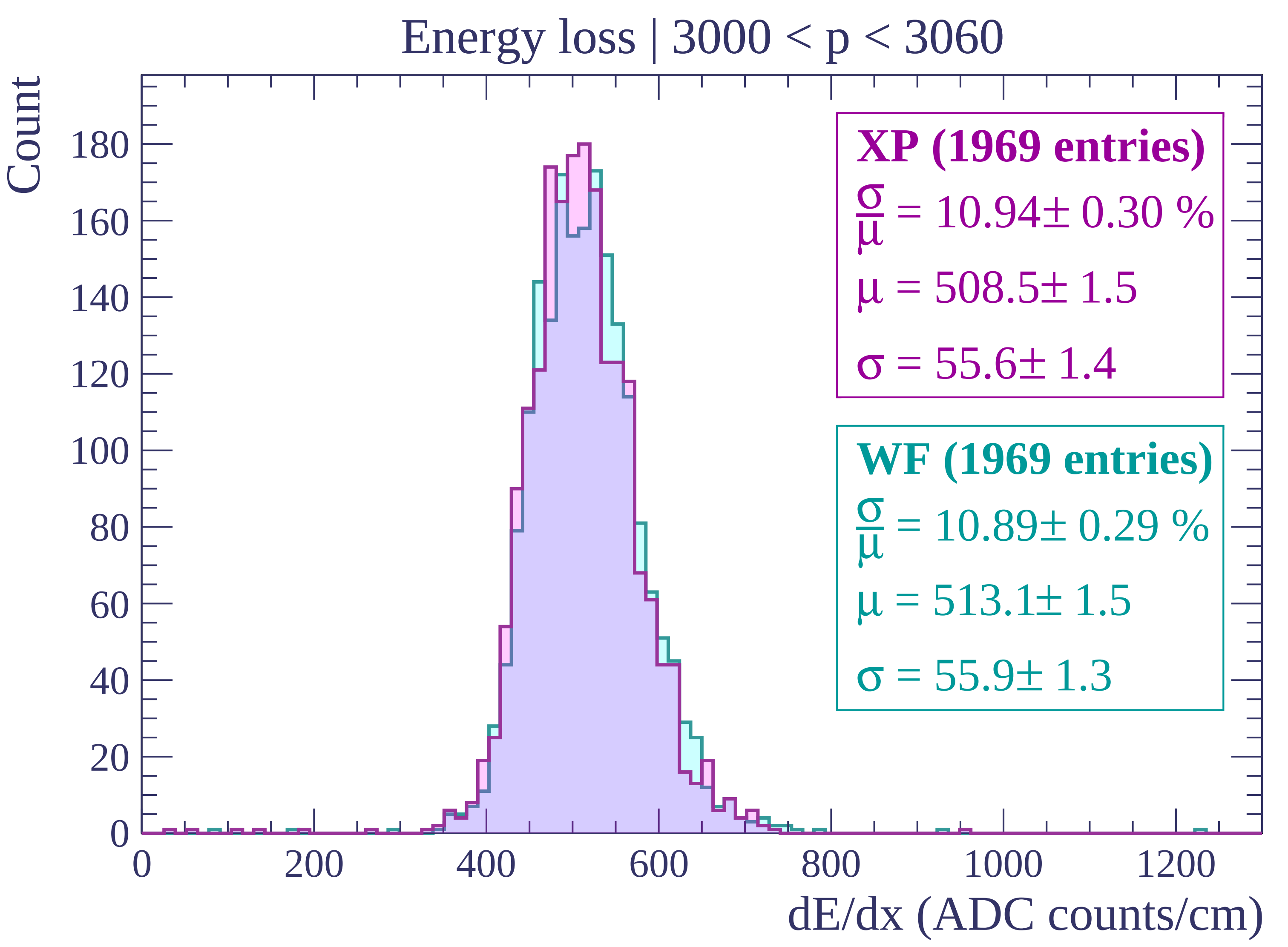
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4254 entries)
$\frac{\sigma}{\mu}$ = 11.42± 0.23 %
μ = 507.5± 1.1
σ = 57.9± 1.0
WF (4254 entries)
$\frac{\sigma}{\mu}$ = 11.20± 0.24 %
μ = 510.4± 1.1
σ = 57.2± 1.1


Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4077 entries)
$\frac{\sigma}{\mu}$ = 11.41± 0.23 %
μ = 508.6± 1.1
σ = 58.0± 1.1
WF (4077 entries)
$\frac{\sigma}{\mu}$ = 11.44± 0.25 %
μ = 512.8± 1.1
σ = 58.7± 1.2

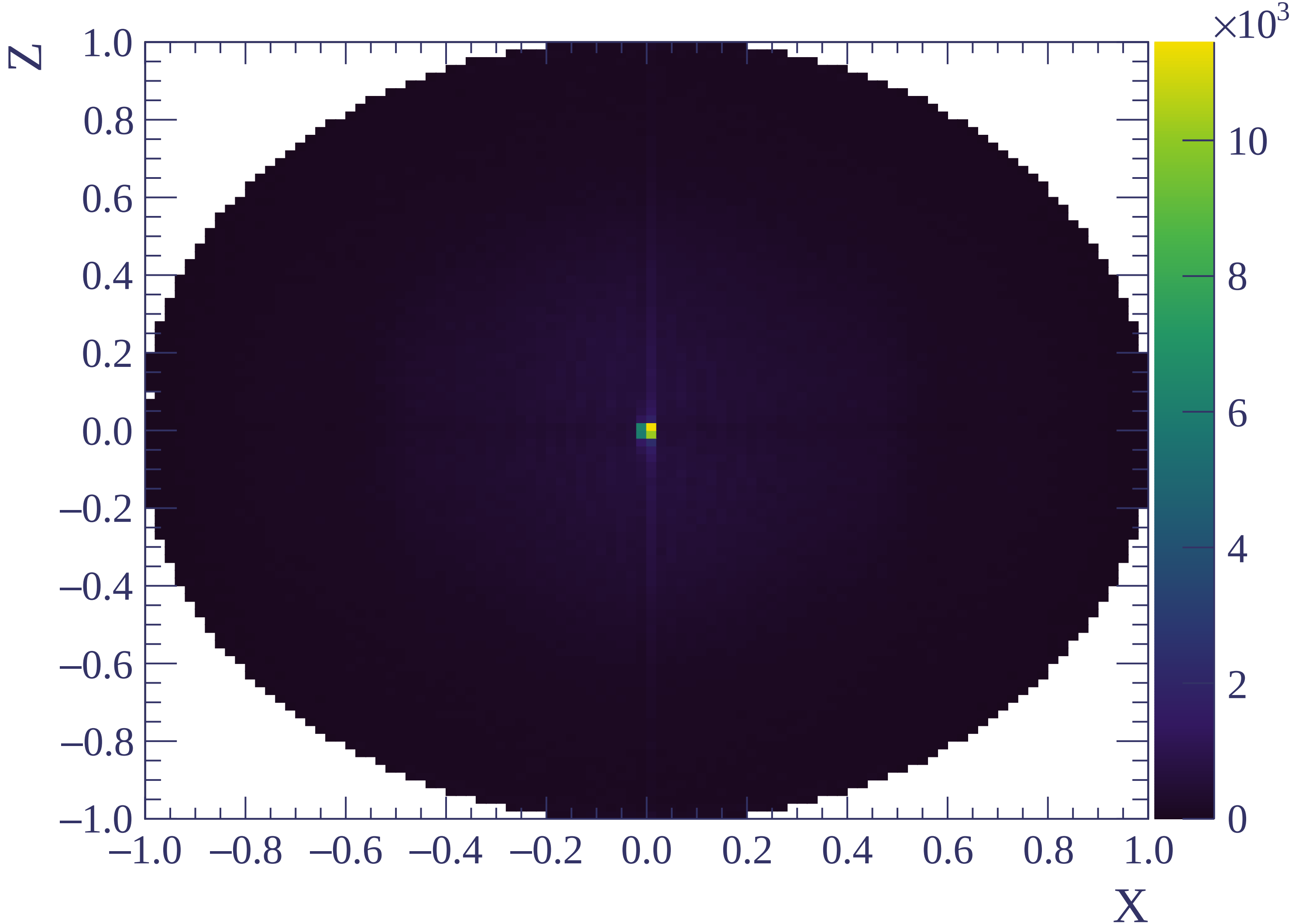

Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
XP (3944 entries)
σ/μ = 11.04± 0.24 %
μ = 507.1± 1.1
σ = 56.0± 1.1
WF (3944 entries)
σ/μ = 11.42± 0.26 %
μ = 512.2± 1.1
σ = 58.5± 1.2
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



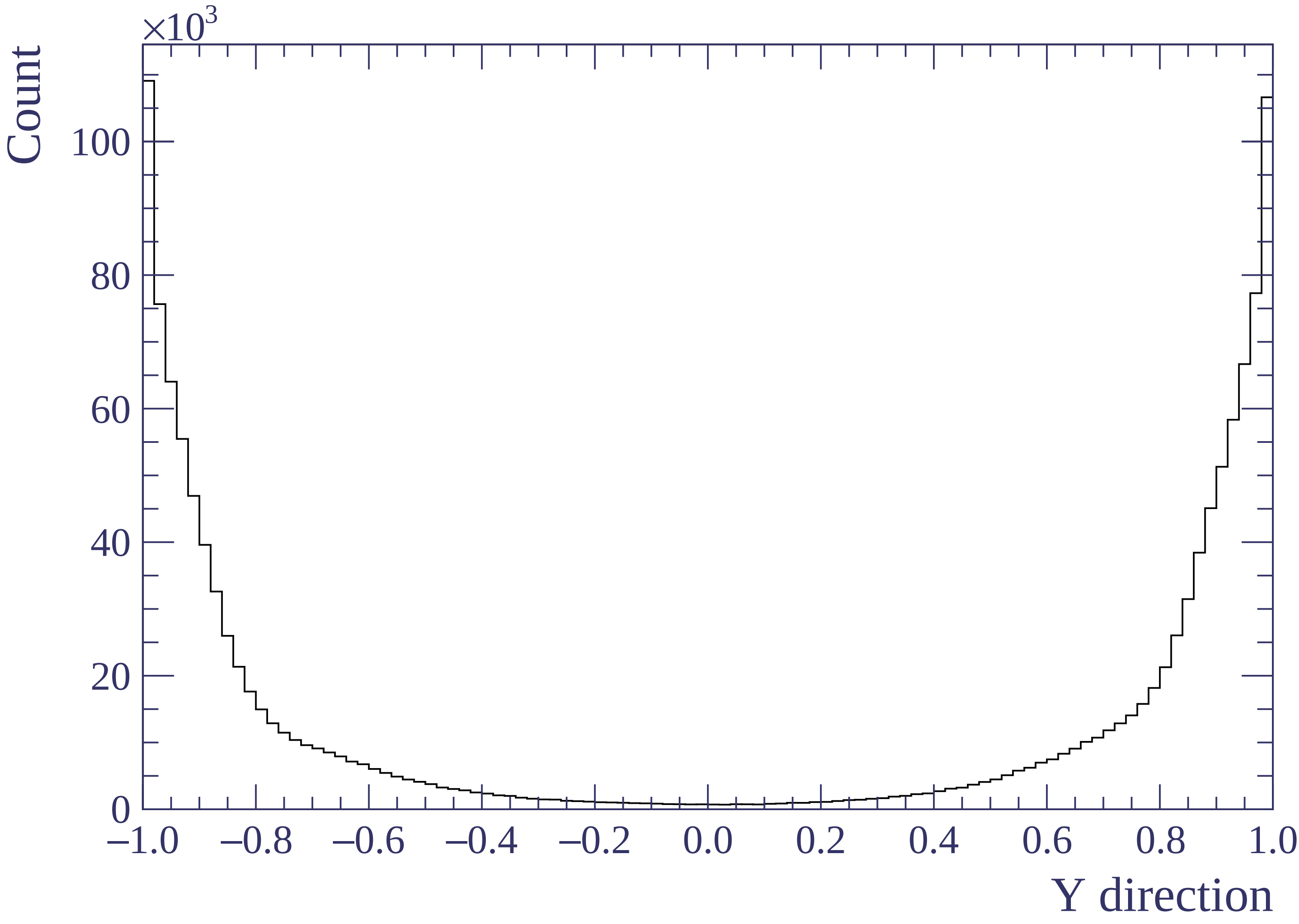
Energy loss | 2940 < p < 3000
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3877 entries)
σ/μ = 10.85± 0.23 %
μ = 507.9± 1.1
σ = 55.1± 1.1
WF (3877 entries)
σ/μ = 11.10± 0.25 %
μ = 512.9± 1.1
σ = 56.9± 1.2



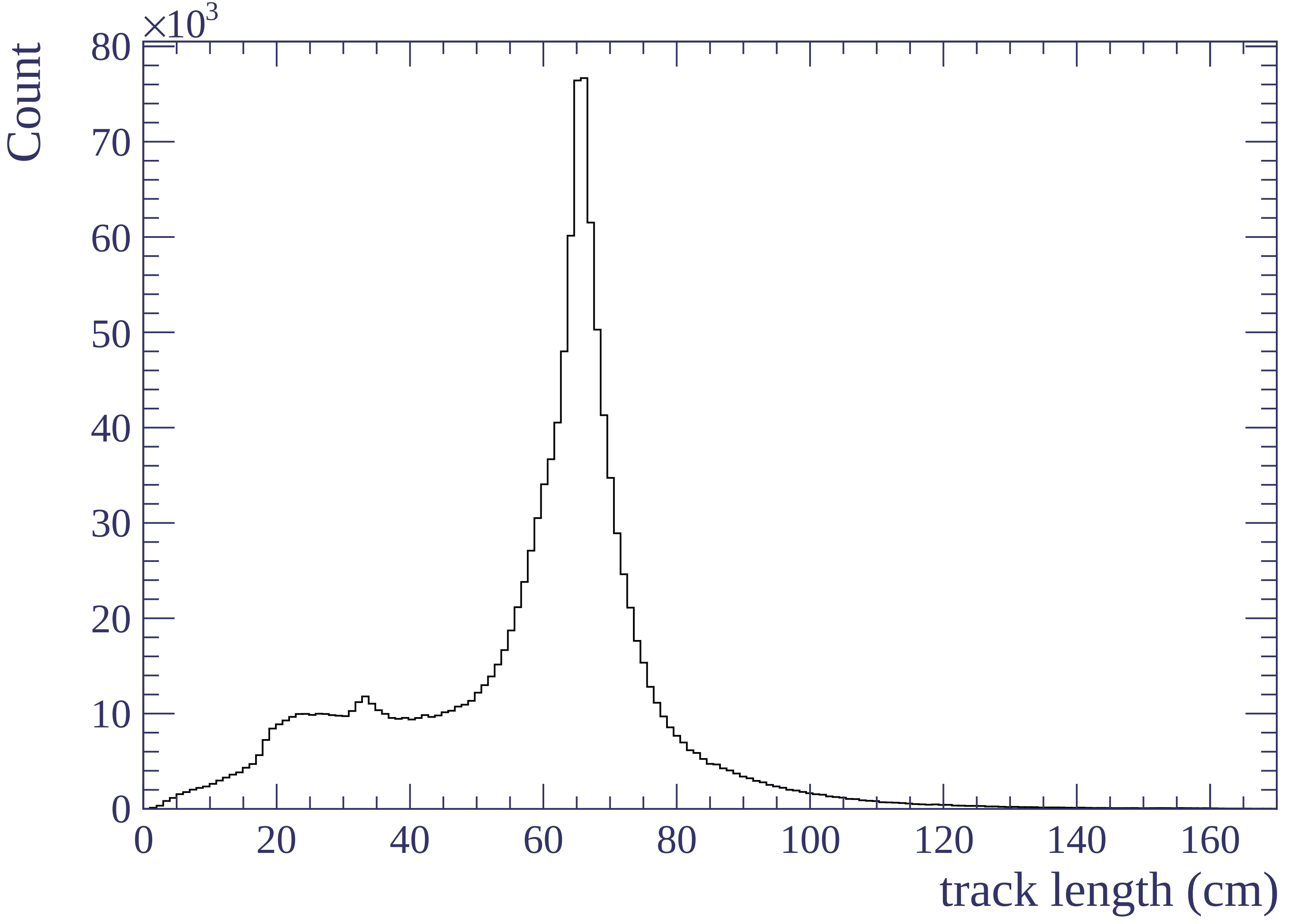
Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1969 entries)
σ/μ = 10.94± 0.30 %
μ = 508.5± 1.5
σ = 55.6± 1.4
WF (1969 entries)
σ/μ = 10.89± 0.29 %
μ = 513.1± 1.5
σ = 55.9± 1.3



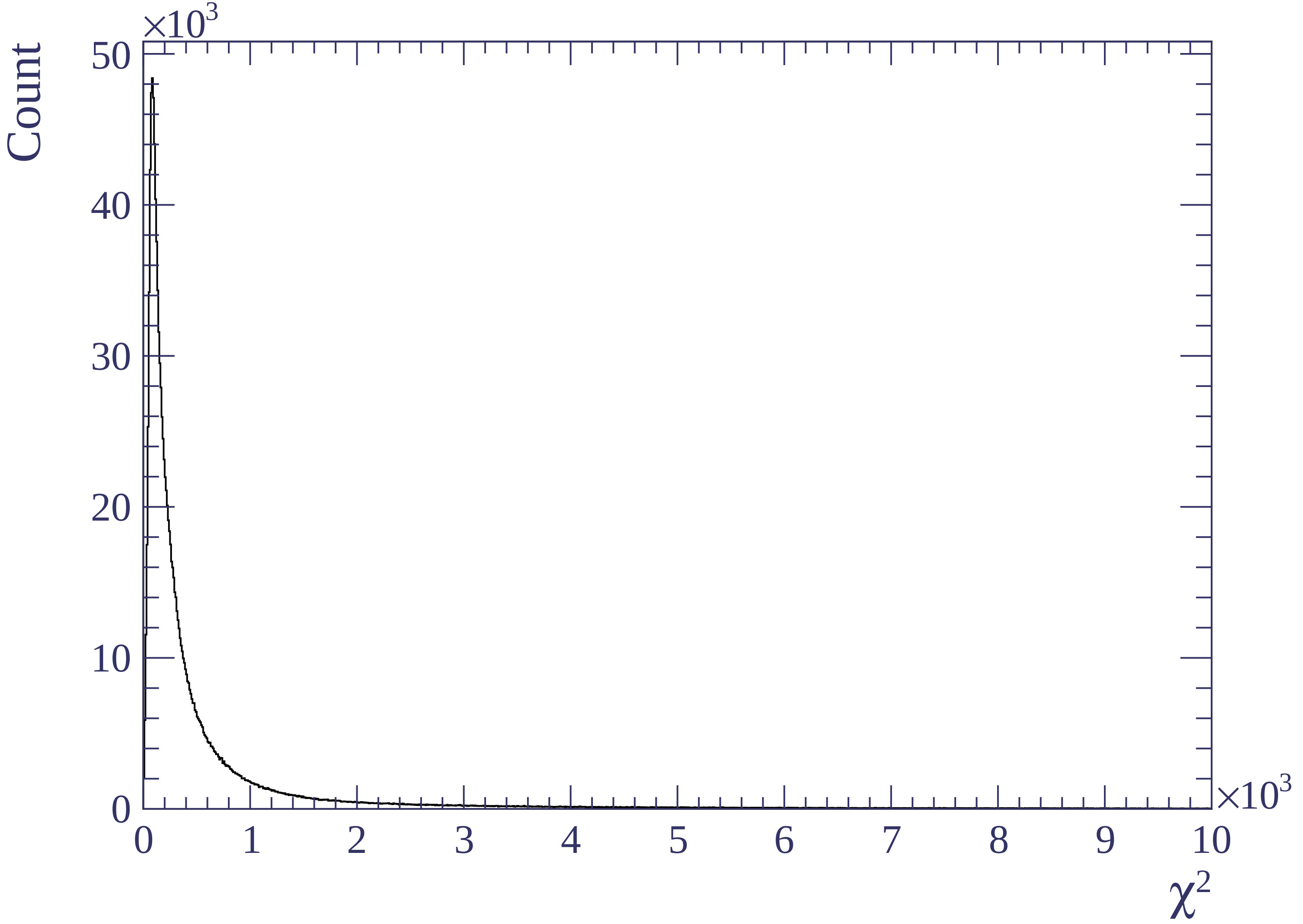
Z
X
×10³
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
10
8
6
4
2
0



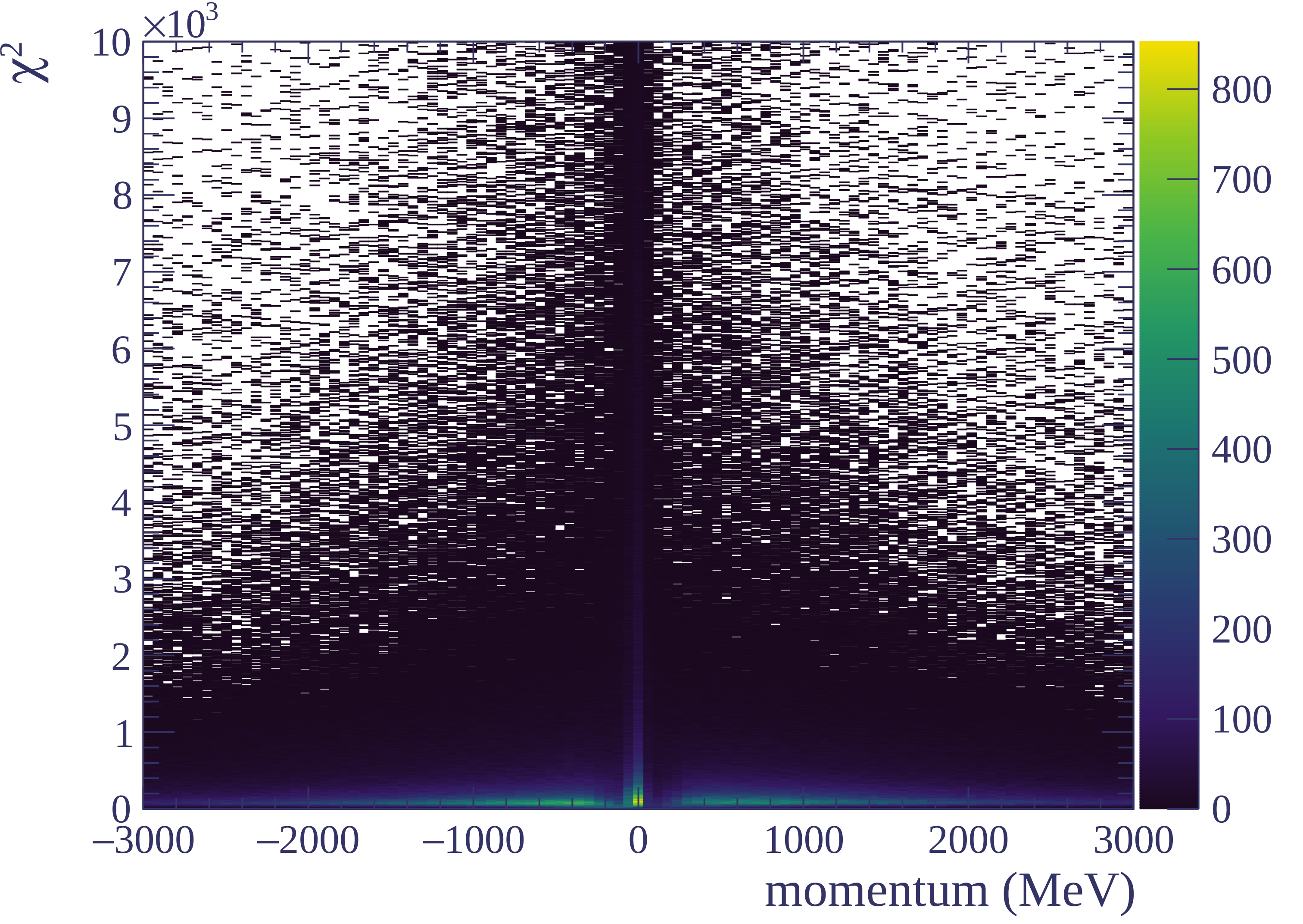
$\times 10^3$
Count
100
80
60
40
20
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



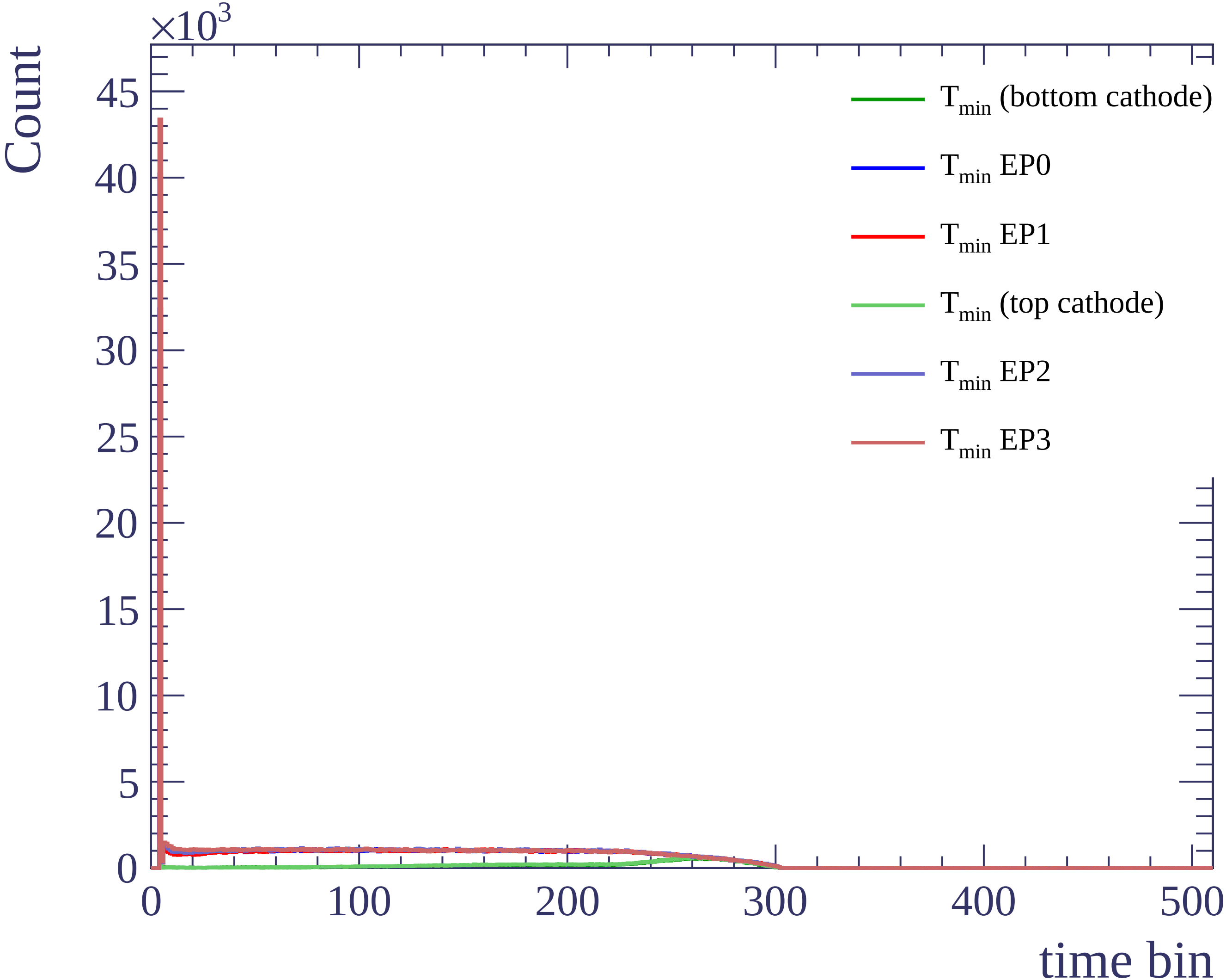
×10³
Count
80
70
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
track length (cm)



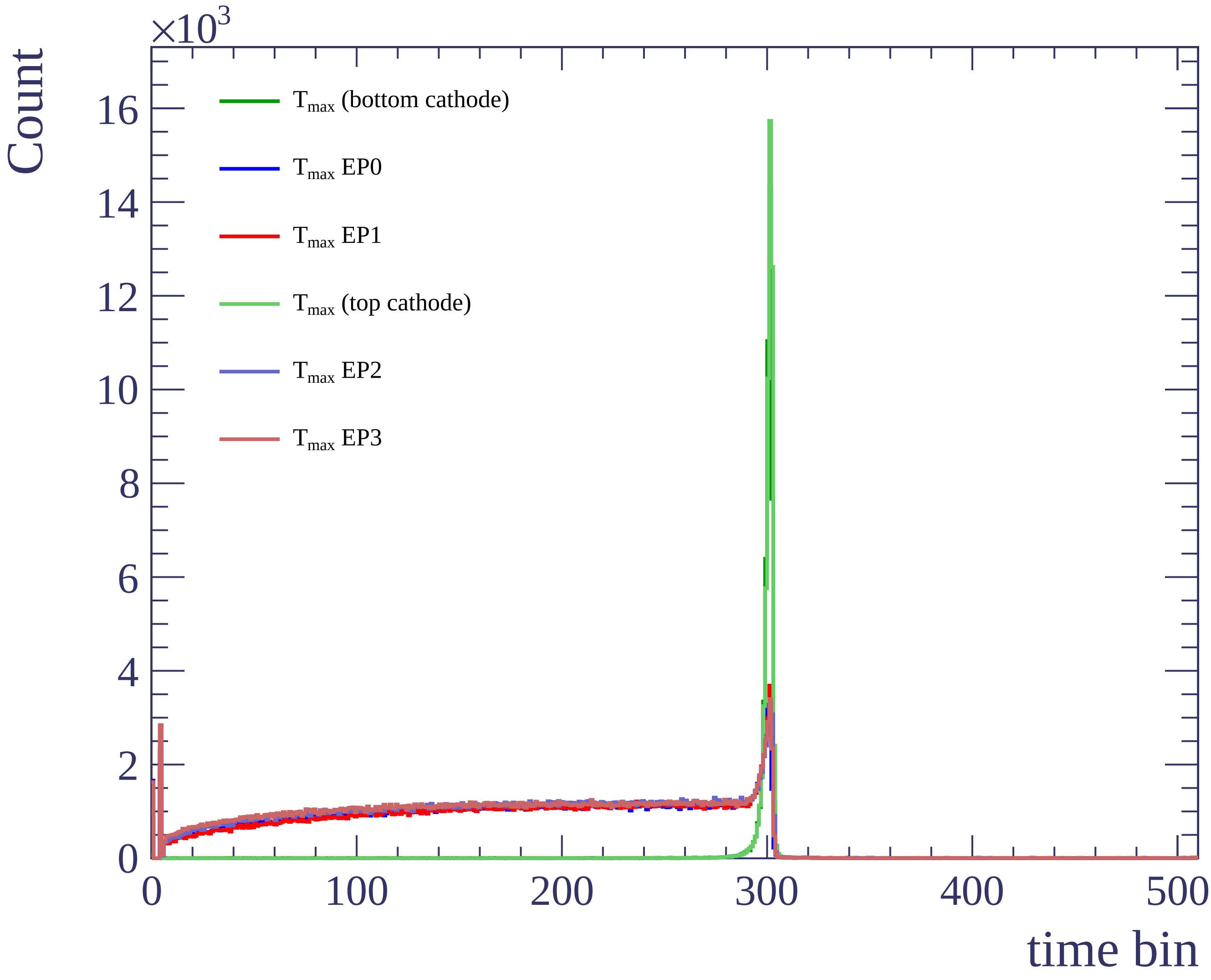
Count
×10^3
50
40
30
20
10
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
800
700
600
500
400
300
200
100
0



$\times 10^3$
Count
45
40
35
30
25
20
15
10
5
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



$\times 10^3$
Count
16
14
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3